July 2024 (solved)

wrong option More Resources Correct Answer

1. Mode of action of

-Lamotrigine: a)Acts on voltage gated sodium channel , b)NMDA Modulation, c) which stabilizes neuronal membranes

-Pregabalin: Acts on voltage gated Calcium channel alpha 2 delta subunit

-Psilocybin: 5HT2A agonist(see April 2024) question 17

-Atomoxetine: Atomoxetine is a highly selective norepinephrine reuptake inhibitor (SNRI), which binds to the NE reuptake transporter on the presynaptic membrane of the nerve and inhibits NE reuptake.

-Bupropion : NDRI, Bupropion Norepinephrine-dopamine reuptake inhibitor (NDRI), and nicotinic acetylcholine receptor antagonist

2)-satiety center: ventromedial hypothalamus, feeding center to the lateral hypothalamus .

3.Legal to keep in UK -anabolic steroids Anabolic steroids

Anabolic steroids are synthetic derivatives of testosterone. They are used by some athletes to enhance performance, and by some men and women to improve physical appearance. They have anabolic (tissue-building) and androgenic (masculinising) properties. The androgenic effect cannot be separated from the anabolic, but purely anabolic steroids have been synthesised to minimise the androgenic effects.

Common anabolic steroids
Nandrolone Stanozolol Oxandrolone Trenbolone

n addition to muscle building properties, testosterone has the following effects:

- muscle protein metabolism
- sexual and cognitive functions
- erythropoiesis
- plasma lipids and bone metabolism

4)Which stages of Erikson

From Nasreen:

Erikson psychosocial development:

Basic conflict:

Trust will automatically initiate industry to identify intimacy generativity and integrity.

Trust vs mistrust: 0-1, Hope

Autonomy vs shame : 1-3, Will.

Initiative vs guilt: 3-6 , Purpose

Industry vs inferiority: 7-11, Competence.

Identify vs confusion : 12-18, Fidelity

Intimacy vs isolation: 19 - 29, Love

Generativity vs stagnation : 34- 64 , care

Integrity vs despair: 65 onward. Wisdom.

Virtue; mnemonic:

Hope will purposely come to find love care and wisdom

5)-Menstrual cycle which cycle

Infradian(more than 24 hours) biological period or cycle longer than 24 hrs .

Circadian within 24 hours.

The infradian cycle refers to biological rhythms that last longer than 24 hours, meaning their frequency is less than one cycle per day. These cycles can span days, weeks, or even months. They are distinct from circadian rhythms, which follow a 24-hour cycle. Some examples of infradian rhythms include:

1. Menstrual Cycle: This is the most well-known infradian rhythm in humans, typically lasting about 28 days, though it can vary between individuals.

2. Seasonal Affective Disorder (SAD): Changes in mood, energy levels, and even physiological functions during specific seasons, often in response to light availability, are linked to infradian rhythms.

3. Hibernation and Migration: In animals, many processes like hibernation in bears or migration in birds are governed by infradian rhythms, often tied to seasonal changes.

4. Lunar Rhythms: Some biological processes align with lunar cycles, such as reproductive cycles in certain marine species.

Infradian rhythms are important in regulating various biological processes, and disruptions to these cycles can have significant impacts on physical and mental health

6-loss of striatum- Huntingtons

7-Hexanucleotide repeat in which FTD

8-Drug with low bioavailability because of wide distribution – chlorpromazine

9-Drug with low bioavailability because of poor absorption: Lurasidone

10-If a person confirms with a group to get accepted what is it

Conformity wasn't in option Compliance Obedience

11-Topiramate side effects: Weight loss

12-Lithium side effects: Worsening of psoriasis

13-What is Ethnicity: Group identity or genetic something option

14-Question on Accommodation :Argyll Robinson pupil,

*Reacts poorly to light & normally to near (accommodation and convergence)

*No reduction in visual acuity

*Associated with tertiary syphilis

*DM

15-Question on Assimilation : *Partial adaptation, without giving up one's culture completely.

16-Equilibrium or centration related EMIs

*Centration: Focusing all attention and disregarding all other aspects

17-question on three mountain task (Piaget) egocentrism

*Pre-operational,*2-7 *Transudative reasoning

EMI options *Modelling , reinforcement , reciprocal inhibition in option of EMIs

Stem was

18-A student gets sexual education at school related to STDs and stigma related to them, if he stops having any sexual relationship, what is this.... Positive punishment

19-. If one student start copying another class mate in Hope of getting romantic relationship like them, what's this: Modelling

20-POMC (Proopiomelanocortin)precursor for ACTH or others (ADH, VMA .....)

*Alpha Melanocyte stimulating hormone & B-endorphin.,ACTH

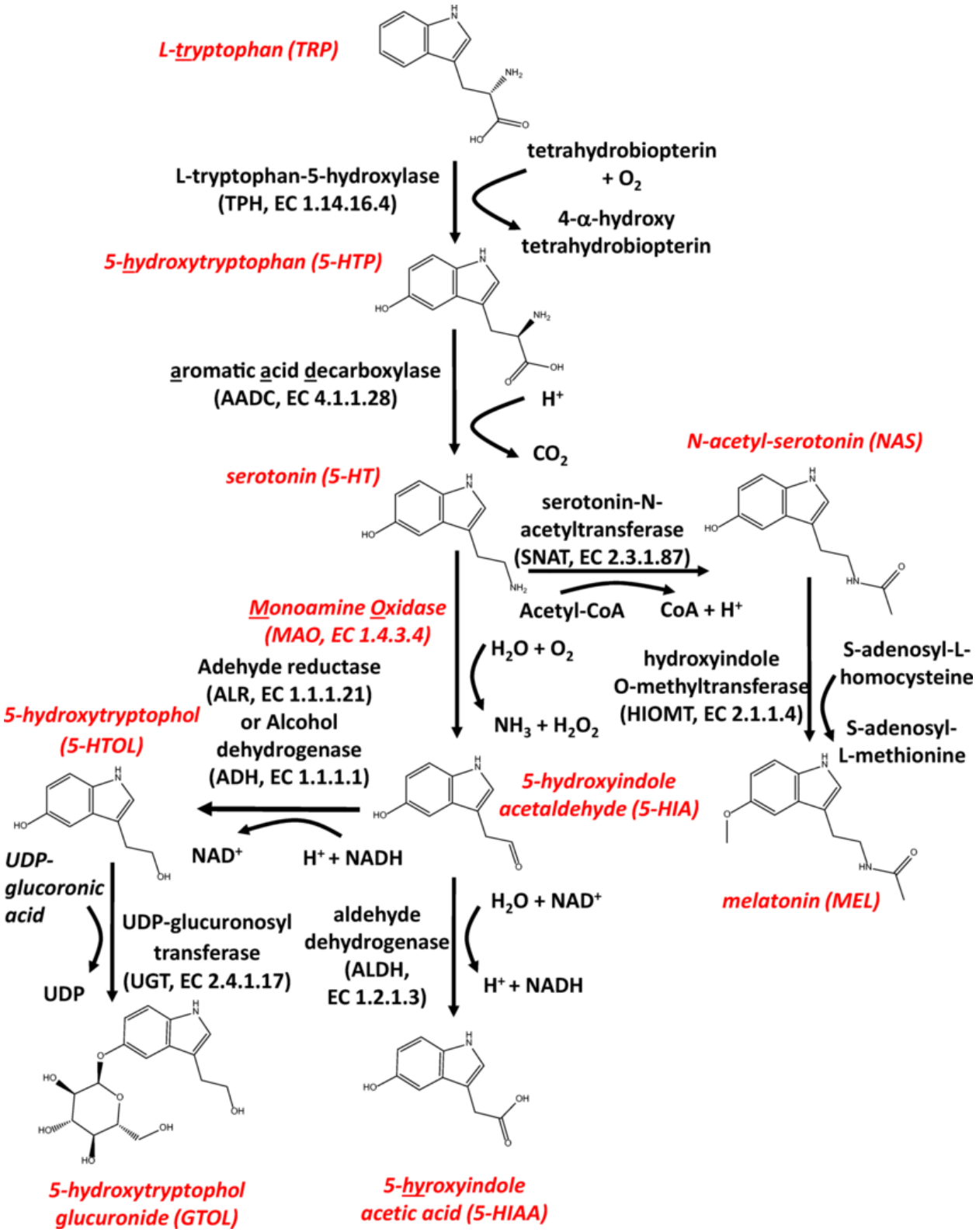
Binds to MC3R & MC4R

Reduces food intake and improves BMR

Lowers blood glucose

Activated by leptins & ant. Pituitary

21-Serotonin is precursor for which monoamine? Melatonin, dopamine, NorE, Epinephrine


L-tryptophan (TRP)
L-tryptophan-5-hydroxylase (TPH, EC 1.14.16.4)
tetrahydrobiopterin + O2
4-α-hydroxy tetrahydrobiopterin
5-hydroxytryptophan (5-HTP)
aromatic acid decarboxylase (AADC, EC 4.1.1.28)
H+
CO2
serotonin (5-HT)
N-acetyl-serotonin (NAS)
serotonin-N-acetyltransferase (SNAT, EC 2.3.1.87)
Acetyl-CoA
CoA + H+
Monoamine Oxidase (MAO, EC 1.4.3.4)
H2O + O2
NH3 + H2O2
hydroxyindole O-methyltransferase (HIOMT, EC 2.1.1.4)
S-adenosyl-L-homocysteine
S-adenosyl-L-methionine
Adehyde reductase (ALR, EC 1.1.1.21) or Alcohol dehydrogenase (ADH, EC 1.1.1.1)
5-hydroxytryptophol (5-HTOL)
5-hydroxyindole acetaldehyde (5-HIA)
NAD+
H+ + NADH
melatonin (MEL)
UDP-glucoronic acid
UDP-glucuronosyl transferase (UGT, EC 2.4.1.17)
UDP
aldehyde dehydrogenase (ALDH, EC 1.2.1.3)
H2O + NAD+
H+ + NADH
5-hydroxytryptophol glucuronide (GTOL)
5-hyroxyindole acetic acid (5-HIAA)

22-L.aminoacid decarboxylase (DOPA Decarboxylase) is involved in which step?Converts L-DOPA (L- Dihyroxyphenyl alanine) to Dopamine

23-Cooing is which stage of language? Pre linguistics

24-Antipsychotics cause postural hypotension thru which receptors Alpha 1 block

25-Amenorrhea because of antipsychotics caused by D2 block in tuberoinfundibular

26-Target for Alzheimer's treatment :Newer treatment for Alzheimer's via the secretase enzyme (Beta amyloid)

Copied from Psych mentor:

Alpha-secretase

Alpha-secretase is hoped to offer a new way of treating/ preventing Alzheimer's disease.

Once amyloid precursor protein (APP) crosses the cell membrane it is cleaved by several different enzymes. Alpha-secretase cleaves the protein in such a way as to produce protein fragments that are non-toxic. In addition to alpha-secretase, there are two other enzymes in the cell that can also cleave APP (beta and gamma-secretase). When these cleave APP the result is shorter, stickier fragments called beta-amyloid, which join together to produce the insoluble amyloid plaques.

Drugs are being developed which either stimulate alpha-secretase or block beta- and gamma-secretase.

27-Scale for change and improvement in psychotic symptoms : BPRS

28-Least risk for manic switch??? Bupropion carries the least ability to switch depression to mania

NB: TCA carries the highest risk.

Phenelzine Tranylcypromine Sertraline Imipramine A SSRis was other option

29-CT over MRI

Better for calcification, bones, bleeding & in emergency

?detects cerebral oedema quickly

30-ARFID(Avoidant Restrictive Food Intake Disorder) difference from anorexia

Intentional weight loss- Anorexia

ARFID Don't have distorted body image. No fear of weight gain .

Not preoccupied by intentional weight loss .

EXPLANATION from SPMM: Individuals with anorexia nervosa (AN), like individuals with avoidant-restrictive food intake disorder (ARFID), present with a pattern of restricted eating and significantly low body weight, with similar health-related consequences. The difference is that in AN, behaviors to establish or maintain an abnormally low body weight are usually explicitly motivated by a desire for thinness or an intense fear of gaining weight. In ARFID, the pattern of eating behavior is not motivated by preoccupation with body weight or shape.Ref: International Statistical Classification of Diseases and Related Health Problems (11th ed. ICD-11; World Health Organization,2020) Weight loss to achieve thinness

3 types:

31 Fear of aversive consequences Sensory

32 ARFID Self assessment questionnaire ; 4 or 5 on at least 6 items. 20 items

33 -Neuroimaging finding in bipolar:Higher bilateral ventricular volume

Lower hippocampal volume, amygdala, thalamic volume

34 Fractional anisotropy related 2 options:

Measure of DTI (Diffusion Tensor Imaging)

Reflects white matter integrity

Corresponds to degree of directionality of water diffusion

35-CPTSD over PTSD

CPTSD- Repeated trauma over time . Severe problem in accepting regulation

PTSD Symptoms relating to single trauma

Relationships difficulty i.e avoid this because of past trauma

EXPLANATION From SPMM: Whereas the ICD-11 diagnostic requirements for complex PTSD include all essential features of PTSD, the diagnosis of complex PTSD also requires the additional essential features of severe problems in affect regulation, persistent negative beliefs about oneself, and persistent difficulties in sustaining relationships.

Ref: International Statistical Classification of Diseases and Related Health Problems (11th ed. ICD-11; World Health Organization,2020)

37-One was icd 11 diagnosis for a veteran who was in war one year ago, now faces re-experience, avoidance of loud notices, nightmares, but good self image and relationships positively maintained.... PTSD is diagnosis

38-Self rated BDI

39-Clinician rated HAMD

40-Fatal toxicity index: Dosulepin aka Dothiepin is 2.7 & Amitriptyline is 12, Doxepin index of 22

41-Fusiform gyrus for face recognition (Prosopagnosia). PCA artery. Occipito-temporal junction

42-auditory hallucinations ?superior temporal gyrus

43-Default mode network part : Anterior cingulate or posterior cingulate Cortex

Default mode MAPPP

Medial prefrontal cortex

Angular gyrus

Precaneous, parietal lobe , posterior cingulate cortex.

Salience network: ACTS

Anterior Insula, Anterior Cingulate cortex,Thalamus,straitum

Central executive: DIP

Dorsolateral prefrontal cortex

Intraparietal sulcus

Posterior parietal cortex

44-A Deletion disorder- prader- will: paternal microdeletion chromosome 15. For angelman : maternal microdeletion of chromosome 15

45-Chromosomal abnormality in Huntington's : 4p16: its chromosome 4 CAG trinucleotides repeat .Anticipation is present in this condition. Has HTT protein. Autosomal dominant. Worse diseases profile with paternal inheritance 35-45 CAG repeat. Pre mutant Huntington for individual with less than 35, has double penetrance. Can't show the disease but can transmit to next generation. Adult onset has 35-45 repeat, childhood >70 repeat with worst prognosis.

46-chromosomal abnormality in FTD: 17q21

Other xomes: C9orf72 [9], PGRN (progranulin) [17], MAPT (tau) [17]

FTLD pathology

The term Frontotemporal Lobar Degeneration (FTLD) refers to the neuropathology of the FTDs. Neuropathologically, FTLD is classified into several subtypes, according to the main protein component of neuronal and glial abnormal inclusions and their distribution.

Currently, three main proteins are associated with FTLD:the microtubule-associated protein Tau in FTLD-Tauthe transactive response DNA-binding protein 43 kD (TDP- 43) in FTLD-TDPthe fused in sarcoma protein (FUS) in FTLD-FUS

47-Kleine depressive position: it's a mental constellation defined by Kleine as central to child's development. Beginning around the 4-6 months of life . xterize by ambivalence, decrease in splitting and perception of objects as a hole. |empathy develops and recognition of others as separate entity develops. Feeling of guilty and fears of loss.

Paranoid schizoid position: splitting, projective identification, idealization, state of fragmentation. Anxiety is persecutory in nature .

Klein posited that a healthy development implies that the infant must split its external world, its objects and itself into two categories: good (i.e., gratifying, loved, loving) and bad (i.e. frustrating, hated, persecutory). This splitting makes it possible to introject and identify with the good.

paranoid-schizoid position is considered the state of mind of children from birth to four or six months of age

48-mirtazapine- serotonin receptor

First introduced in 1996, mirtazapine is of the class of noradrenergic and selective serotoninergic antidepressants (NaSSA).26 May 2018

Noradrenaline and serotonin specific antidepressant (NaSSa) 5HT2A and 5HT2C antagonist, 5HT3 antagonist, H1 antagonist, alpha 1 and alpha 2 antagonist, moderate muscarinic antagonist. Presynaptic alpha 2 antagonist (which stimulates the release of NA and 5HT.

49-group therapy- utilitarianism

utilitarianism, as a teleological theory, seeks 'the greatest good for the greatest number'. This approach is inherently outcome-focused, assessing the morality of actions by the extent to which they contribute to overall well-being or happiness. It is a consequentialist

theory, as it holds that the predicted outcomes (i.e. the consequences) of an action are the most morally important component of that action. NB: Greatest good for greatest number .

50-not prescribing a cardiac toxic drug -non maleficence: Do no harm An example of nonmaleficent action is stopping a medication that is known to be harmful, or refusing to give a medication to a patient if it has not been proven to be effective.

51-several family members blaming one family member -?detouring

Triangulation, circular causality

Triangulation in family therapy is when a third party is involved in a family relationship to address tension, anxiety, or conflict. It can be intentional or unintentional, and can occur in any type of relationship.

Detouring is a psychological defense mechanism, typically seen in family or group dynamics, where conflict or stress between two individuals (often parents) is deflected onto a third party, such as a child or another member of the group. This process allows the original parties to avoid dealing with their own conflict by focusing their attention on the third person, often unconsciously.

Circular causality is a concept often used in systems theory, particularly in fields like family therapy, to describe how events or behaviors are part of ongoing feedback loops, where causes and effects are interdependent rather than linear. In circular causality, behaviors and interactions are seen as mutually influencing each other, rather than following a simple cause-and-effect chain. In family therapy, detouring refers to a dysfunctional pattern in which two family members, often the parents, focus their attention on a third person (usually a child) to avoid addressing their own conflicts or problems. This third person, often referred to as the "scapegoat," becomes the target of blame or concern, distracting from the real issues within the family system.

For example:In a troubled marriage, parents may focus excessively on a child's behavioral problems rather than dealing with their own marital difficulties.

52-understanding family systems or something along those lines- options were same as above : see above

53-tms- severe depression- DLPFC

DBS for depression: Cingulate gyrus

Depression
Repetitive TMS (rTMS) of the left DLPFC is a common treatment for unipolar and bipolar

depression.

Schizophrenia
rTMS of the left DLPFC is a common treatment for negative symptoms of schizophrenia

54-neurotransmitter that maintains ph in brain: GABA, Glutamate , CO2?????Astrocytes regulate brain extracellular pH via a neuronal activity-dependent bicarbonate shuttle.

55-ion channel most implicated schizophrenia -voltage gated calcium ion ? – channel.

56-peduncar hallucinosis- complex visual hallucinations, Peduncular hallucinosis is characterized by striking visual images, highly coloured and mobile, which are recognized by the patient as imaginary. It might be a sign of venous stagnation and infarction of the brain stem [6]. Three cases of peduncular hallucinosis Peduncular hallucinosis (Lhermitte's syndrome) consists of vivid hallucinations that are usually visual but can be auditory, tactile, or mixed. Damage to the brain stem and thalamus and affect visual processing pathway. Causing hallucination including people animals and objects

57-rhesus monkey experiment? attachment theory

58-in vivo neurotransmitter- MRS/ PET

59-Thomas and chess temperament: Good enough mothering, goodness of fit . Slow to warm up 15%, difficult 10%. Easy child 40%, ungrouped/combination 35%

Triats Activity , regularity, initial reaction, adaptability, intensity, mood, distractibility , persistence or attention span, sensory threshold.

Temperament - What is it?
Temperament is a child's emotional and behavioral style of responding to the world. A child displays her temperament style from birth. Alexander Thomas and Stella Chess, researchers, found that temperament is influenced by nine temperament traits: activity, regularity, initial reaction, adaptability, intensity, mood, distractibility, persistence-attention span, and sensory threshold.

60-child wakes up afraid- night terrors

61-child wakes up from bad dream-nightmares

62-which statement is true about lithium induced tremor- inversely related to dose:

63-Propranolol mode of action- Beta 1, 2 adrenergic receptor blocker

64-which is part of the default mode network? See question 43

65-cocktail phenomena- selective auditory attention

66-question on James Lange theory: perception of stimulus leads to bodily changes (skeletal changes). Physiological stimulus.

JLPCE: James Lange physiological changes predicts Emotions.

CBI: canon band is Independent: emotion is independent of physiological changes

SSA: Schachter-singerArousal: arousal is basis of emotion

67-stroop test- selective attention . has attentional bias

68-which is a test of executive function- Tower of London others include:

69-test of abstract thinking- WAIS similarities test

70-brain organoids- 2nd trimester

71-DORAs beneficial for what? Insomnia They are prescribed for insomnia DORA:. Dual Orexin receptor antagonist. They act on orexin 1 and 2 receptors in locus cerelus and dorsa raphe eg. Of Dora include ; daridorexant, Lemborexant and sumborexant.

72-not implicated in delayed language- immigrant status

Risk factors for language delay

being male.

being born prematurely.

having a low birth weight.

having a family history of speech or language problems.

having parents with lower levels of education.

73-question on retrograde amnesia:?????Mock 4: temporal region(hippocampus)

74-what did I eat for breakfast? Episodic

75-what is breakfast? Semantic

76-reverses affects of caffeine- adenosine agonist . caffeine reverses the effect of adenosine

Procedural memory, or 'knowing how', is the unconscious memory of skills and how to do things. It is acquired through repetition and practice.
2) Declarative memory is memory of facts and events that can be consciously recalled (also known as explicit memory).
3) Semantic memory is a structured record of meanings, concepts, facts, knowledge about the world; it's to do with general factual knowledge that is shared with others, independent of personal experience
4) Prospective memory refers to that which needs to be remembered in the future, and may be defined as 'remembering to remember'; it can be either event-based or time-based, and is often triggered by a cue (e.g. finishing breakfast).
5) Episodic memory refers to experiences and specific events in time in serial form, from which we can reconstruct the actual events that took place in our lives; it is the memory of autobiographical events, and has an emotional charge.

77-question on hoovers sign:

Hoover's sign is a neurological test used to differentiate between organic and non-organic leg paresis. It's based on the principle of the crossed extensor reflex, which is when flexing one hip extends the other hip. If the patient has organic paresis, the examiner will feel downward pressure on their hand. If the patient is malingering, no pressure will be felt.

78-Altruism study findings- reduced physical pain

79-K complex, which stage of sleep- Nrem stage 2

80-empathy- insula

81-describes situation related Camberwell interview- criticism

CFI is criticized for being impractical as an effective technique in clinical practice to measure expressed emotions (EE). This criticism is based on the fact that the technique requires highly trained professionals to interview subjects (patients and their family members) and requires approximately 3 hours per assessment including the subsequent coding of the interviews.

82-starting time of strong attachment 12-18m

83-bacteria linked to OCD- strep pyogenes PANDAS is a paediatric disorder that can cause obsessive-compulsive disorder (OCD) in children after a strep infection: PANDAS Stands for Paediatric Autoimmune Neuropsychiatric Disorders Associated with Streptococcal Infections. It's diagnosed when a child develops OCD or tic disorders, or their symptoms worsen, after a strep infection like strep throat or scarlet fever.

EMI

84-Not a side effect of EPSE.Tics, tardive dyskinesia, akathisia, dystonia:

EMI options -concussion syndrome, retrograde amnesia, transient global amnesia, anterograde amnesia

85- person who had an accident and did not remember the accident but remembered after? Retrograde Amnesia

Alpha wave, gamma waves, diffuse slowing, sharp wave

86-EEG finding of dementia: Alpha wave if AlZ dementia

Alzheimer's Dementia Reduced alpha and beta, increased delta and theta

87-Test for executive function: Tower of London Other test for executive functions:continuous Performance Tasks (CPT), Stroop Test, Wisconsin Card Sorting Test (WCST), Tower of London, Animal Fluency Test, Trail Making Test (Part B)

88-Which of the following enzymes are presynaptic : Presynaptic enzymes modify neurotransmitters before they are released from vesicles into the synaptic cleft. One example of a presynaptic enzyme is acetylcholinesterase (AChE), which inactivates acetylcholine by hydrolyzing it into acetate and choline choline acetyl transferase, tyrosine hydroxylase, dopamine beta hydroxylase , glutamic acid carboxylase, tryptophan hydroxylase , MAO, COMT

89-Monoamine oxidase, COMT?????

90-Which of the following are found in the synaptic cleft ???? Acetylcholinesterase (found in synaptic cleft. Monoamine oxidase, COMT(both presynaptic and postsynaptic),

91_ 13-year-old who knows the difference between mass and weight is in what stage of: - Piagets developmental stage Formal operational stage

92-gene implicated in frontotemporal dementia with Parkinson: PGRN (progranulin) [17], MAPT (tau) [17]

93_ By what receptors does mirtazapine increase serotonin and noradrenaline level: it inhibits the central presynaptic alpha 2 adrenergic receptors, hence leading to an increase in serotonin and adrenaline. Adrenergic receptor .NaSSA(Noradrenaline specific serotonergic antidepressant)

5HT1A, 5HT2, alpha1, alpha 2

94_ Which of the following is a full agonist at mew receptorMethadone, Naloxone, naltrexone, buprenorphine

95-Which of the following have the most important active metabolite; lorazepam, temazepam, diazepam,

96-About type b personality: Have apathetic response to stress -Have reduced risk of cardiovascular disease **Type A and Type B personality hypothesis**describes two contrasting personality types. In this hypothesis, personalities that are more competitive, highly organized, ambitious, impatient, highly aware of time management, or aggressive are labeled Type A, while more relaxed, "receptive", less "neurotic" and "frantic" personalities are labeled Type B.

97-CNS Myelination- Oligodentrocytes

Myelination in PNS : schwan cells

98-Macrophages- Microglia

99-Circulation of CSF is maintained by-Ependymal cells

100- Chromosomes in Alzheimer's: PSEN1(CH14, APPch21 . others are ch19(apoE), ch2(PSEN2), ch19(APOE late onset)

101-Locus coeruleus- Nor adrenergic pathway

The main site of norepinephrine release is from the locus coeruleus (aka 'the blue spot) which is in the pons.

102-Parents chose the university for their child, child was told that he does not have a choice but obey- Parenting style- Authoritarian

Authoritarian Parenting This style is characterized by strict rules and punishment if the rules are not adhered to. Explanations behind the rules are often lacking. These parents are focused on status and obedience. Authoritarian parenting styles generally lead to children who are obedient and proficient, but they rank lower in happiness, social competence and self-esteem.

103-Acute presentation of NMS: ??

Fever Diaphoresis Rigidity Confusion Flutuating consciousness Fluctuating blood pressure Tachycardia Elevated CK Leukocytosis Altered LFTs

104-Acute spastic episode- dystonia

105-In Autism- Hypoplastic cerebellar vermis

106-Difference between mania and hypomania- presence of psychotic symptom: 1 week/7days for mania and several days in hypomania <4days

107-Reactive Attachment- ??Minimal seeking of comfort

n ICD-11, which of the following features is characteristic of children with reactive attachment disorder
Select one:
© Cognitive deficits
al communication deficits
• Minimal seeking of comfort
© Fearfulness
• Hyperactivity and inattention
Your answer is incorrect.
EXPLANATION: In ICD-11, reactive attachment disorder is characterised by grossly abnormal behavior: the context of a history of grossly inadequate child care. Even when an adequate primary caregiver is n turn to the primary caregiver for comfort, support, and nurture, rarely displays security-seeking behavi not respond when comfort is offered.
Ref: International Statistical Classification of Diseases and Related Health Problems (11th ed. ICD-11; Wi The correct answer is: Minimal seeking of comfort

108-Dentate gyrus is a part of- Hippocampus

109-Last part of brain to be myelinated- Anterior frontal pole

110-Separation anxiety sequence- Protest, Despair, Detachment

111-Memantine- Non competitive NMDA receptor antagonist

112-The Question was Aromatic amino acid decarboxylase involved in what step and the answer as per google -L.aminoacid decarboxylase is involved in L – Dopamine to Dopamine

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April 2024

1. Which of the following medication predispose to bleeding in the elderly

A. Sertaline

B. Fluoxetine (*longer half-life*)

2. Effect of SSRI on neonate in **late** pregnancy

A. Cleft palate

B. Pulmonary hypertension

C. Pyloric stenosis

D. Ventricular septal defect

3. Which is the following is the quickest to achieve a steady state

A. clozapine *(2-3 days)*

B. olanzapine *(7 days)*

C. aripiprazole *(14 days)*

D. quetiapine *(2-3 days)*

E. sulpiride *(2-3 days)*

4. Direct orexin antagonists (DORA) can be used in the treatment of

A. ADHD

B. OCD

C. Insomnia

D. Generalized anxiety disorder

PS: An orexin receptor antagonist inhibits the effect of orexin by acting as a receptor antagonist of one or both of the orexin receptors, OX1 and OX2. eg Daridorexant and Lemborexant

5. Which of the following cells have phagocytotic effect in the CNS

A. Macrophage

B. Astrocytes

C. Dendrites

D. **Microglia**

E. Oligodendrocyes

6. Which of the following neurons produces glutamate

A. astrocyte

B. glial cells

(PS: Pyramidal cells and granular cells uses glutamate, while its being produced/released from astrocytes)

7. Immunologic therapy in Alzheimer's targets

A. Acetylcholinesterase inhibitors

B. tau protein

C. beta amyloid

D. secretase

8. EMI - Receptors sensitive to changes in pH

A. calcium

B. potassium

C. sodium

D. chloride

E. Aquaporins

F. Photons

9. Receptor responsible for exocytosis of neurotransmitters

A. calcium

B. sodium

C. potassium

D. chloride

E. Aquaporins

F. Photons

10. Which prevents formation of amyloid in Alzheimer's

A. Alpha secretase *(while β & γ secretase supports the formation)*

B. beta secretase

C. gamma secretase

D. cholinesterase

11. Which is true of the side effect of the immunologic therapy of Alzheimers vascular cerebral edema (as seen on ARIA-e)

A. it is duration on treatment dependent

B. it is not dose dependent

C. it depends on amyloid burden

12. According to Yerkson - Dodson stress curve, performance deteriorates when

A. Stress too high or too low

B. Stress too low

MRCPsych Recalls

C. Stress too high

13. Which of the following is self-rated

A. BDI

B. HAMD

C. ZUNG

D. MADRS

14. Bayesian theory based on existing belief is what?

A. **Prior**

B. Posterior (prior multiplied by likelihood)

C. Likelihood (observed sensory data)

D. Prediction error (is generated when there is a disparity between prior expectation and sensory evidence)

15. Which of the following is present in the vision of a newborn

A. **Blink reflex** *(as well as pupillary reflex)*

B. Accommodation (3-4 months)

C. Colour perception (5 months)

D. Fix on distant object (2-3 months)

E. Convergence (2 months)

16. Mirtazapine acts as an antiemetic due to effect on which receptor.

A. NMDA

B. Glutamate

C. Dopamine

D. 5HT3
(PS: Other receptors that contribute to antiemetic property: 5HT2A, H1 receptors)

17. Psilocybin mediates Neuro plasticity via

A. 5HT1A

B. 5HT1B

C. 5HT2A

17. Rivastigmine not affected by

A. Renal impairment

B. Hepatic impairment

PS: Rivastigmine has no hepatic metabolism, while memantine metabolism is primarily non-hepatic. However, in CKD, rivastigmine is advised, while memantine is advised in hepatic impairment.

18. People were asked to state from most excruciating pain and no pain. What scale was used

A. Thurstone scale - *Measures quantitatively attitudes and behaviors: decided by a panel*

MRCPsych Recalls

B. **Semantic differential scale** – *may be subjected to positional response bias*

C. Likert scale – *graded 'agree' to 'disagree' measures.*

19. Which of the following when increased improves cognitive function in schizophrenia

A Dopamine

B Acetylcholine

C Serotonin

D Glutamate

20. Child pugh in pharmacokinetics - EMI

I. To calculate dose in renal impairment

II. To calculate bioavailability

III. To calculate dose in hepatic impairment

IV. To calculate bioequivalence

21. Comparing intravenous and oral administration in pharmacokinetic - EMI

I. To calculate dose in renal impairment

II. **To calculate bioavailability**

III. To calculate dose in hepatic impairment

IV. to calculate bioequivalence

22. Chlorpromazine is low potency antipsychotic yet has high affinity for binding to D2 receptors. What explains this discrepancy

A. Rapid phase metabolism

B. Low plasma protein binding

C. Lowly lipophilic

D. Widely distributed

23. Use of Thurstone scale

A. Measure Qualitatively

B. **Measures quantitatively attitudes and behaviors**

C. Not subject to positional bias

24. Which of the following is an enactive theory of perceptual and mental imagery

A. **Action cycle**

B. Cognitive activation theory

C. Kahnman theory of emotion

D. Variable stage cycle

PS: Action cycle is an enactive theory of the perception and a mental imagery system that is comprised of six

MRCPsych Recalls

modules: Schemata, Objects, Actions, Affect, Goals and Others' Behavior

25. What is Hebbian plasticity?

PS: Hebbian plasticity is widely considered to be the mechanism by which information can be coded and retained in neurons in the brain. "cells that fire together, wire together."

26. What is Vygostky zone of proximal development?

PS: ZPD is the space between what a learner is capable of doing unsupported and what the learner cannot do even with support. It is the range where the learner is able to perform, but only with support from a teacher or a peer with more knowledge or expertise.

27. Cariprazine mechanism of action

A. Partial D1 antagonist

B. Partial D2 antagonist

C. **Partial D3 agonist**

D. D Partial D4

E. Partial D5

PS: A partial agonist at D2, D3, with preference for D3 and partial agonist at 5-HT1A receptors and as an antagonist at 5-HT2A and 5HT2B receptors.

28. EMI - A child has a pet mouse which he spends time with. As part of a psychological experiment, his father creates a loud noise whenever the child is playing with his mouse, and the noise frightens him. This is done a number of times, so that whenever he sees his pet mouse, he feels frightened. What is his response to the rat?

A. Conditioned stimulus

B. Conditioned response

C. Unconditioned stimulus

D. unconditioned response

29. What was his response to the loud bang? Options as above - **Unconditioned response**

PS: The noise is the unconditioned stimulus, while the mouse, is the conditioned stimulus here.

30. Which of the following medication prolong QTc.

A. Methadone

B. Disulfiram

31. Part of brain concerned with thermoregulation. **A. Preoptic nucleus of hypothalamus**

32. Which of the following is part of the Papez circuit:

A. cingulate gyrus

B. Substantia nigra

33. Panksepp theory of emotion has how many basic theory emotions?

A. 3

B. 5

C. **7**

D. 9

E. 11

PS: Panksepp carved out seven biologically inherited primary affective systems called SEEKING (expectancy), FEAR (anxiety), RAGE (anger), LUST (sexual excitement), CARE (nurturance), PANIC/GRIEF (sadness), and PLAY (social joy).

34. EMI - Protest, Despair and Detachment are part of.

A. Bowlby by stages of grief and maternal deprivation

B. Kohlberg theory of moral development

35. Reason why mentally ill people are homeless in a city

A. Alpha male dominance
B. Altruism
C. **Societal schism** — the division or fragmentation within society, often leading to marginalization or exclusion of certain groups.

36. Effect of ECT causes a **reduction** of
A. BDNF in CSF
B. **Acetylcholine**
C. Glutamate
PS: ECT increases (Brain Derived Neurotrophic Factor, dopamine, glutamate, GABA, Noradrenline)

38. At what stage of cognitive development can intellectualization defense mechanism be developed/used
A. Formal
B. Pre-operational
C. sensory motor
D. Formal operational
E. Concrete

39. When a child can perform the 3-mountain test, what does this reveal
A. Animism
B. Conservation
C. Egocentrism
D. Concrete operational stage

40. EMI - Which gene codes for alpha synuclein
A. SCNA
B. ABCD
C. ITIH
D. HERG

41. Which gene codes for IKr potassium - **HERG**
Options same as above

42. EMI - FLU like symptoms, psychological symptoms, seizure and something
A. NMDAR
B. LGI1

43. Another patient has Hyponatremia. Same options as above -

LGI1

44. Animal model of forced swim test
A. Depression
B. Anxiety

45. Which of the following cells release glutamate
A. Medium Spiny motor neurons *(GABAergic)*
B. Pyramidal cells *(uses glutamate)*
C. Astrocytes *(produce glutamate)*

- Test for premorbid intelligence **a. NART**

Test for visuospatial reconstruction **a. Rey-Osterreith complex figure drawing test**

- Test for selective attention - **Stroop test**

- ADHD polygenic risk
A. Attention
B. Verbal
C. Non-verbal

- EMI Imaging findings for Parkinson –
Loss of the normal swallow tail appearance of susceptibility signal pattern in the substantia nigra

- Alzheimer's
SPECT- **Temporal and Parietal hypoperfusion**
MRI- predominantly the hippocampus, the parahippocampal gyrus, and the temporal amygdala, with parietal and frontal lobes being next most severely affected. The occipital lobe is generally spared.

- EMI - Tardive super-sensitivity and Parkinsonism result from
a. High D2 receptor occupancy *(seen in parkinsonism & increased risk of acute EPSE)*
b. Low D2 receptor occupancy
c. **post-synaptic dopamine (D2) receptor super sensitivity** *(Tardive dyskinesia)*

EMI –
EEG **3Hz (Absence Seizures)**
EEG **3 - 6Hz polyspike and waves (Myoclonic seizures)**

EMI neuro imagining of brain in patient on therapeutic dose of antipsychotics a. 70% D2 occupancy at 11'C on PET

EMI - on hyperpolarization, depolarization and resting potential. What Ions and channels are involved

Hyperpolarization- **K+ goes out of the neuron to mediate hyperpolarization (repolarisation below the resting potential). Has something to do with chloride channel**
Depolarisation- **opening of Na ion channels, influx into the neuron**
Resting Potential: -70mV.
Repolarisation, K+ goes out of the neuron

MRCPsych Recalls

Which of the following has the Least sexual side effect antidepressant
A. Trazodone
B. Vortioxetine
C. Amitriptyline
PS: VBAM – antidepressant: vortioxetine | bupropion | agomelatine | mirtazapine
ZOL – antipsychotics: ziprasidone | olanzapine | lurasidone.

- EMI Lung cancer in CT scan, did not inform patient and decided to keep monitoring closely
a. Paternalism
b. Autonomy
c. Non-maleficence
d. Beneficence

- A clinical director has reduced funding and has to decide between cutting funding to rehabilitation funds or early intervention funds. Options same as above
Could be utilitarianism, but most likely Justice

- A refugee who was tortured in his home country by his government. He goes out of his way to avoid CCTV as he thinks he is being monitored by CCTV
A. Overvalued idea
B. Sensitive paranoia
C. Persecutory delusion

- Which part of the Brain is responsible in impulsive behaviour of mania
a. Orbitofrontal cortex
b. Insula

- EMI: Cells producing glutamate **- Astrocytes**
- EMI: Cells responsible for immune action **- Microglia**

- **Pharmacodynamic drug dug interaction**
a. Ocp and St John's
b. Clozapine and Nicotine
c. Phenelzine and tramadol
All three options have significant interactions with each other

- Which of the following is true about musical hallucination: - hallucinosis
A. Common in musicians and former musicians
B. Rapid incomplete musical intonation
C. **Presence of some auditory perceptual impairment (Auditory Charles Bonnet Syndrome)**

- What is true about attribution error.
A. Attribution of one's behavior to one's personal factor
B. **More common in western society**

EMI - on separation anxiety
a. Fear of social situations (20 years)
b. **Begins < 5 years**
c. Lasts > 6 months
d. Specific phobia of going out
e. Situation specific

MRCPsych Recalls

EMI on selective mutism- **Situation Specific**
Options same as above

- Specific for diagnosis of autoimmune encephalitis
  A. MRI Brain
  B. Antibody levels in serum
  C. CRP
  D. Antibody level in CSF

- Treatment of REM sleep behaviour disorder
  A. prazosin
  B. melatonin
  C. phenelzine

71. Brain Lobe responsible for memory consolidation
  a. Frontal
  b. Parietal
  c. Temporal

72. Which route of administration is associated with portal venous system?
  A. **Oral**
  B. Transdermal
  C. Rectal
  D. Inhalational

73. Medium Spiny neurons (spiny projection neurons) deficiency in which condition?
  A. Huntington's
  B. Parkinson's

74. Right lobe dysfunction.
  A. Agnosia
  B. Constructional apraxia gestalt
  C. Constructional apraxia detail

75. Difference between complex PTSD and borderline personality
  A. Unstable relationship
  B. Frantic efforts to avoid triggers
  C. Negative affect despite stable sense of self

76. Which of the following is a first rank symptom of schizophrenia
  A. Thought block
  B. loosening of association
  C. Thought insertion

77. Prazocin is used in what condition:
PTSD (trauma-related nightmares), ADHD

78. Which of the following medication dose need to be adjusted at eGFR 70
A. Lithium

79. Which is the largest structure in the diencephalon
A. Thalamus

B. hypothalamus
C. amygdala
D. cerebral cortex

80. The preoptic nuclei responsible for thermoregulation is found in

A. thalamus
B. hypothalamus *(superior to the optic chiasm)*
C. pituitary

81. Depersonalisation: experiencing the self as unreal or strange
    Derealisation: experiencing the world or other people as unreal or detach from one's surrounding
    Neurobiology of depersonalisation – Insula, Ant. Temporal

82. Partial nicotine agonist: Varenicline *(partial alpha 4 beta 2, safe to use in hepatic dysfunction)*
83. Tobacco acts on nicotine receptor
84. Caffeine acts on adenosine receptor
85. Least likely in NMS – **Myoclonus/hyperreflexia/Bradycardia/Diplopia/Diarrhoea**
86. Ataxic gait – unsteady broad based
87. UMNL – scissoring & stiff legged (cerebral palsy)
88. Frontal lobe – Perseveration
89. AD – alpha secretase (good), B secretase (bad)
90. Target of immunotherapy in AD - **Beta amyloid**
    Vasogenic oedema with passive immunisation – a complication of using monoclonal AB in treating dementia
91. DAT scan can differentiate DLB from AD, SPECT can differentiate between DLB and Parkinsons
92. Deep brain stimulation in OCD – **ant. Limb of internal capsule/vent capsule or striatum**/nucleus accumbent/subthalamic nucleus. Transcranial Magnetic stimulant – targets orbitofrontal cortex
93. Theories of aggression –
94. laxative abuse in Anorexia nervosa –(met acidosis)
95. Vomiting in Bulimia (binge purge type) – EMI (met alkalosis)
96. MRI T2 weighted image – **water**/Hb/Fat/Ca
97. Strongly negatively associated with ADHD – ***attention****/non-verbal mem/verbal mem/working mem*
98. Which is declarative semantic memory – planet names
99. Findings in bipolar studies – Increased grey matter thickness with Lithum
100. Risk of deliquescence – 8 to 10 years
101. Strongest risk for sever domestic abuse
102. Imprinting occur in – Critical phase

MRCPsych Recalls

103. Social learning – modelling/Identification
104. Midlife crisis – Erikson (coined by Elliott Jaques, 1965)
105. Assimilation/Schema/Disequilibrium – Piaget
106. Hebbian synapses potentiation – **Neurons that are wired together fire together.**
107. Yerkes-Dodson Curve - Stress/ Arousal and performance
108. Rosenthal effect – Pygmalion effect (high expectation = high performance and vice versa)
109. Group think reduced by open debate
110. Prejudice reduced by more information or contact with other cultures
111. ADHD in ICD-11 – Neurodevelopmental Disorders
112. Selective mutism in ICD-11 - Anxiety or fear-related disorders
113. Ability to delay gratification is associated with – Stanford Marshmallow experiment **associated with better academic performance**
114. Prelinguistic period – Occurs normally from birth to 12 months. Characterized by crying, babbling, and echolalia.
115. Early onset conduct disorder in children is associated with – Antisocial personality disorder
116. Retrograde Amnesia –
117. Caudate Nucleus is part of the Basal ganglia
118. Two- factor theory- **Singer- Schacther theory**
119. Acute psychotic disorder ICD 11 criteria
120. Theory of perception and mental imagery- ? Gestalt
121. What acts on it's densely packed receptors
122. Phantom Limb pain ??
123. Guanfacine; Non-stimulant, Selective agonist of α2A-adrenergic receptors.
    S.E somnolence (40%), headache (27%), fatigue (18%), abdominal pain (12%)
124. Receptor that Psilocybin acts on; 5HT2A
125. Allosteric Deficiency
126. Wilsons Disease- blood reduced serum ceruloplasmin
    reduced serum copper (counter-intuitive, but 95% of plasma copper is carried by ceruloplasmin)
    increased 24hr urinary copper excretion
127. Mirror Neurons location?? Function ??
    Mirror neurons are a unique type of brain cell discovered in the premotor cortex of monkeys (area F5), which activate both when an individual performs an action and when they observe another performing the same action. This 'mirroring' of behaviour has led many researchers to postulate that they might play a fundamental role in understanding the actions and intentions of others,

MRCPsych Recalls

a basis for empathy, imitation, and even the evolution of language. They offer a model for understanding imitation learning (Rizzolatti 2004).

In humans, mirror neurons and mirror neuron systems are believed to be located in several brain regions:

Inferior Frontal Gyrus (IFG): This is part of Broca's area, located in the frontal lobe, and it plays a significant role in language production. The IFG has been

implicated in action understanding and imitation. In many functional MRI studies on humans observing or imitating actions, activity is detected in the IFG.

Premotor Cortex: Like in monkeys, the human premotor cortex is involved in planning and executing movements. It is also activated during action
observation, suggesting the presence of a mirror system.

Superior Parietal Lobule (SPL): The SPL is involved in integrating sensory information and also shows activation patterns consistent with a mirror
neuron system, especially when tasks involve understanding the intention behind observed actions.

Inferior Parietal Lobule (IPL): Located next to the SPL, the IPL also has areas that become active during both the execution and observation of certain actions.

Posterior Superior Temporal Sulcus (pSTS): This area is believed to process biological motion and is activated when we observe others' actions, especially those that convey social or emotional information.

The functioning of mirror neurons might provide a neurobiological foundation for empathy. Disruptions in this system could potentially contribute to
disorders where empathy or social understanding is compromised, such as autism spectrum disorders.

Imitative behaviours are crucial in learning and socialisation. Dysfunctions in
the mirror neuron system might be linked to challenges in these areas, seen in various developmental and neuropsychiatric conditions.

Added during revision.
Right Answers.
Not sure about the answer.

July 2023 recall

1. Lurasidone mechanism of action. Options **D2, 5HT2A, 5HT7 antagonist**, D2 antagonist 5HT2A, 5HT7 agonist

Antipsychotic	Mechanism
Lurasidone	Dopamine **D2, 5-HT2A, 5-HT7, alpha2A- and alpha2C- adrenoceptor antagonist**, and is a **partial agonist at 5-HT1A receptors**. **Lurasidone does not bind to histaminergic or muscarinic receptors.**

Its absorption is affected by food intake. Doesn't affect QTc

[SPMM] Lurasidone is a relatively new antipsychotic medication. Which of the following is true regarding its absorption?

Select one:
- No effect on food
- Used sublingually
- **Food increases absorption**
- Administered nasally
- Food decreases absorption

Lurasidone is weakly soluble in water; its bioavailability is limited after regular oral ingestion. When administered with a meal of at **least 350 calories**, the **bioavailability increases by 2-3 times.**

Lurasidone has poor absorption **5-15%** orally

2. Least sexual side effects : **Vortioxetine**

BAM
- Anti-depressants with **No** sexual side effect: **Agomelatine, Bupropion, Moclobemide**

MVTR
- Anti-depressants with **Least** sexual side effect: **Mirtazapine, Reboxetine, Trazadone and Vortioxetine**.
- **Citalopram and Fluoxetine** cause **Delayed Ejaculation**
- **Trazadone most common to cause Priapism**

For patients with life-long premature ejaculation, drug treatment is the recommended approach. **Dapoxetine**, a short-acting selective serotonin re-uptake inhibitor, is licensed to be used when required for this condition (not continuous daily use).

Other selective serotonin re-uptake inhibitors (citalopram, fluoxetine, fluvoxamine maleate, escitalopram, paroxetine, sertraline [unlicensed indications] and the tricyclic antidepressant clomipramine [unlicensed indication] have been widely used as regular, daily treatment. Caution is suggested in prescribing selective serotonin re-uptake inhibitors for young adolescents with premature ejaculation, and to men who also have a depressive disorder, particularly when associated with suicidal ideation. Ejaculation delay may start a few days after the start of treatment, but it is more evident after 1 to 2 weeks, since receptor desensitisation requires time to occur.

Added during revision.
Right Answers.
Not sure about the answer.

3. Topiramate prominent side effect: **Weight loss**

Other S/E: Word finding difficulties (dysnomia) and nephrolithiasis, psychosis.

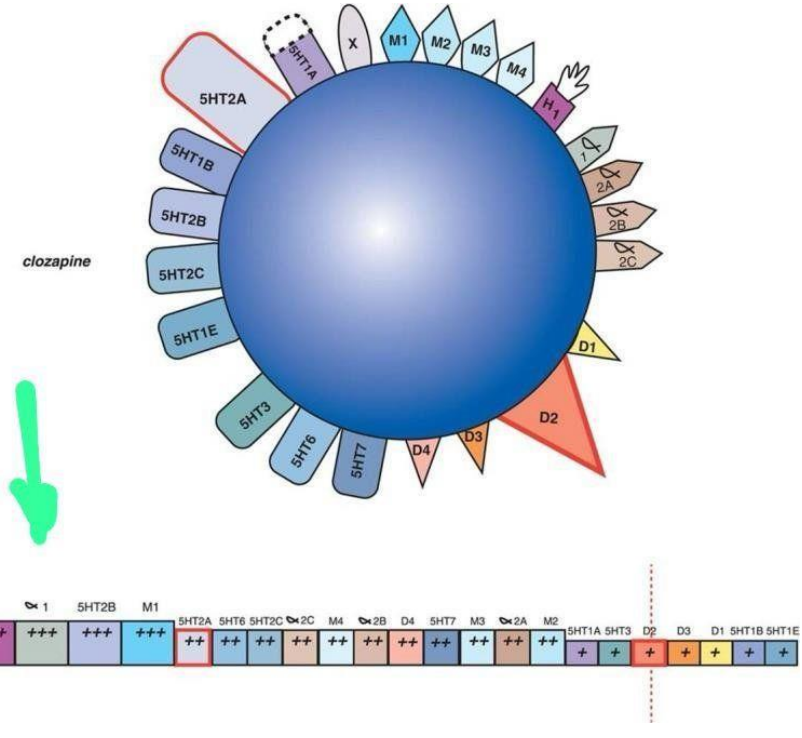
4. Lithium prominent side effect: **Worsening psoriasis**

5. Clozapine most affinity to **5HT2A receptor (depends on the options)**

Receptors from highest to lowest affinity:

H1 > ALPHA 1 > 5HT2B > M1 > 5HT2A > 5HT6 > 5HT2C > Alpha 2C

Hi alpha 1 TB m
26 2c 2c alpha



- **5HT3 antagonism** causing **Constipation**
- **M4** (affected by clozapine metabolites not clozapine itself) **agonism and Alpha 2 antagonism** causing **hyper-salivation and reduces swallowing reflex**

6. Lamotrigine mechanism of action:

- **NMDA receptor modulation,**
- **Some GABA modulation,**
- **Stabilises Na channels.,**
- **Blocks calcium channels.**

7. NMDA most affinity to which receptor (inaccurate recall)

We think Q. might be: Which of the following has high affinity to NMDA: **Glutamate**

Explanation: NMDA receptors are made up of subunits with distinct binding sites for **glutamate**, **glycine**, and **magnesium ion.**

Added during revision.
Right Answers.
Not sure about the answer.

Q. Glycine and d-serine both act as co-agonists at: [SPMM]

Select one:
- Dopamine receptors
- **NMDA receptors**
- GABA receptors
- Adrenaline receptors
- Acetylcholine receptors

Explanation: Glutamate systems are curious in that one of the key receptors for glutamate (the NMDA receptor) requires a co-transmitter in addition to glutamate in order to function. The co-transmitter is either the **amino acid glycine** or another amino acid, closely related to glycine, known as **D-serine**

NMDA is a G-protein coupled receptor

8. Methamphetamine most affinity to: **Dopamine vesicles**/serotonin vesicles

9. MDMA (Ecstasy) most affinity to: Dopamine vesicles /**Serotonin vesicles**

MDMA (3,4-Methylenedioxymethamphetamine) Has 2 isomers: R(-) isomers produce **LSD-like effects** and the S(+) isomers have **amphetamine-like properties**. LSD-like action is mediated via **serotonin release from presynaptic neurons**. In the long term, this can damage serotonergic tracts irreversibly. [SPMM]

10. Molecular Mechanism of NMDA receptor: **Glutamate-gated ion channels that increase Calcium**

11. Molecular Mechanism of SERT transporter:

The serotonin transporter (SERT) terminates serotonin signalling by using **sodium** and **chloride** gradients to drive the reuptake of serotonin into presynaptic neurons and is the target of widely used medications to treat neuropsychiatric disorders.

EMI Q: Types of Autoimmune Encephalitis.

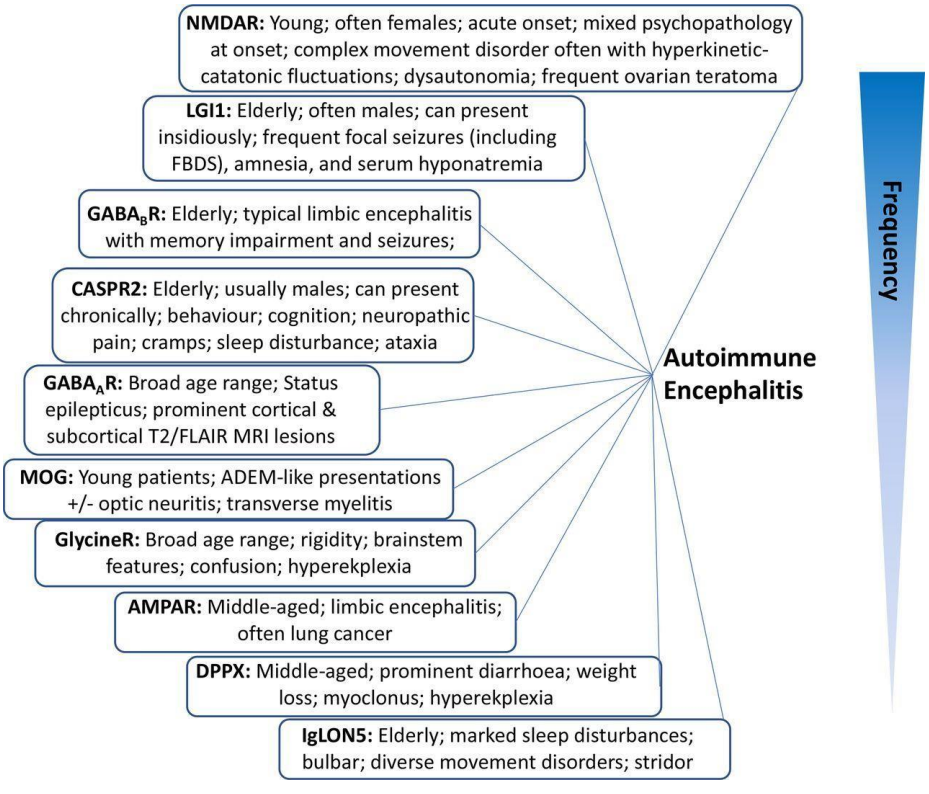
Which antibodies/antigens are implicated in
Options:
- NMDA
- AQP4
- GABA-A
- GABA-B
- more?

12. One with limbic hyperactivity **hyponatremia** and few psychiatric symptoms: **LGI1**

13. 2nd **flu like** prodrome and seizures: **NMDA**

14. 3rd optic nerve and spine inflammation: **Aquaporin 4 (could be MOG as well if it is in the options)**

Added during revision.
Right Answers.
Not sure about the answer.



Previous Exam Question [Psych-Mentor]

EMI

A. NMDA-receptors
B. D2-receptors
C. Voltage-gated calcium channels
D. CASPR2
E. LGI1
F. GAD
G. Glycine receptors

For the presentations of autoimmune encephalitis described, please select the most likely antibodies which might be detected on serological testing?

A 65-year-old male presents with a limbic encephalitis, and impairment in particularly with short-term memory. He has started to develop seizures particularly affecting the face & arms, experiencing several episodes daily which has not responded well to anticonvulsants. He is found to have a **low sodium** upon completing investigations.
Answer: **LGI1**

Explanation: The faciobrachial seizures (brief, dystonic seizures affecting ipsilateral arm and face) and **hyponatremia** are most suggestive of **LGI1** mediated autoimmune encephalitis.

The sister of a 75-year-old man who retired two years ago has noticed that his personality and behaviour has changed gradually over the last year. In particular his memory is notably poor. His initial serological investigations are unremarkable. There is autonomic instability of his

heart rate and blood pressure and he has experienced some falls which he says are due to sudden 'jerks' in his legs associated with pain. His MRI is mostly normal with mildly increased T2-FLAIR signaling noted in the medial temporal lobes.
Answer: **CASPR2**

> **Explanation:** T2-FLAIR: FLAIR stands for 'Fluid-Attenuated Inversion Recovery'. It's a type of MRI sequence that is particularly sensitive to changes in the brain that might be caused by various pathologies. On T2-FLAIR images, fluid (like cerebrospinal fluid) is usually dark, while areas of the brain that are affected by certain conditions (such as inflammation, demyelination, or ischemia) can appear bright or hyperintense. The involvement of the medial lobes suggests a limbic encephalitis.
>
> In this case the **autonomic instability and the 'jerks'** (peripheral nerve hyperexcitability) are suggestive of **CASPR2** mediated autoimmune encephalitis

A 34-year-old woman presents with marked sudden decline in cognition & social withdrawal following a brief period of **viral flu-like** illness which her GP had managed with rest & analgesia. This continues to worsen over the following weeks and the patient develops a reduced level of consciousness, psychosis, repetitive chewing movements of her jaw and intermittent seizures.
Answer: **NMDA-receptors**

> **Explanation:** The psychosis and repetitive chewing (oral dyskinesia) are most suggestive of NMDA-receptor mediated autoimmune encephalitis.

15. Relatively preserved in schizophrenia: short term/long term/**procedural**/semantic

> **Procedural memory is also preserved in normal aging**
> **Working memory, episodic memory and executive functions are markedly affected in schizophrenia.**

16. Test for short term memory: **Digit span**

> **Executive function**: Trail making test type B - Participants connect alternating numbers and letters. It's an extension of Trails A and **evaluates working memory, sequencing, and cognitive flexibility.** Impaired performance might indicate **frontal lobe pathology**. [Psych-Mentor]

17. Test for abstraction , **proverb testing**, cognitive estimates (both in the options)

Frontal lobe function	Test or examination
Abstraction	Proverbs, similarities, cognitive estimates

18. EMI for abstraction

19. Cocktail party phenomenon options had **Selective auditory attention**/ involuntary attention

Added during revision.
Right Answers.
Not sure about the answer.

20. Test for selective attention . **Stroop**

21. Which of the following is defined as the 'meaning of words' in linguistics?

Select one:

- Phonology
- Phenomenology
- **Lexical semantics**
- Orthography
- Syntax

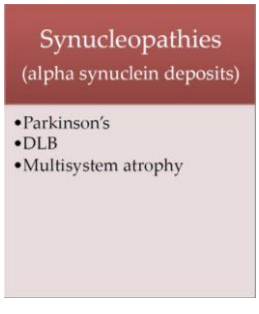
Lexical semantics is the study of word meaning. Phonology, orthography, syntax, and phenomenology are all relevant aspects of linguistics, but they deal with different aspects of language, such as sound patterns (phonology), written symbols (orthography), sentence structure (syntax), and subjective experiences (phenomenology), respectively. Lexical semantics specifically deals with the meaning of words themselves. [SPMM]

22. Pathophysiologic mechanism of Korsakoff's: **Mamillary body alpha ketoglutarate dysfunction**

EXPLANATION: Wernicke encephalopathy (WE) and Korsakoff psychosis, together termed Wernicke-Korsakoff syndrome, are distinct yet overlapping neuropsychiatric disorders associated with **thiamine deficiency**. Deficiency of thiamine can lead to decreased functioning of enzymes involved in the **tricarboxylic acid cycle (alpha-keto glutamate dehydrogenase and pyruvate dehydrogenase)**, pentose phosphate pathway, and branched-chain amino acid metabolism. Mammillary body lesions are the classic autopsy sign in WE patients and are present in almost all cases.

Diagnosis of Korsakoff syndrome correlates best with the presence of lesions in the **dorsomedial thalamus**. But lesions in the **mammillary bodies, the mammillothalamic tract and the anterior thalamus** may be more relevant to memory dysfunction than lesions in the medial dorsal nucleus of the thalamus. [SPMM]

23. Pathophysiologic mechanism of Multisystem atrophy:



Select one:

- **Alpha-synucleinopathy**
- Polyglutamate disease
- Medication-induced parkinsonism
- Hepatolenticular degeneration
- Tauopathy

EXPLANATION: **Multisystem atrophy (MSA)**, a family of **Parkinson-plus diseases, includes acerebellar type (formerly olivopontocerebellar degeneration)** that is notable for ataxia and a subtype with pronounced autonomic dysfunction (formerly Shy-Drager syndrome).There is a **common neuropathological finding of distinctive inclusion bodies in glial cells which contain alpha-synuclein,** leading to the classification of MSA as a synucleinopathy**. No mutations in the gene** for alpha-synuclein have been found in MSA, which appears to exist only as a sporadic disorder.

Added during revision.
Right Answers.
Not sure about the answer.

24. Which gene codes for P-glycoprotein: from google it says **ABCB1 gene located on chromosome 7** And MDR1

25. Which gene codes for alpha synuclein: **SNCA** SNCA Gene on chromosome 4 -Gene for serotonin receptor-SLC6A4

26. Which molecule is component of BBB:

Select one

- **APP** SPMM and PSYCHMENTOR ans
- Huntingtin
- Hypocretin
- ApoE
- Prion protein

If p glycoprotein is an option choose that.

EXPLANATION: Cerebrovascular homeostasis is maintained by the blood-brain barrier(BBB), a highly selective structure that separates the peripheral blood circulation from the brain and protects the central nervous system. Several recent studies have highlighted the importance of the physiological role of the **Beta-amyloid precursor protein (APP)** in endothelial cell homeostasis, suggesting an important role of this protein in maintaining vascular stability.

27. A woman's daughter is undergoing chemotherapy since 3 months. Woman has discussed this with friends and family. It reminds her of her mothers death from cancer.

Options:

- Acute stress
- **Adjustment**
- Complex PTSD
- PTSD

Adjustment disorder is a maladaptive reaction to an identifiable psychosocial stressor or multiple stressors (e.g. divorce, illness or disability, socio-economic problems, conflicts at home or work) that usually **emerges within a month of the stressor**. The disorder is characterized by preoccupation with the stressor or its consequences, including excessive worry, recurrent and distressing thoughts about the stressor, or constant rumination about its implications, as well as by failure to adapt to the stressor that causes significant impairment in personal, family, social, educational, occupational or other important areas of functioning. The symptoms are not better explained by another mental disorder (e.g., Mood Disorder, another Disorder Specifically Associated with Stress) and **typically resolve within 6 months, unless the stressor persists for a longer duration.** **[ICD11]**

28. How many episodes of depression needed for a diagnosis of recurrent depressive disorder: **Two episodes**

29. Consolidation of emotional memories in which stage of sleep: **REM Sleep**

EXPLANATION: Evidence for the role of REM sleep in consolidation of the emotional components of memory is relatively strong. The precise mechanisms by which REM sleep promotes consolidation of emotional memories remains unresolved, although there are some

Normal memory NREM2

Added during revision.
Right Answers.
Not sure about the answer.

studies suggesting that **long-term potentiation processes are activated in REM sleep**. There is a large body of literature suggesting that **theta oscillations,** characteristic of REM sleep, play a role in memory consolidation, particularly in fear conditioning and emotional memories. [SPMM]

30. A man is told he needs to get foot amputation, he is admitted and sees a patient who has had a similar surgery. He breaks equipment and says death is better than this. **Acute stress reaction**/ Adjustment disorder.

31. Difference in Avoidant Restrictive Food Intake and Anorexia. Options were: **Intentional weight loss**, significant functional impairment.

EXPLANATION: Individuals with anorexia nervosa (AN), like individuals with avoidant-restrictive food intake disorder (ARFID), present with a pattern of restricted eating and significantly low body weight, with similar health-related consequences. The difference is that in AN, behaviours to establish or maintain an abnormally low body weight are usually explicitly motivated by **a desire for thinness** or an intense fear of gaining weight. **In ARFID, the pattern of eating behaviour is not motivated by preoccupation with body weight or shape.** [SPMM]

32. Difference between Hoarding Disorder and OCD. Options were: Objects have intrinsic value/ **Absence of obsessional thoughts**

EXPLANATION: Individuals with OCD may accumulate excessive amounts of objects (i.e. compulsive hoarding). However, unlike hoarding disorder, the behaviour is undertaken with the goal of neutralising or reducing concomitant distress and anxiety arising from obsessional content such as aggressive, sexual/religious, contamination, or symmetry/ordering themes. Furthermore, even in individuals affected by OCD who have poor or absent insight, the behaviour is generally unwanted and distressing, whereas in hoarding disorder it may be associated with pleasure or enjoyment. Generally in hoarding disorder, items are hoarded because of their emotional significance, instrumental characteristics, or intrinsic value (e.g. perceived aesthetic qualities). [SPMM]

33. How long after loss event can we diagnose prolonged grief disorder: 1 month/**6 months**/1 year

[ICD11] **Prolonged grief disorder** is a disturbance in which, following the death of a partner, parent, child, or other person close to the bereaved, there is persistent and pervasive grief response characterised by longing for the deceased or persistent preoccupation with the deceased accompanied by intense emotional pain (e.g. sadness, guilt, anger, denial, blame, difficulty accepting the death, feeling one has lost a part of one's self, an inability to experience positive mood, emotional numbness, difficulty in engaging with social or other activities). **The grief response has persisted for an atypically long period of time following the loss (more than 6 months at a minimum) and clearly exceeds expected social, cultural or religious norms for the individual's culture and context**. Grief reactions that have persisted for longer periods that are within a normative period of grieving given the person's cultural and religious context are viewed as normal bereavement responses and are not assigned a diagnosis. The disturbance causes significant impairment in personal, family, social, educational, occupational or other important areas of functioning.

34. Qualifiers for depressive episode in ICD-11. Options were: Biological symptoms/ pseudodementia/ **melancholic**.

MAPS-melancholic, anxiety, panic attacks , seasonal

Others specifiers include: **with prominent anxiety symptoms; with panic attacks; persistent; with melancholia; with seasonal pattern.**

35. Child appears shocked and does not come close when offered help. Answer was: **Reactive attachment disorder. If shock and frozen it will be Disorganised attachment disorder.**

EXPLANATION: In ICD-11, reactive attachment disorder is characterised by grossly abnormal behaviours in early childhood, occurring in the context of a history of grossly inadequate child care. Even when an adequate primary caregiver is newly available, the child does not turn to the primary caregiver for comfort, support, and nurture, **rarely displays security-seeking behaviours towards any adult**, and does not respond when comfort is offered. [SPMM]

36. What feature is seen in reactive attachment disorder.

Options

- Cognitive deficit
- Language deficit
- Hyperactivity
- Fearfulness
- **Minimal seeking of comfort**

Reactive attachment disorder, as the second child-specific stress-related disorder in the ICD-11, is also characterized by highly inadequate childcare. The disorder features an inhibited attachment behaviour of the child.

37. Child goes to day care for first time. Shows no reaction when he is left by parents and neither when they come back.

Options:

A. Reactive attachment disorder
B. Aautism spectrum disorder
C. **Avoidant Attachment**
D. Anxious resistant attachment.

38. Brain organoid represents which age human brain.

Options:

A. 10years
B. 20 years
C. 6 days
D. 2nd trimester.

EXPLANATION: Brain organoids are produced by the differentiation of pluripotent stem cells under three-dimensional culture conditions by adding neurodevelopment- related regulatory signals. They are similar to the cell composition and anatomical structure of the brain, and can reflect the developmental process of the bring, as well as their physiology, pathology, and pharmacology. **In terms of their cellular and molecular composition, the current architecture of brain organoids can mimic the second-trimester human fetal brain.** [SPMM]

39. Prazosin best indicated for which disease.

A. ADHD
B. Anxiety
C. Autism.

It is reported in the group that the question was: **Prazosin is an antihypertensive, it can also be used for which of the following conditions: ADHD was the answer.**

Prazosin used for **PTSD with Nightmares, Raynaud phenomenon, BPH, Hypertension. It is an Alpha 1 blocker. Prazosin NOT used in depression**

One possible medical treatment for **post-traumatic stress disorder (PTSD) nightmares** is the drug **Prazosin**. It can relax the muscles and reduce blood pressure and may decrease the number of nightmares a person has.

As we couldn't find any evidence anywhere that Prazosin is used for ADHD, now we believe Anxiety could me more appropriate answer giving the face that prazosine is known to be used in PTSD with nightmares.

40. Pindolol best indicated for which disorder. ADHD/ Anxiety / autism/ **Depression**.

Beta blocker Pindolol: **Beta blocker with intrinsic sympathomimetic activity. Also 5HT1A antagonism** – tipped to enhance the onset of action of SSRIs through this mechanism.

Pindolol is an antihypertensive medication that has also been used as an augmentation treatment in depression. Which of the following best describes its mechanism of action?

Select one:

- **5-HT1-A antagonist**
- 5- HT1-B agonist
- 5-HT2-A agonist
- 5-HT2-A antagonist
- 5-HT1-B antagonist

Explanation: Pindolol is primarily an adrenergic beta-blocking drug, **which also blocks serotonin 1A auto-receptors**. The co-administration of pindolol with a serotonergic antidepressant could be expected to result in an immediate increase in serotonin neurotransmission, thus eliminating the delay in onset of antidepressant response. As well as being used to speed the onset of antidepressant response, pindolol has also been used to

augment the efficacy of antidepressant drugs in acute-phase non-responders and treatment-resistant depression.

N.B: Guanfacine for ADHD / Propranolol for Akathisia / Prazosin NOT used in depression

41. Guanfacine best indicated for which disorder . **ADHD**/ autism/ anxiety

42. Propranolol best indicated for which disorder. ADHD/ autism/ **Anxiety**.

43. Greatest improvement of cognitive symptoms in schizophrenia with which neurotransmitter Ach/ **Dopamine**/ NE/ Serotonin.

EXPLANATION: Although dopamine dysfunction has mainly been linked to positive psychotic symptoms, striatal dopamine signalling normally has widespread effects on cortical function, and its dysregulation can also cause cognitive symptoms. Many researchers believe that **cognitive and some negative symptoms of schizophrenia may be due to a deficit of dopamine activity** in **meso-cortical projections to the dorsolateral prefrontal cortex (DLPFC).** [SPMM]

44. Abused by exam-taking students: Atomoxetine/Guanfacine/ **Amphetamine**

EXPLANATION: Smart drugs' (also known as 'nootropics') are used by healthy subjects (i.e. exam-taking students and worker) typically to improve memory, attention, learning, executive functions and vigilance. **The most popular drugs are amphetamine-type stimulants (dextroamphetamine, lisdexamfetamine,), methylphenidate, donepezil, selegiline, modafinil, piracetam, benzodiazepine inverse agonists, and uniform analogues.** [SPMM]

N.B: Modafinil is also used for Narcolepsy.

45. Empathy is influenced by DLPFC/ **Insula** Anterior
Right supra marginal gyrus as well [SPMM]

46. Given in REM sleep behaviour disorder: Prazosin/Trazodone/**Melatonin**

Others: Clonazepam, Carbamazepine, Levodopa, Imipramine and Pramipexole.

47. Loss of orexin producing neurons in. **Narcolepsy**

EXPLANATION: Orexin/hypocretin neurons are localised exclusively in certain hypothalamic areas (later hypothalamic area, perifornical area, and posterior hypothalamus). These hypothalamic neurons degenerate in narcolepsy, and loss of these neurons causes the inability of orexin to be produced and released downstream on wake-promoting neurotransmitter centres and thus lack of stabilising wakefulness. [SPMM]

48. Atomoxetine MOA : **Noradrenaline reuptake inhibitor**

EXPLANATION: Atomoxetine is a selective norepinephrine reuptake inhibitor (NRI). Sometimes called NET inhibitors, the selective NRIs have known antidepressant properties.

However, despite its name, Atomoxetine enhances both dopamine and norepinephrine in the frontal cortex, presumably accounting for its therapeutic actions on attention and concentration in ADHD.

49. Nomothetic personality theory, dimensions are divided according to all the following except:

Options were:

- reliability over time
- reliability among cultures
- universalizability in assessment
- **individual variability**

The nomothetic approach in psychology seeks general principles and patterns applicable to groups e.g. Big five model of personality, Hans Eysenck type, while the idiographic approach focuses on understanding individuals in their unique context.

There are two main approaches to conceptualise and describe variations in personality: **nomothetic** and **ideographic. In nomothetic approaches**, personality is seen in terms of **attributes** shared by individuals. **There are two subdivisions: type (or categorical)** approaches - in which there are discrete categories of personality - and **trait (or dimensional)** approaches - in which a limited number of qualities, or traits, account for personality variation. Unlike nomothetic approaches**, ideographic approaches on the other hand, emphasise individuality and seek to understand an individual's personality** by understanding that specific individual and their development, rather than by reference to common factors. Examples include psychoanalytic, humanistic, and cognitive-behavioural approaches. [SPMM]

50. What is the critique on these dimensions. Options were reliability over time/ reliability among cultures?/ universalizability in assessment/ **Individual variability**

51. What is true about categorical classification of disorders. Options were: **Easy collection of epidemiological data**/ allow for accommodation of different levels of severity/?

EXPLANATION: In classical categorical systems, each entity or disorder is distinct and appears only once in a clearly defined place. With a few exceptions, **ICD and DSM are such classical categorical systems.** WHO member states are responsible for reporting health statistics based on ICD to WHO (e.g. causes of morbidity and mortality, prevalence). ICD remains structured as a categorical taxonomic system because this format is necessary for its application as the classification system for global health statistics. Wide use of non-standard categories would not generate valid data for global health statistics. [SPMM]

Dimensional classifies in terms of **S**everity **while C**ategorical is based on making **D**iagnosis and criteria.

52. Bayesian theory: What is an already held belief called: options: evidence/ data/ **prior**/posterior.

EXPLANATION: The account of predictive processing has emerged as an alluring computational model to explain how brain processes information about the world. In the higher brain centers, **expected predictions and pre-existing beliefs (known as priors)** are said to be combined with **observed sensory data (referred to as likelihood)** to compute a **posterior probability (referred to as the posterior).** This posterior probability is said to represent the most probable perception, considering both the prior beliefs and the sensory evidence. **In cases where there is a disparity between the prior expectations and the sensory evidence (likelihood), a prediction error is generated**. This error signal initiates the necessity to revise the brain's current model and adjust its prior beliefs accordingly. To express it formally, a prediction error quantifies the difference between the means of the prior and likelihood distributions, taking into account their respective precisions (inverse variance). [SPMM]

53. Most common inclusion in FTD. Options were:

Select one

- **Tau positive amyloid negative,**
- tau positive amyloid positive,
- amyloid positive,
- tau positive ubiquitin negative,
- tau positive ubiquitin positive

Ua
TATU

EXPLANATION: The classification of frontotemporal dementias is based on the type of protein inclusions observed in neurons as well as the mutational status, if known. **The inclusions contain predominantly or exclusively one of three markers:**

- tau;
- TDP-43, a DNA/RNA-binding protein; and
- FUS (fused in sarcoma).

Atypical frontotemporal lobar degeneration with ubiquitin-positive inclusion is a relatively rare cause of FTD characterised by FUS-positive inclusions.

54. LRRK2 mutation in? It was an option in a question

It is seen in Parkinson disease, Chromosome 12p11.2, other genes are PINK, PARKIN, SNCA

55. How many times does G protein coupled receptor pass through the cell membrane. **Answer**: **7 times**

56. Molecular mechanism of glutamate receptor: **NMDA**

57. Which antipsychotic reaches peak plasma concentration earliest. Options: **Quetiapine**/ Olanzapine/ Sulipride / Aripiprazole.

Peak plasma concentration

- **Quetiapine 1-2 hours**

- Olanzapine **5-8 hours**
- Aripiprazole **3-5 hours**
- Sulipride **3-6 hours**

58. Area under curve denotes: options were: **Exposure to drug over time, (Bioavailability)**

59. In first order kinetics: **Constant fraction of drug eliminated in unit time**/ constant amount eliminated in unit time/ log of plasma concentration plotted against time will show non-linear elimination/ log of plasma concentration plotted against time will show linear elimination

60. Absorption constant is independent of: **Elimination constant**/ form of drug/ composition of drug/ extravascular route of administration/ food taken with drug

61. Patient sticking to treatment is due to advise given by doctor is: referent power/legitimate power/information power/**Expert power**

62. Campaign about harms of smoking: referent power/legitimate power/**information power**/expert power

63. What is most likely to induce retrospective falsification of memory: warning given before misleading/ person being middle aged/ **what is being told aligns with previously held belief**

An unconscious distortion of past experiences to align to a person's need in the present. Retrospective falsification of memory, also known as memory distortion or memory malleability, refers to the phenomenon where our memories of past events can be altered or influenced by various factors, leading to inaccuracies or false recollections. Memory is not a perfect record of our experiences, but rather a reconstruction that can be influenced by a range of factors, including external suggestions, personal biases, and the passage of time.

Several factors can contribute to retrospective memory falsification:

1. Suggestion and misinformation: External suggestions or misinformation can lead to false memories. For example, if someone repeatedly tells you that a certain event occurred, you may eventually come to believe it, even if it didn't happen.

2. Leading questions: The way questions are framed can influence our memory recall. Asking leading questions that imply certain details can lead individuals to incorporate those detail can lead individuals to incorporate those details into their memories, even if they are incorrect.

3. Personal biases and expectations: Our existing beliefs, attitudes, and expectations can shape how we remember events. We tend to fill in gaps in our memory based on what we believe to be true or what we expect to have happened.

4. Source confusion: Sometimes, we may confuse the source of information. For example, if we hear a story from someone else, we may later remember it as our own experience. This can lead to the creation of false memories

Retrospective falsification is more likely to occur when new information aligns with previously held belief, hence individuals interpret current knowledge in light of new knowledge

5. Memory decay and reconstruction: Memories can naturally decay over time, and during the process of recall, we reconstruct them based on fragments of information. This reconstruction process can introduce errors or omissions, leading to memory distortions. It's important to note that **memory falsification is not always intentional or consciously driven**. Our memories are subjective and can be influenced by various factors outside of our control. Additionally, false memories can feel just as real and vivid as true memories, making it challenging to discern between accurate and inaccurate recollections. Researchers have extensively studied memory malleability, particularly in the context of eyewitness testimony and its implications for the legal system. Understanding the limitations and vulnerabilities of memory can help mitigate the impact of false memories and improve the accuracy of information recall.

Retrospective falsification: Memory becomes unintentionally (unconsciously) distorted by being filtered through a person's present emotional, cognitive and experimental state.

64. Incorrect recall is what type of memory issue: cryptomnesia/ primacy effect/ falsification/ proactive falsification/ **Retroactive falsification** Interference

1. **Proactive interference** (pro=forward) occurs when you **cannot learn a new task** because of an old task that had been learnt. When what we already know interferes with what we are currently learning **- where old memories disrupt new memories.**

2. **Retroactive interference** (retro=backward) occurs when **you forget a previously learnt task** due to the learning of a new task. In other words, later learning interferes with earlier learning – **where new memories disrupt old memories.**

Cryptomnesia	This is characterized by having a thought without realizing you have had the thought before (for example, some plagiarists claim they are unaware that they were recounting other people's work)

65. Man is seeing faces as frightening and misshapen. Options **Paraprosopia**/ chromatopsia/ metamorphopsia/ pelopsia/ palinopsia

Paraprosopia: the experience of seeing faces sometimes distorted in grotesque and threatening ways. 👹

Pelopsia: is a vision perception disorder in which object appear nearer than they are actually are. Teleopsia is opposite of pelopsia

Metamorphopsia: is distortion of an object or straight lines.

Metamorphopsia or dysmegalopsia refers to change in the perceived shape of an object. When metamorphopsia affects faces, it is referred to as paraprosopia. Faces generally transform into a grotesque mask (monster, vampire, werewolf). **Auto-paraprosopia** is a term used to describe a phenomenon where an individual perceives a transformation or alteration in their own facial appearance while gazing into a mirror. Note that paraprosopia is a distortion of reality seen in conditions such as schizophrenia. [SPMM]

66. Woman sees a cat and then sees her in different situations throughout the day. Illusion/ hallucination/ **palinopsia**: it is persistent of an image even after removal of the image

Palinopsia: *palin* for "again" and *opsia* for "seeing". It is a visual disturbance that causes images to persist even after their corresponding stimulus has left. It isseen in **LSD** use, **migraine**, **occipital epilepsy**, **head trauma.** It is similar to after image, but colour inversion (usually shadows or distorted colours noted in afterimages) is conspicuously absent.

[SPMM] Despite not having any pets, an elderly woman with visual impairment intermittently sees cats in her apartment, which delight her**: Charles Bonnet syndrome** is a condition in which complex **visual hallucinations** occur in **the absence of any psychopathology** and in clear consciousness. It is associated with either central or peripheral reduction in vision and not surprisingly is **most common in the elderly.**

67. Complex PTSD differentiated from PTSD by: correct option was **poor interpersonal relationships**

Complex PTSD is characterized by:

1. Severe and pervasive problems in **affect** regulation;
2. Persistent beliefs about oneself as diminished, defeated or worthless, accompanied by deep and pervasive feelings of shame, guilt or failure related to the traumatic event; and
3. Persistent difficulties in sustaining relationships and in feeling close to others.

68. Ebstein anomaly is caused by : **Lithium**/Valproate

69. Hypoplasia of which structure in Autism: correct option was **cerebellar vermis**

70. What is true about Purines:

Select one:

- **Are nitrogen-containing compounds**
- Include cytosine and thymine
- Are found only in eukaryotes
- Can only be obtained from the diet
- Include ethanol and nicotine

Explanation: A purine is an **aromatic heterocycle** composed of **carbon and nitrogen**. Eukaryotes, bacteria, and archaea are all able to carry out de-novo biosynthesis of purines. **Purines include adenine and guanine**, which participate in DNA and RNA formation. Purines are also constituents of other important biomolecules, such as ATP, GTP, cyclic AMP, NADH, and coenzyme A. **Other important purines include hypoxanthine, xanthine, theobromine, caffeine, uric acid, and iso-guanine.**

- **Cytosine, thymine, and uracil are pyrimidines.**

71. What is the most biologically plausible genetic basis of schizophrenia: Options **DISC1 and NRG1**

72. Kaiser CDC ACE study. What is true: parental divorce was included as an ACE, **linear relationship between number of abuse categories and depression**, psychological abuse was not included in original study

Added during revision.
Right Answers.
Not sure about the answer.

Which of the following is true of the Kaiser CDC Adverse Childhood Experiences Study? [SPMM]

Select one:

- Parental divorce was included as an adverse childhood experience
- Psychological abuse was not included in the original study
- The study utilised a questionnaire mailed to households in the UK
- Exposure to childhood abuse and household dysfunction showed no relationship to leading causes of adult deaths
- **There was a significant relationship between the number of childhood abuse categories and depression as an adult**

EXPLANATION: The CDC-Kaiser Permanente adverse childhood experiences (ACE) study is one of the largest studies of the long-term relationship of childhood experiences to important medical and public health problems. The ACE study questionnaire was mailed to over 13 000 adults in California, USA, with a 70% response rate. **Seven** categories of adverse childhood experiences were studied: **psychological, physical, or sexual abuse; violence against mother; and living with household members who were substance abusers, mentally ill or suicidal, or ever imprisoned.** Persons who had experienced four or more categories of childhood exposure, compared to those who had experienced none, had **4 to 12-fold** increased health risks for alcoholism, drug abuse, depression, and suicide attempts. The study found a strong graded relationship between the breadth of exposure to abuse or household dysfunction during childhood and multiple risk factors for several of the leading causes of death in adults.

73. Metabolic syndrome is present in what percentage of patients in CATIE trial. Options **41-50**, 31-40, 40.9%, females:51.6%, Male:36%

The prevalence of metabolic syndrome in CATIE cohort was **40.9% and 42.7%,** respectively using the NCEP and AHA derived criteria. In females it was **51.6% and 54.2%** using the NCEP and AHA criteria, compared to 36.0% (p = .0002) and 36.6% (p = .0003), respectively for males.

74. Research replicated finding about altruism: altruism is present at birth, altruism is an exclusively human behavior, **altruism causes reduced physical perception of pain,** altruism doesn't improve wellbeing/ empathy increases altruism for ethnic outgroups?

EXPLANATION: Altruistic behaviours occur among many animals, including humans. It is well-documented that performing altruistic behaviours has a positive effect on psychological well-being and health. There is also evidence that performing altruistic behaviours relieves physical pain.

75. Increased maternal sensitivity during attachment behaviour is due to increased functional connectivity of: hippocampus and putamen, **hippocampus and amygdala,** hippocampus and limbic system (probably incorrect option).

EXPLANATION: Maternal sensitivity consists of the accurate interpretation followed by prompt and appropriate response to the infant's signals. It is a well-documented predictor of attachment. The hippocampus and amygdala are key components of the limbic system that are centrally implicated in socio-emotional functioning. The amygdala plays a salient role in

orienting to and processing emotionally significant information. The hippocampus is central to learning and memory. The hippocampus and amygdala are activated during interactions between infants and their caregivers. [SPMM]

There are several important structures within the limbic system: **the Amygdala, hippocampus, thalamus, hypothalamus, basal ganglia, and cingulate gyrus.** The limbic system is among the oldest parts of the brain in evolutionary terms: it can be found in fish, amphibians, reptiles and mammals.

76. Epigenetic mechanism by maternal attachment affects which neurotransmitter: **Oxytocin**, serotonin, dopamine.

EXPLANATION: Oxytocin is a crucial regulator of human social behaviour. Individual differences within the oxytocin system – assessed through hormonal levels, epigenetic modification, or genetic variation - have been linked to differential sensitivity to social cues, prosocial behaviours, and stress responsiveness in adults. Early experience in maternal care predicts epigenetic change within the infant oxytocin system, levels of which are in turn reflective of infant temperament. [SPMM]

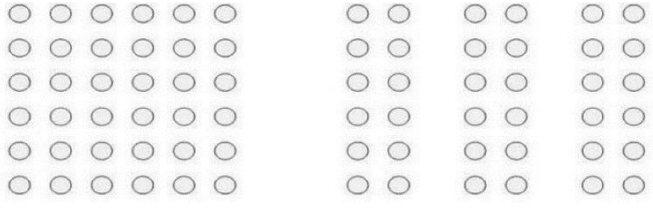
77. Hallucinogenic effects of psychedelics are produced by 5HT1, **5HT2** **(Agonism)**, 5HT3

78. Motor speed assessed by which test: Trail making/ **finger tap test**

79. Non dominant side lesion causes : **constructional apraxia (Gestalt),** constructional apraxia(detail).

Minor details left side big details and layout right gestalt

80. Gestalt principle shown in this figure: proximity/good figure/closure



Select one:

- Closure
- Similarity
- Continuation
- **Proximity**
- Good figure

81. Narrowest therapeutic index: doxepin/ Clomethiazole.

Added during revision.
Right Answers.
Not sure about the answer.

Which of the following drugs has the narrowest therapeutic index? [SPMM]

Select one:

- **Carbamazepine**
- Doxepin
- Clomethiazole
- Zopiclone
- Clonazepam

EXPLANATION: Some drugs have a narrow therapeutic index (a small increase in dose can cause toxicity and a small reduction in dose can cause a loss of therapeutic action). The most commonly prescribed ones are **digoxin, warfarin, theophylline, phenytoin, carbamazepine, and lithium. Also Barbiturates and Clonidine.**

Clomethiazole has narrow therapeutic index as well.

82. Piaget theory. Most important feature of sensorimotor stage: egocentrism/ conservation/ **object permanence**

83. Most biologically important metabolite: diazepam, **oxazepam**, lorazepam **(diazepam and lorazepam are not metabolites)** **If nordiazepam (desmethyl diazepam) in the options then it should be the answer.**

Explanation: Diazepam is metabolized to **nordiazepam (desmethyldiazepam is the major metabolite of diazepam, also named desoxydemoxepam)**, temazepam, oxazepam, temazepam glucuronide (TG), and oxazepam glucuronide (OG).

84. When a person wants to fit in socially and be perceived as a 'nice person' he is exhibiting which level of morality: social order/ **conventional**/ self interest

85. Default mode network involves: **Inferior parietal lobe**/temporal lobe

- **Default (MAPPP)**: **M**edial prefrontal cortex, **A**ngular gyrus, **P**recuneus and **P**osterior cingulate cortex, **P**osterior parietal.
- **Salience (ACTS)**: **A**nterior insular, **C**ingulate cortex (dorsal anterior), **T**halamus and **S**triatum.
- **Central executive DIP**: **D**orsolateral prefrontal cortex, **I**ntraparietal sulcus and **P**osterior parietal cortex.

86. Interviewer rated scale: HADS/ **HAMD**/ PHQ

87. Advantage of using cross sectional method in a study with children: reduced attrition/ **easily accessible data** (fresh question 2023 no 29)

Detouring occurs when parents, rather than directing anger or criticism toward each other, focus the negativity on the child and the parent-child conflict thus serves to distract from the tension in the marital subsystem.Q94

Added during revision.
Right Answers.
Not sure about the answer.

What is an advantage of using a cross-sectional method to study developmental phenomena in children? [SPMM]

Select one:

- High response rates
- Ability to infer causality
- Avoids the confounding eff ect of age
- **Quick data collection**
- Longitudinal insights

EXPLANATION: One advantage of using a cross-sectional method in studying developmental phenomena in children is that it allows for **quick data collection** from participants of different age groups at a single point in time. This method is efficient for obtaining a snapshot of developmental trends across different age cohorts, but it does not provide longitudinal insights into individual development or establish temporal causality. Attrition rates can vary, and the use of this design alone does not inherently improve response rates.

88. Test of premorbid intelligence: **NART** National adult reading test

89. Which is true about alcohol dependence in young people: incidence of binge drinking is lower in 18–20-year-olds, **early onset of drinking is a predictor of later dependence**

90. In vivo measurement of neurotransmitters by which technique: PET, fMRI, MRE, **MRS**

GABA and Glutamate can be detected as well as lithium. Dopamine cannot be detected using MRS because the level is low. MRS uses H1(GABA and Glutamate) and Phosphorous 31

91. [18F]fluorodeoxyglucose (FDG) is used in which technique: **PET**/fMRI

92. Ion channel most implicated in schizophrenia: calcium/ **potassium**/ chloride/ sodium ? calcium activated potassium channel and since they said ion channel we are thinking of potassium

It is reported that the answer should be Potassium (not calcium) even if they ask about the ion, long discussions took place and what we could agree on is the potassium channel as most implicated in schizophrenia. We remain unsure about the ion though.

93. The concept that people's interaction are influenced by each other which in turn are influenced by others - **Circular causality**/ triangulation/ detouring/enmeshment

94. When family members gang up on other family members: Circular causality/ triangulation/ detouring/enmeshment. (We are between Triangulation and Detouring)

Triangulation is when a toxic or manipulative person, often a person with strong narcissistic traits, brings a third person into their relationship in order to remain in control.

Triangulation a child becomes involved in parents' conflictual interactions by taking sides, distracting parents, and carrying messages in order to avoid or minimize conflict between parents

Detouring: detouring scale assesses the degree to which the parents form an **alliance** against the child in a way that functions to create distance between themselves and the child and increase closeness in their own relationship scapegoat syndrome. Detouring occurs when parents, rather than directing anger or criticism toward each other, focus the negativity on the child and the parent-child conflict thus serves to distract from the tension in the marital subsystem.

Circular causality: a sequence of causes and effects that leads back to the original cause and either alters or confirms it, thus producing a new sequence, as in a feedback loop.

Enmeshment: is a trait of family dysfunction that involves poorly defined or non-existent boundaries, unhealthy relationship patterns, and a lack of independence among family members. E.g. This often happens on an emotional level in which two people "feel" each other's emotions, or when one person becomes emotionally escalated and the other family member does as well. A good example of this is when a teenage daughter gets anxious and depressed and her mom, in turn, gets anxious and depressed.

95. Rate limiting step in Alcohol metabolism**: Alcohol dehydrogenase**/ aldehyde dehydrogenase

96. EPS produced at what percentage D2 occupancy: 0.6, **0.8**, 0.2

0.6 to 0.8 is therapeutic effect and 0.8 is EPSE

D2 **Occupancy** – Parkins**O**n's disease (in striatum) [SPMM]. also Akathisia

D2 **Hyper-sensitivity** – Tardive Dyskinesia.

97. Recombination fraction value the means that two genes will be independently assorted:

Options:

A. 0.2,
B. 0.1,
C. **0.5**,
D. 1

RF value of 50 percent or more.The higher the RF value the greater the distance b/w two loci ,high chance of crossing over and chance of independent assortment

N.B.: It varies from 0% to 50%)

98. Metabotropic receptors are : **G protein coupled**

Metabotropic receptors include **muscarinic acetylcholine, Metabotropic glutamate, GABA-B receptors as well as most serotonin receptors (except 5HT-3)**. The **endocannabinoid CB(1) or CB(2) receptors are also metabotropic,** though the endocannabinoids anandamide has been observed to have activating properties at certain voltage gated calcium ion channels.

Metabotropic receptors are indirectly linked to ion channels via signal transduction mechanisms, such as G-proteins

N.B: Ionotropic receptors include: GABA-A, Excitatory Glutamate, Nicotinic Acetylcholine and 5HT-3

Added during revision.
Right Answers.
Not sure about the answer.

Glucocorticoid receptor
Ligand dependent nuclear transcription

July 2023 recall

AMPA/NMDA/Kainate

The following table illustrates the main neurotransmitters and their receptor types:

Neurotransmitter	Ionotropic	Metabotropic
GABA (Gamma-aminobutyric acid)	Yes (GABA-A)	Yes (GABA-B)
Glutamate	Yes (excitatory)	Yes
Glycine	Yes (inhibitory)	-
Dopamine	-	Yes
Norepinephrine	-	Yes
Epinephrine	-	Yes
Serotonin	Yes (5HT-3 only)	Yes (all other than 5HT-3)
Histamine	-	Yes
Acetylcholine	Yes (nicotinic)	Yes (muscarinic)

Metabotropic

Endocannabinoids
CB1 and CB2

99. 1st line treatment in clozapine induced constipation: methylcellulose, senna, lactulose
1st line: Stimulant-senna or Bisacodyl or sodium picosulfate

100. **DBS (Deep Brain Stimulation)** in OCD targets: **anterior limb of internal capsule**

STN, nucleus accumbens

101. Dentate gyrus is a part of: **hippocampus**

102. Which requires longest duration for diagnosis:

- Schizoaffective – **1month**
- Adjustment- **1month**
- Schizophrenia- **1month**
- Delusional disorder- **3month**
- **Schizotypal disorder - 2years**

Post synaptic parasympathetic ganglia is the answer.

103. Muscarinic receptors are mainly found in : parasympathetic ganglia, sympathetic ganglia/ peripheral postsynaptic receptors/ neuromuscular junctions **peripheral parasympathetic system**

104. Aggression is reduced by: **Observing the aggressor receive punishment**, appeasement doesn't have any effect,

> **In Bandura's classic study of imitation learning,** children who saw a model punished for aggressive behaviour tended to exhibit fewer aggressive responses than children who saw the model rewarded for such behaviour, or than those who saw the model neither rewarded nor punished.

105. What is true about pre linguistic communication: **Extended in autistic children,** lost over later life.

106. What is true about adolescent turmoil:

Select one

- self-deprecation and worthlessness is common,
- **Difficulties with parents are common,**
- adolescents with reduced parental contact and social support are underrepresented in group with psychiatric diagnosis,
- most will undergo peer group turmoil

Explanation: According to Hall, most adolescents go through a period of "storm and stress" as they are coping with the transition from childhood to adulthood. This may result in conflicts with parents and authority figures, mood disruptions, and risk-taking behaviors.

107. Bereavement in children:

Select one

- young children more affected than older,
- **Boys more affected than girls**,
- reduced cortisol response in later life, resolved in 3 months,
- school going children revert to magical thinking,
- regression occurs within 3 months.

108. At what stage in Conrad's model does the individual start to find new meaning for psychological processes: Trema, **apophony**, anastrophe, residuum.

Apophenia is the tendency to mistakenly perceive connections and meaning between unrelated things. **Klaus Conrad** defined it as "unmotivated seeing of connections accompanied by a specific feeling of abnormal meaningfulness"**.** He described the early stages of delusional thought as self-referential, over-interpretations of actual sensory perceptions, as opposed to hallucinations. **Trema (stage fright) is the feeling that something very important is about to happen**. Attention is drawn toward irrelevant stimuli, thoughts, and associative connections, which are distressing and unpredictable. **Anastrophe is the feeling that everything revolves around the patient.** [SPMM]

109. Fluent plausible lying with fantastical elements: confabulation/ falsification/ **pseudologia fantastica**

110. Southern blot used for: **DNA**/ protein/ RNA/ DNA binding protein.

SNoW DRoP

Southern blot – DNA

Northern blot – RNA

Western blot – Protein

111. Gregory (1970) theory about perception? **Correct was perception depends on making inferences on past experiences**

Added during revision.
Right Answers.
Not sure about the answer.

Richard Gregory proposed that perception involves a lot of hypothesis testing to make sense of the information presented to the sense organs. **Our perceptions of the world are hypotheses based on past experiences and stored information**

112. Naloxone is mixed with buprenorphine because: **reduces chances of misuse (**by deterring patients from injecting it as it will cause severe withdrawal if injected**)**

113. Many bystanders but no one is likely to help. Options: **Bystander effect** Bystander Apathy

114. Policeman comes to scene of crime, sees a well-dressed man and decides he cannot be the culprit. **Halo effect**/ barnum effect/ pygmalion effect Horn effect when negative opinions

The **Barnum** Effect in psychology, also known as **The Forer Effect**, is when an individual believes that personality descriptions apply specifically to them, for example, reading your horoscope in a newspaper and realising it's surprisingly accurate. We all know that horoscopes are written for the masses, but we can't help feel but it's meant for us

The **Pygmalion** effect, or **Rosenthal** effect, is a psychological phenomenon in which high expectations lead to improved performance

115. Which is first rank symptom: **Man sees clock strike 12 and believes he is Messiah/** voices commanding him/ believing his neighbors are poisoning him Delusional perception

Broadly, **first-rank symptoms** include: auditory hallucinations (including running commentary and voices conversing); somatic hallucinations; thought withdrawal, insertion and interruption; thought broadcasting; delusional perception; and passivity (actions felt to be influenced by external agents. The first rank symptoms include:- Running commentary, Thought echo, Voices heard arguing Thought insertion, Thought withdrawal, Thought broadcast (thoughts are perceived as leaving the subject's head/mind) Delusional perception, Somatic passivity, Made affect (belief that feelings are controlled by an outside force) Made volition (belief that impulses and/or behaviour are controlled by an outside force)

116. Last part of brain to be myelinated: **Anterior frontal pole**/ caudate/ amygdala

117. New information affecting old memories:

Select one:

A. proactive falsification
B. Ganser
C. **Retroactive falsification**
D. falsification
E. paramnesia
F. Cryptamnesia

- **Proactive** - old information interrupts new memories
- **Retroactive** - new information interrupts old memories

Pro new not formed
Retro old can't be recalled.

Added during revision.
Right Answers.
Not sure about the answer.

118. Sherif's experiments on group were for: options about objectives?

SHERIFS experiment is about **GROUP CONFLICT.** It was done in a school camp which studied conflict between groups.

119. Caffeine effect is **DIRECTLY reversed** by: sigma receptors/ **Adenosine Receptors Agonist**

Caffeine works by **blocking of adenosine receptors and inhibition of phosphodiesterase's**. Its action can be reversed by **Adenosine Agonist**

120. Healthcare workers wash hand more when they are being observed, what explains this: **Hawthorne effect**

121. Locus coeruleus is involved in which pathways: serotonergic, ACh, **Noradrenergic**

122. Intellectual disability diagnosis requires IQ to be how many standard deviations away from mean: **2** / 3 / 4 / 5

123. Difference between hypomania and mania: decreased sleep need/ flight of ideas/ **psychotic symptoms**

124. What is the sequence of 3 stages of separation anxiety:

Select one:

A. Protest, detachment, despair
B. **Protest, despair, detachment**.

125. Which pattern of inheritance shown in this pedigree: autosomal dominant/ autosomal recessive/ x-linked dominant/X-linked recessive. Missing the predugree.

126. In Freud's theory, not resolving a stage results in: **fixations**/?

https://www.verywellmind.com/what-is-a-fixation-2795188

127. Mild personality disorder in ICD-11 is equivalent to what in DSM-V: some impairment / no impairment/ severe impairment, **Moderate impairment**

128. Woman says she sees colours when she hears husband's voice or laughter, this is: **Synaesthesia**

[SPMM] **Explanation**: **Synaesthesia** is the experience of a stimulus in one sense modality producing a sensory experience in another (e.g. tasting sounds or hearing colours).

Reflex hallucinations are a **morbid form of synaesthesia**. For example, a patient felt a pain in her head (somatic hallucination) when she heard other people sneeze (the stimulus) and was convinced that sneezing caused the pain.

Functional hallucination, an auditory stimulus causes a hallucination, but the stimulus as well as the hallucination is experienced. In other words, the hallucination requires the presence of another real sensation. (For example, a patient with schizophrenia first heard the voice of God as her clock ticked; later she heard voices coming from the running tap and voices coming from the chirruping of the birds.) Patients can distinguish both features.

Same modality

Added during revision.
Right Answers.
Not sure about the answer.

129. Sample of speech: this is a hospital, hospital is where doctors are trained, doctors are ? Apple keeps doctor away, newton saw apple, newton discovered gravity. What is this: loosening of associations / **flight of ideas.**

> **Flight of ideas** is where your thoughts move very quickly from idea to idea, making links and seeing meaning between things that other people don't**. Loosening of Association** can manifest in speech as an individual moving quickly from one idea to an unrelated one.

Loosening of association, derailment and knights move all same.

> **What is the difference between Knight's move thinking and flight of ideas?**
>
> **Flight of thought** - this is where the patient moves quickly from one idea to another, often half-way through a sentence, with no apparent association between ideas. **Knight's move thinking** (aka **Derailment**) - patient moves from one idea to another with strange illogical associations between the ideas.

130. Parenting with rules but no rigidity and responsiveness present: **Authoritative**

131. Child that follows ruled but has poor self-esteem, what type of parenting: **Authoritarian**

132. Superstition theory of OCD states that:

Select one

- Compulsions formed due to immediate relief from anxiety/
- Behaviour and stimulus are paired by chance/
- **Mental processes by which threatening thoughts are removed (likely to be the answer after long discussion)**

Superstition theory is about magical thinking.

- **Apart from OCD, it is also associated with Schizotypal.**

133. What differentiates oppositional defiant disorder from intermittent explosive disorder: physical aggression / **outburst out of proportion to stimulus**–

> **Intermittent explosive disorder** is characterized by **repeated brief episodes of verbal or physical aggression or destruction of property that represent a failure to control aggressive impulses**, with the intensity of the outburst or degree of aggressiveness being grossly **out of proportion** to the provocation or precipitating psychosocial stressors. The symptoms are not better explained by another mental, behavioural, or neurodevelopmental disorder and are not part of a pattern of chronic anger and irritability (e.g., in oppositional defiant disorder). The behaviour pattern is of sufficient severity to result in significant impairment in personal, family, social, educational, occupational or other important areas of functioning. [ICD11]
>
> **Oppositional defiant disorder** is a **persistent** pattern (e.g., **6 months or more**) of markedly defiant, disobedient, provocative or spiteful behaviour that occurs more frequently than is typically observed in individuals of comparable age and developmental level and that is not restricted to interaction with siblings. Oppositional defiant disorder may be manifest in prevailing, persistent angry or irritable mood, often accompanied by severe temper outbursts

ICD11 what differentiates IED from ODD ?
Significant physical aggression

or in headstrong, argumentative and defiant behaviour. The behavior pattern is of sufficient severity to result in significant impairment in personal, family, social, educational, occupational or other important areas of functioning

134. Inflammatory marker common in depression: IL-4/ **IL-6**/ IL-11/ IL-13

The evidence is most convincing regarding depression where meta-analyses have found increased levels of inflammatory markers, including elevated levels of pro-inflammatory cytokines interleukin-6 **(IL-6)** and tumour necrosis factor-α **(TNF-α)**, as well as c-reactive protein **(CRP)**, in depressed subjects as compared to controls (Santoft, 2020).

135. Which of the following is not seen in serotonin syndrome: nausea/ headache/ tremors/ **diplopia**

		Neuroleptic malignant syndrome	Serotonin syndrome
Precipitated By		Dopamine Antagonists	Serotonergic Agents
Onset		Variable, 1-3 days	Variable, < 12 hours
Identical Features	Vital Signs	Hypertension Tachycardia Tachypnoea Hyperthermia (> 40°C)	Hypertension Tachycardia Tachypnoea Hyperthermia (> 40°C)
	Mucosa	Hypersalivation	Hypersalivation
Overlapping Features	Skin	Diaphoresis Pallor	Diaphoresis
	Mental Status	Variable, stupor, coma, alert	Variable, agitation, coma
	Muscles	'Lead-pipe' rigidity in all muscle groups	Increased tone, especially in lower extremities
Distinct Features	Reflexes	Hyporeflexia	Hyperreflexia Clonus (unless masked by increased muscle tone)
	Pupils	Normal	Dilated
	Bowel Sounds	Normal or decreased	Hyperactive

136. Adding new information to existing information and expanding it is: adaptation/ **assimilation**/ accommodation

[SPMM] According to **Piaget**, a schema is defined as the basic building block or unit of intelligent behaviour. Schemas consist of organized past experiences to understand future experiences. We have physical schema such as bike riding schema and mental schemas such as addition, multiplication and division schema.

Within each developmental stage, functioning is generally internally consistent and stable and thus said to be in equilibrium. Stage-to-stage transformation occurs as a result of interaction with environment, whereas existing schemas cannot solve the environmental realities.

Adaptation is the process of **fitting schemas to environmental information**. Adaptation **can occur either as assimilation or accommodation**. In **assimilation** new information is incorporated into existing schemas without restructuring the schemas. In **accommodation**, the schemas are restructured to 'accommodate' newly learnt information. Equilibration is achieved when all information properly fit into the schemas via either processes of

adaptation. During each developmental stage, the child will experience cognitive disequilibrium, which through adaptation, gets solved, and equilibration results. Each time that equilibration occurs, the child produces more effective schemata or mental structures.

137. Internal Structure of knowledge that humans have to understand a concept is: **schema**/archetype

138. Which of the following medications work by acetylcholine esterase inhibition: **galantamine**/ memantine

139. What is the mechanism of Memantine's action: NMDA antagonism/ **non-competitive NMDA antagonist**

140. Phagocytic cells in the nervous system: astrocytes/ **microglia**/ macrophages

141. Which cells secrete glutamate: pyramidal/ Schwann/ **Astrocyte**/ stellate

Pyramidal cell uses it but astrocyte transports or secrets it

142. GABA is projected onto which cell type in the striatum: pyramidal / **Medium spiny neurons**

143. Which cells responsible for myelination in the CNS: **Oligodendrocyte**/ Schwann

144. Which of the following is seen in pregnancy: **Increased GFR**/ increased binding of drugs to albumin/ decreased volume of distribution of water soluble drugs

Changes in pregnancy:

[SPMM] Pregnancy is associated with several **pharmacokinetic changes**:

- Delayed gastric emptying,
- Decreased GIT motility,
- Increased volume of distribution (5%),
- Decreased drug-binding capacity,
- Decreased albumin level
- Induced liver metabolic pathway,
- Increased GFR & renal clearance.

Psychotropic medication usually passes from the maternal blood to the foetus due to lack of strong barrier, but rate and amount of transfer are variable. Higher doses are associated with higher serum level in the infant.

145. Lyonization is seen in: Down/ **Klinefelter**

146. Test for executive function: clock drawing/ digit span/ stroop/ **Wisconsin card sorting**

Test for executive function WCST, Category test, trail making test, Stroop test. WCST is for set shifting

The following cognitive tests are used in psychiatric practice and commonly come up in the MRCPsych exams.

Primary area of cognition tested	Tests
Premorbid intelligence	National Adult Reading Test (NART)
General Intelligence	Wechsler Adult Intelligence scale (WAIS), Raven's Progressive Matrices
Visual memory / Visio spatial skills	Rey-Osterrieth Complex Figure
Attention	Continuous Performance Tasks
Language	Boston naming test, Animal fluency
Executive function	Stroop test, Wisconsin Card Sorting Test (WCST), Tower of London, Animal fluency, trail making test
Other	Halstead-Reitan battery (for brain damage)

147. Test for set shifting: Stroop/ clock drawing/ **Wisconsin card sort**

Frontal lobe function	Test or examination
Initiation	Verbal and categorical fluency
Abstraction	Proverbs, similarities, cognitive estimates
Problem solving and decision making	Tower of London, Cambridge stockings, gambling tasks
Response inhibition and set shifting	Alternating sequences, go-no-go test, Luria motor test, trail making test, **Wisconsin card sorting test, Stroop test**

148. Reciprocal inhibition is employed in: **Systematic desensitization**

149. What is true about Stupor:

Select one:

- patient appears awake and alert
- It exists on a spectrum of wakefulness to coma
- consciousness is decreased.

Stupor, in the sense used in psychiatry, refers to a condition in which the patient is immobile, mute, and unresponsive, but appears to be fully conscious in that the eyes are usually open and follow external objects. If the eyes are closed, the patient resists attempts to open them.

Reflexes are normal and resting posture is maintained. Stupor may occur in catatonia. Note that in neurology the term 'stupor' is used differently and generally implies an impairment of consciousness. [Shorter Oxford textbook]

150. If **carbamazepine** and **phenytoin** were given together, which malformation is expected to be seen in infants: **Finger nail hypoplasia**

151. Coiled bodies are seen in which condition?

A. **Progressive supranuclear palsy**

[Psych-Mentor] **Explanation**: Progressive supranuclear palsy is rare disorder, which causes problems with gait and balance. It is often associated with alterations of mood and behaviour and also with dementia.

Macroscopic changes include:-

- Pallor of substantia nigra (with sparing of the locus coeruleus)
- Mild midbrain atrophy
- Atrophy of the superior cerebellar peduncles
- Discolouration of the dentate nucleus

tufted astrocytes and the predominance of 4-repeat (4R) tau isoforms in both neuronal and glial tau inclusions,

Microscopic changes include:-

- Gliosis
- Neurofibrillary tangles and tau inclusions in both astrocytes and oligodendrocytes (**coiled bodies**), particularly in the substantia nigra, subthalamic nucleus and globus pallidus.

151. Enzyme deficient in acute porphyria ?

Select one:

A. HMBS
B. galactosidase
C. Aryl sulphatase
D. glucominidase
E. all of the above

Explanation: **Acute intermittent porphyria (AIP) is a rare metabolic disorder that is characterized by partial deficiency of the enzyme hydroxymethylbilane synthase (also known as porphobilinogen deaminase). This enzyme deficiency can result in the accumulation of porphyrin precursors in the body.**

[SPMM] Acute intermittent porphyria (AIP) is one of the groups of disorders of **haem metabolism**, characterised by neurological and psychiatric manifestations without obvious cutaneous markers.

AIP manifests itself by abdomen pain, neuropathies, and constipation, but, unlike most types of porphyria, patients with AIP do not have a rash. It is **an autosomal dominant** disorder with the presentation starting between ages 18 and 40. It is episodic in nature, and the episodes are often triggered by certain medications including estrogens, barbiturates and

Added during revision.
Right Answers.
Not sure about the answer.

benzodiazepines. Diclofenac can precipitate an episode. Psychiatric manifestations include depression, anxiety, delirium and psychosis. Most important lab test is demonstrating increased urinary porphobilinogen during acute attacks. **Treatment is aimed at reducing haem synthesis by administering haemin.**

Porphyria can be precipitated by psychotropics drugs. Common ones include:-

- **Barbiturates**
- **Benzodiazepines**
- **Sulpiride**
- **Certain mood stabilizers**

Personality disorder
ICD 11 to DSM 5
No is none
Difficulty is some.
Mild is moderate
Moderate is severe
Severe is extreme.

I know some concepts for Autoimmune

When it's hyponatremia,old with hand and face seizure it's *Ig1*

When flu flu and seizures🤧 *NMDA*

Abnormal behavior, flair MRI signal ,autonomic instability,jerks *CASPR2* (Casper is an automatic abnormal jerk)

3rd optic nerve and spine inflammation just memorise it's *Aquaporin 4*

Lung cancer ... *AMPAR*

Associated with diarrhea .... *DPPX*

Sleep disturbance in elderly... *IgLONS*

Good luck everyone

Added during revision.
Right Answer/Important points.
Not sure about the answer.

1. Area under the curve – distribution properties of the drug, absorption?

Plasma concentration
AUC is **concentration of drug over time.** The amount eliminated by the body (mass) = clearance (volume/time) - AUC (mass x time/volume)

2. Absorption constant is independent of: Extravascular route of administration, Form of drug (tablets, suspension) **Elimination constant**

3. Bioavailability increased: Poor absorption, extensive pre-systemic metabolism.

Bioavailability is increased through change route of administration e.g. IV will give 100% bioavailability.

4. Which of the following would increase by at least 30% if taken with Lithium: **Amisulpride**, Citalopram:

Lithium is excreted through renal route and alters effects of drugs which are excreted through renal route such as **Amisulpride**

5. EMI – 40year old woman is unable to remember a traumatic experience – **Dissociative amnesia**, dissociative identity disorder, dissociative neurological symptom, possession trance disorder.

6. EMI – A woman is unable to remember personal identity – same options as above: **Dissociative identity disorder**

7. EMI – Regarding type of power, patient stops using a medication following a Dr's advice : **Expert power**, coercive power (if doctor forced the patient), information power (if doctor had shared complete effect and side effects to patient).

8. EMI - Regarding type of power, Group campaign on negative effects of smoking - expert power, coercive power, **Information power**

Legitimate power	The power held by an individual in an organisation
Referent power	The power held by a person with charisma who people gravitate towards
Expert power	The power of a person with specialist skills or knowledge
Reward power	The power of a person in a position to provide rewards
Coercive power	The power of a person who has the ability to apply negative influences such as demotion

Added during revision.
Right Answer/Important points.
Not sure about the answer.

Other terms to be aware of include:

Outcome power - the power of a person to bring about outcomes.
Social power - the power of a person to **influence** others so that they will bring about outcomes.
Counterpower - the power of the oppressed to counterbalance the elite.

9. EMI on medical ethics: Patient on admission was found to have a suspicious mass on routine chest X-ray and the team decides not to tell him: Beneficence, Non-maleficence, Autonomy, **Paternalism** (when a physician or other healthcare professional makes decisions for a patient without the explicit consent of the patient), Utilitarianism (TELEOLOGY).

Paternalism in medical ethics refers to situations in which healthcare professionals make decisions on behalf of patients, often without their full consent, for what is believed to be in the patients' best interests. It involves a degree of control or authority by the healthcare provider to protect or promote the patient's well-being, even if it limits the patient's autonomy or self-determination.

Here's an **example** of paternalism in medical ethics - A patient named Sarah is a 45-year-old woman who smokes heavily, and her physician, Dr. Johnson, has been urging her to quit smoking for years due to the severe health risks associated with it. Sarah has made numerous unsuccessful attempts to quit and is fully aware of the potential health consequences. Paternalistic Action: Dr. Johnson, concerned for Sarah's health, decides to prescribe her a medication to help her quit smoking without discussing this decision with her.

There are three key ethical theories which have dominated medical ethics:

- Utilitarianism
- Deontological
- Virtue based

Utilitarianism:
Founded on the work of **Jeremy Bentham** (1748–1832) and **John Stuart Mill** (1806–1873).

The greatest good for the greatest number. It is a consequentialist theory, as it holds that the predicted outcomes (i.e. the consequences) of an action are the most morally important component of that action. Bentham thought that humans were governed by two factors: 'pleasure and pain', thought it was possible to classify how good an action is by measuring how much pleasure or pain was brought about by that action. He called this process '**felicific calculus**'.

Mill differed, he thought that people should ordinarily stick to moral rules. He is still a utilitarian, because he held that these moral rules should be calculated using the principle of utility. This is what is known as **rule utilitarianism**. For example, lying in general might produce less utility than telling the truth. Therefore, there is a rule that says 'Do not lie!'.

The use of evidence-based medicine to produce guidelines on cost-effectiveness by organizations such as the National Institute for Health and Clinical Excellence (NICE) is utilitarian, based on ideas of what promotes the greatest good for the greatest number.

Added during revision.
Right Answer/Important points.
Not sure about the answer.

Deontological:

Deontology covers those theories that emphasize moral duties and rules, rather than consequences (from the Greek deon , meaning 'duty'). A good example is the Ten Commandments. The General Medical Council's code of conduct, Good Medical Practice , is essentially deontological, founded on the duties of a doctor, and can be seen as a rulebook.

Deontology is associated with **Immanuel Kant** (1724–1804). He believed that morality was not dependent on how much happiness resulted from particular actions. Rather, he thought morality was something humans imposed upon themselves because they are rational beings.

Kant argued that we can find out which moral rules to obey by using our powers of reason. He said that by seeing whether our desires can be applied universally, we can tell whether or not they follow rational moral principles. This '**universalizability**' test is called the '**categorical imperative**'. It states:

'act only on that maxim through which you can at the same time will that it should become a universal law'

This means that we should behave in such a way that we can imagine everyone can behave. For example, if our 'maxim' or 'desire' is to 'steal other people's things when we want them', we need to consider whether or not this maxim could be held for everyone.

Virtue based:

In ethical assessment, **utilitarians** look at the **consequences** of a particular act, and **deontological** ethicists at the **conformity** of the act with pre-defined rules. **Virtue ethics stands in a different position:** its foundation is the ethical characteristics of a person.

Certain personal characteristics are recognized as virtues and others as vices. **Honesty, generosity, compassion, courage, justice, fidelity, and veracity are virtues that characterize an ethical person,** while deception, selfishness, cruelty, infidelity, and disingenuousness are some of the many characteristics that denote a vicious person.

Virtue ethics is associated with the concept of a good, happy, flourishing life - in Greek 'eudaimonia'.

The four key principles (Principlism):

More recent frameworks have attempted to reconcile different theories and values. In the late 1970s, two Americans, **Tom Beauchamp and James Childress**, introduced the idea of the 'four principles' or 'principlism'. The approach, developed in the United States, is based on four common, basic prima facie moral commitments.

*Note: 'Prima facie' (translates as '**at first glace')** is a term introduced by the English philosopher W D Ross, means that the principle is binding unless it conflicts with another moral principle - if it does we have to choose between them. The idea is that our moral duties are conditional duties. Our actual duty in any situation will depend on the context.*

Basic ethical principle	Description
Autonomy	refers to a patient's right to make their own decisions. This is affected by their capacity.
Beneficence	refers to the expectation that a doctor will act in a way that will be helpful to the patient
Non-maleficence	refers to the fact that doctors should avoid harming their patients. Associated with the Latin phrase 'primum non nocere' or 'above all, do no harm'.
Justice	refers to the expectation that all people should be treated fairly and equally. There is, however, no single answer as to what constitutes fair and equal distribution.

10. EMI on medical ethics – Patient developed agranulocytosis following Clozapine, Dr decides not to prescribe same medication as a re-challenge – **Non-maleficence**

11. EMI on medical ethics – Man with depression, made an advanced decision not to have ECT. When admitted, the decision was respected – **Autonomy**

12. Which of the following describes a "flash bulb" experience: heightened visual stimulation, **intense emotion**

13. Which of the following was listed in more than 1 category in ICD 11:

Select one:

A. Depression,
B. **Dissociative Disorder** (primary due to psychiatry and secondary due to organic reason), Anxiety Disorder,
C. Factitious disorder.

In addition catatonia is also mentioned in more than 1 category

14. Rapid Catalytic rate with substrate – GABA A, COMT, Glutamate

15. Which of the following has categories in ICD 11 and cannot have further subtypes – Depression, Dissociative Disorder, Anxiety Disorder, **Bipolar 1**

16. What differentiates Complex PTSD (on top of PTSD self-hate, interpersonal relation disturbed, diminished self-worth, and affective instability) from PTSD: **Repetitive, intense, severe traumatic experiences**, continuing threats from the memory of trauma, psychotic symptoms, suicidal ideation.

Complex PTSD is characterized by:

1. Severe and pervasive problems in **affect** regulation;
2. Persistent beliefs about oneself as diminished, defeated or worthless, accompanied by deep and pervasive feelings of shame, guilt or failure related to the traumatic event; and

Added during revision.
Right Answer/Important points.
Not sure about the answer.

3. Persistent difficulties in sustaining relationships and in feeling close to others.

[SPMM] As defined in ICD-11, which of the following best differentiates complex PTSD from PTSD?

Select one:

A. Avoidance of memories of traumatic events
B. Re-experiencing of traumatic events
C. Dysphoria
D. Persistent difficulties sustaining relationships
E. A series of continued traumatic events

EXPLANATION: Whereas the ICD-11 diagnostic requirements for complex PTSD include all essential features of PTSD, the diagnosis of complex PTSD also requires the additional essential features of severe problems in affect regulation, persistent negative beliefs about oneself, and persistent difficulties in sustaining relationships.

17. Which of the following drugs do not cross the Blood-brain barrier: Aripiprazole, Lithium, **Amisulpride** (**it is transferred through active transport**), Risperidone, Olanzapine

18. Which of the following drugs has a poor absorption:

- Risperidone 75% PO,
- **Lurasidone (5-15% PO),**
- Asenapine (35% sublingual)

19. Which of the following drugs increases the risk of Autism – Carbamazepine, Amisulpride, Levetiracetam, **Sodium Valproate**

Benzodiazepine intake during early pregnancy may be teratogenic, and their intake during late pregnancy may be associated with **neonatal withdrawal syndrome and 'floppy infant' syndrome**

Fetal alcohol syndrome is caused by alcohol
Serotonin withdrawal syndrome (Neonatal adaption syndrome 30%)

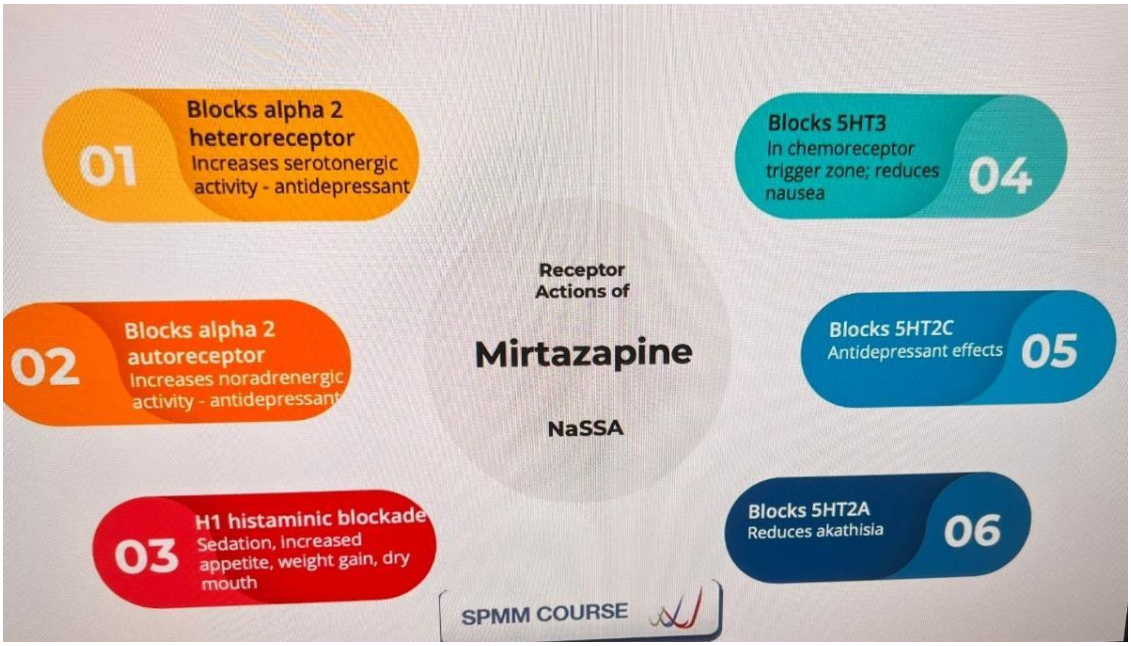
20. Action of Mirtazapine is mediated through what receptor: (We believe the question is asking about the Antiemetic action of Mirtazapine)

Select one:

A. 5HT2A
B. **5HT3**
C. H1
D. 5HT6
E. 5HT7

5HT3 Antagonism.

Antidepressant	Mechanism
Mirtazapine	Noradrenaline and serotonin specific antidepressant (NaSSa) 5HT2A antagonist, 5HT3 antagonist, H1 antagonist, alpha 1 and alpha 2 antagonist, moderate muscarinic antagonist



21. How long does IM Paliperidone monthly depot take to reach maximum plasma concentration: 3 days, **13 days**, 23 days, 33 days, 43 days

22. By what mechanism does Lithium-induced polyuria occur – Increased sensitivity to ADH, Changes in chloride channel and signalling, By increasing sodium excretion leading to hyponatremia, **reduced expression of aquaporins** (collecting tubule distal tubule ADH **aquaporin expression decreased leads to diabetes insipidus**)

23. EMI – Likely changes to be seen in **Vomiting** in Anorexia nervosa (binge eating/purging type) – gastric ?, acute pancreatitis (refeeding after anorexia), High T3, decreased colonic motility(In laxative type), Low GH and Cortisol, **Parotid swelling (in vomiting type)**

24. EMI – Likely changes to be seen in **laxative** abuse in Anorexia nervosa – same options as above **decreased colonic motility(In laxative type)**

Added during revision.
Right Answer/Important points.
Not sure about the answer.

25. Attention difficulty in a teenager, part of brain responsible – orbitofrontal cortex, dorsomedial prefrontal cortex, insula.

***Attention related to DLPFC** (executive function, planning and goal-directed, attention)

Basal ganglia structures increases whilst rest of cortical structures and grey matter decrease in volume

26. EMI on Radiological finding in Antipsychotic use in a schizophrenia patient – **reduced volume of striatum**, dorsolateral prefrontal cortex

(Although further study is required, first-generation antipsychotic (FGA) exposure appears to correlate with **enlargement of the caudate over time**, while **cumulative antipsychotic exposure may be associated with Gray matter loss over time.** The reduction of grey matter progresses with the duration of illness, especially in the **temporal** lobe, and seems to be associated with antipsychotic treatment. **However, even drug-naïve patients show volume reductions (albeit not as pronounced as treated patients), especially in the caudate nucleus and thalamus.)** (Cerebral asymmetry is also reduced in both affected individuals and healthy relatives, particularly in the planum temporale which shows diminished left–right asymmetry in patients with schizophrenia compared with healthy controls)

27. Part of brain involved in Auditory hallucinations – **Superior temporal lobe (superior temporal gyrus**), fusiform gyrus

The most consistent finding of structural neuroimaging studies of patients with auditory hallucinations is **reduced grey matter volume in the superior temporal gyrus, including the primary auditory cortex**

28. Antipsychotics with low availability in the brain- **Sulipride, Amisulpride** (don't cross BBB)

Substituted Benzamides

29. Which one is activated during mentalisation in adolescents more than in adults: medial orbitofrontal cortex, lateral orbitofrontal cortex

Following further discussion it seems that **Dorsomedial prefrontal cortex is the possible answer not the amygdala.**

Amygdala is more activated in mentalisation in adolescents.

- In adults the pictures activate the **amygdala and regions of the frontal cortex** (frontal lobes) in the brain. In adolescents, the same images activate **the amygdala, but not the frontal lobes**)
- In adolescents limbic system matures before prefrontal cortex
- Conjunction and contrast analyses revealed that children and adolescents (N = 479) recruit similar but fewer regions within core mentalizing regions. Subgroup analyses revealed an early continuous engagement of middle **medial prefrontal cortex,** precuneus and right temporoparietal junction in younger children (8–11 years) and

Added during revision.
Right Answer/Important points.
Not sure about the answer.

adolescents (12–18 years). Adolescents additionally recruited the left temporoparietal junction and middle/inferior temporal cortex.

Overall, the observed engagement of the **medial prefrontal cortex, precuneus and right temporoparietal junction during mentalizing** across all ages reflects an early specialization of some key regions of the social brain.

30. What explains "cocktail party phenomenon"? – **selective auditory attention**, focused attention, clouding of consciousness

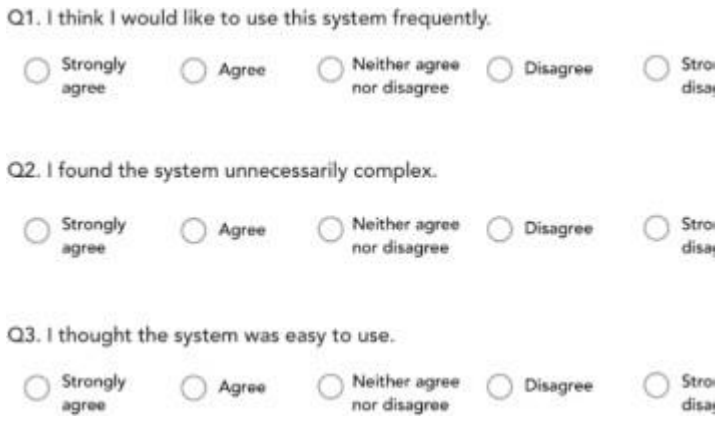
31. EMI Genetic antigen??

32. Which of these is true about Likert scale? – free from bias, similar to visual analogue, …. Differential scale

Likert Scale

Here, respondents are asked to indicate a degree of agreement or disagreement with each of a series of statements. Each scale item has 5 response categories ranging from strongly agree to strongly disagree.

QS



33. Which scale measures "no pain at all" to "excruciating pain" – Likert, **Semantic**, nominal, Thurstone scale Semantic Differential scale

34. Which of these is true about Purines- **gotten primarily from diet** (also synthesized in the body) -not synthesized in the body, made up of cytosine and thymine, only present in eukaryotic cells.

35. What is true of Prejudice – **obtaining more information about a class of people will reduce prejudice**

36. Q on stigma – "Courtesy stigma" is from being with someone who is stigmatised, "Felt stigma" is the feeling that one will be stigmatised by others due to a condition or experience

Added during revision.
Right Answer/Important points.
Not sure about the answer.

Stigma type	Description
Discredited and discreditable stigma	The discredited are individuals who have a stigma that is predominantly visible such as race/ethnicity, gender, or physical disability. In contrast, the discreditable are individuals who have a stigma that is predominantly concealable such as mental illness, HIV infection, or sexual minority status. Thus, these terms refer to the visual conspicuousness of the stigmatized attribute.
Felt stigma	Felt stigma (**internal** stigma or self-stigmatisation) refers to the shame and expectation of discrimination that prevents people from talking about their experiences and stops them seeking help Might be stigmatized
Enacted stigma	Enacted stigma (**exte rnal** stigma, discrimination) refers to the experience of unfair treatment by others
Courtesy stigma	The stigma that attaches to those who are merely associated with a stigmatized person

37. EMI on **GABA A receptor** – heterodimer (NMDA), homodimer ,**heteropentomer**, heterotetramer, serpentine (G-protein)

[SPMM] **Other Pentameric receptors:**

- NMDA
- NICOTINIC
- 5HT3
- GLYCINE

NMDA is a heterotetramer

Metabotropic glutamate are heterodimers while ionotropic are heterotetramer

N.B. Ionotropic glutamate - tetrameric

38. EMI on **Glutamate** receptor – same options as above **Heterodimer** (The ionotropic glutamate receptors are multimeric assemblies of four and are subdivided into three groups (AMPA, NMDA and Kainate receptors) based on their pharmacology structural similarities.

39. EMI on types of gait seen in Neurological disorders – UMN lesion (spastic/scissor)

Gait	Description	Associated conditions
Festinating (Shuffling)	Posture is stooped forward. Gait initiation is slow and steps are small and shuffling	Parkinson's disease
Ataxic	Gait is wide-based with truncal instability and irregular lurching steps which results in lateral veering and if severe, falling	Cerebellar disease, e.g. Wernicke's
Antalgic	Stance phase of gait is abnormally shortened relative to the swing phase, usually done to minimise pain	Lower limb trauma
Spastic (scissor or diplegic)	Rigidity and excessive adduction of the leg in swing , plantar flexion of the ankle, flexion at the knee, adduction and internal rotation at the hip, and contractures of all spastic muscles	Cerebral palsy
Steppage (neuropathic or equine)	High-stepping gait so as to prevent scraping of the toe on the ground	Foot drop
Myopathic (waddling)	A broad-based gait with a duck-like waddle to the swing phase, the pelvis drops to the side of the leg being raised with forward curvature of the lumbar spine, and a marked body swing	Proximal myopathy

Added during revision.
Right Answer/Important points.
Not sure about the answer.

Gait	Description	Associated conditions
Pigeon	Toe(s) point(s) inwards when walking	Structural abnormalities
Trendelenburg	During the stance phase, the weakened abductor muscles allow the pelvis to tilt down on the opposite side. To compensate, the trunk lurches to the weakened side to attempt to maintain a level pelvis throughout the gait cycle	Poliomyelitis or muscular dystrophy
Stomping	Bilateral high steppage due to lack of proprioception	Friedreich's ataxia
Magnetic	Feet seem as if magnetically attracted to the floor	Normal pressure hydrocephalus
Choreiform (hyperkinetic)	Irregular, jerky, involuntary movements in all extremities. Walking may accentuate their baseline movement disorder	Sydenham chorea, Huntington's
Sensory	In an effort to know when the feet land and its location, the patient will slam the foot hard onto the ground in order to sense it	proprioceptive loss e.g. diabetes, B12 deficiency
Hemiplegic	Unilateral weakness on the affected side, arm flexed, adducted and internally rotated. Leg on same side is in extension with plantar flexion of the foot and toes. When walking, the patient will hold his or her arm to one side and drags his or her affected leg in a semicircle (circumduction) due to weakness of leg flexors and extended foot	Stroke

**Apraxic gait- can't initiate gait (tumors, hydrocephalus)*

40. Gait seen in Normal Pressure Hydrocephalus (**magnetic**)

41. Gait seen in Cerebellar disease (**ataxic**)

42. What part of the brain is more active in teenager than an adult: Limbic-system, amygdala

(The dorsomedial prefrontal cortex (dmPFC) is part of the mentalizing network, a set of brain regions consistently engaged in inferring mental states)

Following further discussion we think it is **dorsolateral prefrontal cortex** but in PubMed it is mentioned that **Dorsomedial prefrontal cortex.** Hence if dorsolateral prefrontal cortex is not in option we can go for **Dorsomedial prefrontal cortex.**

43. Q. on maternal attachment with infant – hippocampus and limbic system, amygdala and hippocampus, hippocampus and dorsolateral prefrontal cortex

44. What receptor does Cocaine act on – **Dopamine Transporter**, GABA, Acetylcholine

Cocaine (modafinil and methylphenidate) **directly blocks the dopamine transporter (DAT)** (Massaro 2002), meaning that dopamine cannot be pumped back into the cell. This results in increased levels of dopamine in the synaptic cleft. The increase in

extracellular dopamine relies primarily on normal synaptic release, which is more amenable to physiological feedback regulation.

Illicit drugs can be classified according to their mechanism of action.

Mechanism	Examples
Drugs which interfere with ionotropic receptors or ion channels	Alcohol, nicotine, benzodiazepines, ketamine
Drugs which interfere with G-coupled receptors	Opioids, cannabinoids, y-hydroxybutyrate (GHB)
Drugs that target monoamine transporters	Amphetamine, ecstasy, cocaine

45. Receptor acted on by use of psychedelic substances- ? NMDA, **5HT2A** Agonism

Receptor	Action
$5HT_{1A}$	Antidepressant (agonist), anxiolytics (partial agonist)
$5HT_{1B}$	Aggression
$5HT_{1D}$	Antimigraine (antagonist)
$5HT_{2A}$	Antipsychotic (antagonist); hallucinogens (agonist / partial agonist); implicated in working memory; also seen in platelets and smooth muscles
$5HT_{2B}$	Stimulation may produce cardiac valvular fibrosis (dexfenfluramine)
$5HT_{2C}$	Anxiogenic and anorexic effect (agonists)
$5HT_{3}$	Antiemetic (antagonist)
$5HT_{6}$	Possible antipsychotic/antidepressant action (antagonism)
$5HT_{7}$	Regulation of circadian rhythm

46. "Khat" has pharmacological effect similar to which substance – **Amphetamine**, Cocaine, MDMA, Ketamine.

Active components (**cathine**, **cathinone**)

47. Hayling Test is for – intelligence, language, **Executive function**, verbal memory.

(The **Hayling** and Brixton **tests** are neuropsychological **tests** of executive function created by psychologists)

The Hayling and Brixton tests are designed to assess executive function.

Added during revision.
Right Answer/Important points.
Not sure about the answer.

The Hayling Sentence Completion Test consists of two sets of 15 sentences each having the last word missing. In the first section the examiner reads each sentence aloud and the participant has to simply complete the sentences, yielding a simple measure of **response initiation speed.**
The second part, the Hayling requires subjects to complete a sentence with a nonsense ending word (and suppress a sensible one), **giving measures of response suppression ability and thinking time**. It provides a measure of basic task initiation speed as well as performance on a response suppression task. Performance on such tests has been repeatedly associated with frontal lobe dysfunction and dysexecutive symptoms in everyday life.

The Brixton Test is a rule detection and rule following task. Impairments on such tasks are commonly demonstrated in individuals with dysexecutive problems.

48. Part A of trail making test assesses? – **Pe**ntelligence, visual memory, verbal memory

Trail making test part A:

- Visual attention
- Tracking
- Processing Speed

Trail making test part B:

- Set shifting
- Attention
- Visual Sketch
- Cognitive flexibility
- Switch attention

49. Short term memory stores information for – 10 secs, **30secs,** 1 minute, 15minutes

Short-term memory (STM), also referred to as short-term storage, or primary or active memory indicates different systems of memory involved in the retention of pieces of information (memory chunks) for a relatively short time, usually up to **30 seconds.**

Echoic memory deals with auditory information, holding that information for **1 to 2 seconds**. Iconic memory deals with visual information, holding that information for **0.5-1 second, Haptic- 2 sec.**

- Iconic **less than one second** 0.5 to 1 second.
- Echoic **1-2 secs**
- Haptic **2 sec**
- Short-term memory 30 secs

50. Which is true about bullying:
    - Commoner in girls,
    - **bullies prefer same type of victim,**
    - Commoner in primary than secondary school,
    - **bullies have less social maturity**

51. Q on institutional racism:

- Not common,
- overt so it can be easily identified,
- last option likely answer?

We believe none of these options correct as institutional racism is covert and common.

For many years there has been concern that institutional racism might explain the over representation of Black patients in mental health settings. Based on the 2001 UK Census, ethnic minorities form just 9% of the UK population. The 'Count me in Census' was introduced in 2005 to keep track of the ethnic variation in use of psychiatric services. The 2010 census found that: 23% of all inpatients and those on CTOs were from Black and minority ethnic groups (that is, not White British). The reasons for these findings are unclear and are hypothesized to fall into two groups:

1. Patient factors:

Higher rates of mental illness have been found in Black minority groups. Black Caribbean and Black African groups in the UK have consistently been observed to have the highest incidence of psychoses, with conservative estimates suggesting a risk between four- and sixfold that of the White British population. Perceptions of Black and minority ethnic patients being at greater risk

2. System factors:

Suggestion of inherent racism within psychiatry.

52. Which is true about Clozapine in pregnancy:

- contraindicated,
- **Dose needs increasing due to enzyme induction,**
- Dose needs increasing due to increased GFR,
- Dose reduction required due to enzyme inhibition by Oestrogens

Changes in pregnancy:

Pregnancy is associated with several **pharmacokinetic changes**:

- ✶ Delayed gastric emptying,
- ✶ Decreased GIT motility,
- ✶ Increased volume of distribution (5%),
- ✶ Decreased drug-binding capacity,
- ✶ Decreased albumin level
- ✶ Induced liver metabolic pathway,
- ✶ Increased GFR & renal clearance.

Psychotropic medication usually passes from the maternal blood to the fetus due to lack of strong barrier, but rate and amount of transfer are variable. Higher doses are associated with higher serum level in the infant.

Added during revision.
Right Answer/Important points.
Not sure about the answer.

Sex steroid levels increase during pregnancy and increase the activity of enzymes in the cytochrome P450 (CYP) family and Phase 2 metabolic enzymes such as uridine diphosphate glucuronosyltransferase (UGT). This induction of activity may result in clinically significant reductions in active drug levels.

53. Test for abstract reasoning: Go/No go test, Luria test, **Similarities**, Clock drawing, Tower of London

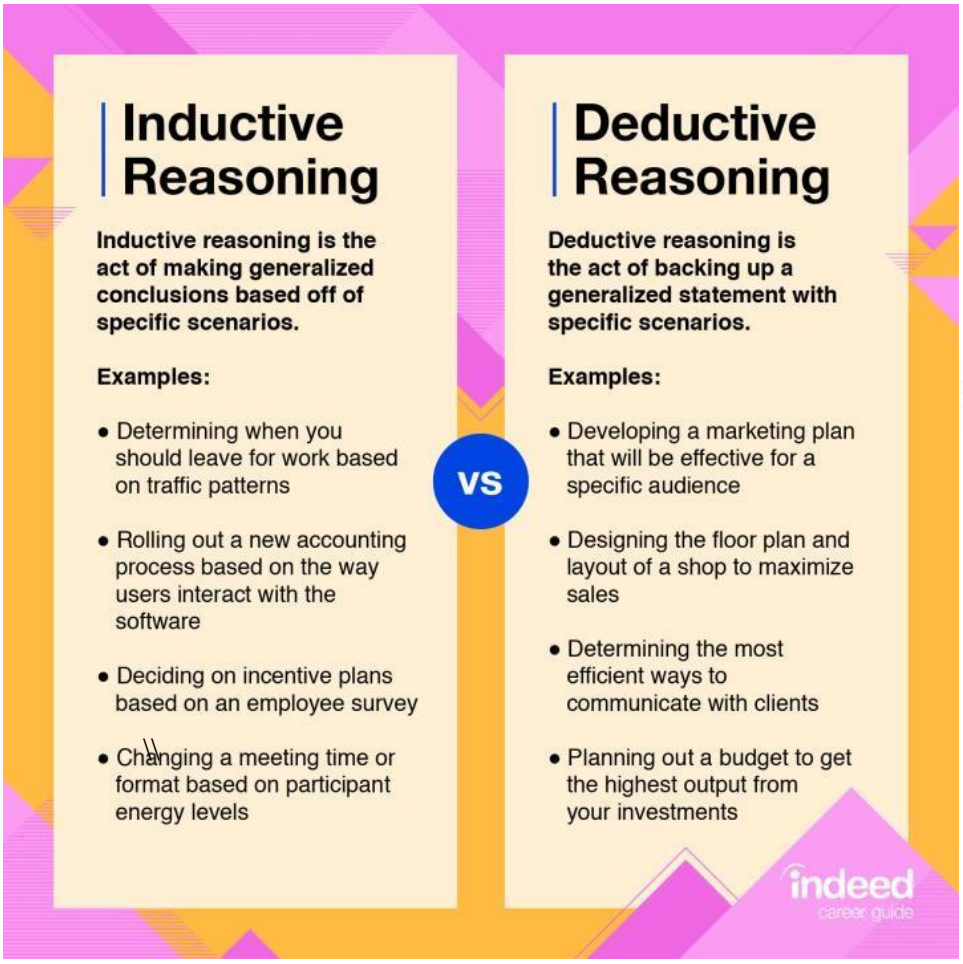
Others: Cognitive estimate and proverb

54. Test for executive function: Go/No go test, Luria test, Similarities, Clock drawing, **Tower of London**

55. Which of the following organism is implicated in the development of OCD – CMV, Varicella, Staph aureus, **Strept Pyogenes**, Clostridium botulinum

Patients with **OCD, Turner S, or tic disorder** were more likely than controls to have had prior streptococcal infection (OR: 2.22; 95% CI: 1.05, 4.69) in the 3 months before onset date. The risk was higher among children with multiple streptococcal infections within 12 months (OR: 3.10; 95% CI: 1.77, 8.96). Having multiple infections with group A beta-hemolytic streptococcus within a 12-month period was associated with an increased risk for TS (OR: 13.6; 95% CI: 1.93, 51.0).

56. Which stage is Conservation seen? **Concrete operational**





Added during revision.
Right Answer/Important points.
Not sure about the answer.

57. Which stage does mid-life crises occur in? 40-45 years
In SPMM they said between **40 and 60**

Stage
Pre-adulthood stage (age 0 – 22)
Early adulthood stage (age 17 – 45)
Middle adult stage (age 40 – 65)
Late adulthood stage (age 60 – 85)
Late late adult stage (age 80 plus)

58. Drug with active metabolite – **sertraline**(**Desmethylsertraline** the active metabolite of sertraline, has a half-life of 56–120 hours and reaches peak plasma concentrations 8–10 hours following administration)

https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3770982/table/T1/?report=objectonly

59. A drug whose active metabolite may be an anxiogenic – **Fluoxetine**

Unadjusted and adjusted hazard ratios and their 95% confidence intervals (CIs) for anxiety development (anxiety indicated by either antianxiety medicine or anxiety diagnosis) (N=328,888)

	Unadjusted		Adjusted	
	Hazard ratio	95% CI	Hazard ratio	95% CI
Antidepressants				
Bupropion	1.09*	(1.02, 1.18)	0.94	(0.87, 1.01)
Citalopram	1.09**	(1.04, 1.17)	1.10*	(1.04, 1.16)
Fluoxetine	1.03	(0.97, 1.11)	0.97	(0.90, 1.04)
Mirtazapine	1.54**	(1.39, 1.71)	1.42**	(1.28, 1.58)
Paroxetine	1.28**	(1.20, 1.37)	1.25**	(1.17, 1.34)
Venlafaxine	1.30**	(1.17, 1.44)	1.16*	(1.05, 1.28)

60. Drug that does not act as a prodrug – **Desipramine** (it is **active metabolite** of **Imipramine** and **lofepramine**)

A prodrug is a medication or compound that, after administration, is metabolised into a pharmacologically active drug. **Desipramine** is the metabolite of **imipramine**. All the others listed below are pro-drugs: [SPMM]

- Lisdexamphetamine
- Fosphenytoin
- Oxcarbazepine
- Prednisone

In punishment and obedience orientation focus is on consequence of action.

61. MI - Theory that believes consequences of an action determines whether the action is good or bad – social contract, law and order, punishment

Utilitarianism

Added during revision.
Right Answer/Important points.
Not sure about the answer.

62. Kohlberg's theory that is Authority based? **Authority based/social order(2nd substage of conventional)**

Level	Age	Stage	Characteristics
Preconventional (level 1)	0-9 years	Obedience and punishment orientation	Focus is on direct consequence of action and unquestioning deference to power (not in terms of respect for an underlying moral order supported by punishment and authority)
		Self-interest orientation (aka Instrumental-relativist orientation)	Right behaviour defined purely by what is in the individual's own interest (you scratch my back and I'll scratch yours, and what is important is what I want)
Conventional (level 2)	9-20 years	Interpersonal accord and conformity (good boy - nice girl orientation)	Focus is on how the individual will appear to others Behaviour should accord with a consensus view on what is good.
		Authority and social order obedience driven (the law and order orientation)	What is lawful is judged to be morally right. Right behaviour is dictated by societal rules. There is a greater respect for social order and the need for laws.
Postconventional (level 3)	20 and onwards	Social contract (aka legalistic) orientation	Individual rights determine behaviour. The individual views laws and rules as flexible tools for improving human purposes. That is, given the right situation, there are exceptions to rules. When laws are not consistent with individual rights and the interests of the majority, they do not bring about good for people and alternatives should be considered.
		Universal ethical principles orientation	The right action is the one that is consistent with abstract reasoning using universal ethical principles

Social rules can be broken of universal morality is not upheld, actions are guided by self chosen principles

Regarding Kohlberg stages Age:

- SPMM says **7-12, 13-16yr and 16-20**
- Psych-Mentor it says 0-9, 9-20 and >20,

It was reported that **TRONs** uses Psych-Mentor values

63. Which of the following is true about the Big Five model of personality – personality changes with time, **biological characteristics**,

Personality can be divided into 5 **broad** categories.

- Openness to experience
- Conscientiousness
- Extraversion (aka surgency)
- Agreeableness
- Neuroticism (aka emotional stability)

*This can be remembered by the mnemonic **OCEAN**.*

Added during revision.
Right Answer/Important points.
Not sure about the answer.

McCrae and his colleagues found that the Big 5 personality traits are remarkably universal. One study that looked at people from more than 50 different cultures found that the five dimensions could be accurately used to describe personality.

Based on this research, many psychologists now believe that the five personality dimensions are not only **universal** but that they also have **biological origins**. Psychologist David Buss has proposed an evolutionary explanation for these five core personality traits, suggesting that they represent the most important qualities that shape our social landscape.

Research suggests that both biological and environmental influences play a role in shaping our personalities. Twin studies suggest that both nature and nurture play a role in the development of each of the five personality traits.[2]

One study of the genetic and environmental underpinnings of the five traits looked at 123 pairs of identical twins and 127 pairs of fraternal twins. The findings suggested that the heritability of each personality trait was **53% for extraversion, 41% for agreeableness, 44% for conscientiousness, 41% for neuroticism, and 61% for openness.**

Longitudinal studies also suggest that these big five personality traits tend to be relatively stable over the course of adulthood. One four-year study of working-age adults found that personality changed little as a result of adverse life events.

Studies show that maturation may have an impact on the five personality traits. **As people age, they tend to become less extraverted, less neurotic, and less open to an experience. Agreeableness and conscientiousness, on the other hand, tend to increase as people grow older.**

64. Forced swim test is related to which condition (learned helplessness)– **Depression**, OCD, Anxiety

65. Psychopathology of illusions – perception and cognitive, cognitive and conceptual, conceptual and perceptual, **perceptual and signal processing**(bottom up)(i hope so)

66. "Top-down" theory of perception/illusion – Cognition and perception

What Is **Top-Down Processing?** In top-down processing, perceptions begin with the most general and move toward the more specific. These perceptions are heavily influenced by our expectations and prior knowledge. **Put simply, your brain applies what it knows to fill in the blanks and anticipate what's next.**

Top-down processing helps simplify our understanding of the world. It allows us to quickly make sense of all the information our senses bring in. As you begin to take in more information about your environment, your initial impressions (which are based on previous experiences and patterns) influence how you interpret the finer details.

This type of processing can be useful when we are looking for patterns in our environment, but these predispositions can also hinder our ability to perceive things in new and different ways.

Top-down processing involves perceiving things based on your prior experiences and knowledge. In other words, you use what you already know to make sense of the new

Top down is cognitive and conceptual
Bottom up is sensory and perceptive

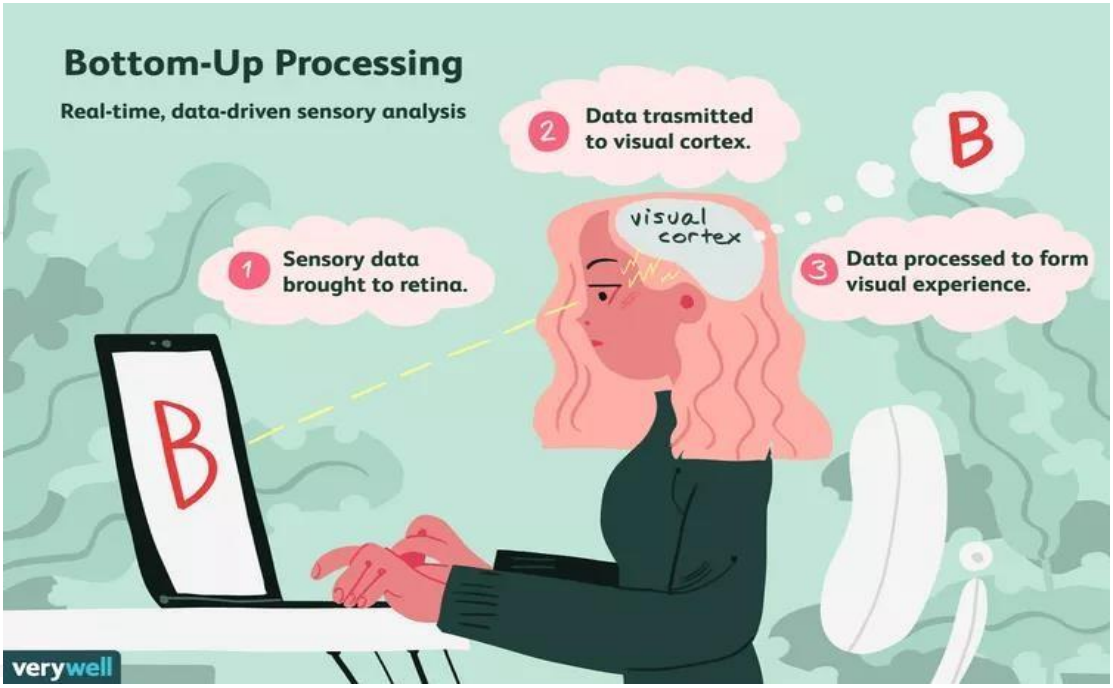
information you encounter. It refers to how our brain utilizes existing knowledge and expectations to interpret new sensory information

Bottom-up processing is an explanation for perceptions that start with an incoming stimulus and work upward until a representation of the object is formed in our minds. This process suggests that our perceptual experience is based entirely on the sensory stimuli that we piece together using only data that is available from our senses.

In order to make sense of the world, we must take in energy from the environment and convert it to neural signals, a process known as sensation. It is in the next step of the process, known as perception, that our brains interpret these sensory signals.

The theory of bottom-up processing was introduced by psychologist E. J. Gibson, who took a direct approach to the understanding of perception. Rather than being dependent upon learning and context, Gibson felt that perception was a "what you see is what you get" process.[1] He argued that sensation and perception are the same things.

Because Gibson's theory suggests that processing can be understood solely in terms of environmental stimuli, it is referred to as the **ecological theory of perception.**



67. Which is true about Agomelatine – MT antagonist, half-life of 12 hours, increases serotonin in the synapses, **extensively metabolised by CYP1A2**,

Half-life of Agomelatine is 2-3hours
Agomelatine undergoes extensive first-pass metabolism with a low bioavailability. It is **extensively protein bound (95%)** and has a **half-life of 2.3 hours**. It is mostly metabolized by CYP1A2 (90%) and CYP2C9 (10%)

68. Drug with high affinity for Histamine (H1) receptors: **Mirtazapine**, TCAs

Central

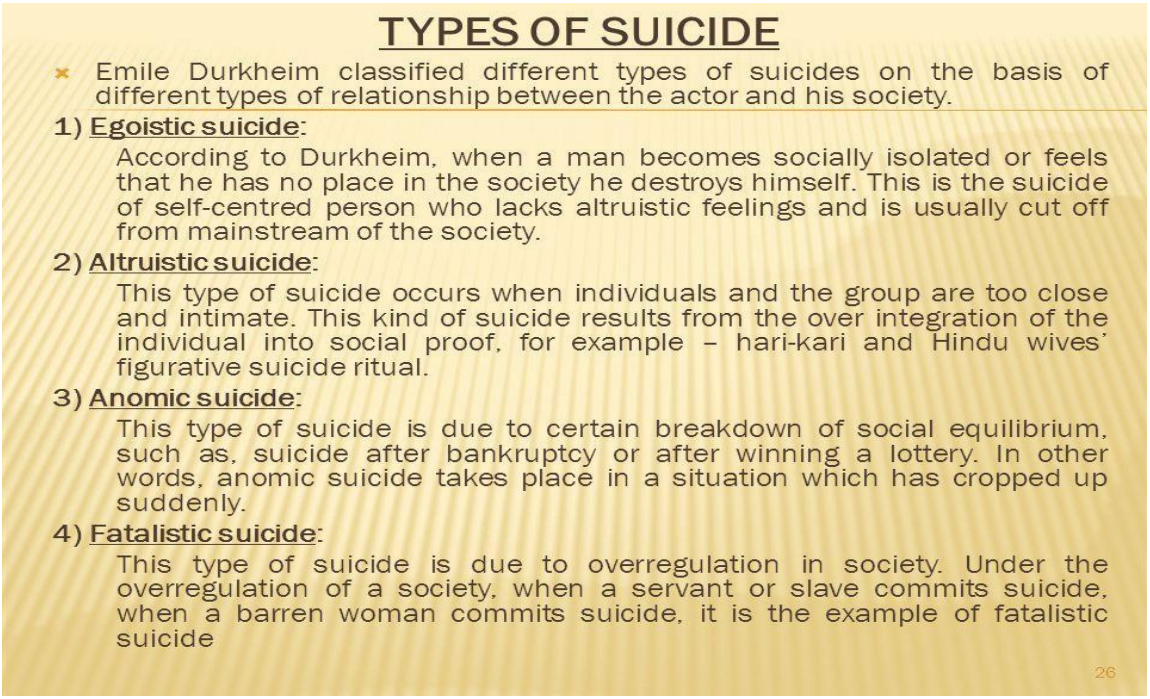
- Most **noradrenergic**: **Desipramine**,
- Most **anti-histaminergic**: **Doxepin**,

- Most **inhibit serotonergic uptake: Clomipramine** and **Imipramine**,
- Most toxic in overdose: **Amitriptyline** and **Dothiepin**,
- Most anticholinergic: **Amitriptyline and Imipramine**

69. Anomie suicide according to Durkheim theory – lack of close relationship
After thorough discussion we agreed that that egoistic suicide is lack of close relationship

Emile Durkheim was a French sociologist who challenged the assumption that social processes can be understood in terms of the actions of individuals and argued that even suicide, apparently the ultimately individualistic act, cannot be fully explained at the purely individual level. In his book Suicide , Durkheim used survey data purporting to show that egoistic suicide, resulting from feelings of self-reproach and sense of failure, arises from a lack of social cohesion; that altruistic suicide, that altruistic suicide, carried out for the benefit of others, arises from a sense of failure of society; and that anomic suicide, arising from a sense that life is pointless, arises from an absence of social norms that he called anomie

- **Altruistic suicide** Individual is overly attached to social norms and dies for the sake of the society (i.e. for others in the society) e.g. self-molestation among Buddhist monks in Tibet
- **Egoistic suicide** Excessive individualism, but low social integration. No cohesive group attachment
- **Fatalistic suicide** Society's control on the individual is very strong such that it interferes with moral values and personal goals
- **Anomic suicide** Individual feels that he has no guidance or regulations from the societal system; feels disillusioned e.g bankruptcy



Added during revision.
Right Answer/Important points.
Not sure about the answer.

70. Exaggerated and catastrophic misinterpretation of body signs and symptoms – **hypochondriasis**, bodily distress disorder

Parent

- Disorders of bodily distress or bodily experience

Show all ancestors

Description

Body integrity dysphoria is characterised by an intense and persistent desire to become physically disabled in a significant way (e.g. major limb amputee, paraplegic, blind), with onset by early adolescence accompanied by persistent discomfort, or intense feelings of inappropriateness concerning current non-disabled body configuration. The desire to become physically disabled results in harmful consequences, as manifested by either the preoccupation with the desire (including time spent pretending to be disabled) significantly interfering with productivity, with leisure activities, or with social functioning (e.g. person is unwilling to have a close relationships because it would make it difficult to pretend) or by attempts to actually become disabled have resulted in the person putting his or her health or life in significant jeopardy. The disturbance is not better accounted for by another mental, behavioural or neurodevelopmental disorder, by a Disease of the Nervous System or by another medical condition, or by Malingering.

71. Bright light therapy is used in – **circadian rhythm**, diurnal rhythm,

Bright Light Therapy

Bright light therapy is used to manage circadian rhythm disorders such as delayed sleep phase syndrome (DSPS), a condition which shifts the normal sleeping pattern outside what is considered the social norm. People who have DSPS generally don't fall asleep until several hours after midnight and have trouble waking up in the morning, making adhering to a normal work or school schedule difficult. However, if allowed to follow their own schedules, their quality of sleep is otherwise normal.

circadian rhythm controlled by **5HT7 and Suprachiasmatic Nucleus**

72. Which is part of the Papez circuit – insula, **cingulate cortex**, angular gyrus, **hippocampus**

James Papez described a neural pathway in the brain that mediates the process of emotion. It is referred to as the 'Papez circuit' and was one of the first descriptions of the limbic system.

The Papez circuit is **bilateral, symmetrical and is located on the medial surface of the brain**. It links the cortex to the hypothalamus.

Papez suggested that information about emotion passes through the:

Fornix Mamillothalamic tract

Hippocampus → the mamillary bodies of the hypothalamus → the anterior nucleus of the thalamus → Cingular cortex → the entorhinal cortex and finally → hippocampus again thus forming a circuit.

Added during revision.
Right Answer/Important points.
Not sure about the answer.

73. Psychiatric Screening tool: **EPND**, MADRS, Hamilton, Montgomery scale, PHQ, GHQ, HADS, (0-3)

Edinburg postnatal depression scale

Caseness: when a psychological rating scale is used for screening an operational definition of 'caseness' (the threshold for identifying a case) must be specified. The 'caseness' usually refers to a numerical value (cutoff) above which a respondent/ participant is considered to be a 'positive' (case).

74. Advantage of research in children using a Cross-sectional method – reliability, validity, **Easily Accessible data**

EXPLANATION: One advantage of using a cross-sectional method in studying developmental phenomena in children is that it **allows for quick data collection** from participants of different age groups at a single point in time. This method is efficient for obtaining a snapshot of developmental trends across different age cohorts, but it does not provide longitudinal insights into individual development or establish temporal causality. Attrition rates can vary, and the use of this design alone does not inherently improve response rates.

75. Test of premorbid intelligence – **National Adult Reading Test** (words that you have to pronounce correctly), Wechsler reading.

76. Condition whose phenotype is related to Schizophrenia – **DiGeorge**, Fragile X, Prader Willi

77. Autosomal Recessive condition – Autism, Fragile X, Prader Willi, **Wilson's dx**

- **Autism**: SHANK 3.
- **Fragile**: X linked dominant.
- **Prader Willi:** Deletion in chromosome 15 in paternal.
- **Wilson:** Autosomal Recessive.

78. Which is true about alcohol dependence in young people – incidence of binge drinking is lower in 18–20-year-olds**, Early onset of drinking is a predictor of later dependence**

According to the National Institute of Alcohol Abuse and Alcoholism (NIAAA), binge drinking **is "a pattern of drinking alcohol that brings blood alcohol concentration (BAC) to 0.08 percent . . . or higher."**

For adults, **binge** drinking means drinking about "**4 or more drinks (female), or 5 or more drinks (male) in about 2 hours**." In youth and teens, binge drinking could be "only 3 drinks for girls, and 3 to 5 drinks for boys, depending on their age and size."

The NIAAA also states that rates of binge drinking are generally high among college-aged youth aged 18-22. Among non-college students, binge drinking rates are about 27.7% and 33.0% among full-time college students

young people who drink alcohol might be more susceptible to addiction than older adults. Teens and young adults have more intense emotions and are more sensitive to the effects of alcohol. Young people are more prone to risk-taking, which includes substance use.

Added during revision.
Right Answer/Important points.
Not sure about the answer.

Brain development does not end until young adulthood, by around age 25. Impulse control, planning, and decision-making are some of the last areas of the brain to develop.

79. Which of the following **do not** cause delayed language development – **migration**, maternal smoking, lack of cognitive stimulation, large family size, maternal alcohol use

A number of risk factors are associated with delayed speech and language (Nelson 2006):-

- Family history
- Male gender
- Twins
- Lower maternal education
- Childhood illness
- Being born late in the family order
- Young mother at birth
- Low socioeconomic status
- maternal smoking
- large family size

80. Pathological finding in Alzheimer's – Alpha synuclein, **Beta Amyloid,** Ubiquitin

81. Neurodegenerative disorder linked to Schizophrenia – **Huntington**, Muscular Dystrophy, Friedreich's Ataxia???

We believe Huntington is likely the answer

Google**: Friedreich's Ataxia** is the most common form of **progressive spinocerebellar ataxia** with mixed sensory and cerebellar components and is **inherited via an autosomal-recessive gene**. Cognitive impairment, depression, emotional lability, and schizophrenia- like psychosis are the psychiatric manifestations of the disease.

Schizophrenia-like psychosis occurs in HD disease with an estimated frequency of 6% to 25%, the paranoid form being apparently the most common type [1,2,12-14,16]. Patients with an early age at onset of HD seem to have a greater risk of developing psychosis

We identified positive and significant correlation between schizophrenia and ALS at genetic (correlation 0.22; 95% CI: 0.16–0.28; p = 4.00E-04) and transcriptomic (correlation 0.08; 95% CI: 0.04–0.11; p = 0.034) levels. We further demonstrated that schizophrenia- and ALS-inferred gene expression overlap significantly in four tissues including skin, small intestine, brain cortex and lung, and highlighted three genes, namely *GLB1L3*, *ZNHIT3* and *TMEM194A* as potential mediators of the correlation between schizophrenia and ALS. Our findings revealed overlapped gene expression profiles in specific tissues between schizophrenia and ALS, and identified novel potential shared genes. These results provided a better understanding for the pleiotropy of schizophrenia, and paved way for future studies to further elucidate the molecular drivers of schizophrenia.

Friedreich's ataxia (FRDA) is the most common autosomal recessive ataxia, in the vast majority of patients, due to homozygous expansion of a guanine–adenine–adenine (GAA) trinucleotide repeat in intron 1 of the frataxin gene on chromosome 9q13 [1]

82. Which is a **human** Prion Disease:
Select one
- **Fatal Familial Insomnia**,
- Progressive Focal Leukoencephalopathy,
- Friedreich's Ataxia

Fatal familial insomnia is a very rare and invariably fatal **autosomal dominant** neurodegenerative prion disease caused by a mutation of the prion protein (PRNP) gene.

Human Prion Diseases:

1. Creutzfeldt-Jakob Disease (CJD)
2. Variant Creutzfeldt-Jakob Disease (vCJD)
3. Gerstmann-Straussler-Scheinker Syndrome
4. Fatal Familial Insomnia
5. Kuru

In animals, examples include **bovine spongiform encephalopathy** (BSE or "mad cow disease"), **scrapie in sheep**, and **chronic wasting disease (CWD) in deer and elk.**

83. 2 EMI questions on **Ideomotor and Ideational(multistep)** apraxia

Ideomotor apraxia - an inability to follow out learned tasks when given the necessary objects e.g. if given a hairbrush they try to write with it. Ideomotor apraxia (IMA) is the impaired ability to perform a skilled gesture with a limb **upon verbal command and/or by imitation.** It can be shown for both meaningful motor acts that do not imply objects and gestures that imply object use

Ideational apraxia - an inability to follow a sequence of actions in the correct order e.g. Take a match out a box and strike it with your left hand.

Fail to carry out ideal sequence of steps of doing something

Apraxia	Deficit
Ideomotor	Deficit in pantomiming tool use and gestures specifically. Knowledge of tasks is still present.
Limb-kinetic	Loss of hand and finger dexterity generally contralateral to the lesion. Mainly affects "manipulative" movements.
Ideational	Failure to carry out a series of tasks using multiple objects for an intended purpose. Tools are identifiable, but no coherent action is made.
Conceptual	Loss of tool knowledge and inappropriate use of tools and objects.
Verbal–motor dissociation	Inability to respond properly to verbal commands to make movements.
Tactile	Disruption of use of the hand as a sense organ, in which object exploration and manipulation are impaired.

84. Deep Brain stimulation (DBS) is most useful for – **Depression**, Psychosis, ADHD

- DBS: In OCD → stimulate **anterior limb of internal capsule and nucleus accumbens**
- DBS: Depression → stimulate **cingulate gyrus**

Deepbrain stimulations uses electrode in brain which are connected to a device beneath clavicle

EMI - EEG findings:
- atypical absence seizures,
- juvenile myoclonic epilepsy,
- typical absence seizures,
- infantile seizure)

85. 3-4Hz polyspike and slow wave: **Juvenile myoclonic epilepsy,**

86. 3Hz generalized slow wave: **Encephalopathy**

Seizure type	Pattern during the seizure (ictal EEG)
Typical absence (aka petit mal)	Generalised 3 Hz spike-wave
Atypical absence	Slow (<2.5 Hz) generalized spike-and-wave
Focal (previously known as partial)	Focal spikes
Myoclonic	Generalized 3-6 Hz polyspike and wave discharge
Generalised tonic-clonic	EEG often obscured by artifact (movement). Generalized fast rhythmic spikes are seen in the tonic stage. Bursts of spikes and after-coming slow waves are synchronous with clonic jerks. A postictal period of irregular slow activity follows
Atonic (drop attack)	Generalized spike-and-wave is typical, with atonia at the time of the slow wave

	vCJD	CJD
MRI	Pulvinar sign	Bilateral anterior basal ganglia high signal
EEG	Generalised slowing	Biphasic and triphasic waves 1-2 per second

Origin	Infected meat products	Genetic mutation (bad luck)
Age of onset	Younger people (age 25-30)	Old people (age 55-65)

Findings on BZDs effects on sleep architecture confirm an increase in stage 2 of NREM sleep and a decrease in time of stages 3 and 4 of NREM sleep with a reduction in time of REM sleep during the nocturnal sleep

Condition	EEG findings
CJD (sporadic only, does not apply to variant)	Early on there is non specific slowing, later periodic biphasic and triphasic synchronous sharp wave complexes superimposed on a slow background rhythm
Huntington's	Low voltage EEG, in particular no alpha (flattening)
Delirium	Diffuse slowing, decreased alpha, increased theta and delta
Delirium tremens	Hyperactive trace, fast
Alzheimer's	Reduced alpha and beta, increased delta and theta
Normal aging	Diffuse slowing, which can be focal or diffuse, if focal most commonly seen in the left temporal region
Encephalopathy	Diffuse slowing

- **Triphasic wave:** hepatic encephalopathy
- **DTABG:** Delta, Theta, Alpha, Beta, Gamma **from lowest to highest frequency.**
- Mu is between theta and Alpha with wave length:7-11

Medication	Alpha	Beta	Delta	Theta
Clozapine	Decrease	Varied	Increase	Increase
Lithium	Increase	Increase	Increase	Increase
Lamotrigine	Decrease	Decrease	Decrease	Decrease
Valproate	Decrease	Increase	Inconclusive	Inconclusive

Added during revision.
Right Answer/Important points.
Not sure about the answer.

Effect of drugs on EEG

Psychotropics	
Antipsychotics	Slowing of beta activity with increase in alpha, theta and delta activity
Antidepressants	Slowing of beta activity with increase in alpha, theta and delta activity
Lithium	Slowing of alpha or paroxysmal activity
Anticonvulsants	No effect on awake EEG

Primarily sedating drugs – decrease alpha	
Barbiturates	Effects are opposite to that of alcohol. Increased beta activity upon intoxication; generalized paroxysmal activity and spike discharges (even without overt fits) in withdrawal states.
Benzodiazepines	Increased beta; decreased alpha. Overdose leads to diffuse slowing
Opioids	Decreased alpha activity; increased voltage of theta and delta waves; in overdose, slow waves are seen.

Primarily recreational drugs – increase alpha	
Alcohol	Increased alpha activity; increased theta activity. Withdrawal increases beta. Delirium tremens has beta (fast) wave activity – other deliria have increased slow waves.
Marijuana	Increased alpha activity in frontal area of brain; overall slow alpha activity
Cocaine	Same as marijuana; longer lasting.
Nicotine	Increased alpha activity; in withdrawal, marked decrease in alpha activity
Caffeine	In withdrawal, increase in amplitude or voltage of theta activity

87. Patient with Hyperprolactinemia on Risperidone – increased?? by adding Aripiprazole, mechanism is by blockage D2 Receptors in the Hypothalamus, other options???

Effect on prolactin	**Antipsychotics**
Prolactin sparing (or very rare increase)	Clozapine Aripiprazole Asenapine Quetiapine
Low risk / minor change	Lurasidone Olanzapine Ziprasidone
High risk / major change	All the typical Antipsychotics Risperidone Amisulpride Paliperidone Sulpiride

88. Definition of Neurulation – **formation of neural tube**

Neurulation refers to the process of transforming the flat neural plate into a neural "tube". This process results in formation of the brain cranially, and the spinal cord caudally. It appears there are multiple initiation sites along the neural plate. Disruption of the process cranially results in anencephaly or encephalocele, whereas failure of neural tube closure caudally results in spina bifida . Day 18-21

Added during revision.
Right Answer/Important points.
Not sure about the answer.

89. Memory affected in Korsakoff syndrome: Implicit memory, semantic memory, **Episodic memory**

90. Memory preserved in Normal aging: working memory, speed of processing, attention, executive function, **Procedural memory**

91. Which is true in Confabulation: other options same as previous recalls

Options

- **Autobiographical memory is affected**.
- There is subjective awareness of memory.
- Can only occur in Korsacoff syndrome
- May occur in clouded consciousness
- There is deliberate falsification of facts

Conclusion: We think the highlighted answer is the right one, it is not ALWAYS affected though, Also sometimes there might be subjective awareness of confabulation so we decided to go for autobiographical memory first, if not an option then subjective awareness – please feel free to correct us if you think otherwise.

Confabulation

Confabulation is the falsification of memory occurring in clear consciousness in association with organic pathology. It manifests itself as the filling-in of gaps in memory by imagined or untrue experiences that have no basis in fact. Some of the statements may be contradictory yet no attempt is made to correct them. The confabulation diminishes as the impairment worsens. Two broad patterns emerge (Bonhoeffer, 1901), the embarrassed type in which the patient tries to fill in gaps in memory as a result of an awareness of a deficit and the fantastic type in which the lacunae are filled in by details exceeding the need of the memory impairment such as descriptions of wild adventures. Overall the embarrassed type is much more common and it may represent real memories displaced in time. Some schizophrenics confabulate and provide detailed descriptions of fantastic events that have never happened. Some suggest that confabulation is a misnomer since these memories are fixed and unchanging. The term pictorial thinking is used instead by some, while others call them memory hallucinations, a term rejected by Fish as 'not very suitable'. They may best be termed retrospective delusions (see below). Lethologia, the temporary inability to remember names or proper nouns, is common and generally not indicative of any pathology.

60

DISORDERS OF MEMORY

eudologia fantastica

Pseudologia fantastica or fluent plausible lying (pathological lying) is the term used, by convention, to describe the confabulation that occurs in those without organic brain pathology such as personality disorder of antisocial or hysterical type. Typically the subject describes various major events and traumas or makes grandiose claims and these often present at a time of personal crisis, such as facing legal proceedings. Although it seems that the person with pseudologia believes their own stories and there is a blurring of the boundary between fantasy and reality, when confronted with incontrovertible evidence these individuals will admit their lying. Minor

92. Brain Organoid – age corresponds with **second** trimester fetus

EMIs on hormones for satiety and hunger:

93. Hormone secreted in Gut - **CCK**

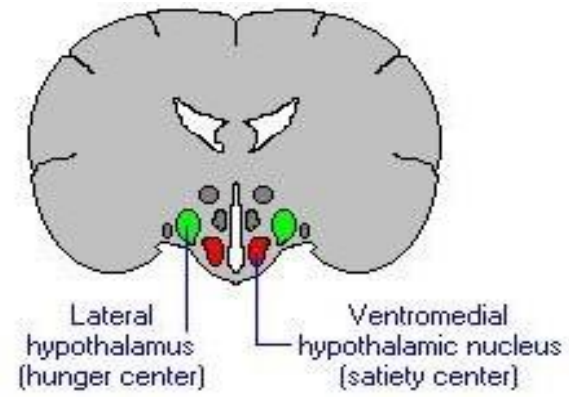
Added during revision.
Right Answer/Important points.
Not sure about the answer.

94. Hormone secreted in Adipose tissue - **Leptin**
95. Hormone secreted in Hypothalamus – **Neuropeptide Y**

- **Neuropeptide Y** is produced by the hypothalamus and **increases appetite**
- **Leptin is produced** by adipose tissue and **reduces appetite**
- **Ghrelin is produced** mainly by the gut and **increases appetite**
- **Cholecystokinine (CCK) is** produced mainly by the gut and **reduces appetite**

Gastrin	Secreted by cells in the stomach and duodenum	Stimulates the exocrine cells of the stomach to secrete gastric juice (a mixture of hydrochloric acid and the proteolytic enzyme pepsin)
Somatostatin	Secreted by cells in the gastric glands of the stomach	On the stomach (thus a paracrine effect) it inhibits the release of gastrin and hydrochloric acid On the duodenum it inhibits the release of secretin and cholecystokinin On the pancreas it inhibits the release of glucagon
Secretin	Secreted by cells in the duodenum when they are exposed to the acidic contents of the emptying stomach	Stimulates the exocrine portion of the pancreas to secrete bicarbonate into the pancreatic fluid (thus neutralizing the acidity of the intestinal contents)
Cholecystokinin (CCK)	Secreted by cells in the duodenum and jejunum when they are exposed to food	On on the gall bladder stimulating it to contract and force its contents of bile into the intestine On the pancreas stimulating the release of pancreatic digestive enzymes into the pancreatic fluid On vagal neurons leading back to the medulla oblongata which give a satiety signal
Ghrelin	Secreted by endocrine cells in the stomach	Acts on the hypothalamus to stimulate feeding

L.H (Lateral is hunger centre) lesions are opposite
Anterior helps you cool down posterior warms up



Added during revision.
Right Answer/Important points.
Not sure about the answer.

Gastrointestinal regulatory substances

REGULATORY SUBSTANCE	SOURCE	ACTION	REGULATION	NOTES
Gastrin	G cells (antrum of stomach, duodenum)	↑ gastric H^+ secretion ↑ growth of gastric mucosa ↑ gastric motility	↑ by stomach distention/alkalinization, amino acids, peptides, vagal stimulation via gastrin-releasing peptide (GRP) ↓ by pH < 1.5	↑ by chronic PPI use ↑ in chronic atrophic gastritis (eg, *H pylori*) ↑↑ in Zollinger-Ellison syndrome (gastrinoma)
Somatostatin	D cells (pancreatic islets, GI mucosa)	↓ gastric acid and pepsinogen secretion ↓ pancreatic and small intestine fluid secretion ↓ gallbladder contraction ↓ insulin and glucagon release	↑ by acid ↓ by vagal stimulation	Inhibits secretion of various hormones (encourages somato-stasis) Octreotide is an analog used to treat acromegaly, carcinoid syndrome, and variceal bleeding
Cholecystokinin	I cells (duodenum, jejunum)	↑ pancreatic secretion ↑ gallbladder contraction ↓ gastric emptying ↑ sphincter of Oddi relaxation	↑ by fatty acids, amino acids	Acts on neural muscarinic pathways to cause pancreatic secretion
Secretin	S cells (duodenum)	↑ pancreatic HCO_3^- secretion ↓ gastric acid secretion ↑ bile secretion	↑ by acid, fatty acids in lumen of duodenum	↑ HCO_3^- neutralizes gastric acid in duodenum, allowing pancreatic enzymes to function
Glucose-dependent insulinotropic peptide (incretins)	K cells (duodenum, jejunum)	Exocrine: ↓ gastric H^+ secretion Endocrine: ↑ insulin release	↑ by fatty acids, amino acids, oral glucose	Also called gastric inhibitory peptide (GIP) Oral glucose load ↑ insulin compared to IV equivalent due to GIP secretion
Motilin	Small intestine	Produces migrating motor complexes (MMCs)	↑ in fasting state	Motilin receptor agonists (eg, erythromycin) are used to stimulate intestinal peristalsis
Vasoactive intestinal polypeptide (VIP)	Parasympathetic ganglia in sphincters, gallbladder, small intestine	↑ intestinal water and electrolyte secretion ↑ relaxation of intestinal smooth muscle and sphincters	↑ by distention and vagal stimulation ↓ by adrenergic input	VIPoma—non-α, non-β islet cell pancreatic tumor that secretes VIP; associated with Watery Diarrhea, Hypokalemia, Achlorhydria (WDHA syndrome)
Nitric oxide		↑ smooth muscle relaxation, including lower esophageal sphincter (LES)		Loss of NO secretion is implicated in ↑ LES tone of achalasia
Ghrelin	Stomach	↑ appetite ("ghrowlin' stomach")	↑ in fasting state ↓ by food	↑ in Prader-Willi syndrome ↓ after gastric bypass surgery

96. In children with ADHD which is better to help them focus on tasks – ignore negative actions, **use of small immediate rewards**, use of large, delayed rewards, small immediate punishments, use of large, delayed punishment

> Small-to-medium effect sizes emerged for both paradigms, confirming that participants with ADHD choose small immediate over large delayed rewards more frequently than controls. Moderation analyses show that offering real rewards in the SCP almost doubled the odds ratio for participants with ADHD.

97. HIV-associated neurocognitive disorder in patient who has been on ART for a long time, what is seen?

- High CD4 count (Initially low and later Normalize after ART)
- **High CD8 count (not normalize even with ART)**
- Neuronal loss

> During the ART (antiretroviral therapy) era, observations of cortical, subcortical, gray matter, and white matter atrophy have continued, **with decreases in cortical gray matter and brain parenchymal volume observed even in the first year of HIV infection.**
>
> **Studies have reported that individuals initiating ART at higher counts have their CD4+ cell count return to nearly normal or normal (>500 cells/mm3) than those who initiated at lower counts (< 200 cells/mm3)**
>
> **CD8+** T-cell counts are elevated during HIV infection and **do not normalize** despite long-term cART. **Low CD8+ T-cell counts are associated with increased AIDS-related mortality.**
>
> Untreated HIV infection drives the CD4/CD8 ratio lower. In some populations, initiation of ART can increase the ratio; however, early and continuous treatment is essential.
>
> normal CD4/CD8 ratio is greater than 1.0, with CD4 lymphocytes ranging from 500 to 1200/mm 3 and CD8 lymphocytes ranging from 150 to 1000/mm 3. If your ratio is higher than 1, it means your immune system is strong and you may not have HIV. **If your ratio is less than 1, You may have HIV**
>
> HIV does not directly invade nerve cells (neurons) but puts their function at risk by infecting cells called **glia** that support and protect neurons. HIV also triggers inflammation that may damage the brain and spinal cord (central nervous system) and cause symptoms such as: Confusion and forgetfulness.
>
> Macrophage and microglia in brain
> peripheral CD4 cells
>
> **Cells least likely to be affected by HIV in brain: Neurons.** Theoretically all cells are affected including astrocyte, oligodendrocyte, perivascular macrophages, microglia and even neurons

98. Definition of avoidant/restrictive food intake disorder: **Not preoccupied by weight just excludes certain food groups from diet**

99. What is used in T2 weighted MRI – Calcium, **water**, haemoglobin

- **T1**-weighted MRI enhances the signal of the **fatty** tissue and suppresses the signal of the water.

- **T2**-weighted MRI enhances the signal of the **water**. Consideration of all the information provided by these modalities is conducive to MRI image analysis and diagnosis.

100. Ion implicated in Schizophrenia: **Calcium**, potassium.

Channel involved : Potassium
Ion involved: Calcium

KCNN3, encoding the small conductance calcium-activated potassium channel SK3, is located at 1q21, a region reproducibly and strongly implicated in schizophrenia.

Potassium channels are widely distributed in the CNS and thought to have a role in modulating electrical excitability in neurons, regulating action potential duration and neurotransmitter release. Therefore, it is no surprise that some potassium channel genes have been considered candidate for susceptibility to schizophrenia.

101. What acts as intracellular second-messenger system: **Calcium**, potassium.

Other messengers include: IP3, DAG, Cyclic AMP, Cyclic GMP.

Examples of second messenger molecules include cyclic **AMP, cyclic GMP, inositol triphosphate, diacylglycerol, and calcium**. First messengers are extracellular factors, often hormones or neurotransmitters, such as epinephrine, growth hormone, and serotonin.

EMI: Family System Theory:
Options:
- Triangulation
- Detouring
- Circular casuality
- Enmeshment

102. family members ganging up against one family member: **Detouring**.

103. Being aware that people's interaction are influenced by each other which in turn are influenced by others - **Circular causality**

Triangulation is when a toxic or manipulative person, often a person with strong narcissistic traits, brings a third person into their relationship in order to remain in control.

Detouring: detouring scale assesses the degree to which the parents form an alliance against the child in a way that functions to create distance between themselves and the child and increase closeness in their own relationship scapegoat syndrome. Detouring occurs when parents, rather than directing anger or criticism toward each other, focus the negativity on the child and the parent-child conflict thus serves to distract from the tension in the marital subsystem.

Circular causality: a sequence of causes and effects that leads back to the original cause and either alters or confirms it, thus producing a new sequence, as in a feedback loop.

Enmeshment: is a trait of family dysfunction that involves poorly defined or non-existent boundaries, unhealthy relationship patterns, and a lack of independence among family members. E.g. This often happens on an emotional level in which two people "feel" each other's emotions, or when one person becomes emotionally escalated and the other family member does as well. A good example of this is when a teenage daughter gets anxious and depressed and her mom, in turn, gets anxious and depressed

104. Q on Rogerian (person centred therapy): reflecting on how intended outcomes are affected by current behaviours: Cognitive Therapy, **Motivational interview**, Cue therapy

Motivational interviewing: is a counselling method that helps people resolve ambivalent feelings and insecurities to find the internal motivation they need to change their behaviour. It is a practical, empathetic, and short-term process that takes into consideration how difficult it is to make life changes.

Person-cantered therapy, also referred to as non-directive, client-centered, or Rogerian therapy, was pioneered by Carl Rogers in the early 1940s. This form of psychotherapy is grounded in the idea that people are inherently motivated toward achieving positive psychological functioning.

What is an example of a Rogerian theory? For example, if a client reported feeling stressed about the fact that a co-worker was receiving credit for a project the client worked on, the Rogerian therapist might say, "So, it sounds like you're upset because your boss isn't recognizing your contributions." In this way, the Rogerian therapist attempts to give the client an environment to explore their own thoughts and feelings and decide for themselves how to bring about positive change.

105. **Q.** on alcohol addiction: **Cue therapy**

Cue therapy in addiction, also known as cue exposure therapy, is a type of psychological treatment used to help individuals with substance use disorders (addictions) reduce their cravings and problematic behaviors related to substance abuse. This therapy focuses on identifying and managing the specific cues or triggers that lead to substance cravings and use.

106. What setting has the highest likelihood of obedience, Presence of:
Options:
- **legitimate authority in prestigious setting,**
- legitimate authority in mundane setting,
- uniform-wearing person in mundane setting,
- telephone instruction from legitimate authority

Milgram found that subjects were more likely to obey in some circumstances than others. Obedience was highest when:

- Commands were given by an authority figure rather than another volunteer
- The experiments were done at a prestigious institution
- The authority figure was present in the room with the subject
- The learner was in another room
- The subject did not see other subjects disobeying commands

In everyday situations, people obey orders because they want to get rewards, because they want to avoid the negative consequences of disobeying, and because they believe an authority is legitimate. In more extreme situations, people obey even when they are required to violate their own values or commit crimes. Researchers think several factors cause people to carry obedience to extremes:

- People justify their behaviour by assigning responsibility to the authority rather than themselves.
- People define the behaviour that's expected of them as routine.
- People don't want to be rude or offend the authority
- People obey easy commands first and then feel compelled to obey more and more difficult commands. This process is called entrapment, and it illustrates the foot-in-the-door phenomenon.

107. Infected neuron with cytokines in the surrounding milieu is suggestive of: **Necrosis**

Apoptosis is described as an active, programmed process of autonomous cellular dismantling that avoids eliciting inflammation.

Necrosis has been characterized as passive, accidental cell death resulting from environmental perturbations with uncontrolled release of inflammatory cellular contents

108. When does stranger anxiety develop? – 3 months, **9 months**, 6 years, 12 years

initially, babies show a positive response to strangers (between 14-18 weeks), this is **normally replaced by stranger anxiety at around 8 months.** When an infant is separated from their main carer it is common for them to become anxious (aka separation anxiety). **'Separation anxiety' is most common when an infant is 10 - 18 months and usually diminishes by year three.**

109. Neurotransmitter which is easily disintegrated:
Options
- **Acetylcholine (ACHE present right in synaptic cleft),**
- Dopamine
- Serotonin

110. Disulfiram acts on what ? **Aldehyde dehydrogenase**
Alcohol dehydrogenase is rate limiting step

111. Q on Imprinting?

imprinting is a term used to describe the phase-sensitive attachment seen in some animals and in humans.

The classic demonstration of imprinting was performed by Konrad **Lorenz**. He was working with Greylag geese, and showed that imprinting behaviour consists of three parts:-

- At some time after hatching the gosling will start to follow the first conspicuous object it sees (the following response)

- After following the object for a number of hours the hatchling will start to develop an attachment for the object
- Once an animal has imprinted, it develops a sexual preference, when it is mature, for the object on which it imprinted

Lorenz also observed the following:-

- Imprinting is **rapid** (sometimes occurring within minutes)
- Imprinting is **unreinforced** (no reinforcer is necessary for the response to occur)
- Imprinting occurs during a clearly defined period (known as the critical period)
- Imprinting is **irreversible**

112. Which neuroimaging modality is more **11c labelling** approach **PET**, SPECT, fMRI,

Technetium99,xenon123 ,Iodine 123 for SPECT

PET and SPECT scans use radiopharmaceuticals to create 3D images, but the tracers used differently. SPECT scans measure gamma rays, and the radiotracers used with PET scans produce small particles called positrons.

113. Which neuroimaging technique to detect 70% D2 occupancy **DAT scan,** other options same as above

- DAT IS DONE WITH SPECT SCAN
- DAT stands for **dopamine active transporter.** Dopamine is a chemical which is important in areas of the brain that help control movement. A DAT brain scan is a nuclear medicine test that looks at the function of dopamine transporters in your brain. **It is often used to help distinguish Parkinson's disease from other causes of tremor or difficulties with movement.**

114. Potent inhibitor of DAT:

Options:

- Fluoxetine
- **Sertraline**
- Citalopram
- Trazodone (weak inhibitor)
- Vortioxetine

Table 1

Pharmacological profile of selective serotonin reuptake inhibitors (SSRIs) and vortioxetine [13]

	SERT inhibition	5HT2C antagonism	NET inhibition	DAT inhibition	σ1 agonist	M1 antagonism	NOS inhibition	H1-antagonis
Citalopram	+							+
Escitalopram	+							
Fluoxetine	+	+	+ (only at very high doses)					
Fluvoxamine	+				+			
Paroxetine	+		+			+	+	
Sertraline	+			+	+			
Venlafaxine	+		+					
Vortioxetine	+							

Open in a separate window

SERT serotonin transporter, *5-HT* 5-hydroxytryptamine, *DAT* dopamine transporter, *H1* histamine 1 receptor, *M1* muscarinic acetylcholine 1 receptor, *NET* norepinephrine transporter, *NOS* nitric oxide synthetase, *SERT* serotonin transporter, *DAT* dopamine transporter, *σ1* sigma 1 receptor, *M1* muscarinic acetylcholine 1 receptor, *NOS* nitric oxide synthetase, *H1* histamine 1 receptor, *σ1* sigma 1 receptor

115. A man who migrated to another country has adopted the new culture while maintaining his culture of origin – **Integration**, Assimilation (New culture adopted and home culture forgotten), Marginalisation (No culture present), Separation (Only home culture)

116. What bias is expected if individuals are asked to fill a self-questionnaire about their personality traits – **Social desirability**, factor analysis

Social desirability bias is the tendency to underreport socially undesirable attitudes and behaviors and to over report more desirable attributes. One major theory of social desirability bias by Paulhus (1984) suggests two components

117. A child is given a star sticker every time he takes his medicine, after collecting 10 stickers he can get a fruit. What type of reinforcement is achieved with the fruit?
Options
- **primary reinforcement,**
- positive conditioning,
- secondary reinforcement

What is a token economy? A token economy is a behavior-management system in which targeted behavior is reinforced with tokens (secondary reinforcers) and later exchanged for rewards (primary reinforcers). The method is based on operant conditioning and is widely used in elementary classrooms.

118. Which of these is part of the basal ganglia? **Putamen**, stria terminalis.

The basal ganglia are a cluster of subcortical nuclei deep to cerebral hemispheres. The **largest component of the basal ganglia is the corpus striatum** which contains the caudate and lenticular nuclei, the subthalamic nucleus (STN), and the substantia nigra (SN)

Lenticular nuclei consist of: **P**utamen**, G**lobus **P**allidus externus, and internus **(LPGP)**
Striatum: **caudate + putamen**

119. Which of these is a Metabotropic receptor? **5HT2A**, AMPA, GABA A, NMDA

Examples of metabotropic receptors include:

- glutamate receptors,
- muscarinic acetylcholine receptors,
- GABA-B receptors,
- most serotonin receptors, **Except 5HT-3**
- receptors for norepinephrine,
- epinephrine,
- histamine,
- dopamine,
- neuropeptides, and
- endocannabinoids.

120. After how many half-lives is a drug eliminated?

A. 2,
B. 3,
C. 4,
D. 5,
E. 10

Added during revision.
Right Answer/Important points.
Not sure about the answer.

121. Piaget's stage where a child can understand the pain of another child – **Formal operational**, Preoperational, Sensorimotor

122. Question about auto-immune encephalitis and the antibodies of this? **NMDA**, DPPX, IGCLON.

Stage	age	Features
Sensorimotor	birth to 18-24 months	Inborn motor and sensory reflexes Primary Circular Reactions 1 to 4 months Secondary Circular Reactions 4 to 10 months Use of familiar means to obtain ends Tertiary Circular Reactions and discovery through active experimentation 12 to 18 months Insight and object permanence
Preoperational	2-7 years	Deferred imitation Symbolic play Transductive Reasoning Graphic imagery (drawing) Mental imagery Semiotic function (ability to use symbols) Language Egocentrism
Concrete operational	7-11 years	Egocentric thought replaced by operational thought (child now deals with information outside them) Syllogistic reasoning (logical conclusions formed from two premises). Conservation (Watanabe, 2017) of number (age 6-7), length (age 6-7), mass (age 7-8), area (age 8-9), weight (age 9-10), volume (age 11-12) Classifies objects by several features
Formal operational	11 years to adolescence	Thinks logically and tests hypotheses

Added during revision.
Right Answers.
Not sure about the answer.

December 2022

1. EMI: Half-life more than 50 hrs:
   Options
   - **Single dose diazepam (20-100 hrs),**
   - Clonazepam (20-50h),
   - Pregabalin (6 hrs)

2. SE Paroxetine: **Sweating**, amenorrhea, QTC prolongation

SSRI	Half life	Main side effects	Major interactions
Paroxetine	22 hours	Nausea, vomiting, dyspepsia, Abdominal pain, diarrhoea, rash, sweating, agitation, anxiety, headache, insomnia, tremor, sexual dysfunction, hyponatremia, cutaneous bleeding disorders	Potent inhibitor of CYP2D6/3A4. Increases levels of some antipsychotics and tricyclics Never use with MAOIs Avoid St John's Wort

N.B. from SSRI only Citalopram cause QTc prolongation

3. SE Lithium **Worsening psoriasis**

   Most **important** S/E of lithium - Worsening of psoriasis
   Another question on SPMM – Most **common**: polyuria/tremors

4. Topiramate **weight loss**

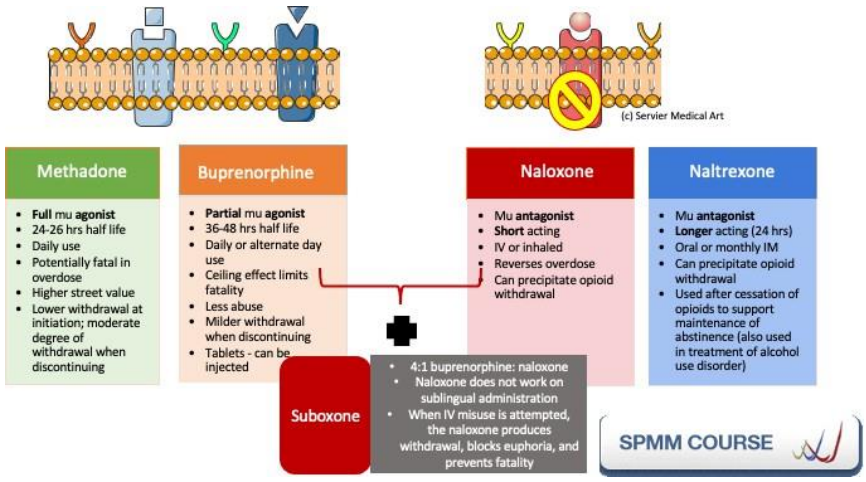
   Other side effects include: word finding difficulties (**dysnomia**), psychosis, affect acid-base balance (low bicarbonate), nephrolithiasis, arrythmia, angular closure glaucoma.

   Topiramate can cause nausea, appetite loss, and weight loss. It blocks voltage-sensitive sodium channels by an unknown mechanism. Topiramate inhibits the release of glutamate and potentiates activity of GABA.

5. Selective D2 antagonist: Amisulpride, Asenapine**, Chlorpromazine**, Risperidone,

   Amisulpride is D2 and D3 antagonist, while **Sulpride** is D2 antagonist only **(Chlorpromazine is also D2 antagonist only)**

6. Partial Mu agonist **buprenorphine**



7. Pregabalin **alpha 2 delta subunits of voltage ca channel**

8. Lamotrigine **Voltage Na**

Lamotrigine mechanism of action: [Psych-Mentor]

- **NMDA receptor modulation,**
- **Stabilises Na channels.**
- **Blocks calcium channels,**
- **Some GABA modulation.**

9. Inhibit aldehyde dehydrogenase **Disulfiram**

10. Zopiclone **alpha subunit GABA-A receptor** Alpha 1 and omega 1

11. **Glycine** at NMDA receptor **antagonist** Glycine is a co-agonist at NMDA receptors and it's presence is a prerequisite for channel activation by glutamate or NMDA.

12. EMI **Ideational apraxia**: an inability to follow a sequence of actions in the correct order e.g. Take a match out a box and strike it with your left hand

13. EMI **Ideomotor apraxia:** an inability to follow out learned tasks when given the necessary objects e.g. if given a hairbrush they try to write with it

14. EMI Unawareness one side of body: **Hemisomatagnosia**

[SPMM] **Lesions in the non-dominant parietal lobe result in anosognosia, hemisomatagnosia, dressing apraxia and prosopagnosia. Gerstmann syndrome is associated with lesions of the dominant parietal lobe.**

15. EMI : A Rogerian psychotherapy developed by incorporating the Beck's analytic therapy

RBC pneumonic

A. Cognitive therapy
B. Behaviour therapy ???
C. Analytic therapy
D. DBT
E. Rational emotive therapy
F. Group therapy
G. Cue therapy
H. 12-step program
I. Alcohol Anonyms

After many discussions around that question we agreed on cognitive therapy as the answer when the question asks about Rogerian developed by INCORPORATING BECK'S.

If the question about Rogerian only then answer will be Motivational interviewing.

16. EMI: During substance abuse therapy, Alcohol or substance signifier is used in controlled conditions?

A. Cognitive therapy
B. Behavior therapy
C. Analytic therapy
D. DBT
E. Rational emotive therapy
F. Group therapy
G. Cue therapy.
H. 12-step program
I. Alcohol Anonymous

17. EMI: During substance abuse treatment, people believe that they are helpless, hopeless and powerless?

A. Cognitive therapy
B. Behavior therapy
C. Analytic therapy
D. DBT
E. Rational emotive therapy
F. Group therapy
G. Cue therapy.
H. 12-step program
I. Alcohol Anonymouss

18. EMI Incongruence between action and gesture – **Paramimia** (disorganized facial expressions)

19. EMI a misuse of gestures in expressing thoughts or words that produce an appearance of inappropriateness of affect, **Parathymia** (diminished affect modulation)

Example of Paramimia: Imagine a person with paramimia who is trying to express happiness. They might be talking about something joyful but have a neutral facial expression, no smile, making it difficult for others to perceive their emotional state.

Suppose someone with **Parathymia** is discussing a tragic event like the death of a loved one. Instead of using a sad or mournful tone, they might speak with an inappropriate, cheerful tone or with a complete lack of emotional inflection.

Mimia is about gestures and expressions with face or hand, **Thyme** is the mood / emotion

20. EMI Postural hypotension **alpha 1 antagonist**

Postural hypotension can be also caused by Alpha 2 agonism. E.g. Clonidine and Guanfacine

21. What makes people eat less released by duodenum/jejenum - **CCK**.

22. What is released by hypothalamus and stimulates appetite- **Neuropeptide Y**.
Options: POMC, CKK, gherlin, leptin, insulin, neuropepride y PYY

23. Motor cycle accident paralyzed one limb **Conversion (Dissociative neurological symptom disorder in ICD11)**

    N.B: A **superior quadrantanopia** results from an insult to the **optic radiation** inferiorly in the **temporal** lobe, resulting in a 'pie in the sky' type of visual field defect (Figure 1d), while an **inferior** quadrantanopia is caused by damage to the **parietal** lobe optic radiation.

24. Female gynecological, Gastro-symptoms multiple Dr convinced missing something:
    Options
    - Hypochondriasis disorder,
    - **Somatization disorder (bodily distress disorder according to ICD11)**
    - Conversion

25. Trinucleotide repeat:

    - **Fragile X: CGG**
    - **Huntington: CAG,**
    - **Friedrich's Ataxia: GAA,**
    - **Myotonic dystrophy: CTG**

26. Purine Lesh Nhan syndrome **X linked recessive, HGPRT deficiency**

    - Inherited genetic disorder characterized by **impaired purine salvage pathway**, resulting in an **overproduction of uric acid**
    - Inheritance: **X-linked recessive**
    - Pathophysiology: defect in the enzyme **hypoxanthine-guanine phosphoribosyltransferase (HGPRT)** → impaired conversion of hypoxanthine to IMP and guanine to GMP → excess uric acid and ↑ de novo purine synthesis

27. Multifactorial: **Autism and Schizophrenia (Complex non-Mendelian pattern of inheritance)**

28. **Tau mutation**: (options missing)

    Select one:
    A. Familial AD,
    B. sporadic AD,
    C. Familial Frontotemporal dementia

    Tauopathies include: **APCD all squared**

    - Alzheimer's disease
    - Argyrophilic granular dx
    - Picks dx
    - Progressive supranuclear palsy
    - CJD
    - Corticobasal degeneration
    - Down syndrome

- Dementia pugilistic a

[Psych-Mentor] **Alzheimer's** disease is a progressive degenerative disease of the brain accounting for the majority of dementia seen in the UK

Genetics:

- Most cases are **sporadic.**
- **Around 5% of cases are inherited as an autosomal dominant** trait.
- Mutations in the **amyloid precursor protein** (chromosome **21**), **presenilin 1** (chromosome **14**) and **presenilin 2** (chromosome **1**) genes are thought to cause the inherited form.
- Apoprotein E allele E4 - encodes a cholesterol transport protein

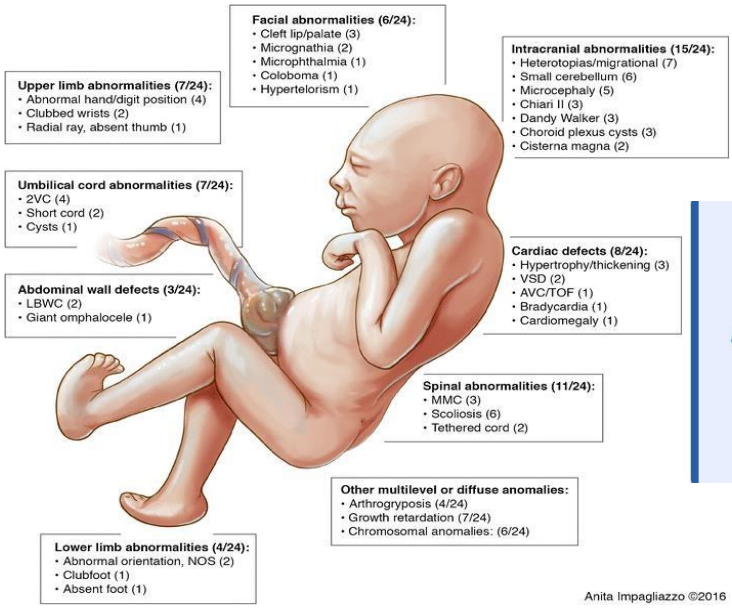
29. Tardive dyskinesia due to: **D2 super-sensitivity**

- D2 **Occupancy** – Parkinson's disease (in striatum) , also Akathisia
- D2 **Over-sensitivity** – Tardive Dyskinesia.

Schizophrenia genes
Dysbnidin (chromosome 6) NRG(chromosome 8)
Neur-EIGHT-gulin
SIX-bindin
Disc-ONE

30. Trisomy 18: **Edward Syndrome**

Severe intellectual disability, babies often die soon after birth. **Kidney malformations, upturned nose, webbing of second and third toes, and clubbed feet (rocker bottom)**



PRINCE Edward turned **18**: **P**rominent occiput, **R**ocker-bottom feet, **I**ntellectual disability, **N**ondisjunction (in meiosis), **C**lenched fists, low-set **E**ars, and chromosome **18**.

31. Use glutamate as neurotransmitter: Astrocytes, **pyramidal neurons**

32. Efferent(out) to cortex: **Pyramidal neurons**

33. Maternal deprivation theory of Bowlby: **Protest-despair-detachment**

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34. Preconventional conventional post conventional: **These are the 3 levels of moral development by Lawrence Kohlberg**

    Preconventional: obedience and punishment, self-interest(instrumental-relativist)
    Conventional: good boy: nice girl and authority (law and order)
    Post conventional: social contract and universal ethical principles.

    0-9/7-12
    9-20/13-16
    >20/16-20

35. Using saline pads - Someone stated that this question was asking about the procedure where we use saline pads, options may include:
    - DBS [deep brain stimulation],
    - ECT,
    - rTMS [Repetitive Transcranial Magnetic Stimulation],
    - **tDCS [transcranial direct current stimulation]**

    Performing a Stimulation Session
    Here we describe a standard **tDCS (transcranial direct current stimulation)** set-up, using a target and a reference electrode. First, the desired locations of where the electrodes will be positioned need to be ascertained (further details of localization techniques are in section Localizing Electrode Placement). Prior to attaching the electrodes to the scalp, the Experimenter should ensure that there is no damaged or broken skin. If **saline** is being used as a conductive substance, the electrodes may be placed in sponge holding bags, saturated so that they are sufficiently damp but not dripping. However, it is **becoming increasingly common to use conductive paste or EEG gel to affix the** electrodes to the scalp, which may control the distribution of the current more effectively than saline. The participant's hair should be parted to ensure good contact between scalp and electrode. Saline should not run down the scalp or spread over the hair. Electrodes are then attached to the stimulator using wires connected to corresponding anodal/cathodal ports. Once the electrode is placed over the target region it should be secured using a cap, rubber bands or elastic tubular netting. The reference electrode should then be secured in the same manner.

    Transcranial Direct Current Stimulation (tDCS) passes current through the scalp, skull, and meninges in order to stimulate the brain, generally undetectably. A small electric current (1–2 mA) is delivered to the scalp by a battery-driven device connected to two saline-soaked surface electrodes. The current is not sufficient to generate action potentials but instead alters neuronal resting membrane potentials. Like rTMS, tDCS can alter cortical excitability. tDCS used in **depression.**

Stimulate cranial nerve 10

Vagus nerve stimulation works by applying electrical impulses to the vagus nerve. The stimulation of the vagus nerve can be performed in two different ways: a direct invasive stimulation, which is currently the most frequent application and an indirect transcutaneous non-invasive stimulation. Invasive VNS (iVNS) requires the surgical implantation of a small pulse generator subcutaneously in the left thoracic region. Electrodes are attached to the left cervical vagus nerve and are connected to the pulse generator by a lead, which is tunneled under the skin. The generator delivers intermittent electrical impulses through the vagus nerve to the brain. **It is postulated that these electrical impulses exert antiepileptic, antidepressive, and anti-inflammatory effects** by altering the excitability of nerve cells. In contrast to iVNS, transcutaneous VNS (tVNS) allows for a non-invasive stimulation of the vagus nerve without any surgical procedure. Here, the stimulator is usually attached to the auricular concha *via* ear clips and delivers electrical impulses at the subcutaneous course of the afferent auricular branch of the vagus nerve. **A pilot study that examined the application of VNS in 60 patients with treatment-resistant depressive disorder showed a significant clinical improvement in 30–37% of patients and a high tolerability**. Five years later, the stimulation of the vagus nerve for the treatment of refractory depression was approved by the U.S. Food and Drug Administration (FDA) (79). Since then, **the safety and efficacy of VNS in depression has been demonstrated in numerous observational studies** as can be seen below. **In contrast, there is no randomized, placebo-control clinical trial that reliably demonstrates antidepressant effects of VNS**.

36. Rogers and beck CBT techniques combined:
    - group therapy.
    - **Cognitive therapy**
    - motivational interview (increase cognitive dissonance)

37. ICD11 adult lady who was abused and keeps thinking about the past when she goes home. complex pTSD, from the information given we think it is normal PTSD

Post-traumatic stress disorder (PTSD) is a syndrome that develops following exposure to an extremely threatening or horrific event or series of events that is characterized by all of the following:

1) **Re-experiencing** the traumatic event or events in the present in the form of vivid intrusive memories, **flashbacks**, or **nightmares**, which are typically accompanied by strong and overwhelming emotions such as fear or horror and strong physical sensations, or feelings of being overwhelmed or immersed in the same intense emotions that were experienced during the traumatic event;

2) **Avoidance of thoughts and memories** of the event or events, or avoidance of activities, situations, or people reminiscent of the event or events; and

3) **persistent perceptions of heightened current threat**, for example as indicated by hypervigilance or an enhanced startle reaction to stimuli such as unexpected noises. The symptoms must persist for at least several weeks and cause significant impairment in personal, family, social, educational, occupational or other important areas of functioning.

Complex post-traumatic stress disorder (Complex PTSD) is a disorder that may develop following exposure to an event or series of events of an extremely threatening or horrific nature, **most commonly prolonged or repetitive events** from which escape is difficult or impossible (e.g., torture, slavery, genocide campaigns, prolonged domestic violence, repeated childhood sexual or physical abuse). The disorder is characterized by the **core symptoms of PTSD**; that is, all diagnostic requirements for PTSD have been met at some point during the course of the disorder. In addition, Complex PTSD is characterized by

1) **severe and pervasive problems** in **affect** regulation;
2) **persistent beliefs about oneself as diminished, defeated or worthless**, accompanied by deep and pervasive feelings of **shame, guilt or failure** related to the traumatic event; and
3) **persistent difficulties in sustaining relationships** and in feeling close to others. The disturbance causes significant impairment in personal, family, social, educational, occupational or other important areas of functioning.

38. Child 18m in care institution doesn't seek out when injured: **Reactive attachment disorder**

Reactive attachment disorder is characterized by grossly abnormal attachment behaviours in early childhood, occurring in the context of a history of grossly inadequate child care (e.g., severe neglect, maltreatment, institutional deprivation). Even when an adequate primary caregiver is newly available, the **child does not turn to the primary caregiver for comfort, support and nurture, rarely displays security-seeking behaviours** towards any adult, and does not respond when comfort is offered. **Reactive attachment disorder can only be diagnosed in children,** and features of the disorder develop within the **first 5 years of life**. However, the **disorder cannot be diagnosed before the age of 1 year (or a developmental age of less than 9 months),** when the capacity for selective attachments may not be fully developed, or in the context of Autism spectrum disorder.

39. Five year old child who abused, neglect running to social worker and sitting on knee, asking to go home with other parent: **-Disinhibi ted social engagement disorder**

Disinhibited social engagement disorder is characterised by grossly abnormal social behaviour, occurring in the context of a history of grossly inadequate child care (e.g., severe neglect, institutional deprivation). The child approaches adults indiscriminately, lacks reticence to approach, will go away with unfamiliar adults, and exhibits overly familiar behaviour towards strangers. **Disinhibited social engagement disorder can only be diagnosed in children, and features of the disorder develop within the first 5 years of life. However, the disorder cannot be diagnosed before the age of 1 year** (or a developmental age of less than 9 months), when the capacity for selective attachments may not be fully developed, or in the context of Autism spectrum disorder.

Added during revision.
Right Answers.
Not sure about the answer.

EMI about Parenting style:

40. High demands with no rigidity and parents are responsiveness: **Authoritative**.
41. Low self-esteem low competence. **Uninvolved (permissive) (lower in self-esteem and competency than authortrian)**

	Demanding	Undemanding
Responsive	*Authoritative*/Propagative	Indulgent (*Permissive*)
Unresponsive	*Authoritarian*/Totalitarian	Neglectful

Authoritarian Parenting [Psych-Mentor]

This style is characterized by strict rules and punishment if the rules are not adhered to. Explanations behind the rules are often lacking. These parents are focussed on status and obedience. Authoritarian parenting styles generally lead to **children who are obedient and proficient, but they rank lower in happiness, social competence and self-esteem.**

Authoritative Parenting

Authoritative parents are similar to authoritarian parents except they tend to be more responsive to their children. There are strict rules but explanations behind them are given. When children fail to meet expectations they are nurtured rather than punished. The focus here is on setting standards but also being supportive. Authoritative parenting styles tend to result in **children who are happy, capable and successful.**

Permissive Parenting

These parents rarely discipline their children. They avoid confrontation and allow their children to self-regulate. Permissive parents prefer to take the role of a friend rather than a disciplinarian.

Permissive parenting often results in children who rank low in happiness and self-regulation. **These children are more likely to experience problems with authority and tend to perform poorly in school.**

Uninvolved Parenting

This style of parenting is characterised by little involvement and few demands.

Uninvolved parenting styles rank lowest across all life domains. These children tend to lack self-control, have **low self-esteem and are less competent than their peers.**

EMI about Philosophy principle

42. Choosing group therapy over individual therapy because it will help a larger group: **Utilitarianism**.
43. Pt on clozapine agranulocytosis so the Dr decide not to take it again- **non-maleficence.**
44. Pt lung cancer after X ray continue treatment without informing the patient- **paternalism**

Utilitarianism Group therapy preferred more than individual.

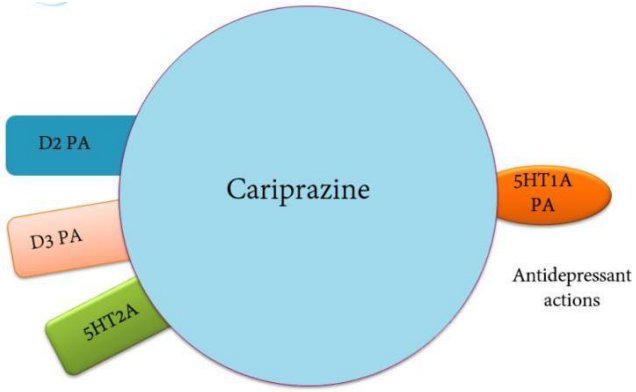
MCQ

45. Pluripotent stem cell at what age
    - **2nd trimester** (6 months intrauterine)
    - 3 month old age
    - 10 teenage
    - 20 adult child

    N.B. Pluripotent stem cell comes from embryo that is 3-5days old.

46. Cariprazine: **partial D3 agonist**

    Cariprazine acts primarily as a **D3 and D2 receptor partial agonist** with preference for D3 receptor.
    Also, **partial agonist at 5-HT1A and antagonist at 5-HT2B and 5-HT2A receptors** with **high selectivity for D3 receptor.**



47. Least sexual SE: **Vortioxetine**, Amitiptuline, Fluoxetine

    - Anti-depressants with **No** sexual side effect: **Agomelatine, Bupropion, Moclobemide**
    - Anti-depressants with **Least** sexual side effect: **Mirtazapine, Vortioxetine**, **Reboxetine, and Trazadone.**
    - **Citalopram and Fluoxetine** cause **Delayed Ejaculation**
    - **Trazadone most common to cause Priapism**

48. Most concentrated in breast milk:
    - Carbamazepine
    - **Fluoxetine**
    - Lamotrigine
    - Valproate

    Explanation
    - Fluoxetine is variable and can be more than 10% of maternal plasma level and highest reported in SSRI's
    - Lamotrigine undetectable to 48% of maternal plasma level.
    - Lithium is undetectable to 57% of maternal plasma level

Added during revision.
Right Answers.
Not sure about the answer.

49. Mirtazapine **antiemetic** effect through which receptor: **5HT3** antagonism (ondansetron is a 5HT3 antagonist too)

Antidepressant	Mechanism
Mirtazapine	Noradrenaline and serotonin specific antidepressant (NaSSa) **5HT2 antagonist, 5HT3 antagonist, H1 antagonist, alpha 1 and alpha 2 antagonist, moderate muscarinic antagonist**

50. Mechanism of action of Atomoxetine: **selective noradrenaline NaRI**

Drug	Primary mechanism (hypothesised)	Contraindications	Very common side-effects*	Interactions
Atomoxetine	Increasing NA levels in the synaptic cleft	Use of MAO inhibitors Glaucoma Pheochromocytoma Significant cardiovascular disorders Cerebrovascular disorders	- appetite decreased - headache - somnolence - abdominal pain (includes epigastric discomfort) - vomiting - nausea - blood pressure increased - heart rate increased	Metabolised by CYP2D6

51. MOA Memantine: **Non competitive NMDA antagonist**

DRUG	MECHANISM
Donepezil, Galantamine, Rivastigmine	Cholinesterase Inhibitors. The act by inhibiting acetyl cholinesterase enzyme that breaks down acetylcholine centrally. Rivastigmine inhibits both the acetyl and butyl-cholinesterase while donepezil and galantamine are acetyl specific. Galantamine also has nicotine agonistic properties.
Memantine	Blockade of N-methyl-d-aspartate (NMDA) glutamate receptors. Unlike ketamine, which is a high-affinity noncompetitive blocker, memantine is a non-competitive blocker with low affinity and binds only to actively open NMDA channels. Its receptor dissociation rate is relatively fast, and so it does not accumulate and interfere with normal NMDA activity.

Characteristic of cognitive enhancer	Donepezil	Rivastigmine	Galantamine	Memantine
Primary mechanism	AChE-I (selective and reversible)	AChE-I (reversible, non-competitive inhibitor)	AChE-I (competitive and reversible)	NMDA receptor antagonist (non-competitive)
Other mechanism	None	BuChE-I	Nicotine receptor agonist	5-HT3 receptor antagonist
Starting dose	5 mg daily	1.5 mg bd (oral) (or 4.6 mg/24 hours patch)	4 mg bd solution or 8 mg XL daily	5 mg daily
Usual treatment dose (and max dose)	10 mg daily	3-6 mg bd (oral) or 9.5 mg/24 hours patch	8-12 mg bd solution or 16-24 mg XL daily	20 mg daily or (10 mg bd)
Adverse effects	Diarrhoea, nausea, and headache most common	Anorexia, dizziness, nausea, vomiting, and diarrhoea most common	Nausea and vomiting most common	Dizziness, headache, fatigue, diarrhoea, and gastric pain most common
Half-life (hours)	~70 (permitting once daily dosing)	~1 (oral) – active for 10 3.4 (patch)	7-8 (oral solution) 8-10 (XL capsules)	60-100
Metabolism	CYP3A4 CYP2D6 (minor)	Minimal involvement of CYP isoenzymes	CYP3A4 CYP2D6	Primarily non-hepatic
Drug-drug interactions	Yes	Interactions unlikely	Yes	Yes
Effect of food on absorption	None	Delays rate and extent of absorption	Delays rate but not extent of absorption	None
Other	Absorbed completely from the GI tract Fairly well-tolerated and widely used	More likely to cause GI adverse effects than donepezil due to more peripheral activity	More likely to cause GI and neuropsychiatric adverse effects than donepezil	Generally better tolerated than cholinesterase inhibitors

Pseudo irreversible

52. Long acting injection need test dose
    - Olanzapine
    - **zuclopenthixol**
    - risperidone
    - paliperidone

Medication	Test dose
Haloperidol decanoate	25mg*
Zuclopenthixol decanoate	100mg
Flupentixol decanoate	20mg
Pipothiazine palmitate	25mg
Fluphenazine decanoate	12.5mg

N.B: Paliperidone needs Loading dose and not testing dose

53. G protein (metabotropic) passes through the cell membrane **7 times**/serpentine

54. Long acting opioid receptor antagonist: **Naltrexone**

55. Caffein action: (no Question mentioned)
    - Caffeine is a central nervous system stimulant that belongs to a class of compounds called **xanthines**. Its primary mechanism of action is as an **antagonist of adenosine receptors**. Adenosine is a neurotransmitter that promotes sleep and relaxation by acting on its receptors, **mainly the A1 and A2A subtypes**. Caffeine works primarily by blocking these receptors, preventing adenosine from binding to them. This antagonistic action results in increased alertness and wakefulness. [Psych-Mentor]
    - Toxic dose is **250** mg, Lethal blood levels are typically above 80 to 100 mg/L, although one case occurred at 15 mg/L.
    - Caffeine withdrawal is not recognized in DSM but is a likely diagnosis in a student who has just appeared for an exam (case 1). DSM research criteria lists headache, marked fatigue or drowsiness, marked anxiety or depression, nausea or vomiting as the prominent features that can be seen even at ow doses (from 100mg/day), beginning within 12 hours, peaking at 24-48 hours and lasting up to 1 week.
    - **Caffeine, the most widely used psychoactive compound, is an adenosine receptor antagonist**

56. BNZ with important metabolite:
    - **Diazepam**
    - lorazepam
    - oxazepam
    - alprazolam

Added during revision.
Right Answers.
Not sure about the answer.

Diazepam is metabolized to **Nordiazepam (desmethyl diazepam is the major metabolite of diazepam, also named desoxydemoxepam)**, temazepam, oxazepam, temazepam glucuronide (TG), and oxazepam glucuronide (OG).

Another question from different recall: **Which is Most biologically important metabolite**

A. diazepam,
B. **Oxazepam,**
C. lorazepam**,**

N.B. We believe if Nordiazepam (desmethyldiazepam) in the options then it should be the answer.

57. Lithium with HTN medication interact the least:
    - **Atenolol**
    - furosemide
    - benothyazide

    If **beta blocker an option, it is safer than furosemide**, but if beta blocker is not in the option we choice furosemide. **Amiloride** is used in lithium induced diabetics insipidus and not for hypertension

58. Drug absorption constant is independent of: food ,form of drug, extravascular route, **Elimination constant**

59. Muscle rigidity , sweating, confusion in women just started medication for psychotic depression haloperidol: **NMS**

60. What medication partial agonist on nicotine:
    - **Varenicline**
    - bupropion
    - buprenorphine
    - methadone

    Varenicline is a selective nicotine-receptor partial agonist (higher affinity but lower intrinsic efficacy than nicotine) used as an aid for smoking cessation, with a high affinity for the alpha 4 beta 2 nicotinic receptor. **It is recommended by NICE for smoking cessation. It both reduces cravings for and decreases the pleasurable effects of cigarettes and other tobacco products**. It undergoes very little (approx 10%) hepatic metabolism and so is safe to use in cases of liver dysfunction.

61. Erickson 22yrs: **intimacy vs isolation**

62. Kohlberg 12y old: punishment, obedience, deindividuation exchange, **social order,** universal ethical.

    SPMM
    - Pre-conventional- 7-12yrs
    - Conventional-13-16yrs
    - Post conventional-16-20yrs

Added during revision.
Right Answers.
Not sure about the answer.

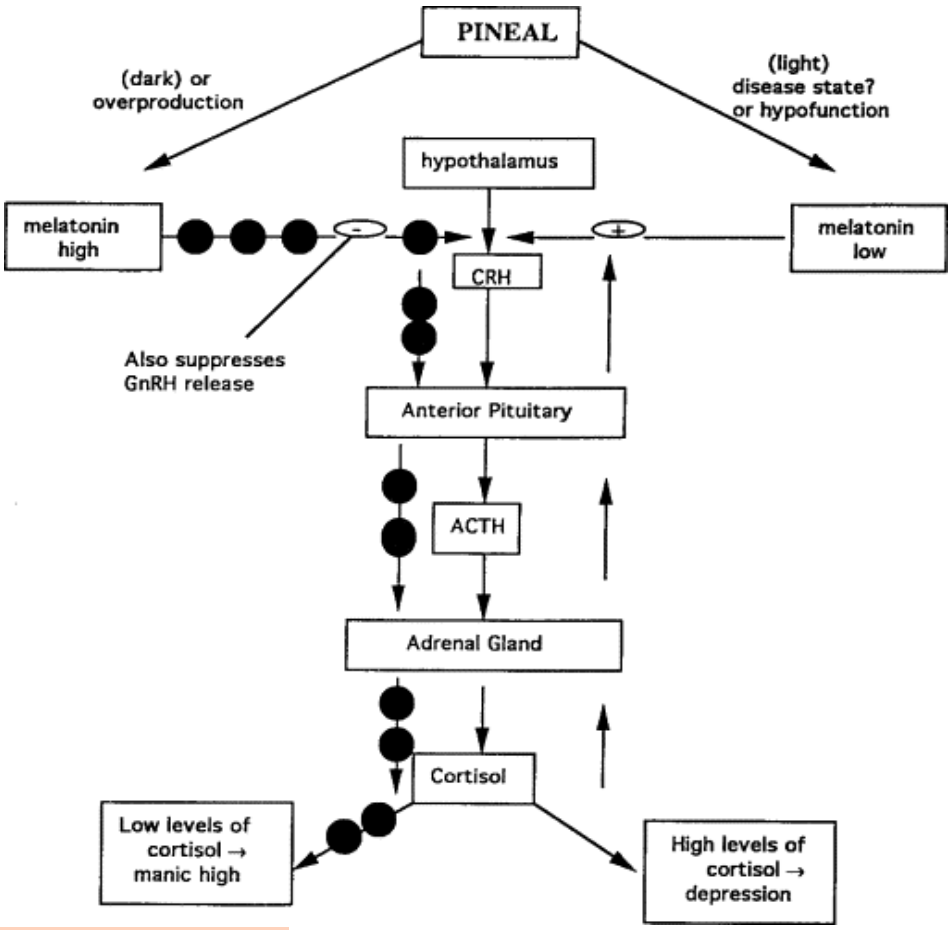
PSYCHMENTOR/TRONS

- Preconventional- 0-9yrs
- Conventional- 9-20
- Post conventional- >20yrs

63. **Werneck's** aphasia: Fluent agrammatical, comprehension not intact.
    - Wernicke is wordy, Broca's is broken
    - Wernicke - fluent aphasia with impaired comprehension and repetition
    - Broca's - non fluent aphasia , with impaired repetition but intact comprehension.

64. Locus's Coeruleus: **Noradrenergic**

65. Hormone increase with REM sleep: adrenaline, **ACH (melatonin is suppressed so cortisol increases)**



66. Neurotransmitter in multitasking
    - adrenal cortisol
    - Dopamine
    - serotonin
    - GABA

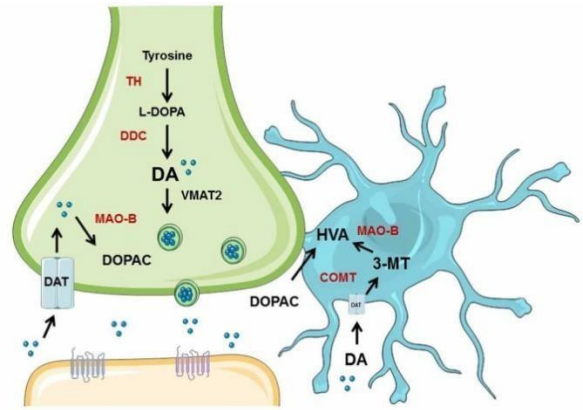
Added during revision.
Right Answers.
Not sure about the answer.

→ **Cortisol production is increased in multitasking and dopamine is released in multitasking**

67. What is hydroxylation in production of dopamine: **tyrosine**, L-Dopa, COMT.

68. What degrade dopamine in presynaptic?

- **MAO**,
- COMT,
- tyrosine hydroxylase,
- dopamine decarboxylase



69. Causes of Inattention blindness:

Examples include:
Use satellite navigator,
Drinking alcohol more than limited,
Eating sandwich,
Text mobile and call mobile

Inattentional blindness, also known as perceptual blindness, refers to the phenomenon where an individual fails to notice or perceive an unexpected stimulus or event in their visual field, even though it may be clearly visible. It occurs when attention is focused on a specific task or object, leading to a lack of awareness of other stimuli that may be present. The term "inattentional blindness" was coined to describe this phenomenon by psychologists Arien Mack and Irvin Rock in the late 1990s.

70. Test need interviewer: **Hamilton**

Self-rated	Observer rated
Beck's Depression Inventory	Dementia scales
Zung Depression Inventory	PANSS
Symptoms checklist SCL	BPRS
GHQ	HAMD
Lunser's (extrapyramidal)	MADRS
Edinburgh postnatal depression scale	YBOCS
	SCID

71. Screening tool **EPDS**, **'caseness', score of above 12/13 is suggestive.**

72. Gene implicated in ADHD: **DRD4>DRD5.**

Others: **DAT1, DRD4, DRD5, 5HTT, HTR1B, and SNAP25**
DRD2- is implicated in **schizophrenia Tourette, PTSD, affective disorder.**

73. Familial frontotemporal dementia: tau positive, ubiquitin negative/**tau negative ubiquitin positive.**

74. Autism: **hypoplasia of cerebellar vermis**

Added during revision.
Right Answers.
Not sure about the answer.

75. ApoE4 **homozygous 10-30 times**, ApoE4 heterozygous 3 times increased risk

Gene	Chromosome
APP	21
PSEN-1	14
PSEN-2	1
APOE	19
SORL1	11

76. **Human prion disease**?

1. Creutzfeldt-Jakob Disease (CJD)
2. Variant Creutzfeldt-Jakob Disease (vCJD)
3. Gerstmann-Straussler-Scheinker Syndrome
4. Fatal Familial Insomnia
5. Kuru

In **animals**, examples include **bovine spongiform encephalopathy** (**BSE** or "mad cow disease"), scrapie in sheep, and chronic wasting disease (CWD) in deer and elk. **BSE occurs in cow and if infected meat is eaten, it causes variant CJD.** Scrapie seen in goat and sheep, no evidence of infection in human if infected meat is eaten.

77. Schizophrenia: reduction in volume of hypothalamus, grey mater cortex, White matter striatum, amygdala

Structural and functional changes to the brain: **Enlarged lateral and third ventricles**, Decreased symmetry **Decreased volume of the limbic system, prefrontal cortex**, and **thalamus. Decrease Volume of the hippocampus and amygdala.** Hypoactivity of the frontal lobes and hyperactivity of the basal ganglia. There is also increase in Striatum volume in Schizophrenia

Neuroimaging findings in schizophrenia
Ventricular enlargement
Loss of grey matter – especially insular cortex, anterior cingulate (medial prefrontal cortex) and medial temporal lobe
Progressive loss of brain volume in first few years of diagnosis
fMRI reveals poor DLPFC activation in executive tasks
Decreased NAA (N-Acetyl aspartate) in PFC (neuronal loss) in MRS
Widespread reduction in DTI (diffusion tensor) – fractional anisotropy: frontal and corpus callosum – more in chronic treated patients

78. PTSD most implicated **amygdala** (fear response, consolidation of emotional memories)

Reduced **bilateral amygdala volume and left hippocampal volume**

79. Empathy: **Insula**, DLPC, occipital cortex, amygdala, **Supramarginal gyrus particularly on the right**

The supramarginal gyrus (plural: supramarginal gyri) is a portion of the **parietal** lobe of the brain. It is one of the two parts of the inferior parietal lobule, the other being the angular gyrus. It plays a role in phonological processing (i.e. of spoken and written language) and emotional responses.

80. **Confabulation:**

Select one:

A. Can only occur in Korsacoff syndrome
B. May occur in clouded consciousness
C. There is deliberate falsification of facts
D. Autobiographical memory if affected
E. There is subjective awareness of memory impairment

Confabulation is a falsification of memory occurring in clear consciousness associated with organic states. Suggestibility is a prominent feature of confabulation. It is often described in Korsakoff syndrome. There can either be confabulation of embarrassment or of fantastic nature.

- Confabulation has been known to occur among clients with brain damage, Fetal Alcohol Spectrum Disorder (FASD), traumatic-brain injury (TBI), and Wernicke-Korsakoff syndrome (WKS).
- Does not occur in clouded consciousness
- **Autobiographical memory** is a memory system consisting of episodes recollected from an individual's life, based on a combination of episodic and semantic memory. It is thus a type of explicit memory.
- **Conclusion:** We think the highlighted answer is the right one, it is not ALWAYS affected though, Also sometimes there might be subjective awareness of confabulation so we decided to go for autobiographical memory first, if not an option then subjective awareness.

81. Different between complex PTSD&PTSD: include:

- **Decreased self-worth,**
- **Affective symptoms,**
- **poor interpersonal relationships, (first option to choose if more than right answer in options)**
- **repetitive exposure to trauma.**

82. Different between food restrictive disorder & anorexia. [ICD11]

Essential (Required) Features: Avoidance or restriction of food intake that results in either or both of the following:

- The intake of an insufficient quantity or variety of food to meet adequate energy or nutritional requirements that has resulted in significant weight loss, clinically significant nutritional deficiencies, dependence on oral nutritional supplements or tube feeding, or has otherwise negatively affected the physical health of the individual.
- Significant impairment in personal, family, social, educational, occupational or other important areas of functioning (e.g., due to avoidance or distress related to participating in social experiences involving eating). The pattern of eating behaviour is not motivated by preoccupation with body weight or shape.
- Restricted food intake and consequent weight loss (or failure to gain weight), or other impact on physical health or related functional impairment, are not due to unavailability of food; are not a manifestation of another medical condition (e.g., food allergies, hyperthyroidism) or mental disorder; and are not due to the effects of a substance or medication, including withdrawal effects.

Individuals with Anorexia Nervosa, like individuals with Avoidant-Restrictive Food Intake Disorder, present with a pattern of restricted eating and significantly low body weight, with similar health-related consequences. The difference is that in Anorexia Nervosa, behaviours to establish or maintain an abnormally low body weight are usually explicitly motivated by **a desire for thinness** or an **intense fear of gaining weight**.

83. ICD11 criteria difference between mania and hypo mania:

Mania **affects functioning**, might need hospital admission, and could have **psychotic** features, is **7 days minimum** vs hypomania is **several** days minimum.

A hypomanic episode is a persistent mood state lasting at least **several** days characterised by mild elevation of mood or increased irritability and increased activity or a subjective experience of increased energy, accompanied by other characteristic symptoms such as rapid speech, rapid or racing thoughts, increased self-esteem, an increase in sexual drive or sociability, decreased need for sleep, distractibility, or impulsive or reckless behaviour. The symptoms are not severe enough to cause marked impairment in occupational functioning or in usual social activities or relationships with others, does not necessitate hospitalization, and there are no accompanying delusions or hallucinations.

84. Reciprocal inhibition: **2 opposing emotions cannot coexist** vs systemic desensitization **graded exposure and preventing response**

85. Parkinson's disease true: **Frequency of tremor 4-6 Hz (5 Hz)**

86. **Make person find interpretation to psychosis**

Select one

A. Trema
B. **Apophenia (delusional mood/perception),**
C. Anastrophe
D. consolidation

Conrad proposed five stages involved in the **formation of delusions:**

- **Trema**: Delusional mood representing a total change in the perception of the world
- **Apophany**: A search for, and the finding of **new meaning for psychological events**
- **Anastrophe**: **Heightening** of the psychosis, the feeling that everything revolves around the patient.
- **Consolidation**: Forming of a new world or psychological set based on new meaning
- **Residuum**: Eventual autistic state

87. Preserved in schizophrenia: **procedural memory**, semantic, working, long term

Numerous neuropsychological studies demonstrate a broad range of measurable cognitive deficits in Schizophrenia including **impairments in attention, working memory (WM), episodic memory, processing speed and executive functions**. Among these traditional cognitive domains, **episodic memory has frequently been highlighted as showing the largest effect sizes among cognitive deficits in Schizophrenia**

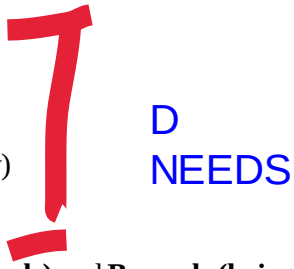
88. Anti-inflammatory cytokine involved in depression: **IL6 (IL1Beta increased)**

DEPRESSION, SCHIZOPHRENIA, IL-**1 ,** IL-**6** along with **CRP and (TNF-α),** 1L-**2,6, 8**

89. Component of ORIGINAL Maslow hierarchy of needs: **Self-esteem** (deficiency need 4thlevel)

At the bottom of the hierarchy are the most basic needs. He proposed that a person would only become concerned with the needs of a particular level when all the needs of the lower levels had been satisfied. Hence a person will not be motivated to address their social needs if they have outstanding physiological needs such as the need for food and water.

- **Level 5** (most advanced) - Self-actualization (quest to achieve full potential, e.g. needs for truth and meaning)
- **Level 4** - Esteem needs (e.g. Social recognition, personal worth)
- **Level 3** - Social needs (e.g. Feelings of love and belonging)
- **Level 2** - Safety needs (e.g. Roof over your head, financial security)
- **Level 1** (most basic) - Physiological needs (e.g. Air, water)



Maslow made a fundamental distinction between **D-needs (deficiency needs)** and **B-needs (being needs). B-needs allow us to reach our full potential** but they **cannot be fulfilled until the D-needs have first been satisfied. D-needs relate to the bottom 4 levels**.

Later in life (1970), Maslow expanded upon the model and included **cognitive, aesthetic, and transcendence needs.** This resulted in an **8 staged model (see below):**

- Cognitive needs - knowledge and understanding, curiosity, exploration, need for meaning and predictability.
- Aesthetic needs - appreciation and search for beauty, balance, form, etc.
- Transcendence needs - A person is motivated by values which transcend beyond the personal self (e.g., mystical experiences and certain experiences with nature, aesthetic experiences, sexual experiences, service to others, the pursuit of science, religious faith, etc.).

90. **Concrete operation:** know the size of 2 ball although at different distance- the concept of **conservation** in **Concrete operational stage (7-11yrs)**

91. Rat walking on regions free of electric shock what learning is: **Avoidance learning a form of operant conditioning.**

92. What can be done to reduce group think? **open debate.**

Several interventions can help reduce groupthink:

- Leaders should allow each member to challenge ideas and present objections
- Members should talk about and solicit ideas with people outside the group
- Outside Experts should be invite to attend meetings

- Avoid expressing opinions about the preferred outcome
- Assign **'Devil's Advocate'** at all meetings to challenge any and all ideas

93. Popularity more ? Early Mature female mature elderly more advanced

94. Male with their peer, less confident, family birth order (youngest):

The present study reveals that middleborn students have been more represented than the firstborns (Table 2). Specifically, Adler's (1964) study has been supported by the present study's findings that **firstborns have normally higher self-esteem** than other siblings in families. The higher self-esteem aids them in getting good academic performance and enviable work professionalism. They have high confidence which emanates from parents' direction and the tutor effect to their younger siblings (Adler, 1964).
This phenomenon denotes the nature of interactions students have in their families. Later born students end up observing and imitating what the firstborn students do. Such imitation limits the later born students from having high self-esteem. Parent-child relationship, parental support and care accelerate students' self-esteem development, and in all these, firstborns are the first beneficiaries (Collins, 2006). In social and career studies, the middle-born students are very social, interactive and like negotiations and are compassionate (Adler 1964). Further, the later born students do not follow the rules and ethics of a given place.

95. Last to mature: **Ant frontal pole**

- **Frontal lobe matures at 25**

96. **Heritability:** Genetic contribution to phenotypic variation in a **POPULATION**

97. Incidence of schizophrenia in African Caribbean migrant: **9 times more**

98. Least to cause Delay speech and language ? **Immigration**

A number of risk factors are associated with delayed speech and language (Nelson 2006):-

- Family history
- Male gender
- Twins
- Lower maternal education
- Childhood illness
- Being born late in the family order
- Young mother at birth
- Low socioeconomic status

The most reliable risk factor is thought to be a positive family history.

Large family size and smoking in pregnancy are strongest factors.

99. Absorption constant is independent of:

- **Elimination constant**
- form of drug
- composition of drug
- extravascular route of administration
- food taken with drug

100. Area under curve (plasma concentration): **Exposure to drug overtime.**

101. In first order kinetics:

- **Constant fraction of drug eliminated in unit time**
- constant amount eliminated in unit time/
- log of plasma concentration plotted against time will show non-linear elimination/
- log of plasma concentration plotted against time will show linear elimination

102. People need leader

103. Pedigree AD (EOFAD 1-6% autosomal dominant)

104. Research volunteering in old age: **Reduced risk of dementia in old people when they volunteer**

Volunteering for 100 h or more per year was associated with older adults' more positive and less negative self-perceptions of aging in the subsequent wave (i.e., 4 years later), which in turn predicted **fewer depressive symptoms**.

105. Stroop test: **selective attention**

106. Cocktail party phenomena: **selective auditory** - visual, Focused attention ,change attention

107. Temperament sensitivity anger impulsivity

108. Big five personality traits

- (Openness to experience
- Conscientiousness
- Extraversion (aka surgency)
- Agreeableness
- Neuroticism (aka emotional stability)

This can be remembered by the mnemonic OCEAN.

109. Bullying correct bullying **increase in transition from primary to secondary**–
Other : bullying more in female bullying on the teacher ?

110. Which is true about bullying:

- Commoner in girls,
- **bullies prefer same type of victim,**
- commoner in primary than secondary school,
- **bullies have less social maturity (most likely the answer)**

111. How does aggression **decrease**:

- **When the observer to aggressors punishing**
- Challenging the thoughts of the aggressor
- nonaggressive responses to a situation ?

In Bandura's classic study of imitation learning, children who saw a model punished for aggressive behaviour tended to exhibit fewer aggressive responses than children who saw the model rewarded for such behaviour, or than those who saw the model neither rewarded nor punished.

112. Most important pathophysiology for Alzheimer's: **Beta amyloid (extracellular) and tau (intracellular)**

113. **Default mode network post cingulate cortex**

- **Default mode network** - Medial prefrontal cortex, medial parietal lobe, lateral parietal lobe, inferior parietal lobe, Angular gyrus, Posterior cingulate cortex, Precuneus (**MAPPP**) – During Rest
- **Salience Network** - Anterior Insula, Dorsal anterior cingulate cortex, thalamus, striatum (**ACTS**) - Switch between the two – **aPFC – Frontal opreciulum**
- **Central executive network** - Dorsolateral prefrontal cortex, Intraparietal sulcus, Posterior parietal cortex (**DIP**) – Active

114. Congenital malformation risk of valproate:

Congenital anomalies (8-12%)
Developmental delays (31-40%)

115. HIV with Encephalitis: A.chase, memantine, CBT, **Anti-retroviral**

HIV encephalitis is a complication of HIV infection, which presents with neurocognitive dysfunction of varying severity. Almost half of HIV-infected patients suffer from this condition during their illness. Prompt **administration of antiretroviral therapy** is the most effective treatment choice for this condition

116. Most reliable indicator with myocarditis due to clozapine:

- **Eosinophilia (most specific)**
- Tropnin T
- creatine kinase

  - Most **reliable / specific : Eosinophilia**
  - Most Sensitive is Troponin only if more than twice above the upper limit.

[SPMM] Miss. Amelie Toutou is a 27-year-old woman with a diagnosis of treatment resistant schizophrenia. When started on Clozapine she develops symptoms suggestive of myocarditis. Which of the following is **a specific** indicator of clozapine-induced myocarditis**?**

Select one:

- CRP
- ECG changes
- Chest x-ray
- **Eosinophilia**
- Troponin T

Myocarditis is notoriously difficult to definitively diagnosis, with **Troponin and Creatine Kinase levels often showing poor sensitivity for the disorder.** In a review conducted of 47 cases of Clozapine induced myocarditis, the development of eosinophilia suggesting IgE mediated hypersensitivity reaction has been documented in ~66% of cases, though interpreting isolated eosinophilia in the absence of other markers is controversial. The Australian treatment guide recommends monitoring for potential myocarditis through baseline and weekly (for the first month) assessments, including vitals, laboratory evaluation of peripheral eosinophils, CRP, ECG, and troponin levels**. In the presence of suggestive symptoms of myocarditis, troponin more than twice the upper limit of normal or CRP over 100 mg/L is associated with 100% sensitivity.**

The correct answer is: Eosinophilia
N.B. following discussions regarding Clozapine induced myocarditis we believe answer will be most probably Eosinophilia. As Troponin is only sensitive if twice above normal limit. If the question most URGENT test then answer Troponin.

117. Adding new information in existing schema:

- **Assimilation**,
- accommodation (adjusting the schema to new info)
- equilibrium

118. Moving to the UK practice English in work return home retain culture, religious, family matter: **Biculturalism/Integration**

119. His culture dissolved and take up the dominant culture: **Assimilation**.

- **Marginalization**: reject both culture
- **Separation**: retained your culture and rejected the host culture.

120. Sally Anne test what: **Theory of mind**
Theory of mind starts developing at **3years and developed completely at 4years**. It is deranged in ASD

121. Attachment: Attachment occurs in all cultures

122. Question on association study: GWAS-genome wide association studies. genetic association studies. The goal of genetic association studies is to investigate the effects of genotypic variation on disease risk. These designs are built on a concept of a genetic cause that closely resembles the concept of a risk factor in epidemiology. Moreover, they correspond well to designs previously articulated by risk factor epidemiologists.)

123. Rate limiting enzyme in alcohol metabolism - **Alcohol dehydrogenase.**

124. During interview a doctor offered handshake to a schizophrenic patient. Patients move forward and then withdraw hand for 20 times - **Ambitendency**.

125. **Hawthorne effect**: observer effect: people tend to perform well when they are being observed.

Halo effect: The halo effect is a type of cognitive bias in which our overall impression of a person influences how we feel and think about their character. It is a cognitive bias whereby the perception of one characteristic of a person or object is influenced by the perception of another. **An example would be judging a good looking person as more intelligent.**

The term "the halo effect" was coined in the early 1920s by an American psychologist called Edward Thorndike. He astutely noted that 'conventionally-attractive' people tend to be perceived as being competent and successful. Essentially, one redeeming aspect of a person leads us to cast their whole character in an inherently positive light. As a result, any negative characteristics are therefore overshadowed, such poor punctuality and communication skills, or frequently missing deadlines

Horn effect is also called the reversed halo or devil's effect. The horn effect is essentially the complete opposite of the halo effect. The horn effect is a cognitive process in which we immediately ascribe negative attitudes or behaviours to someone based on one aspect of their appearance or character. A common example of this is overweight people, who unfortunately are often stereotyped as being lazy, slovenly or irresponsible. Whether it's how someone looks, speaks or even their body language, hiring managers may quickly have a 'bad feeling' about them.

126. patient can be positioned in uncomfortable postures, with resistant tone which are maintained for a considerable period. **Waxy flexibility/cerea flexibilitas/catalepsy**

127. Advantage of prospective study ?? no retrospective bias

a) **Recall bias was minimized**
b) **It was possible to estimate the population at risk**

- **Disadvantage**

Added during revision.
Right Answers.
Not sure about the answer.

The results may be biased if a substantial number of cohort members were lost to follow-up: **attrition bias**.

128. According to ÆSOP study, what accounts for the nine-fold increased risk of schizophrenia in patients of African-Caribbean descent: **Immigration**

129. Depression learned helplessness: **Forced swim test**.

The forced swim test is a rodent behavioral test used for evaluation of antidepressant drugs, antidepressant efficacy of new compounds, and experimental manipulations that are aimed at rendering or preventing depressive-like states. Mice are placed in an inescapable transparent tank that is filled with water and their escape related mobility behavior is measured. The forced swim test is straightforward to conduct reliably and it requires minimal specialized equipment. Successful implementation of the forced swim test requires adherence to certain procedural details and minimization of unwarranted stress to the mice.

130. Mother behavior interaction: **Oxytocin**, progesterone, estrogen, Testosterone

[SPMM] **EXPLANATION**: Oxytocin is a crucial regulator of human social behaviour. Individual differences within the oxytocin system – assessed through hormonal levels, epigenetic modification, or genetic variation - have been linked to differential sensitivity to social cues, prosocial behaviours, and stress responsiveness in adults. Early experience in maternal care predicts epigenetic change within the infant oxytocin system, levels of which are in turn reflective of infant temperament.

131. Tract white matter track **DTI** MRI PET

T2 weighted MRI .

- **DTI (Diffusion Tensor Imaging)** – for **white matter detection**.
- **PET** – measurement of regional brain **metabolism** by tagging organic compounds in metabolic pathway i.e. gives direct info about neuronal metabolism.
- **MRS** (Magnetic resonance spectroscopy) – dynamic measurement of resonance hydrogen and phosphate ions – odd number of protons - **Neurotransmitters**
- **SPECT** – measurement of receptor binding and distribution using radiolabeled receptor
- **fMRI** - tasks-related changes in blood oxygenation level
- **Magnetoencephalography** – measurement of static magnetic activity of brain
- **DAT** (dopamine active transporter) – differentiate between LBD and Alzheimer's, **DAT is done with SPECT**
- Best to visualize damage in basal ganglia due to Lewy DAT

Reduced dopamine uptake for LBD in basal ganglia

132. When used MRI what 's signal?

Signal Intensity: refers to the brightness or darkness of an area on an MRI image. Signal intensity can be used to differentiate between normal and abnormal tissues and help diagnose certain conditions.

MRI capitalizes on the inherent magnetic properties of tissues. First a large magnetic field is applied that aligns all the hydrogen atoms in the same direction. Then a radio frequency is applied that realigns the atoms (Larmour frequency). Once the radio frequency is turned off the atoms realign and give of a signal in the process. This signal is detected by the MRI machine to produce an image. **T1 and T2 weighting refers to the way atoms realign after the radio frequency is removed.**

- **T1-weighted MRI** enhances the signal of the **fatty** tissue and suppresses the signal of the water.
- **T2-weighted MRI** enhances the signal of the **water**. Consideration of all the information provided by these modalities is conducive to MRI image analysis and diagnosis.

133. Alcohol more than limited dose craving: Alcohol use disorder, alcohol dependence syndrome, harmful use

Bad than harmful use

ICD11 **Alcohol dependence:** is a disorder of regulation of alcohol use arising from repeated or continuous use of alcohol. The **characteristic feature is a strong internal drive to use alcohol**, which is manifested by impaired ability to control use, increasing priority given to use over other activities and persistence of use despite harm or negative consequences. These experiences are often accompanied by a subjective sensation of urge or craving to use alcohol. Physiological features of dependence may also be present, including tolerance to the effects of alcohol, withdrawal symptoms following cessation or reduction in use of alcohol, or repeated use of alcohol or pharmacologically similar substances to prevent or alleviate withdrawal symptoms. **The features of dependence are usually evident over a period of at least 12 months but the diagnosis may be made if alcohol use is continuous (daily or almost daily) for at least 3 month.**

Harmful pattern of use of alcohol: A pattern of alcohol use that has caused damage to a person's physical or mental health or has resulted in behavior leading to harm to the health of others**. The pattern of alcohol use is evident over a period of at least 12 months if substance use is episodic or at least one month if use is continuous**. Harm to health of the individual occurs due to one or more of the following: (1) behaviour related to intoxication; (2) direct or secondary toxic effects

Essential (Required) Features: ICD11

A pattern of continuous, recurrent, or sporadic use of alcohol that has caused clinically significant damage to a person's physical health or mental health (e.g., Alcohol-Induced Mood Disorder), or has resulted in behaviour leading to harm to the health of others. Harm to health of the individual occurs due to one or more of the following: **(1)** behaviour related to intoxication; **(2)** direct or secondary toxic effects on body organs and systems; or **(3)** a harmful route of administration.
Harm to health of others includes any form of physical harm, including trauma, or mental disorder that is directly attributable to behaviour related to substance intoxication on the part of the person to whom the diagnosis of Harmful Pattern of Use of Alcohol applies.
The pattern of alcohol use is evident over a period of at least 12 months if alcohol use is episodic or at least 1 month if use is continuous.
Harm to health is not better accounted for by another medical condition or another mental disorder, including another Disorder Due to Substance Use (e.g., Substance Withdrawal).

Alcohol use disorders is a broad category which encompasses dependence syndrome, harmful use , withdrawal syndrome, intoxication and alcohol induced psych disorders

134. DSM5 schizophrenia subtypes removed, also removed in ICD 11. (Residual, negative, paranoid, hebephrenic/disorganized, catatonic)

135. Criteria for Delusional Disorder: **No persistent hallucination**

ICD11 **Delusional disorder** is characterized by the development of a delusion or set of related delusions, typically persisting for at **least 3 months in ICD11 (only one month in DSM-V)** and often much longer, in the absence of a Depressive, Manic, or Mixed mood episode. The delusions are variable in content across individuals, but typically stable within individuals, although they may evolve over time. **Other characteristic symptoms of Schizophrenia** (i.e. clear and persistent hallucinations, negative symptoms, disorganized thinking, or experiences of influence, passivity, or control) **are not present**, although various forms of perceptual disturbances (e.g. hallucinations, illusions, misidentifications of persons) thematically related to the delusion are still consistent with the diagnosis. Apart from actions and attitudes directly related to the delusion or delusional system, affect, speech, and behavior are typically unaffected. The symptoms are not a manifestation of another medical condition (e.g., a brain tumour) and are not due to the effect of a substance or medication on the central nervous system (e.g. corticosteroids), including withdrawal effects (e.g. alcohol withdrawal)

136. In ICD11 : **No age criteria needed.**

137. Bodily distress disorder replaced **Somatization**.
fear and anxiety obsessive compulsive stress

138. ICD11 most accurate Gaming disorder:

Gaming disorder is characterised by a pattern of persistent or recurrent gaming behaviour ('digital gaming' or 'video-gaming'), which may be online (i.e., over the internet) or offline, manifested by: **1. impaired control over gaming** (e.g., onset, frequency, intensity, duration, termination, context); **2. increasing priority given to gaming** to the extent that gaming takes precedence over other life interests and daily activities; and **3. continuation or escalation of gaming despite the occurrence of negative consequences**. The pattern of gaming behavior may be continuous or episodic and recurrent. The pattern of gaming behaviour results in marked distress or significant impairment in personal, family, social, educational, occupational, or other important areas of functioning. **The gaming behaviour and other features are normally evident over a period of at least 12 months** in order for a diagnosis to be assigned, although the **required duration may be shortened if all diagnostic requirements are met and symptoms are severe**.

Physical consequence online friends at least 12 month duration (might be related to the previous question about gaming disorder)

139. **Superstition** theory of OCD:, thoughts DM to decrease it, **Options** from July 2023 recall include:

- Immediate relive anxiety after compulsion displacement
- Behaviour and stimulus are paired by chance
- Mental processes by which threatening thoughts are removed **(likely the answer)**

The term "superstitious theory" might be a colloquial or layperson's way of describing the belief that individuals with OCD engage in certain compulsive behaviors due to irrational or superstitious beliefs or fears **(Magical thinking).**

One of the earliest, and better, psychological explanations for the acquisition and elimination of OCD symptoms is Mower's two-stage theory (Mower O H 1947). In the first stage classical conditioning occurs.

Mary's case is a good example of the first stage of Mower's theory on the acquisition of obsessions. She recalled that as a seven-year-old child she developed a fear of contracting cancer from an aunt with terminal carcinoma who came to live with her family. The aunt had a favorite chair she always sat in that Mary associated with her fear of cancer, so that eventually simply the site of the chair provoked dread in the child. Thus, by means of classical conditioning the chair became a conditioned stimulus for fear and anxiety.
However, in OCD, the conditioning does not stop with just one association of a neutral and feared stimulus. It continues by means of higher order conditioning to include more and more situations, objects, and events. Continuing with Mary's example — other people sat in the aunt's chair; and by means of higher order conditioning they too became conditioned stimuli for fear. Furthermore, the aunt sat on other furniture and touched other objects in the house. So a kind of fear conditioning contagion took place that transformed many non-fearful situations, events, circumstances, people, etc. into conditioned stimuli for fear.
By the time Mary started treatment as a 51-year-old single woman she felt contaminated whenever she ventured from her apartment into the outside world that she now considered teaming with triggers for cancer. So to prevent cancer immediately upon coming home just inside the doorway she practiced the compulsion of completely undressing depositing her clothes in a special container and then sponging down her entire body with alcohol to "decontaminate herself," then she felt "comfy." If she didn't wash, she experienced intense anxiety until she did.

Her bathing is an example of the second stage of Mower's two-stage theory. In Mower's second stage, operant conditioning occurs. This type of conditioning happens when behaviors become more frequent if followed by satisfying consequences, but less frequent if followed by aversive consequences. Everyday examples include awarding bonuses to employees for increased production or docking employees for missing deadlines. In OCD people motivated to relieve fear learn by trial and error to perform certain compulsive behaviors or mental activities that result in the satisfying consequences of diminished anxiety — but only until the next cycle of obsession and compulsions. Nevertheless, the immediate reward from anxiety relief is positive enough to perpetuate ongoing compulsions in response to continuing obsessions. Once the rituals are established they become entrenched as the only means of coping with fear and thereby prevent habituation. That is, if the person sits with the fear it naturally burns itself out, as the physiological resources for "fight or flight" are depleted and the fear is extinguished.

Added during revision.
Right Answers.
Not sure about the answer.

Fortunately, both forms of conditioning can be eliminated through the process of extinction. This requires exposure to conditioned fear triggers (e.g. things that stimulate the fear of contracting cancer) in the absence of the unconditioned stimulus (e.g actually contracting cancer), which neutralizes the conditioned fear triggers, rendering them incapable of provoking further distress. For OCD this means repeatedly exposing the person to the conditioned fear triggers only. The unconditioned fear is always absent, because the dangers posed by obsessions are always unreal or highly unlikely (e.g. contracting cancer from exposure to the outside world). As a result the fear triggering content of the obsessions is extinguished. Having been unaware of the need to experience the fear instead of avoiding or blocking it, patients have distracted themselves by using avoidance and rituals and therefore have not experienced habituation the natural elimination of anxiety from exposure to obsessions not followed by rituals. In other words, instead of avoiding contact with fear it must be faced. With the elimination of the conditioned fear the obsessions fade away, and with no obsessions there is no need for further compulsions.

140. Standard deviation for intellectual disability in ICD11:

- One more
- **two more**
- three more
- four more
- five more

Disorders of intellectual development are a group of etiologically diverse conditions originating during the developmental period characterized by significantly below average intellectual functioning and adaptive behavior that are approximately **two or more standard deviations below the mean** (approximately less than the 2.3rd percentile), based on appropriately normed, individually administered standardized tests. Where appropriately normed and standardized tests are not available, diagnosis of disorders of intellectual development requires greater reliance on clinical judgment based on appropriate assessment of comparable behavioral indicators.

141. D2 receptor occupancy in one Research in Extrapyramidal symptom: 0.2 / 0.4 / 0.5 / 0.6 / **0.8**

EPSE produced when D2 receptor occupancy is **>80% in nigrostriatal pathway.**
For **therapeutic** efficacy it is **60-80%**

142. CATIE study in metabolic syndrome in patients taking antipsychotics: **41-50**, 50-55, 30-40

The prevalence of Metabolic Syndrome at baseline in the CATIE group was 40.9%. **By gender this was 51.6% in females and 36% in males**. **Male** patients were **twice** as likely to have MS than matched controls, and **female** patients were **three** times as likely compared to matched controls.

In the CATIE trial, metabolic syndrome prevalence was 41% and 43%, respectively, using the National Cholesterol Education Program (NCEP) and AHA (fasting glucose threshold of 100mg/dl) derived criteria.

Added during revision.
Right Answers.
Not sure about the answer.

1. Pregabalin MOA. Options: **Calcium**, NMDA, AMPA

- potent ligand for the alpha-2-delta subunit of voltage-gated **calcium** channels
- Pregabalin does not bind to plasma proteins and is excreted virtually unchanged by the kidneys
- It is not subject to hepatic metabolism and does not induce or inhibit liver enzymes such as the cytochrome P450 system

2. A man unable to follow order - **Ideational Apraxia**

an inability to follow a sequence of actions in the correct order e.g. Take a match out a box and strike it with your left hand

Apraxia is the inability to carry out learned voluntary movements (assuming everything works and they are not paralyzed).
The main types of apraxia are as follows:-

- **Limb kinetic apraxia** - an inability to make fine/ delicate movements
- **Ideomotor apraxia** - an inability to follow out learned tasks when given the necessary objects e.g. if given a hairbrush they try to write with it
- **Constructional apraxia** - an inability to copy a picture or combine parts of something to form a whole
- **Ideational apraxia** - an inability to follow a sequence of actions in the correct order e.g. Take a match out a box and strike it with your left hand
- **Oculomotor** - an inability to control eye movement

3. A man unable to use hammer, uses it later - **Ideomotor Apraxia**
↑ Above Q

4. Which neuronal cell that releases glutamate: **Astrocyte**

5. Man had accident 5 years previously, now has bilateral coarse tremors and eye not focusing (reduced gaze) , sleeping tablets made his tremors better; which diagnosis:

- PTSD,
- Adjustment disorder,
- Post concessional syndrome Post concussional syndrome can persist or recur years after the original injury.

? 5 years so No concussion PTSD less likely

6. Where can you find the Dentate gyrus? -
Hippocampus → Episodic Memory

7. Receptor matching:
Memantine: Non-competitive NMDA antagonist
Buprenorphine:- partial opioid agonist
Naltrexone Reversible competitive antagonist at μ and κ receptors (δ receptor antagonist to a lesser extent)

Added during revision.
Right Answers.
Not sure about the answer.

8. Imprinting
   - phase-sensitive attachment
   - Lorenz
   - Imprinting is rapid
   - Unreinforced
   - occurs during a clearly defined period (known as the critical period)
   - Irreversible.

9. Neuroimaging findings in: PTSD**,** Neuronal axon damage, Antipsychotic effect, vCJD

   PTSD: Increased **amygdala** activation and smaller amygdala volumes are most consistently reported, Smaller hippocampal volume commonly appear repeatedly

   vCJD: Pulvinar Sign, Bilateral anterior basal ganglia high signal and thamalus

   Antipsychotic effect: increased ventricular volume and reduced frontal and medial temporal lobe grey matter over time. Basal ganglia volume may increase. **Increase in Corpus striatum with long term use of typical anti-psychotics.**

   [google] **Neuronal axon Damage** : CT Scan - Shows areas of punctate haemorrhage

   MRI shows High flair signal and in the months that follow, brain volume reduction is seen.

10. Tetrabenazine receptor of highest affinity - ?specific SERT
    VMAT-2* inhibitor, Reversible

    Tetrabenazine is an inhibitor of synaptic vesicular monoamine transporter 2 (VMAT2), the inhibition of which causes a depletion of neuroactive monoamines (**serotonin, norepinephrine and particularly dopamine)** in nerve terminals. The reduction in these active neurotransmitters results in a decrease in **spontaneous jerk-like movements** of the extremities, trunk, face and neck (chorea) that are typical of patients with degenerative neurologic conditions such as Huntington disease.

11. Ketamine: Non-competitive NMDA Antagonist

    Ketamine is a rapid-acting general anesthetic. It produces an anesthetic state characterized by profound analgesia, normal pharyngeal-laryngeal reflexes, normal or slightly enhanced skeletal muscle tone, cardiovascular and respiratory stimulation, and occasionally a transient and minimal respiratory depression. Ketamine's mechanism of action is that it binds to the phencyclidine receptor of the NMDA channel noncompetitively and thus, **inhibits activation of NMDA receptors by glutamate**.

12. Form of therapy where ppl admit feeling helpless/powerless
    - Learned Helplessness ,
    - CBT
    - 12-step therapy for AA

Added during revision.
Right Answers.
Not sure about the answer.

13. **EMI:** EEG Waves – Polyspikes 3-6 hertz, 2-4 hertz

Seizure type	Pattern during the seizure (ictal EEG)
Typical absence (aka petit mal)	Generalised 3 Hz spike-wave
Atypical absence	Slow (<2.5 Hz) generalized spike-and-wave
Focal (previously known as partial)	Focal spikes
Myoclonic	Generalized 3-6 Hz polyspike and wave discharge
Generalised tonic-clonic	EEG often obscured by artifact (movement). Generalized fast rhythmic spikes are seen in the tonic stage. Bursts of spikes and after-coming slow waves are synchronous with clonic jerks. A postictal period of irregular slow activity follows
Atonic (drop attack)	Generalized spike-and-wave is typical, with atonia at the time of the slow wave

Condition	EEG findings
CJD (sporadic only, does not apply to variant)	Early on there is non specific slowing, later periodic biphasic and triphasic synchronous sharp wave complexes superimposed on a slow background rhythm
Huntington's	Low voltage EEG, in particular no alpha (flattening)
Delirium	Diffuse slowing, decreased alpha, increased theta and delta
Delirium tremens	Hyperactive trace, fast
Alzheimer's	Reduced alpha and beta, increased delta and theta
Normal aging	Diffuse slowing, which can be focal or diffuse, if focal most commonly seen in the left temporal region
Encephalopathy	Diffuse slowing

14. Exposure of a patient to alcohol and drug misuse without actually taking the substances Behavior therapy, **Cue therapy**

Cue Exposure Therapy (CET) is a behavioristic psychological approach to treating substance use disorders (SUD) whereby individuals are exposed to relevant drug cues to extinguish conditioned responses. The procedure for CET: exposes alcohol-dependent patients to the sight and smell of alcohol without allowing consumption in an attempt to weaken the CS-US link via extinction and by extension attenuate cravings.

15. Duloxetine use: Licenced for **GAD** (Maintenance **60mg** od can go up to **120** mg max), Pain disorders

    Duloxetine is also used for **Major depressive disorder** 60mg once daily, **diabetic neuropathy** 60mg OD can go up to 120 mg – to be discontinued after 2 months if no response, **Moderate to severe stress urinary incontinence** 40 mg BD

    [SPMM] Duloxetine, along with venlafaxine, is sometimes termed as a **dual** action antidepressant (acts on both **serotonin** and **noradrenaline**). The elimination half-life of **duloxetine is approximately 12 hours**. Usual dosage ranges are 40-60 mg/day in 1-2 doses for depression, 60mg once daily for diabetic peripheral neuropathy and fibromyalgia, and 40mg twice daily for stress urinary incontinence. Studies have not demonstrated increased efficacy of duloxetine beyond 60mg/day. Some patients may require up to or more than 120mg/day, but clinical experience is quite limited with high dosing. In neuropathic pain and fibromyalgia, doses above 60mg/day have been associated with increased side effects without an increase in efficacy. Plasma levels of duloxetine can be reduced by up to 50% in smokers and 42% lower after gastric bypass surgery. Higher dose may be necessary for these patients to achieve therapeutic benefits.

16. A man whose friend died in a motor cycle accident and now has a paralyzed arm despite doctors not finding any neuro issues with the arm - ? **Conversion disorder** (it is now called dissociative neurological symptom disorder in ICD11)

    Dissociative neurological symptom disorder is characterised by the presentation of motor, sensory, or cognitive symptoms that imply an involuntary discontinuity in the normal integration of motor, sensory, or cognitive functions and are not consistent with a recognised disease of the nervous system, other mental or behavioural disorder, or other medical condition. The symptoms do not occur exclusively during another dissociative disorder and are not due to the effects of a substance or medication on the central nervous system, including withdrawal effects, or a Sleep-Wake disorder.

    6B60.6 with paresis or weakness Characterized by a difficulty or inability to intentionally move parts of the body or to coordinate movements that is not consistent with a recognized Disease of the Nervous System, other mental disorder, or another medical condition and does not occur exclusively during another Dissociative Disorder.

17. Child waking up agitated

    Sleep terrors are characterized by a sudden arousal from slow wave sleep with a piercing scream or cry, accompanied by autonomic and behavioural manifestations of intense fear.

    Sleep terrors manifest as a severe autonomic discharge, with tachycardia, tachypnea, flushing of the skin, diaphoresis, mydriasis, decreased skin resistance, and increased muscle tone. The patient usually sits up in bed, is unresponsive to external stimuli, and, if awakened, is confused and disoriented. Amnesia for the episode occurs, although sometimes there are reports of fragments or very brief vivid dream images or hallucinations. **Stage 3 is the stage when sleepwalking, night terrors, and bedwetting occurs.**

Added during revision.
Right Answers.
Not sure about the answer.

Different from **sleep nightmares**
Nightmares are frightening dreams that usually awaken the sleeper from REM sleep. The nightmare is almost always a long, complicated dream that becomes increasingly frightening toward the end. The long, dreamlike feature is essential in making the clinical differentiation from sleep terrors. The awakening occurs out of REM sleep.

18. The woman with multiple gynaecological, gastric pain…for 10 years , all tests normal, convinced drs missed something-?

- Delusional disorder, ?
- **Somatisation(Bodily distress disorder in ICD11)**

Bodily distress is about symptoms and hypochondriasis is about a specific illness.

Hypochondriasis (under OCD and related disorder in ICD11)is characterised by persistent preoccupation or fear about the possibility of having one or more serious, progressive or life-threatening **illnesses**. The preoccupation is accompanied by either: 1) repetitive and excessive health-related behaviours, such as repeatedly checking of the body for evidence of illness, spending inordinate amounts of time searching for information about the feared illness, repeatedly seeking reassurance (e.g. arranging multiple medical consultations); or 2) maladaptive avoidance behaviour related to health (e.g. avoids medical appointments). The symptoms result in significant distress or significant impairment in personal, family, social, educational, occupational or other important areas of functioning.

Bodily distress disorder is characterised by the presence of bodily symptoms that are distressing to the individual and excessive attention directed toward the symptoms, which may be manifest by repeated contact with health care providers. If another health condition is causing or contributing to the symptoms, the degree of attention is clearly excessive in relation to its nature and progression. Excessive attention is not alleviated by appropriate clinical examination and investigations and appropriate reassurance. Bodily symptoms are persistent, being present on most days for at least several months. Typically, bodily distress disorder involves multiple bodily symptoms that may vary over time. Occasionally there is a single symptom—usually pain or fatigue—that is associated with the other features of the disorder. The symptoms and associated distress and preoccupation have at least some impact on the individual's functioning (e.g. strain in relationships, less effective academic or

19. Area of brain affected in auditory hallucinations-

Language (**Wernicke's area**), and sensation of sound. Reduced gray matter volumes in the **superior temporal gyrus (STG)** have been reported in patients with schizophrenia

Added during revision.
Right Answers.
Not sure about the answer.

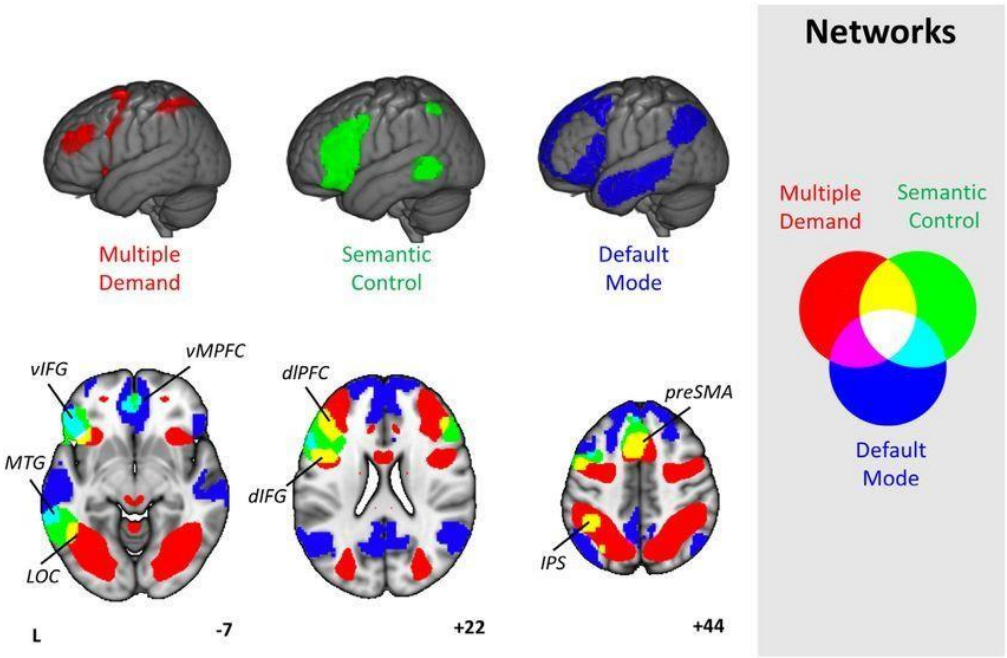
20. Area of brain involved in Default mode network

1. Default mode network - Medial prefrontal cortex, medial parietal lobe, lateral parietal lobe, inferior parietal lobe, Angular gyrus, Posterior cingulate cortex, Precuneus (MAPPP) Rest

Active when a person is not focused on the outside world and the brain is at **wakeful rest**, such as during daydreaming and mind-wandering. Also **Active thinking about themselves, remembering the past, and planning for the future**

2. **Salience Network** - Anterior Insula, Dorsal anterior cingulate cortex, thalamus, striatum (ACTS) switch between the two

3. **Central executive network** - Dorsolateral prefrontal cortex, Intraparietal sulcus, Posterior parietal cortex (DIP) active



21. Example of activity that involves the use of default mode network
See above question

Added during revision.
Right Answers.
Not sure about the answer.

22. Piaget, Kohlberg, Eisenberg, Erikson, Bowlby theories

Piaget

Piaget developed four stages of development that describe how children think and acquire knowledge.

Stage	age	Features
Sensorimotor	birth to 18-24 months	• Inborn motor and sensory reflexes • Primary Circular Reactions • Secondary Circular Reactions • Use of familiar means to obtain ends • Tertiary Circular Reactions and discovery through active experimentation • Insight and object permanence
Preoperational	2-7 years	• Deferred imitation • Symbolic play • Graphic imagery (drawing) • Mental imagery • Semiotic function (ability to use symbols) • Language
Concrete operational	7-11 years	• Egocentric thought replaced by operational thought (child now deals with information outside them) • Syllogistic reasoning (logical conclusions formed from two premises). • Conservation (Watanabe, 2017) of number (age 6-7), length (age 6-7), mass (age 7-8), area (age 8-9), weight (age 9-10), volume (age 11-12)

A 'circular reaction' is a behavior that produces an event (thumb brushes mouth, kicking moves a rattle) that leads to repetition of the behavior.

Primary circular reactions involve only the infants own body.

Secondary circular reactions are like primary, only the behavior involves external objects (e.g. shaking a rattle).

Tertiary circular reactions involve experiments with the different kinds of effects the same behavior can have (e.g. bouncing a ball down stairs then bouncing lots of other things down stairs).

Added during revision.
Right Answers.
Not sure about the answer.

23. Locus coeruleus and the neurochemical pathway
Noradrenergic projection originates at the locus coeruleus (pons) and ascends to most of the cortex via medial forebrain bundle. noradrenergic projections also descend to the Spinal cord.

24. Causes of Postural hypotension as a side effect and which receptor Alpha one antagonist. (antipsychotics), **Alpha 2 agonist** also causes postural hypotension – **Clonidine and guanfacine**

25. Alpha 2 blockade

Agitation and Impotence

2. Mechanism of adverse effects

Side effect	Receptor
Agitation	α_2 blockade, 5HT2A/2C stimulation, DRI
Akathisia	D2 blockade, 5HT2A stimulation (hence some data on mirtazapine, 5HT2A antagonist, reducing akathisia)
Delirium	Anticholinergic effect (antimuscarinic)
EPSE	D2 blockade reduces with 5HT2A antagonism
Hyperthermia	Antimuscarinic action, in serotonin syndrome, may be mediated via 5HT2A/2C
Insomnia	α_1 stimulation, 5HT2Astimulation (hence SSRIs cause insomnia)
Amnesia (memory defects)	Anticholinergic effects, $GABA_A$ stimulation
Hyperprolactinaemia	D2 blockade, 5HT1A stimulation
Disrupted slow wave sleep	SWS is maintained by 5HT2A inhibition; 5HT2A stimulation disrupts sleep architecture
Sweating	Cholinergic effect and increases with noradrenaline reuptake inhibition
Postural hypotension	α 1 antagonism
Appetite loss	5HT2A stimulation (antihistaminics can increase appetite)
GI discomfort, nausea, vomiting	5HT3 stimulation
Weight gain	5HT2C antagonism and antihistaminic effects
Anticholinergic effects	Blurred vision, exacerbation of narrow-angle glaucoma, delirium, and photophobia due to mydriasis, dry secretions, constipation, tachycardia, decreased sweating, urinary retention and hyperthermia.
Anorgasmia	α 1 antagonism, 5HT2A/2C stimulation (delayed ejaculation in SSRIs). Retrograde ejaculation due to α 1 block, anticholinergic and antihistaminic effects.
Tardive dyskinesia	Supersensitivity of dopamine receptors, which develops because of prolonged therapy with dopamine-blocking drugs
Impotence	α 2 blockade, 5HT2A/2C stimulation. 5HT2A/2c stimulation can also reduce libido.
Priapism	α 1 blockade
Obsessions	5HT1D stimulation can induce obsessions. 5HT1A and 2A stimulation reduce OCD.
Pathological gambling	Habituation of dopamine receptors on repeated use of dopamine agonists (e.g. levodopa) leading to dopamine dysregulation syndrome (DDS)

Added during revision.
Right Answers.
Not sure about the answer.

26. 5HT2A

Receptor	Action
$5HT_{1A}$	Antidepressant (agonist), anxiolytics (partial agonist)
$5HT_{1B}$	Aggression
$5HT_{1D}$	Antimigraine (antagonist)
$5HT_{2A}$	Antipsychotic (antagonist); hallucinogens (agonist / partial agonist); implicated in working memory; also seen in platelets and smooth muscles
$5HT_{2B}$	Stimulation may produce cardiac valvular fibrosis (dexfenfluramine)
$5HT_{2C}$	Anxiogenic and anorexic effect (agonists)
$5HT_{3}$	Antiemetic (antagonist)
$5HT_{6}$	Possible antipsychotic/antidepressant action (antagonism)
$5HT_{7}$	Regulation of circadian rhythm

27. Antidepressant with discontinuation syndrome risk – **Venlafaxine**?, Fluoxetine?

Others: Paroxetine, TCA (Amitriptyline, Imipramine), MAO inhibitors
The shorter the half-life the more common it will cause discontinuation syndrome/

Discontinuation symptoms are common in patients stopping antidepressants of all types. About 20% of patients develop antidepressant discontinuation syndrome following an abrupt stoppage of or marked reduction in the dose of an antidepressant taken continuously for one month (Gabriel, 2017).

The onset is usually **within 5 days** of stopping but varies depending on half-life. Most antidepressant discontinuation reactions are of short duration, resolving spontaneously between 1 day and 3 weeks after onset.

The mean duration of SSRI discontinuation symptoms is **5 days**. Clinical experience is that discontinuation symptoms usually resolve fully within 24 h if the original antidepressant is recommenced. Tapering antidepressants at the end of treatment, rather than abrupt stoppage, is recommended as standard practice by several authorities and treatment guidelines.

Common symptoms include:

- Flu-like symptoms
- Dizziness
- Insomnia
- Vivid dreams
- Irritability
- Crying spells

Added during revision.
Right Answers.
Not sure about the answer.

- Sensory symptoms (e.g. sensations resembling electric shocks in the arms, legs, or head)

Discontinuation symptoms seem least likely with Fluoxetine.

TCA discontinuation
Common symptoms include:

- Flu-like symptoms
- Insomnia
- Excessive dreaming

MAOI discontinuation

Common symptoms include:

- Agitation/ irritability
- Ataxia
- Movement disorders
- Insomnia
- Vivid dreams

All MAOIs are associated with prominent discontinuation symptoms.

If a discontinuation reaction does occur, explanation and reassurance are often all that is required. If this is not sufficient, and for more severe reactions, the antidepressant should be restarted (or an alternative with a longer half-life) and tapered more slowly.

28. Agomelatine mode of action/receptor
agonist at melatonin M1 and M2 receptors and as an antagonist at 5HT2C receptors.

Agomelatine is a relatively new drug used for treating depression. It acts as an agonist at melatonin M1 and M2 receptors and as an antagonist at 5HT2C receptors. The melatonin effects appear to promote sleep whereas the 5HT2C antagonism leads to the release of dopamine and norepinephrine in the frontal cortex. Interestingly serotonin levels appear not to be affected.

29. Which is extensively metabolized by CYP450, and the answer given is **CYP2D6**

We think the question might be

Which of the CYP450 types are used in metabolism of most psychotropic drugs?
CYP2D6

Cytochrome P450 system [Psych-Mentor]

- **CYP2D6** shows the largest phenotypic variation amongst the cytochromes
- **CYP1A2** is involved in metabolism of clozapine. Tobacco induces CYP1A2 and so lower clozapine levels. Grapefruit juice inhibits CYP1A2 and therefore can increase clozapine levels.

Added during revision.
Right Answers.
Not sure about the answer.

- **Grapefruit juice inhibits CYP3A4 and CYP1A2**
- The Cytochrome P450 system is an evolved mechanism that deals with exogenous toxins. The system comprises of about fifty separate enzymes. The system is essential in the metabolism of drugs.

Do not be confused by the name, The P450 part relates to the wavelength of light it absorbs.

The enzymes are located on the **endoplasmic reticulum** of cells and are mainly found in the liver and small intestine. The Cytochrome system metabolizes drugs by altering the functional groups on the parent molecule (e.g. Oxidation). Drug interactions involving the system generally result from one of two processes, enzyme induction, or enzyme inhibition.

Notable inducers of the system include: smoking, alcohol, barbiturates, carbamazepine, Phenytoin, and St John's Wort. Notable inhibitors include: chlorpromazine, SSRI's, and grapefruit juice.

The main ones to be aware of are CYP2D6, CYP1A2, and CYP3A4.

CYP2D6 is important as it exhibits genetic polymorphism (it varies in its expression from person to person). Caucasians show less activity of CYP2D6 (6-10% poor metabolisers) than their Asian (2% poor metabolisers) counterparts. In African Americans and Africans, the frequency of poor metabolizers is greater than for Caucasians. This is also important when the effect of a drug relies on an active metabolite as in the case of codeine.

The CYP3A subfamily is the most abundant. They are most commonly involved in interactions involving induction and inhibition. Grapefruit juice specifically inhibits CYP 3A4.

Grapefruit juice also inhibits CYP 1A2 causing an increase in clozapine levels. Tobacco smoking causes an induction of CYP 1A2.

Note: Fluoxetine is metabolised by 2D6 but also inhibits activity of this enzyme.

30. Hormone produced by the hypothalamus that regulates appetite – **NeuropeptideY**, PPY, PPC, Cholecystokinin

31. Hormone produced by the adipose tissue that regulates appetite – **Leptin**, Neuropeptide Y

- **Neuropeptide Y** is produced by the hypothalamus and increases appetite
- **Leptin** is produced by adipose tissue and reduces appetite
- **Ghrelin** is produced mainly by the gut and increases appetite
- **Cholecystokinin** (CCK) is produced mainly by the gut and reduces appetite

Added during revision.
Right Answers.
Not sure about the answer.

32. Which pro-inflammatory interleukin is elevated in depression- IL 11?, IL 4?, **IL 6**

    The evidence is most convincing regarding depression where meta-analyses have found increased levels of inflammatory markers, including elevated levels of pro-inflammatory cytokines interleukin-6 **(IL-6)** and **tumour necrosis factor-α (TNF-α)**, as well as **c-reactive protein (CRP),** in depressed subjects as compared to controls.

33. Altruism: ? innate, ?**volunteering makes life better for old people**
    Vicarious but constructive and instinctually gratifying service to others.

    It includes benign and constructive reaction formation, philanthropy, and well repaid service to others. Altruism differs from projection and acting out in that it provides real , not imaginary, benefit to others and from reaction formation that leaves the person using the defence at least partly gratified

34. MOA Cocaine, MDMA

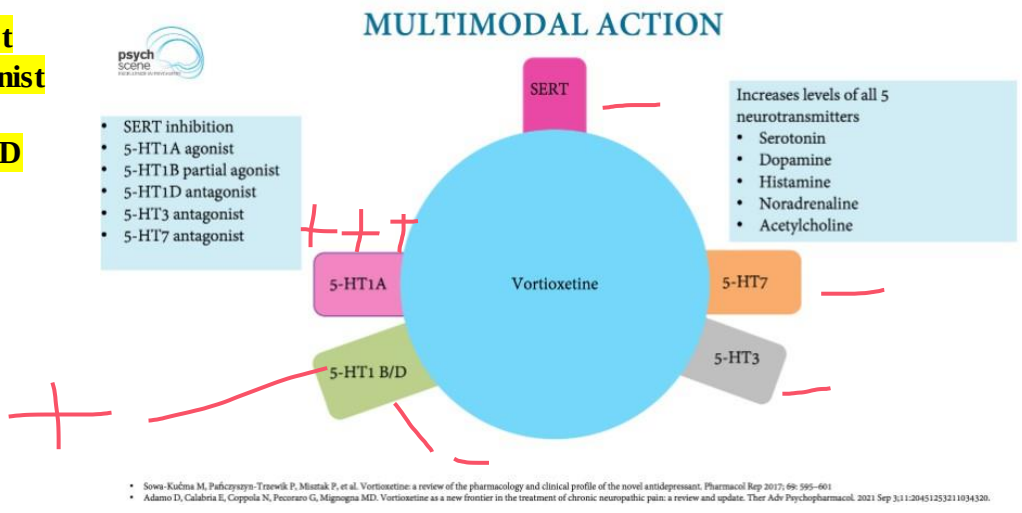
    Cocaine (and methylphenidate) directly blocks the dopamine transporter (DAT)

    MDMA Blocks SERT Monoamine Transporter

35. Hospital management prioritizing early intervention unit for funding over the rehabilitation unit cos more pts use it. This is an example of which principle? – Autonomy?, Beneficence?, Non-maleficence?, **Utilitarian**

    The greatest good for the greatest number.

36. Vortioxetine receptors
    5HT1A Full agonist
    5HT1B partial agonist
    Antagonist at
    5HT3, 5HT7,5HT1D



37. Antidepressant for a patient with hyponatremia
Bupropion , Mirtazapine, (Mianserine as well)

Drug / group*	Risk of hyponatremia	Level of supporting evidence
SSRIs	High	Strong
SNRIs	High	Strong
Tricyclics	Moderate	Strong
MAOIs	Low	Weak
NaSSAs	Low	Strong
Bupropion	Low	Moderate
Agomelatine	Low	Weak

38. Ion implicated in schizophrenia: **Calcium** Ion and channel both are calcium

If question is asking about channel then **K channel.**

These alterations cause the most important inhibitory current in the brain, called $GABA_A$, to become excitatory. Inhibiting the transporter activity with bumetanide, an FDA-approved diuretic, rescued cognitive symptoms in mouse models of schizophrenia. The polarity of the $GABA_A$-mediate current is determined by the balance of two chloride transporters: NKCC1 and KCC2

It is reported that the answer should be Potassium (not calcium) even if they ask about the ion, long discussions took place and what we could agree on is the potassium channel as most implicated in schizophrenia. We agreed to choose potassium as ion implicated in schizophrenia as well.

39. Brain organoid of pluripotent cell is at what age?, what are the drawbacks? **2nd trimester (6+ months gestation)**

Drawback: **lack of reproducibility**

From google: **However, existing brain organoids still have limitations: most notably, they do not contain vascular cells, immune cells and other non-neural cells.**

Despite the promising features of organoids, their broad utility is tempered by a variety of limitations yet to be overcome, including **lack of high-fidelity cell types, limited maturation, atypical physiology, and lack of a realization**, features that may limit their reliability for certain applications.

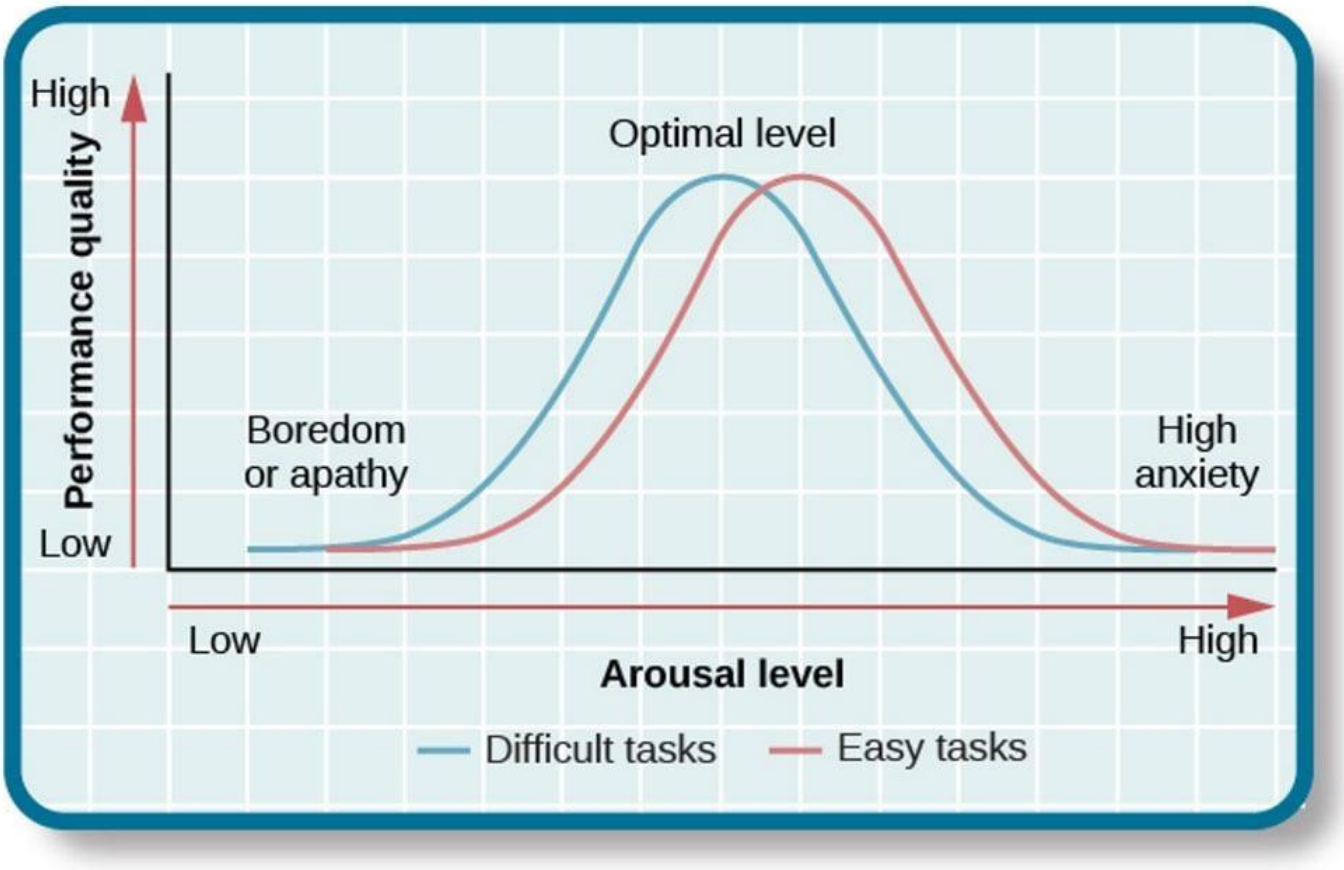
40. Area under the curve means? **plasma concentration of a drug versus time (Bioavailability)**

41. The graph on arousal and performance

The Yerkes-Dodson law is a model of the relationship between stress and task performance. **It proposes that you reach your peak level of performance with an intermediate level of stress, or arousal.**

Yerkes-Dodson Law: An inverted U-shaped curve relates the level of arousal with the performance of an act. Optimum arousal (moderate) is required for best performance; too low or too high arousal proves to be a hindrance e.g. sexual performance. But it is demonstrated that the relationship is not as simple as proposed as task difficulty varies highly. So, difficult or intellectually demanding tasks may require a lower level of arousal (to facilitate concentration).

But tasks demanding stamina or persistence may be performed better with higher levels of arousal (to increase motivation). Because of task differences, the shape of the curve can be highly variable.



42. Piaget's assimilation, accommodation and equilibrium

- **Assimilation** – The process of taking new information into an existing schema.
- **Accommodation** – Altering a schema in view of additional information.
- Under standard condition one is at **equilibrium**, when a new information arises you assimilate/accommodate to return everything back to equilibrium.

Added during revision.
Right Answers.
Not sure about the answer.

43. Differentiating vascular dementia from Alzheimer's: No Options given

From google

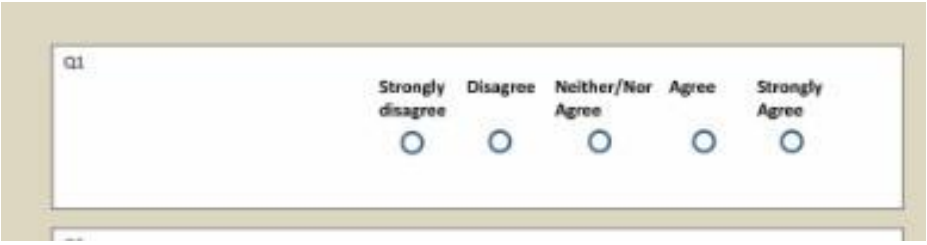
	Alzheimer	Vascular dementia
Risk factors	APOE, diabetes	Hypertension, diabetes, dyslipidemia, atheromatosis
Etiology	Amyloid plaques and neurofibrillary tangles	Vascular lesions
Clinical evolution	Progressive	Stepwise
Neuropsychological profile	Memory impairment (encoding)	Executive/frontal and semantic dysfunction
Neuroimaging	Medial temporal lobe atrophy, amyloid PET positivity, FDG hypometabolism in parieto-temporal regions	White matter lesions, lacunae, leukoaraiosis

44. Scale that has 'no pain' on one end and 'excruciating pain ' on the other end:

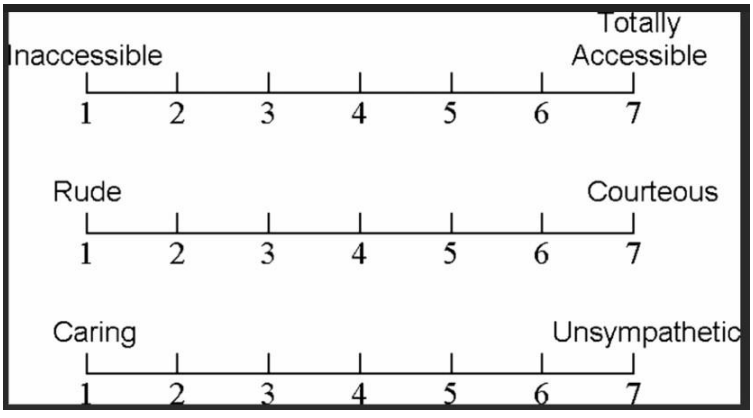
- Likert scale — agreement
- **Semantic** — extremes
- Nominal — exact answers
- Ordinal

Ordinal(this is not mentioned in TRONS)
RCPsych Syllabus 1.2.1 Attitudes: components and measurement by Thurstone, Likert and semantic differential scales. Attitude change and persuasive communication. Cognitive consistency and dissonance. Attitude behaviour relationships.

Likert

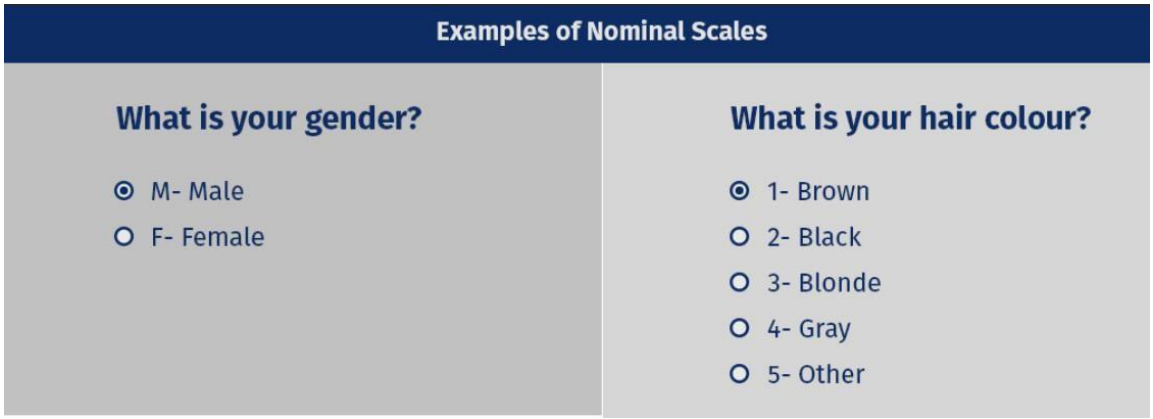


Semantic

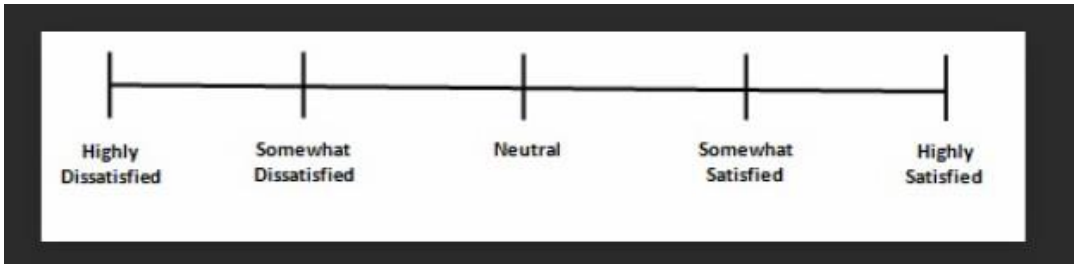


Weak	o	o	o	o	o	o	o	Strong
Ugly	o	o	o	o	o	o	o	Beautiful
Cool	o	o	o	o	o	o	o	Warm
Amateur	o	o	o	o	o	o	o	Professional

Nominal



Ordinal



45. Psilocybin receptor: **5 HT2A agonist**

46. Purine properties (adenine, guanine): **Aromatic**

- **Pyrimidines are aromatic as well (thyramine, cytosine, uracil)**
- **Purines** contain two carbon-nitrogen rings and include adenine and guanine1234.
- **Pyrimidines** contain one carbon-nitrogen ring and include thymine, cytosine, and uracil1234.
- **Purines** are larger and have a more complex structure than pyrimidines3.
- **Purine** bases are degraded into uric acid in humans, which can lead to conditions like gout when present in high levels. Pyrimidines degrade into simpler compounds like beta-alanine and are generally less problematic in terms of accumulation and degr

Added during revision.
Right Answers.
Not sure about the answer.

[SPMM] A purine is an **aromatic heterocycle** composed of **carbon and nitrogen**. Eukaryotes, bacteria, and archaea are all able to carry out de-novo biosynthesis of purines. **Purines inclu de adenine and guanine**, which participate in DNA and RNA formation. Purines are also constituents of other important biomolecules, such as ATP, GTP, cyclic AMP, NADH, and coenzyme A. **Other important purines include hypoxanthine, xanthine, theobromine, caffeine, uric acid, and iso-guanine.**

47. Child language abilities at age 1,2,13 years

Stages of Language Development	age
1. Crying	Birth
2. Cooing	6 weeks
3. Babbling	6 months
4. Intonation patterns	8 months
5. One-word utterances	1 year
6. Two-words utterances	18 months
7. Inflectional words	2 years
8. Negative and Interrogative	2 ¼ years
9. Rare and complex constructions	5 years
10. Mature speech	10 years

Communication & Language Development Milestones 2-24 months		
Milestone	Average Age of Attainment (months)	Developmental Implications
Smiles in response to face/voice	1.5	More active social participation
Babbles	6	Experimentation with sound
Inhibits to "no"	7	Response to tone
Follows one step command when said with gesture	7	Response to gesture
Follows one step verbal command when said without gesture	10	Verbal receptive language
Monosyllabic- mama/dada	10	Expressive language
Points to object	10	Interactive communication
Speaks first real word	12	Beginning of labeling
Speaks 4-6 words	15	Learns names of object and people
Speaks 10-15 words	18	Learns names of object and people
Makes 2 words sentences	19	Grammatization, 50+ words vocabulary

48. What not to take with fluoxetine - ? **Aspirin, ?NSAIDs**

- NSAIDs: NICE guidelines advise 'do not normally offer SSRIs', but if given co-prescribe a proton pump inhibitor
- Warfarin / heparin: NICE guidelines recommend avoiding SSRIs and considering mirtazapine
- Triptans: avoid SSRIs: TRIPTANS PLUS SSRIS CAN CAUSE **SEROTONIN SYNDROME**

49. Indicator for clozapine induced myocarditis

[Psych-Mentor] Myocarditis is rarely seen in clozapine use but carries with it a high mortality.

Studies have estimated the risk of clozapine-induced myocarditis to be 1 in 500 to 1 in 10,000 (Hass, 2007). The true risk is likely closer to the 1 in 500 rate, as often only a minority of adverse medication-related events are diagnosed and reported.

Most cases of myocarditis occur early in the course of clozapine treatment. The condition was detected within 16 days (median) of initiating clozapine in a study of 116 cases (Hass, 2007). Approximately 80 percent of cases of clozapine-induced myocarditis occur within four weeks of drug initiation, and 90 percent occur within eight weeks.

Postmortem examination has most commonly revealed damaged myocytes and eosinophilic infiltration suggesting a **type I Ig E-mediated acute hypersensitivity** reaction.

Some patients will have a mild and temporary increases in inflammatory markers and eosinophils at the beginning of treatment without progression to myocarditis.

Subsequent use of clozapine in cases with clear clozapine-induced myocarditis leads to recurrence of myocarditis in most cases when the drug is restarted.

Features include:

- Fever
- Chest pain
- Tachycardia
- Dyspnoea
- Flu-like symptoms
- Elevated eosinophil count
- Elevated cardiac enzyme levels

The gold standard for diagnosis of myocarditis is an endomyocardial biopsy, but this is not a practical initial approach.

Cardiac biomarkers, including **creatine kinase and troponin,** are routinely measured in patients with suspected myocarditis. Creatine kinase has been shown to be inferior to troponin for assessing myocardial injury, and is not useful for screening because of its low predictive value. **Eosinophilia is the most specific indicator**

Outcomes of clozapine-induced myocarditis are poor, with mortality rates as high as 50%

50. Fetal malformation in Sodium Valproate

Congenital malformations (8-12%), facial dysmorphism, cleft lip and palate, craniostenosis (skull contractions), cardiac, renal and urogenital defects, limb defects (including bilateral aplasia of the radius).
Developmental delays (31-40%)

51. Which part of the frontal lobe is prominent in teenagers? Options

- Dorsomedial prefrontal
- Dorsolateral prefrontal ??
- Medial orbitofrontal
- Lateral orbitofrontal

Answer might be **DLPFC**

During adolescence, various parts of the prefrontal cortex continue to develop, but one specific region that undergoes significant development during this period is the dorsolateral prefrontal cortex (DLPFC). The DLPFC is responsible for higher-order cognitive functions, such as complex decision-making, reasoning, working memory, and

impulse control. It plays a crucial role in executive functions, which are essential for planning, problem-solving, and regulating behavior.

52. EMI: When one's schema is corresponding with one's experiences. Options

- Accomodation
- Assimilation
- **Equilibrium**
- Organisation
- Centration

- Piaget defined assimilation as the cognitive process of fitting new information into existing cognitive schemas, perceptions, and understanding. Overall beliefs and understanding of the world do not change as a result of the new information.
- Assimilation occurs when the new experience is not very different form previous experiences of a particular object or situation we assimilate the new situation by adding information to a previous schema.
- This means that when you are faced with new information, you make sense of this information by referring to information you already have (information processed and learned previously) and try to fit the new information into the information you already have.
- For example, a 2-year-old child sees a man who is bald on top of his head and has long frizzy hair on the sides. To his father's horror, the toddler shouts "Clown, clown"
- For example, a baby learns to pick up a rattle he or she will then use the same schema (grasping) to pick up other objects.

- Accommodation
- **Accommodation**: when the new experience is very different from what we have encountered before we need to change our schemas in a very radical way or create a whole new schema.
- Psychologist Jean Piaget defined accommodation as the cognitive process of revising existing cognitive schemas, perceptions, and understanding so that new information can be incorporated. This happens when the existing schema (knowledge) does not work, and needs to be changed to deal with a new object or situation.
- In order to make sense of some new information, you actual adjust information you already have (schemas you already have, etc.) to make room for this new information.
- For example, a baby tries to use the same schema for grasping to pick up a very small object. It doesn't work. The baby then changes the schema by now using the forefinger and thumb to pick up the object.
- Also, a child may have a schema for birds (feathers, flying, etc.) and then they see a plane, which also flies, but would not fit into their bird schema.

- In the "clown" incident, the boy's father explained to his son that the man was not a clown and that even though his hair was like a clown's, he wasn't wearing a funny costume and wasn't doing silly things to make people laugh.
- With this new knowledge, the boy was able to change his schema of "clown" and make this idea fit better to a standard concept of "clown".

- **Equilibration**
- Piaget believed that all human thought seeks order and is uncomfortable with contradictions and inconsistencies in knowledge structures. In other words, we seek 'equilibrium' in our cognitive structures.
- Equilibrium occurs when a child's schemas can deal with most new information through assimilation. However, an unpleasant state of disequilibrium occurs when new information cannot be fitted into existing schemas (assimilation).
- Piaget believed that cognitive development did not progress at a steady rate, but rather in leaps and bounds. Equilibration is the force which drives the learning process as we do not like to be frustrated and will seek to restore balance by mastering the new challenge (accommodation).
- Once the new information is acquired the process of assimilation with the new schema will continue until the next time we need to make an adjustment to it.
- **Equilibration** is a regulatory process that maintains a balance between assimilation and accommodation to facilitate cognitive growth. Think of it this way: We can't merely assimilate all the time; if we did, we would never learn any new concepts or principles.
- Everything new we encountered would just get put in the same few "slots" we already had. Neither can we accommodate all the time; if we did, everything we encountered would seem new; there would be no recurring regularities in our world. We'd be exhausted by the mental effort!

53. Innate ability to gather information and integrate available info into a system of knowledge.
Same options as Q52
Assimilation
As Above

54. Stimulated part of the brain in Visual hallucinations, Auditory hallucinations

- **Visual hallucinations** were associated with **reduced volume in bilateral occipital regions, right supramarginal gyrus and left fusiform gyrus, bilateral dorsolateral prefrontal cortex, frontal pole and the middle portion of the left cingulate gyrus**.
- **Auditory Hallucination** : **left SUPERIOR temporal lobe, in the peri-Sylvian region.**

55. Progranulin gene: **FTD (chromosome 17) / MAPT also 17**

56. Confabulation . Options :

Added during revision.
Right Answers.
Not sure about the answer.

- Can only occur in Korsacoff syndrome
- May occur in clouded consciousness
- There is deliberate falsification of facts
- Autobiographical memory is affected
- **There is subjective awareness of memory impairment**

Confabulation is a falsification of memory occurring in clear consciousness associated with organic states. Suggestibility is a prominent feature of confabulation. It is often described in Korsakov syndrome. There can either be confabulation of embarrassment or of fantastic nature.

Conclusion: We think the highlighted answer is the right one, it is not ALWAYS affected though, Also sometimes there might be subjective awareness of confabulation so we decided to go for autobiographical memory first, if not an option then subjective awareness.

57. FTD

- ✓ **Familial FTD: Tau -ve, Ubiquitin +ve**
- ✓ **Pick's Disease: Tau +ve, Ubiquitin +ve**
- ✓ **FTD in general: Most common is:** UA TATU
  - **Ubiquitin positive , Amyloid negative**
  - **Tau positive , Amyloid negative THEN**
  - **Tau positive, Ubiquitin positive**

58. Which is not a risk for SSRI induced hyponatremia ? Options:

- Male sex
- Low body mass
- Low sodium level

Table 4: Risk factors for development of hyponatremia with psychotropics

Demographic variables: Increasing age, female gender
Physical characteristics: Low body weight
History: History of hyponatremia in the past
Baseline serum sodium levels: Low baseline levels
Environmental factors: Summer season
Nature of psychiatry disorder: Early-onset psychiatric illnesses, longer duration of psychiatric disorder, prolonged admission
Co-morbid medical conditions: Diabetes mellitus, hypertension, hypothyroidism, COPD, heart failure, head Injury, stroke, liver cirrhosis and malignancies
Concomitant medications: Diuretics, antihypertensives, antidepressants, cytochrome P450 inhibitors
Dose of psychotropic: Inconclusive (antidepressants and antipsychotics); higher dose in case of carbamazepine
Duration of treatment: during the initial phase of treatment

COPD: Chronic obstructive pulmonary disease

Added during revision.
Right Answers.
Not sure about the answer.

59. One of these is not a risk factor for delayed speech impairment. Options:

- Immigration status
- Maternal alcohol intake
- Maternal smoking in pregnancy
- Large families
- Reduced cognitive stimulation

A number of risk factors are associated with delayed speech and language :

- Family history
- Male gender
- Twins
- Lower maternal education
- Childhood illness
- Being born late in the family order
- Young mother at birth
- Low socioeconomic status

60. Which of these drugs do not cross the blood brain barrier? Options:

- **Amisulpride**
- Olanzapine
- Quetaipine
- Clozapine
- Risperidone

Sulpride and Serotonin don't cross BBB.

L-DOPA can pass through the blood brain barrier via a specialized transmembrane transporter called L-type amino acid transporter. Molecules can pass through the blood brain barrier if they are small, lipophilic (un-ionised), a gas or if they have a specific transporter.

61. Risk factors for QTc prolongation. Options:

- Male sex
- Sinus tachycardia
- **Hypokalemia**

62. Tardive dyskinesia- **D2 hypersensitivity**

The cause of TD: one of the most common theories is that chronic dopamine blockade of dopamine D2 receptors results in a compensatory behavioral **super-sensitivity** of the post synaptic dopamine receptors **in the striatum** due to up regulation of dopamine receptor number and/or responsiveness.
for positive symptoms --mesolimbic

Added during revision.
Right Answers.
Not sure about the answer.

for negative--mesocortical

63. Orthostatic hypotension- **Alpha 1 adrenergic blocker**

64. A cell was infected by toxins and inflammatory markers seen around the mileu. What process is this? Options:

    - Apoptosis
    - **Necrosis**
    - Entosis
    - Excitoxicity

65. Rate limiting step in alcohol metabolism. Options:

    - Acetaldehyde dehydrogenase
    - **Alcohol dehydrogenase**
    - Alcohol hydroxylase

66. How to reduce groupthink? - **open debate**

67. ADHD – **DRD4**

    Others: DAT1, DRD4, DRD5, 5HTT, HTR1B, and SNAP25.

68. Ketamine MOA – NMDA **antagonist** — non-competitive

69. Cocaine receptor/MOA: DAT

    The rewarding effect of cocaine is related to increase in dopamine concentration in limbic regions due to either dopamine transporter blockade or direct occupancy of D2 receptors. D1 is implicated in conditioned response to cocaine.

70. Southern blot- **specific DNA sequence**

    SNoW- DRoP
    Southern blot- DNA
    Northern blot- RNA
    Western blot- Protein
    PCR - abnormal trinucleotide expansions

71. Locum 1 hr extra – **fixed interval**

72. Gambling- **variable ratio**

Added during revision.
Right Answers.
Not sure about the answer.

73. Reversal of Caffeine inhibited by **Adenosine antagonist readily cross BB barrier , it is methyl xanthine derivative and completely reverses the effect of adenosine**

74. Thomas scale for infants. Options:

- Anger
- Impulsivity
- Control

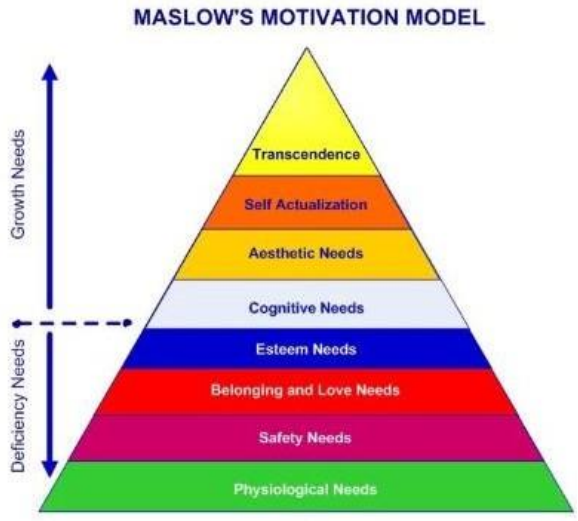
Thomas and Chess assessed children on Nine traits / aspects of behaviour.

- activity level
- biological regularity (sleep, appetite)
- adaptability
- approach / withdrawal (reactions to new people)
- sensitivity threshold (sensitivity to stimuli such as sounds, crowds etc)
- intensity of emotional response
- distractibility
- quality of mood
- persistence / attention span

- **Easy child (40%).** These obtained high scores on regularity (degree to which they fall into a schedule), approachability, adaptability, mild intensity of reactivity, and predominance of positive mood.
- **Difficult child (10%).** These children were the polar opposite of easy children.
- **Slow to warm up child (15%).** These children are somewhere in the middle. They often have moderately negative reactions to new things but warm to the idea eventually.

75. Which is part of Maslow's original hierarchy of needs – Esteem





In Extended 8 levels – the following were added: PS BE CAST

Added during revision.
Right Answers.
Not sure about the answer.

Cognitive needs - knowledge and understanding, curiosity, exploration, need for meaning and predictability. **(Above self-esteem).**

Aesthetic needs - appreciation and search for beauty, balance, form, etc. **(Above cognitive needs)**

Transcendence needs - A person is motivated by values which transcend beyond the personal self. (e.g., mystical experiences and certain experiences with nature, aesthetic experiences, sexual experiences, service to others, the pursuit of science, religious faith, etc.). **(Above self-actualization)**

Monotropy attachment to single figure mainly mother **According to Bowlby, the strong innate tendency to attach to one adult female is seen – this is monotropy**

76. Changes in pregnancy. Options:

- Increase in gastric PH
- **Increase in GFR**
- Increase in gastric motility

If both an option pick **GFR**

Pregnancy is associated with several **pharmacokinetic changes**:

- Delayed gastric emptying,
- Decreased GIT motility,
- Increased volume of distribution (5%),
- Decreased drug-binding capacity,
- Decreased albumin level
- Induced liver metabolic pathway
- Increased GFR & renal clearance

77. Monotropy attachment to single figure mainly mother **According to Bowlby, the strong innate tendency to attach to one adult female is seen – this is monotropy**

Added during revision.
Right Answers.
Not sure about the answer.

Paper A december 2021 recalls

1) EEG Wave form in Delirium Tremens- Beta activities unlike normal delirium that has diffuse slowing, some alpha waves and some generalised increased delta and theta waves.
Alcohol effect on EEG:

- Increased alpha activity.
- Increased theta activity.
- Withdrawal increases beta.
- Delirium tremens has beta (fast) wave activity –
- other deliria have increased slow waves.

2) Hormone secreted by hypothalamus that regulates hunger- Neuropeptide Y
- Increasing appetite (Orexigenics): **ghrelin** (stomach) and **neuropeptide Y and orexin** (hypothalamus) **.**
- Mediators of satiety (Anorexigenics): **Leptin** (Adipose cells), **cholecystokinin** (small intestines & brain) and **serotonin** (raphe nuclei)**.**
- Food and food cues **increase dopaminergic activity** in **nucleus accumbens (reward centre).** In obesity**, D2 receptors are reduced in the striatum.**

3) Hormone secreted by adipocytes for regulation of food intake- **Leptin**
lacks of this hormone is implicated in bulimia.

4) What reduces group thinking:
a) Appointing a devil's advocate.
b) Acknowledging the presence of group thinking.
c) Encouraging open debate.
d) Reserving leader's opinions until after the group's discussion has been completed
e) Seeking external opinion
f) Splitting the group into smaller units for discussion.
g) Holding last chance meeting to encourage challenges.

5) A patient who thinks he has cancer despite negative clinical findings:
- Hypochondriasis
- Men and women are equally affected by hypochondriasis.
- In hypochondriasis patient looks for diagnosis, **not symptom relief;**
- He/she names the suspected disorder and may be more or less convinced about having the disorder.

Fear of Cancer is now under **"symptoms, Signs and clinical findings not elsewhere specified"** in **ICD11** and **not under hypochondriasis.**

6) Question on somatization disorder **(6C20 Bodily distress disorder in ICD-11):**

Added during revision.
Right Answers.
Not sure about the answer.

a) At least 2 years (ICD-11; most days for at least several months) of multiple and variable physical symptoms (Musculo-skeletal symptoms most common) for which no adequate physical explanation has been found.
b) Persistent refusal to accept the advice and reassurance of several doctors regarding the absence of a physical illness.
c) Notable impairment of social and family functioning due to the symptoms and the illness behaviour.

*Note:

- The term **Briquet Syndrome** or **St. Louis Hysteria** is sometimes applied to denote somatisation disorder.
- Family history of alcohol use and antisocial personality are common in women with somatisation disorder.

Conversion disorders; functional neurological deficits:

a) Loss of function (negative)
b) Paralysis, blindness, loss of balance, etc.
c) Neurological
d) Monosymptomati

7) Nucleotide most affected by methylation (inactivation of nucleotide)- **cytosine** (The majority of DNA methylation occurs on cytosines that precede a guanine nucleotide or CpG sites)

- **Methylation**: a chemical reaction in the body in which a methyl group gets added to **DNA, proteins, or other molecules.**
- DNA methylation is essential for silencing retroviral elements, regulating tissue-specific gene expression, genomic imprinting, and X chromosome inactivation.
- DNA methylation is **an epigenetic mechanism** involving the transfer of a methyl group onto the C5 position of the cytosine to form 5-methylcytosine

8) Nucleotide repeat in Huntington- CAG

- CAG trinucleotide repeat on the short arm of chromosome 4 (4p16.3).
- Expansions of 40-55 repeats in adults HD
- Expansions (>70 repeats) are seen in childhood-onset HD

9)Hypoplasia part of brain in ASD- Cerbellar Vermis

10) What induces CYP1A2(which metabolises Clozapine)- **Smoking.**

Important CYP 1A2 inducers: Smoking, Carbamazepine, Barbiturates and (Cruciferous vegetables & Grilled meat).

Potent CYP 1A2 inhibiters: Caffeine, Fluvoxamine, Ciprofloxacin, Enoxacin and Cimetidine

Important CYP 1A2 Substrates: Theophylline, Clozapine, Olanzapine, caffeine, Melatonin, Mirtazapine, Ropinirole and Tacrine.

CYP enzyme	Main substrates	Main inhibitors	Main inducers
CYP1A2	Clozapine Olanzapine Asenapine Agomelatine Melatonin Theophylline Mirtazapine Ropinirole Tacrine Caffeine	Fluvoxamine Grapefruit Caffeine Ciprofloxacin	Barbiturates Smoking Carbamazepine
CYP2D6 **uscd i→ mctabolism of most psQckotíopic díugs**	Amphetamines Atomoxetine Aripiprazole Clomipramine Fluoxetine* Olanzapine Risperidone Mianserin Imipramine Venlafaxine Zuclopenthixol	Fluoxetine* Paroxetine Bupropion Asenapine Amitriptyline	Not known
CYP3A4	Aripiprazole Clonazepam Donepazil Fentanyl Methadone Mirtazapine Quetiapine Zopiclone	Fluoxetine Paroxetine	Carbamazepine Phenobarbital Phenytoin St John's Wort

12) What is the basis of modern-day psychiatry diagnostic schedule--**Operationalized**

[SPMM] Various terms are used to describe the characters of classificatory systems. The concept of operationalized criteria, atheoretical approach, hierarchical organisation and multi-axial classification are important for MRCPsych Paper A exam and are described below.

Operationalised approach: In DSM-III operationalised diagnosis was first introduced. Operational criteria include the use of precise clinical description of disorders, together with predefined exclusion and inclusion criteria and details of the number and duration of symptoms required for diagnosis. It enables algorithm-based clinical diagnosis using intensity, duration of the symptoms and impairment tests. This more or less equates to using a checklist for diagnosis, but some rules are necessary while some are optional for a diagnosis.

ICD 10 is hierarchical and categorical, ICD 11 is life span/dimensional approach, DSM categorical non- hierarchical

Added during revision.
Right Answers.
Not sure about the answer.

13) Diagnostic criteria of delusional disorder....**3 months of delusion for ICD 11**, for **DSM V- 1 months durations** (with bizarre delusions allowed)

14) Persistence of visual image/sensation after cessation of stimuli ...**Palinopsia**

Palinopsia: *palin* for "again" and *opsia* for "seeing". It is a visual disturbance that causes images to persist even after their corresponding stimulus has left. It is seen in LSD use, migraine, occipital epilepsy, head trauma. It is similar to after image, but colour inversion (usually shadows or distorted colours noted in afterimages) is conspicuously absent.

15) Durkheim theory of suicide- **suicide was linked to lack of society cohesion**- suicide rates increased during economic decline and reduced in war.

He described altruistic suicide:-

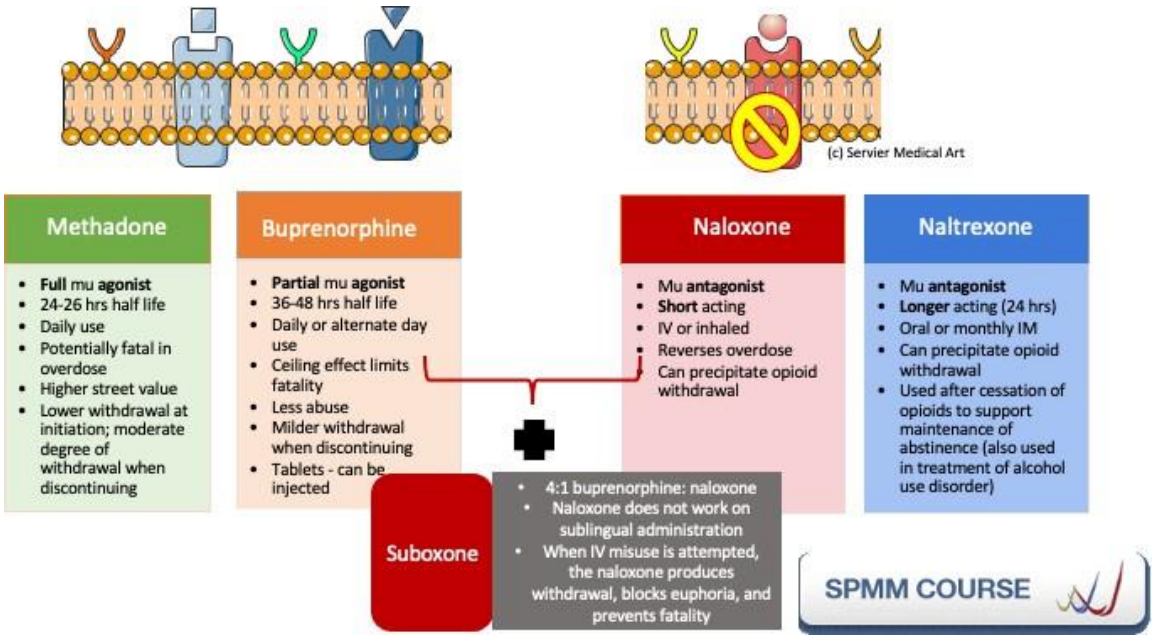
Altruistic suicide Individual is overly attached to social norms and dies for the sake of the society (i.e. for others in the society) e.g. self-molestation among Buddhist monks in Tibet

Egoistic suicide Excessive individualism, but low social integration. No cohesive group Attachment

Fatalistic suicide Society's control on the individual is very strong such that it interferes with moral values and personal goals

Anomic suicide Individual feels that he has no guidance or regulations from the societal system; feels disillusioned

16) Buprenorphine MOA- **Partial mu agonist**



Added during revision.
Right Answers.
Not sure about the answer.

17) Vygotsky model of learning - **collaborative learning theory**

Collaborative learning theory (Vygotsky): Development is not entirely private, child acts as an apprentice in social surroundings rather than a scientist. Parents and teachers carry out the role of **scaffolding** to introduce familiarity for the child to develop its own expertise (**collaborative learning**).

Zone of proximal development : refers to functions that are not yet fully achieved but are in the process 'pipeline' whose development is aided by scaffolding.

18) When does theory of the mind develop- 3 years (3-4 years).

Experiment done in preschool aged children; Theory of mind absent in Autistic children

Theory of mind refers to the ability to understand that other individuals have mental processes similar to self, leading to appropriate behaviour and conversation e.g. taking turns while conversing (as others also think and so want to speak). This is **deficient in the development of autistic** children and **can become acutely deficient (but develops normally) in schizophrenia** during psychotic episodes. This can explain some pragmatic errors in FTD.

Theory of Mind (ToM) develops around age 3 ½ to 4 years. ToM refers to the understanding that other persons do have mental processes similar to self; in this context it forms an essential part of the social attribution process. Lack of development of the theory of mind (trait related) could explain the apparent lack of empathy seen in autism. In acute psychosis, state related deficits in ToM are noted i.e. the deficit is not pervasive but seen only when relapsing into positive symptoms. **Poor ToM in association with reduced empathic ability is also demonstrated in conduct disorder and in antisocial personality disorder.**

19) What causes distress in the child in strange situation experiment- Read Attachment theory

	Secure attachment	Anxious-resistant (or ambivalent)	Anxious-avoidant
Separation anxiety	Distressed when mother leaves	Infant shows intense distress	No distress when mother leaves
Stranger anxiety	Avoids stranger when alone but friendly if mother present	Infant avoids stranger and shows fear	Infant is ok with the stranger
Reunion behaviour	Positive and happy when mother returns	Child approaches mother but resists contact, may even push her away	Infant shows little interest when mother returns
Other	Uses parent as safe base to explore environment	Infant cries more and explores less than the other 2 types	Mother and stranger able to comfort infant equally well
% of infants who fit this category	70%	15%	15%

Anxious resistant can be called ambivalent

Strange situation attachment style	Adult attachment style
Secure	Autonomous
Avoidant	Dismissing
Resistant	Preoccupied
Disorganised	Unresolved

20) Most common feature of Wernicke aphasia- **Fluent unintelligible jargon aphasia**

Broca's expressive aphasia is broken-non fluent aphasia also called agrammatism

Wernike aphasia – word salad / Broca's – Broken

The nature of conversational speech in **jargon dysphasia** is
Select one:

- Non-fluent and intelligible
- **Fluent and unintelligible**
- The rhythm of speech is not retained
- Non-fluent and unintelligible
- Fluent and intelligible

Jargon aphasia is a form of fluent aphasia characterised by a copious flow of unintelligible speech. Jargon-aphasic speech may be sub divided into **semantic jargon**, in which ordinary words are strung together to form unintelligible utterances, and **phonemic jargon** involving neologisms.

Transcortical sensory aphasia involves damage to specific areas of the **temporal** lobe of the brain, resulting in symptoms such as poor auditory comprehension, relatively intact repetition, and fluent speech with semantic paraphasias present. It is a **fluent aphasia similar to Wernicke's, with the exception of a strong ability to repeat words and phrases.**

In **Broca's aphasia** individuals have great difficulty forming complete sentences with generally only basic content words (leaving out words like "is" and "the").

In **conduction aphasia** individual can comprehend what is being said and is fluent in spontaneous speech, but they **cannot repeat** what is being said to them. **Lesion in arcuate fasciculus**

Transcortical motor aphasia is characterized by non-fluent speech output, **with good comprehension and repetition.**

Broca is broken speech that is non-fluent, while Wernicke's is wordy that is fluent

Type of aphasia	Fluency	Repetition	Comprehension	Naming
Wernicke's sensory aphasia	Intact	Lost	Lost	Lost
Broca's motor aphasia	Lost	Lost	Intact	Lost
Conduction aphasia	Intact	Lost	Intact	Lost
Transcortical sensory aphasia	Intact	Intact	Lost	Lost
Transcortical motor aphasia	Lost	Intact	Intact	Lost

21) **Ideational apraxia**- can perform actions but cannot perform them in sequence, can obey single commends

22) **Ideomotor apraxia**- disorder of goal directed movement, patient knows what to do but not how to do, like using finger to brush teeth, impaired ability to imitate unfamiliar actions, **from dominant parietal lobe damage**

Apraxia is the inability to carry out learned voluntary movements (assuming everything works and they are not paralysed).

The main types of apraxia are as follows:-

- **Limb kinetic apraxia** - an inability to make fine/ delicate movements
- **Ideomotor apraxia** - an inability to follow out learned tasks when given the necessary objects e.g. if given a hairbrush they try to write with it
- **Constructional apraxia** - an inability to copy a picture or combine parts of something to form a whole
- **Ideational apraxia** - an inability to follow a sequence of actions in the correct order e.g. Take a match out a box and strike it with your left hand
- **Oculomotor** - an inability to control eye movements

Ideational apraxia was thought to result from disruption of action representations in the left parietal lobe and limb-kinetic apraxia from disruption of "kinaesthetic-innervatory engrams" in the left frontal lobe

Ideational and Ideomotor both are dominate partial lobe lesions.

23) Inability to identify landmark: **Topographical agnosia (visuospatial agnosia)/Topographanosia/ or land mark agnosia. Lesion in lingual gyrus in occipital lobe.**

It is inability to recognize salient environmental stimuli such as landmarks. Patients are able to draw the maps & visualize places familiar to them before illness.

24) What is preserved in normal aging: spatial abilities?, (working memory?) **procedural/implicit/ non declarative**

25) Function of Stroop test:

The Stroop test involves naming the colour of a word, which may or may not match with the word itself (if it's not a match, it is a different colour). **It tests selective attention and inhibitory control.**
Other tests: trail making test.

26) Serotonin Auto-receptor: 5HT2 and 5HT3 antagonist **5HT1A PRESYNAPTIC INHIBITORY**
From other recall : Alpha 2 autoreceptor,5HT2 and 5HT3 antagonist – **Mirtazapine**

27) Antisocial personality Feature: **Orbitofrontal dysfunction causing impulsivity.**

Personality changes are most likely after a head injury to the orbitofrontal lobe or anterior temporal lobe. [SPMM]

28) Patient describes teaching hospital, as young doctors training place **circumstantiality/metonymy.**

Metonymy is a figure of speech in which a concept is referred to by the name of something closely associated with that thing or concept.

Metonymy is a speech disturbance common in schizophrenia in which the affected person uses a word or phrase that is related to the proper one but is not the one ordinarily used (for example, the patient refers to losing the 'piece of string' of the conversation rather than the 'thread' of the conversation)

29) A disorder characterized by forgetting name, personal identity and episodic memory. **Fugue**

30) Features of developing adolescent brain myelination and synaptogenesis:

Q. D2 receptors seen in children before 5 years is greater than adult level and later regressed, this is called**: - Pruning**

Synaptogenesis peaks around 34th week of gestation in (~ 40,000 new synapses formed per second) and continues in postnatal life alongside active synapse elimination; **eventually the net number of synapses begins to decrease at puberty (pruning).** Synaptic pruning cannot be observed directly with neuroimaging, but a prominent progressive cortical thinning of frontal and parietal cortices is noted in MRI studies during adolescence and is attributed to synaptic pruning. **An indirect indicator of synaptic removal is the glucose metabolism measured using positron emission tomography (PET).**

Synaptogenesis occurs very rapidly from the second trimester through the first ten years of life. The peak of synaptogenesis occurs within the first 2 postnatal years. By mid-childhood, more neurons and cellular processes are established than required for adult's brains. Thereafter a process of pruning or synaptic elimination takes place to select and preserve the most useful while eliminating the unnecessary neuronal connections in the adult's brain. This **synaptic pruning** continues through the early teen years.

Myelination begins prenatally at around **4th gestational month**; it is largely complete in early childhood (by 2 years), but does not reach its full extent especially in association cortices until late in the third decade of life.

Myelination of the visual cortical white matter begins prenatally (**last trimester**); by 9 months of postnatal life, myelination extends to frontal cortex (posterior-to-anterior maturation starting with sensory pathways motor pathways, and finally the higher-order association areas).

31) GABA B amino butyrate: **dissociative anaesthetic**

GHB (gamma-Hydroxybutyric acid) is medically used as a general anaesthetic, but is also used recreationally. Which of the following receptors does it act on?

Select one:

- GABA-A
- Glutamate
- Glycine
- GABA-A and -B
- **GABA-B**

GHB acts on the specific GHB receptors that are abundant in hippocampal dentate gyrus, median raphe nucleus as well as nucleus accumbens, mediating its effect on memory and sleep but also has weak agonistic action at the GABA-B receptor, mediating addictive properties and GABA-A receptor, mediating anxiolytic and hypnotic effects.
Ref: Kamal RM, et al. The neurobiological mechanisms of gamma-hydroxybutyrate dependence and withdrawal and their clinical relevance: a review. Neuropsychobiology. 2016;73(2):65-80.

The correct answer is: GABA-B

32) DSM-V has **no subtype for schizophrenia.**

ICD-11 as well has no subtypes for Schizophrenia

33) Melanie Klein **depressive position/ paranoid schizoid position**

Soon after birth, fear of annihilation is present. This cannot be tolerated by the child and projects this destructive impulse to external objects.

Projection of both bad and good impulses occurs followed by **splitting** of the external world into good and bad, Cannot unify these elements into one. Bad objects include non-gratifying bad breasts (parts). This leads to persecutory anxiety, and the child is said to be in **Paranoid –schizoid position.**

Klein defence mechanisms:
SIPDOG: Splitting,
Introjection,
Projection Identification,
Denial,
Omnipotence,
Grandiosity

Later the child realizes that both good and bad things emanate from the unified single object (whole). At same time weaning occurs – perceived as a loss. Subsequent guilt develops for

having destructive impulses against the mother. **Depressive position** – fear of loss of the love of object.

Reparation phase – creativity emanates as an attempt to repair damage done by 'destructive impulse'. Continues lifelong. In the absence of reparation, a maladaptive defence called manic defence can emerge characterized by denial of reality (refusal to take guilt), omnipotence and grandiosity.

[SPMM] Which of the following is NOT a Kleinian defence?

Select one:

A. Isolation

B. Projective identification

C. Splitting

D. Introjection

E. Denial

Isolation is not a Kleinian defence. It is characterised as a mental process involving the creation of a gap between an unpleasant or threatening cognition, and other thoughts and feelings. By minimising associative connections with other thoughts, the threatening cognition is remembered less often and is less likely to affect self-esteem or the self-concept. The other defences listed are more primitive and were addressed more by Klein in her work with children.

34) Amygdala helps in **storing emotionally salient memories.**

N.B Ant. Insula and Right Supramarginal gyrus for Empathy

35) What are the risk factors of prolonged QT:

ELECTROCARDIOGRAM:

- The major use of ECG in a psychiatric ward, apart from emergency needs, is to measure QT interval when treating patients using antipsychotics.
- Prolonged QT can predispose to fatal ventricular arrhythmias such as torsades de pointes (polymorphic ventricular tachycardia).
- QTc is QT corrected for heart rate. While valuable for classifying risk groups, it is not a precise predictor of torsade de pointes as it has low positive predictive value.
- There are different methods to arrive at QTc from QT – these give markedly different values.
- As a clinical measure, the risk is said to increase if QTc is beyond normal limits (440 ms for men; 470 ms for women) – anything more than 500 ms is clearly an increased risk.
- QT varies with **gender, time of day, food intake, alcohol intake, menstrual cycle, ECG lead used.**
- **Risk factors for prolonged QTc include:**

  • Congenital long Q-T syndrome,
  • Underlying heart disease, bradycardia, heart failure, and ischemic disease
  • Female gender,
  • Extremes of age.

- Presence of liver disease,
- Electrolyte abnormalities (hypokalaemia, hypocalcaemia and hypomagnesemia),
- Illicit drug use (principally stimulants).
- Starvation or anorexia.
- High physical exertion (agitation),
- High dosages of the drug contributing to the lengthened Q-T interval, and
- Rapid infusion of teratogenic drugs.

36) Side effects of carbamazepine: **Agranulocytosis**

Frequency	Side-effect(s)
Very common	**leucopenia** ataxia, dizziness, somnolence vomiting, nausea urticaria fatigue
Common	**Thrombocytopenia**, **eosinophilia** Oedema, fluid retention, weight increase, hyponatremia and blood osmolarity decreased due to an antidiuretic hormone (ADH)-like effect, leading in rare cases to water intoxication accompanied by lethargy, vomiting, headache, confusional state, neurological disorders diplopia and accommodation disorders (e.g. blurred vision), headache dry mouth

37) Most **sensitive** finding in clozapine induced myocarditis – **Eosinophilia/ Troponin**

[SPMM]Miss. Amelie Toutou is a 27-year-old woman with a diagnosis of treatment resistant schizophrenia. When started on Clozapine she develops **symptoms suggestive of myocarditis**. Which of the following is **a specific** indicator of clozapine-induced myocarditis**?**

Select one:

- CRP
- ECG changes
- Chest x-ray
- **Eosinophilia**
- Troponin T

Myocarditis is notoriously difficult to definitively diagnosis, with Troponin and Creatine Kinase levels often showing **poor sensitivity for the disorder**. In a review conducted of 47 cases of Clozapine induced myocarditis, the development of eosinophilia suggesting IgE mediated hypersensitivity reaction has been documented in ~66% of cases, though interpreting isolated eosinophilia in the absence of other markers is controversial. The Australian treatment guide recommends monitoring for potential myocarditis through baseline and weekly (for the first month) assessments, including vitals, laboratory evaluation of peripheral eosinophils, CRP, ECG, and troponin levels. **In the presence of suggestive symptoms of myocarditis, troponin more than twice the upper limit of normal or CRP over 100 mg/L is associated with 100% sensitivity.**

The correct answer is: Eosinophilia

38) Screening tool among: (**EPDS**, MADRS, HAMD)

39) Semantic differential scale

Semantic Differential Scale

The semantic differential technique involves presenting pairs of bipolar, or opposite, adjectives at either end of a series of scales, such as the following:

Weak	o	o	o	o	o	o	o	Strong
Ugly	o	o	o	o	o	o	o	Beautiful
Cool	o	o	o	o	o	o	o	Warm
Amateur	o	o	o	o	o	o	o	Professional

The semantic differential scale is a visual analogue scale (7 points) with two polarized adjectives at either extremes separated by a line. The subjects mark their attitudes between the two. It may be subject to **positional response bias.**

Q. The semantic differential scale may be subject to which of the following problems?

Select one:

- Performance bias
- **Positional response bias**
- Recall bias
- Selection bias

40) Likert scale

Likert scale includes graded 'agree' to 'disagree' measures. This is one of the most popular and statistically more reliable measures. It is easy to construct, and no assumptions are made about the equality of intervals.

41) Remembering what breakfast is - **Semantic memory**

42) Remembering what you had for breakfast - Episodic /**declarative**

[SPMM] **Q.** Match the following descriptions with the correct type of memory described

1. Remembering how to eat breakfast.
2. Remembering what you ate for breakfast.
3. Remembering what breakfast is.
4. Remembering after you have eaten breakfast that you need to pick up more cereal on your way home from work.
5. Remembering that after breakfast you took your bike out and cycled your usual route to

work, and this felt tedious.

Explanation:

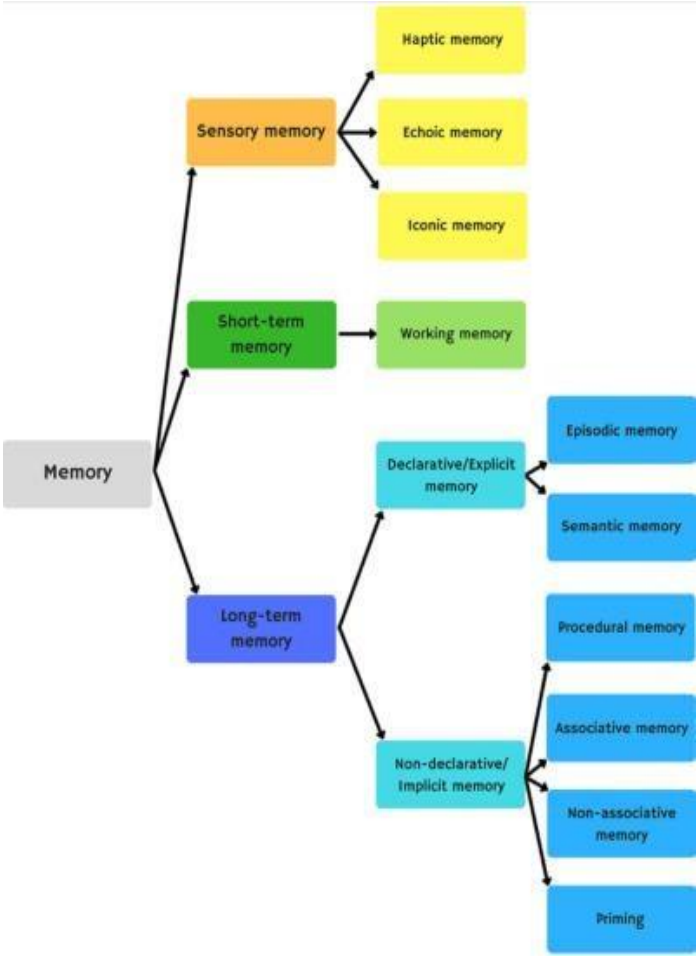
1) **Procedural memory,** or 'knowing how', is the unconscious memory of skills and how to do things. It is acquired through repetition and practice.

2) **Declarative memory** is memory of facts and events that can be consciously recalled (also known as explicit memory).

3) **Semantic memory** is a structured record of meanings, concepts, facts, knowledge about the world; it's to do with general factual knowledge that is shared with others, independent of personal experience.

4) **Prospective memory** refers to that which needs to be remembered in the future, and may be defined as 'remembering to remember'; it can be either event-based or time-based, and is often triggered by a cue (e.g. finishing breakfast).

5) **Episodic memory** refers to experiences and specific events in time in serial form, from which we can reconstruct the actual events that took place in our lives; it is the memory of autobiographical events, and has an emotional charge.



43) Ion most implicated in schizophrenia: ca?, k?, mg? - (mg) **calcium is ion and potassium is the channel in schizophrenia**

44) Anti-inflammatory cytokine involved in depression? IL-4 / **IL-6** / IL-10.

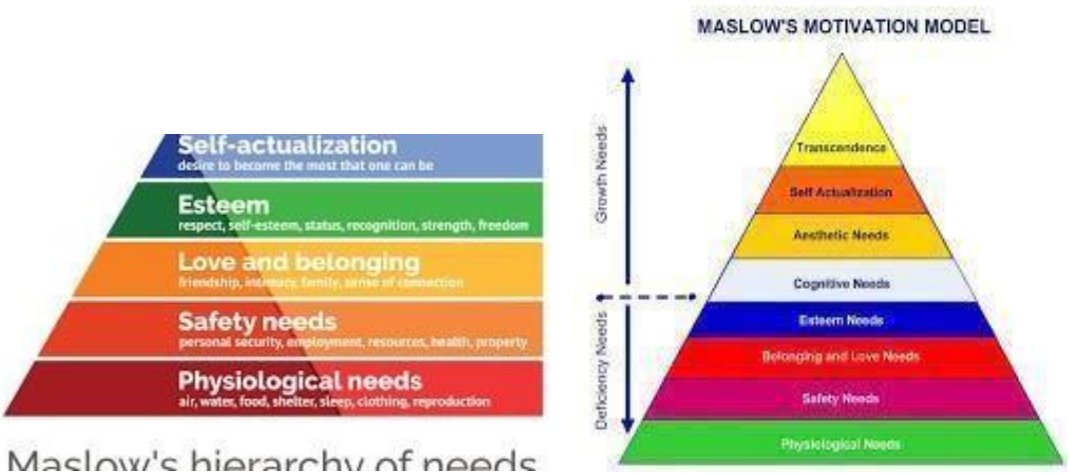
The evidence is most convincing regarding depression where meta-analyses have found increased levels of inflammatory markers, including elevated levels of pro-inflammatory cytokines interleukin-6 **(IL-6)** and tumour necrosis factor-α (**TNF-α**), as well as **c-reactive protein (CRP)** in depressed subjects as compared to controls.

IL-1 , IL-6 DEPRESSION,

1L-2,6, 8 FOR SCHIZOPHRENIA

Added during revision.
Right Answers.
Not sure about the answer.

45) Component of **original** Maslow hierarchy of needs – **Esteem**



In Extended 8 levels – the following were added:

- **Cognitive needs** - knowledge and understanding, curiosity, exploration, need for meaning and predictability. (**Above self-esteem**)

- **Aesthetic needs** - appreciation and search for beauty, balance, form, etc. (**Above cognitive needs**)

- **Transcendence needs** - A person is motivated by values which transcend beyond the personal self. (e.g., mystical experiences and certain experiences with nature, aesthetic experiences, sexual experiences, service to others, the pursuit of science, religious faith, etc.). (**Above self-actualization**)

46) DSM-V classification of eating where individuals don't take enough of what is required nutritionally, shows preference for enteral feeding and supplements: avoidance/resistant feeding disorder, **restrictive feeding disorder**

47) Not comfortable if not at the centre of attention - **Histrionic personality**

48) Miserly with money and doesn't spend on self or others, saves for future purpose - **obsessive compulsive personality (anankastic)**

[SPMM] Mr Whitaker is berated by some of his acquaintances for holding onto money very tightly. He claims he has to hold onto it for future catastrophise, and will never offer to pay for someone else's drink. He is regarded as miserly and uptight

49) Inflated sense of self - **Narcissistic**

50) ADHD mutated chromosome - **DRD4**

51) When does concrete operational begin - **7years (7-12)**

Piaget's stages of cognitive development

Stage	Age
Sensorimotor	0-2 years
Preoperational	2-7 years
Concrete operational	7-11 years
Formal operational	11 and onwards

52) Not a type of **FTD (Fronto-temporal degeneration)**: types are MND (ALS), Pick's disease, FTLD, Primary progressive non fluent aphasia**, PSP (progressive supranuclear palsy)**

From another version of this recall: Not a type of FTD – **Huntington**, Progressive non fluent aphasia, Pick's Disease, Semantic dementia, PSP.

(5 clinical subtypes of FTD: Behavioural variant FTD, primary progressive aphasia (pick's disease), progressive non fluent aphasia, semantic dementia, logopenic progressive aphasia)

PSP is no more classified under Fronto-temporal degeneration in ICD11, it is now under **Atypical Parkinson (Parkinson's plus** in other classifications**) - however, it overlaps with FTDs and still mentioned under it.**

53) Gambling - **variable ratio**

54) Locum 1h - **fixed interval**

Locum in general is variable interval

55) Payment for consultation at a place where processing takes an unpredictable amount of time - **variable interval**

56) Identity moratorium question: **Only exploration.**

57) Identity foreclosure question: **Decides without exploration**

58) Delay of gratification helps to measure - **Long term academic performance**/Disruptive behaviour?

Marshmallow experiment

59) Flumazenil helps to manage toxicity caused by – **Zopiclone/benzo** – Barbiturates – Baclofen.

Flumazenil is a selective **GABA-A** receptor **antagonist** that is used for the complete or partial reversal of the sedative effects caused by benzodiazepines.

Competitive

It exerts its action through competitive interaction, by **blocking the effects of substances acting via the benzodiazepine receptor.** It can also reverse the binding of benzodiazepines to benzodiazepine receptors.

It is administered intravenously.

Its half-life is short (about **60 minutes**). The action of flumazenil is usually shorter than that of benzodiazepines and sedation may possibly recur.

Non-benzodiazepine-agonists, like cyclopyrrolones (**zopiclone**) are also blocked by flumazenil. It has **no effect on other drugs such as barbiturates, ethanol, or other GABA- mimetic agents** (unless they act on the benzodiazepine receptor site).

The hypo-sedative effects of benzodiazepines are blocked rapidly (within 1-2 minutes) after intravenous administration.

The duration range is from 20 minutes to 50 minutes.

60) Neuroimaging used for measurement of neurotransmitters – PET (for metabolism) or **MRS**

PET – measurement of regional brain metabolism by tagging organic compounds in metabolic pathway i.e. gives direct info about neuronal metabolism.

MRS – dynamic measurement of resonance hydrogen and phosphate ions – odd number of protons

SPECT – measurement of receptor binding and distribution using radiolabelled receptor

fMRI - tasks-related changes in blood oxygenation level

Magnetoencephalography – measurement of static magnetic activity of brain

DAT (dopamine active transporter) – differentiate between LBD and Alzheimer's

DTI (Diffusion Tensor Imaging) – for white matter detection.

61) Archetype answer to a definition

Carl Jung

• Introduced the concept of the persona (mask) which is the part of the ego presented to other people. The other (more hidden) part of the self is the 'shadow'.

• Differentiated between the personal unconscious (which contains an individual's personal memories) and collective unconscious (a set of memories and ideas that is shared amongst all of humanity).

• **Talked of archetypes (symbolic images in the collective unconscious)**. Important archetypes are anima (female principle), animus (male principle), the shadow, and the self.

Added during revision.
Right Answers.
Not sure about the answer.

62) Important kind of learning in the critical period - **Imprinting**

Imprinting is a term used to describe the phase-sensitive attachment seen in some animals and in humans.

The classic demonstration of imprinting was performed by **Konrad Lorenz**. He was working with Greylag geese, and showed that imprinting behaviour consists of three parts:-

- At some time after hatching the gosling will start to follow the first conspicuous object it sees (the following response)
- After following the object for a number of hours the hatchling will start to develop an attachment for the object
- Once an animal has imprinted, it develops a sexual preference, when it is mature, for the object on which it imprinted

Lorenz also observed the following:-

- Imprinting is **rapid** (sometimes occurring within minutes)
- Imprinting is **unreinforced** (no reinforcer is necessary for the response to occur)
- Imprinting occurs during a clearly defined period (known as the critical period)
- Imprinting is **irreversible**

[SPMM] Explanation: **Imprinting** is a form of rapid learning that takes place during a critical period of development immediately or shortly after birth and is extremely resistant to extinction, the most familiar example being the behaviour of newly hatched ducklings that become imprinted on, and subsequently follow around, virtually any moving object to which they are exposed during the critical period following hatching. Imprinting was first studied in detail by the zoologist **Konrad Lorenz.**

63) Drug with relative D2/D3 receptor activity – **Amisulpride**
Sulpride is pure D2 antagonist only.

Antipsychotic	Mechanism
Amisulpride	D2/D3 selective antagonist (low affinity selective antagonist of 'D2 like' receptors (D2=D3>D4) it has little affinity for D1 like' receptors (D1 and D5) or non dopaminergic receptors (serotonin, histamine, adrenergic, and cholinergic)

64) Non selective long acting opioid antagonist - **Naltrexone**

Naloxone and naltrexone are competitive antagonist

65) What is likely observed by family members who show high expressed emotions to patient during interview: **Critical comments**? Hopelessness in the treatment, Doubt a positive outcome

1. Critical comments
2. Positive remarks

3. Emotional over involvement
4. Hostility
5. Emotional warmth

66) What does Camberwell assessment of needs? **Measure social and health needs of people with mental health problems**

67) Question on family system theory viz, anorexia???

-Enmeshment/protectiveness
-Triangulation Triangulation or triangling is defined in the AAMFT Family Therapy Glossary as the "process that occurs when a third person is introduced into a dyadic relationship to balance either excessive intimacy, conflict, or distance and provide stability in the system.
-Detouring
-Circular causality

Triangulation: is when a toxic or manipulative person, often a person with strong narcissistic traits, brings a third person into their relationship in order to remain in control.).

Detouring: Detouring scale assesses the degree to which the parents form an alliance against the child in a way that functions to create distance between themselves and the child and increase closeness in their own relationship, scapegoat syndrome. Detouring occurs when parents, rather than directing anger or criticism toward each other, focus the negativity on the child and the parent-child conflict thus serves to distract from the tension in the marital subsystem.

Circular causality: a sequence of causes and effects that leads back to the original cause and either alters or confirms it, thus producing a new sequence, as in a feedback loop.
(e.g This often happens on an emotional level in which two people "feel" each other's emotions, or when one person becomes emotionally escalated and the other family member does as well. A good example of this is when a teenage daughter gets anxious and depressed and her mom, in turn, gets anxious and depressed)

Enmeshment: is a trait of family dysfunction that involves poorly defined or non-existent boundaries, unhealthy relationship patterns, and a lack of independence among family members.

68) Question on Erikson stages of psychosocial development - **Identity and role confusion can affect sexual orientation**

69) Which Hormone is expressed in a child whose mother show good interaction with – **Oxytocin**

70) To make a diagnosis of Bipolar 1 in DSM5 - **you need at least one episode of mania**

Bipolar 1: one episode of Mania at least
Bipolar 2: one episode of hypomania and depression

Bipolar disorder is a category that includes three different condition--bipolar I, bipolar II, and cyclothymic disorder.

Bipolar I disorder is a manic-depressive disorder that can exist both with and without psychotic episodes

Bipolar II disorder consists of depressive and manic episodes which alternate and are typically less severe and do not inhibit function

Cyclothymic disorder is a cyclic disorder that causes brief episodes of hypomania and depression for **2 years.**

71) Neurochemical basis for tardive dyskinesia - **D2 supersensitivity**

72) Parkinsonism - **High D2 occupancy in the striatum**

SPMM: Antipsychotic induced:

Parkinsonism → High D2 receptor occupancy in striatum and basal ganglia.
Tardive dyskinesia→ High Dopamine receptor sensitivity,
SSRI-induced nausea → Serotonin 5HT3 activation

73) Lyonization **is associated with Klinefelter**

Lyonization: The X chromosome contains over 1,000 genes that are essential for proper development and cell viability. Females carry two copies of the X chromosome, resulting in a potentially toxic double dose of X-linked genes. To correct this imbalance, by way of the process called Lyonization (X-inactivation), females transcriptionally silence one of their two Xs. The inactivated X chromosome, then condenses into a compact structure called a Barr body, which is maintained in a silent state.

The process of X-inactivation is random. There is no preference for the inactivation of paternal or maternal X chromosomes. It occurs very early in embryogenesis, soon after the fertilization of the egg by the sperm (when the dividing conceptus is about 16-32 cells big).

It occurs in all somatic cells of women (but not in germ cells - those involved with forming gametes). X inactivation affects most but not all genes on the X chromosome. If a cell has more than two X chromosomes then the extra Xs are also inactivated.

74) Psychedelics mechanism of action e.g. magic mushrooms - **5HT2A Agonism**

75) Drug absorption constant is independent of: food **(vortioxetine)** ,form of drug, extravascular route, **Elimination constant**

(zero order - alcohol, modified release, phenytoin, depots, salicylates, aspirin, high dose fluoxetine, omeprazole)

76) Antihypertensive helpful for patient on lithium - **atenolol/propanolol** beta blocker / furosemide is safe also

Go for beta blocker (atenolol/propranolol) if both in the options.
(Amiloride for diabetes insipidus; not hypertension)

77) Drug that follows non-linear, kinetic at high dose – **Fluoxetine**

78) Patient at risk of hyponatremia will benefit from: amitriptyline? citalopram? sertraline? **Mirtazapine**? fluoxetine? Nortriptyline**,**
agomelatine, bupropion, MAO inhibitor

Drug / group*	Risk of hyponatremia	Level of supporting evidence
SSRIs	High	Strong
SNRIs	High	Strong
Tricyclics	Moderate	Strong
MAOIs	Low	Weak
NaSSAs	Low	Strong
Bupropion	Low	Moderate
Agomelatine	Low	Weak

79) Most important S/E of lithium - **worsening of psoriasis**
Another question on SPMM – most common: polyuria/tremors

80) Topiramate - weight loss/ name finding difficulties, psychosis, affect acid-base balance (low bicarbonate) nephrolithiasis, arrythmia, **Anomia**, angular closure glaucoma.

Topiramate can cause nausea, appetite loss, and weight loss. It blocks voltage-sensitive sodium channels by an unknown mechanism. Topiramate inhibits the release of glutamate and potentiates activity of GABA.

81) Why do we increase slowly when giving Lamotrigine – **SJS (Steven's Johnosn Syndrome).**
Very sensitive to dose changes. Rash is a common side effect and not always means SJS.

82) Which AED is more implicated in aggression - **Topiramate (2-10%)? Levetiracetam (13%) and vigabatrin (LVAT – Levetiracetam>Vigabatrin>Topiramate).**

83) Empathy is principally mediated by: DLPF? Hippocampus? Amygdala, **anterior insula**, ACC, medial prefrontal cortex **(supramarginal as well)**

84) Question on association study.

Q. what the exact disease-causing genes are: **association study**

Linkage analysis – location of culprit gene

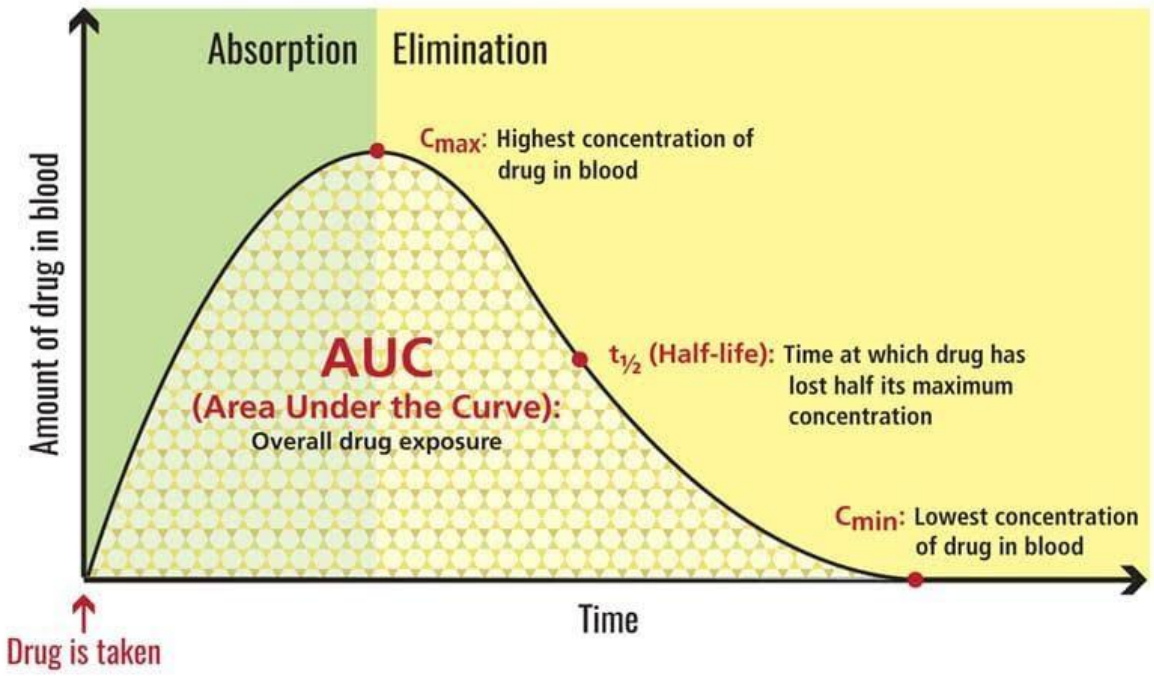
Association studies are more straightforward to carry out than linkage studies. Here a case control design is often adapted, and a sample of cases affected by a disorder is compared with controls. The frequency of alleles at the marker locus is then compared in the two groups. This method, though increasingly used, cannot make strong causal inferences. The locus chosen for study must predispose to illness. Thus, loci chosen for association studies are often known as **candidate genes**. If the locus does not predispose to illness, then the results of an association study should be negative. However, false positive results can occur if the two populations are not carefully matched for ethnic background. One alternative control group is the parents or relatives of affected individuals (the alleles *not* transmitted to the affected child compose the "control group"—this is known as the **Transmission Disequilibrium Test** or TDT). In **Genome Wide Association Studies** (GWAS), 'candidate gene' approach is not used. Instead, several thousands of single nucleotide polymorphisms are assayed in thousands of individuals. This is the new'hot' study technique in psychiatric genetics.

Questions	Most appropriate method
Is the phenotype familial?	Family study
What is the relative contribution of genetic and environmental factors? (Heritability)	Twin studies, adoption studies
What is the mode of transmission?	Segregation analysis
Where might be the 'culprit' genes?	Linkage analysis (known ancestries)
What are the actual genes responsible?	Association analysis (population level)

85) Area under curve in pharmacokinetics: **Drug distribution, bioavailability.**

Plotting plasma concentration against time, for a given dose, provides oral bioavailability. **The area under the curve (AUC)** after a single dose is proportional to the amount of drug in plasma and allows determination of the fraction of the dose absorbed-the bioavailability. The area under the curve obtained for orally administered drug divided by the area under the curve obtained for intravenous administration of the same dose gives the **bioavailability fraction**. It is determined by three factors:

1. Absorption
2. Distribution
3. Elimination (metabolism and or excretion).



86) Question on first order kinetic - options were on clearance of the drug and their linearity- **Clearance is dose independent**

[Psych-Mentor] **Note**: **Elimination and clearance are not the same.** Elimination is the 'irreversible removal of the drug from the body'. Consider: in first order kinetics, elimination rate is proportional to dose. The higher the dose, the greater the rate of elimination. However, clearance rate remains **dose-independent**: it is a totally theoretical volume of blood which is cleared of the drug per every unit of time, a measure which has nothing to do with the drug dose or concentration.

[SPMM] Drugs can undergo two different types of clearance (similar to absorption) when administered. When a constant fraction of drug is cleared per unit time, it is called as first order kinetics. This means that when the amount of drug in plasma or dose of administered drug increases, the clearance proportionately increases as a stable fraction of plasma concentration. In other words, the higher the amounts of a drug present, the faster the elimination. When represented graphically, first-order elimination follows an exponential decay versus time. Using this exponential curve, the time to eliminate 50% of a given amount (or time to achieve a decrease in plasma level to 50% of original) is the elimination half-life (t1/2). For example, if t1/2 is 2 hours for a drug A then the plasma concentration changes as follows

100mg/ml (2hours) →50mg/ml (2hours) → 25mg/ml (2hours) →12.5mg/ml

Most psychotropic drugs follow first order kinetics. In first order kinetics, the rate depends only on the drug concentration. It is not dependent on any other rate-limiting step. When the system facilitating such clearance of drugs gets saturated, drugs follow zero-order kinetics. Here a constant amount, not a fraction, of the drug is cleared per unit time. This means that irrespective of the amount of drug in plasma or dose of drug administered, the body clears only a fixed unit of the drug. As such, increasing dose might result in serious toxicity in this case. Certain drugs have propensity to undergo zero order kinetics even at therapeutic dose levels. Here the concept of half-life does not hold true as 'half life' depends on the dose administered.

Added during revision.
Right Answers.
Not sure about the answer.

87) Question on Crisper- GENE EDITING STUFF/

CRISPR systems have been used to study the roles of different proteins in directing neurodevelopment, so as to better understand the function of pathways that regulate genes and their expression.

CRISPR is a technique that is used to make specific changes in the DNA of plants and animals. CRISPR stands for

Select one:
- Cutting Recurring Introns and Selectively Pasting Repeats
- Cutting Regularly Interspaced Small Purposeless Repeats
- Clustered Regularly Interspaced Short Protein Rings
- Cleaving Randomly Interleaved Silencing Protein Repeats
- Clustered Regularly Interspaced Short Palindromic Repeats

CRISPR is the commonly known acronym for the genome editing technique Clustered Regularly Interspaced Short Palindromic Repeats. Ref: Heidenreich M, Zhang F. Applications of CRISPR–Cas systems in neuroscience. Nature Reviews Neuroscience. 2016 Jan;17(1):36.

The correct answer is: Clustered Regularly Interspaced Short Palindromic Repeats

CRISPR-associated protein system Cas is best described as

(quiz/review.php?attempt=886654&cmid=754[24/07/2022 9:13:33 PM]
W* December 2018: Attempt review

Select one:
- an innate sequence defending against epigenetic changes
- a viral vector delivery system
- an innate microbial defense mechanism against pathogens
- DNA-guided RNA editing sequence
- a RNA dependent RNA polymerase

The major function of CRISPR-associated (cas) genes for the bacteria that carry them is to provide innate immunity against the invasion of foreign nuclei acids such as plasmids and phages (viruses). Upon the first exposure to such foreign agents, short fragments of the foreign DNA (called protospacers) get incorporated into the CRISPR gene's repeat-spacer array (short palindromic repeats) as new spacers (usually of 20 nucleotides length) . This becomes the genetic memory of the prior invasion. Transcription of this array results in short mature CRISPR RNAs (crRNAs) that carry the tag (the RNA complement of the protospacer). Upon second exposure, the crRNA spacer 'sticks' to the target sequence of the foreign body and triggers sequence-specific destruction of theinvader using endonucleases. There are several types (1 to 5) of CRISPR systems identified to date. OF these, the type II CRISPR systems (Cas9) has a seed region called protospacer-adjacent-motif (PAM) that consists of 2 to 5 basepairs at the 3' end of the 20-nucleotide long spacer sequence. PAM is critical for the target selection and degradation. Ref: Heidenreich M, Zhang F. Applications of CRISPR–Cas systems in neuroscience. Nature Reviews Neuroscience. 2016 Jan;17(1):36.

The correct answer is: an innate microbial defense mechanism against pathogens

Which of the following elements of CRISPR gene editing produces hindrance to DNA expression without causing a degradation?

(quiz/review.php?attempt=886654&cmid=754[24/07/2022 9:13:33 PM]
W* December 2018: Attempt review

Select one:
- spacer
- dead Cas9
- protospacer
- PAM
- short Cas9

While the Cas9 gene used in routine CRIPSR experiments is a nuclease that cleaves DNA, CRISPR can also be used for gene silencing (i.e. reversible repression), gene activation (inducing expression). When the 2 catalytic domains of Cas9 are inactivated, this produces dead Cas9 (dCas9). dCas9 does not cleave DNA but produces stereological hindrance, resulting in a phenomenon known as CRISPR interference (CRISPRi).

The correct answer is: dead Cas9

Which of the following is a major challenge for the use of CRISPR technique to alter neuronal gene expression in human adult brains?

Select one:
- Ischemic response to the use of Cas gene in humans
- Blood-Brain Barrier
- Autoimmune inflammatory reaction to Cas
- Epigenetic effects on the human genome
- Lack of knowledge regarding human genome sequences

The major obstacle for employing CRISPR as a gene therapy in neuroscience is our inability to safely overcome the blood brain barrier (BBB) and reach the specifc cells that are affected by a genetic aberration. Approaches such as the use of adenovirus vehicles have fallen short to date as the Cas gene and the other biological components required for gene editing through CRISPR constitute a much larger baggage than what adenoviruses can effectively carry past the BBB. Ref: Senís E, Fatouros C, Große S, Wiedtke E, Niopek D, Mueller AK, Börner K, Grimm D. CRISPR/Cas9-mediated genome engineering: an adeno-associated viral (AAV) vector toolbox. Biotechnology journal. 2014 Nov;9(11):1402-12.

The correct answer is: Blood-Brain Barrier

The most commonly used system for CRISPR based genome editing is

Select one:
- System I
- System II
- System III
- System V
- System IV

There are several types (1 to 5) of CRISPR systems identified to date. Of these, the type II CRISPR systems (Cas9) has a seed region called protospacer-adjacent-motif (PAM) that consists of 2 to 5 basepairs at the 3' end of the 20-nucleotide long spacer sequence. PAM is critical for the target selection and degradation. Ref: Heidenreich M, Zhang F. Applications of CRISPR–Cas systems in neuroscience. Nature Reviews Neuroscience. 2016 Jan;17(1):36.

The correct answer is: System II

You are reviewing a graduate student who is seeking help for the treatment of his longstanding anxiety. He has recently come across procedures that allow brains to be "grown in a dish" through a novel stem-cell technology. Which of the following refers to cells that are reprogrammed to an embryonic stem cell-like state?

Select one:
- hiPSC
- zinc-finger cells
- PAM
- trans-activating CRISPR
- Cas9

Well-differentiated adult somatic cells such as fibroblasts from the human skin can be obtained from a donor, and reprogrammed to become pluripotent cells. These human induced pluripotent stem cells (hiPSC) have properties similar to those of embryonic stem cells. They can be further differentiated into any other cell type. Furthermore, gene editing can be performed on these hiPSC, providing a means to observe the effect of genetic mutations across various cellular lineages. While cells such as neurons are difficult to grow in a dish, hiPSCs have enabled the culture of neurons much more accessible.

The correct answer is: hiPSC

- Major challenge for use of CRISPER to alter neuronal gene expression: **Blood Brain Barrier**
- CRISPR-associated protein system Cas is best described as: **an innate microbial defence mechanism against pathogens**

Added during revision.
Right Answers.
Not sure about the answer.

- The most commonly used system for CRISPR based genome editing is: **System II**
- Elements of CRISPR gene editing that produce hindrance to DNA expression without causing a degradation: **dead Cas9**
- Which of the following is an advantage of CRISPR technology compared to other gene editing tools: **multiple genetic targets can be edited**.
- Nuclease based gene targeting techniques are = CRISPR, ZFN, TALEN

88) Question on pluripotent stem cell in the study of schizophrenia. The draw back and what age group brain they resemble equal to **6 months (2nd trimester) intrauterine life.**

89) Diagram MOA of SERT and **VMAT2**????m,o,a... atpase?G coupled? Etc Valbenazine and tetrabenazine in movement disorders, amphetamine Inhibition of VMAT 2 .

> Amphetamine acts via two separate ways (Saunders et al 2000). Firstly, It binds to DAT and is then taken up into the presynaptic axon terminal. Once inside the presynaptic terminal, amphetamine increases the amounts of monoamine neurotransmitters in the cytosol through the inhibition of vesicular monoamine transporter 2 (VMAT2) as well as through disruption of the electrochemical gradients necessary for vesicular transporter function.

90) How many turns do D receptor take in the cell - **7 times**

91) Neurasthenia ICD10 classification - Other neurotic disorders / **ICD 11 bodily distress disorders**

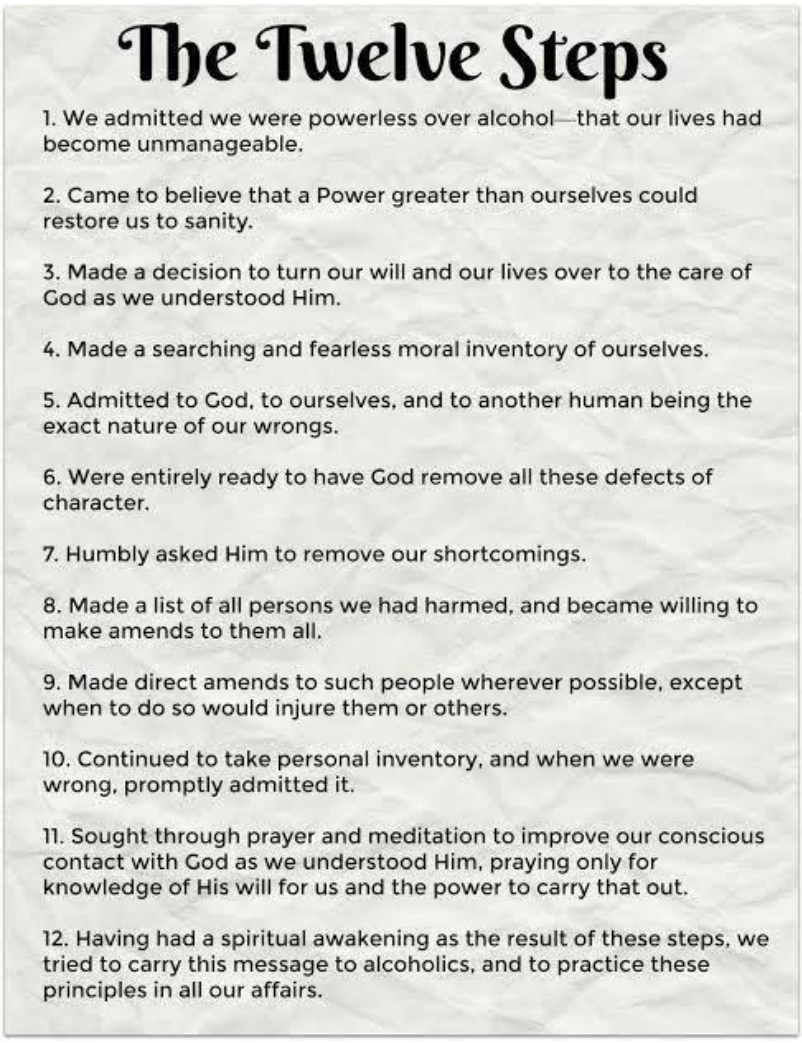
92) Ganser ICD10 classification: dissociative disorder ICD11 - **dissociative and conversion disorder**. ICD11 **6B6Y Other specified dissociative disorders**

93) Question about - Cue exposure alcohol anonymous.

Cue exposure – conditioned,
12steps – powerless, hopeless (in the context of AA rehabilitation)

From Dec 22 recall / EMI: During substance abuse therapy, Alcohol or substance signifier is used in **controlled conditions**?

A. Cognitive therapy
B. Behavior therapy
C. Analytic therapy
D. DBT
E. Rational emotive therapy
F. Group therapy
G. **Cue therapy**
H. 12-step program
I. Alcohol Anonymouss



The Twelve Steps

1. We admitted we were powerless over alcohol—that our lives had become unmanageable.

2. Came to believe that a Power greater than ourselves could restore us to sanity.

3. Made a decision to turn our will and our lives over to the care of God as we understood Him.

4. Made a searching and fearless moral inventory of ourselves.

5. Admitted to God, to ourselves, and to another human being the exact nature of our wrongs.

6. Were entirely ready to have God remove all these defects of character.

7. Humbly asked Him to remove our shortcomings.

8. Made a list of all persons we had harmed, and became willing to make amends to them all.

9. Made direct amends to such people wherever possible, except when to do so would injure them or others.

10. Continued to take personal inventory, and when we were wrong, promptly admitted it.

11. Sought through prayer and meditation to improve our conscious contact with God as we understood Him, praying only for knowledge of His will for us and the power to carry that out.

12. Having had a spiritual awakening as the result of these steps, we tried to carry this message to alcoholics, and to practice these principles in all our affairs.

EMI: During substance abuse treatment, people believe that they are helpless, hopeless and powerless?
H. **12-step program**

94) Trisomy 18 - **Edward syndrome**

95) Rate limiting enzyme in alcohol metabolism: **Alcohol dehydrogenase**

96) Precursor of serotonin: **5hydroxy tryptophan**

97) What undergoes hydroxylation in the synthesis of dopamine: **L-tyrosine**

98) Which receptor does, ketamine act on: **Non-competitive NMDA antagonist with high affinity located on GABA interneurons causing disinhibition of GABA interneurons effect and increases glutamate.**

The glutamate model of depression implicates alterations to glutamate- and γ aminobutyric acid (GABA)-related synaptic function (4); stress-related dysfunctional glutamate cycling is thought to result in excitotoxicity and neuronal atrophy. **N-methyl-D-aspartate (NMDA)-receptor-mediated inhibition of GABA interneurons, resulting in reduced inhibitory control over prefrontal glutamate neurons and subsequent glutamate burst**, are thought to facilitate ketamine's antidepressant effects (5). Accompanying α-amino-3-hydroxy-5-methyl-4-isoxazolepropionic acid (AMPA)-receptor agonism activates second messenger pathways implicated in neuroplasticity (6). Thus, assessing brain glutamate and GABA levels provides insight into ketamine's antidepressant mechanisms of action.
https://www.frontiersin.org/articles/10.3389/fpsyt.2020.549903/full

99) Memantine MOA: **Non-competitive NMDA antagonist with low affinity and 5HT3 antagonist**

98) Aura for **frontal** lobe epilepsy: **Cephalic sensory disturbances like numbness, funny feeling in head, something crawling in head or pressure in head, dizziness, vertigo, light headedness.**

99) For **Temporal** lobe epilepsy: **Epigastric like butterflies in stomach.**

100) For **parietal** seizures: **Somatosensory like body image distortion, hyperschemezia**

Electrical sensations , tingling

Epilepsy and aura[Psych-Mentor]

An aura is a subjective sensation and is a simple partial seizure. Typically, it is of brief duration, rarely lasting longer than seconds. Determining the type of aura present help in identifying the site of cortical onset. Eight different varieties are recognized: somatosensory, visual, auditory, gustatory, olfactory, autonomic, abdominal, and psychic.

Auras precede temporal lobe seizures in approximately 80% of cases. The most common auras in temporal lobe seizures are **abdominal** (a rising epigastric sensation) and **psychic** aura (fear, déjà vu, jamais vu).

Parietal lobe seizures may begin with a contralateral sensation, usually of the positive type (electrical sensation, tingling). **Occipital** lobe seizures may begin with contralateral **visual** changes, usually of the positive type, such as colored lines, spots, or shapes, or even a loss of vision. Temporal-parietal-occipital seizures may produce more formed auras.

Complex partial seizures, are defined by impairment of consciousness. This implies decreased responsiveness and awareness of ones self and surroundings. Usually, during a complex partial seizure, a patient is unresponsive and does not remember events that occurred.

111) Drug used in ADHD that causes hypotension: **Guanfacine**

Some more common side effects: Anxiety; appetite decreased; arrhythmias; asthenia; constipation; depression; diarrhoea; dizziness; drowsiness; dry mouth; gastrointestinal discomfort; headache; **hypotension**; mood altered; nausea; skin reactions; sleep disorders; urinary disorders; vomiting; weight increased (BNF).

Clonidine is not a popular option for treating tics/ADHD due to high rates of hypotension associated with it (SPMM, Adverse Reactions).

112) Which area of the brain is stimulated during fMRI in

Auditory hallucination: Auditory hallucinations were associated with activation in the inferior frontal/insular, anterior cingulate, and **temporal** cortex bilaterally (with **greater responses on the right**), the right thalamus and inferior colliculus, and the left hippocampus and parahippocampal cortex. **Greater right frontotemporal activation**

Visual hallucination: occipital lobe.

fMRI in common conditions:

In **schizophrenia**, fMRI shows poor dorsolateral pre-frontal cortex activation in executive tasks, while decreased N-acetyl aspartate in prefrontal cortex is seen in spectroscopy (EMI Neurosciences part 2, Q 26)

In **AD**, Neuroimaging findings in Alzheimer's disease include ventricular enlargement, reduced medial temporal lobe volume - especially hippocampus and a reduction in parieto-temporal fMRI activation and SPECT blood flow.

Neuroimaging findings in psychiatry:
Neuroimaging findings in depression
Periventricular and deep WM hyperintensities Subcortical – thalamic and striatal hyperintensities Decreased frontal and basal ganglia volumes Decreased metabolism in prefrontal cortex, Anterior cingulate & amygdale Higher prefrontal metabolism (esp. anterior cingulate) predict better treatment response Higher 5HT2A receptor density – higher dysfunctional negative thoughts Increased MAO-A activity (especially women) Elevated D2 binding in untreated depression – psychomotor retardation Therapeutic dose of SSRIs- 80% 5HT transporters occupied
Neuroimaging findings in schizophrenia
Ventricular enlargement Loss of grey matter – especially insular cortex, anterior cingulate (medial prefrontal cortex) and medial temporal lobe Progressive loss of brain volume in first few years of diagnosis fMRI reveals poor DLPFC activation in executive tasks Decreased NAA (N-Acetyl aspartate) in PFC (neuronal loss) in MRS Widespread reduction in DTI (diffusion tensor) – fractional anisotropy: frontal and corpus callosum – more in chronic treated patients
Neuroimaging findings in Alzheimer's
Ventricular enlargement Loss of temporal lobe volume – especially hippocampus Decreased parieto-temporal fMRI activation and SPECT blood flow
Neuroimaging findings in OCD
Both reduced and increased volumes of caudate nuclei reported. Higher caudate blood flow due to increased metabolism. This reduces after effective treatment of the OCD.

Functional MRI (fMRI) takes advantage of the fact that blood with different levels of oxygenation produces different signals. It therefore is a measure of oxygen consumption. This technique is called blood level-dependent **(BOLD)** contrast (PsychMentor)

115) Test for executive function: **Tower of London**

Primary area of cognition tested	**Tests**
Premorbid intelligence	National Adult Reading Test (NART)
General Intelligence	Wechsler Adult Intelligence scale (WAIS), Raven's Progressive Matrices
Visual memory / Visio spatial skills	Rey-Osterrieth Complex Figure
Attention	Continuous Performance Tasks
Language	Boston naming test, Animal fluency
Executive function	**Stroop test, Wisconsin Card Sorting Test (WCST), Tower of London, Animal fluency, Trial making test**
Other	Halstead-Reitan battery (for brain damage)

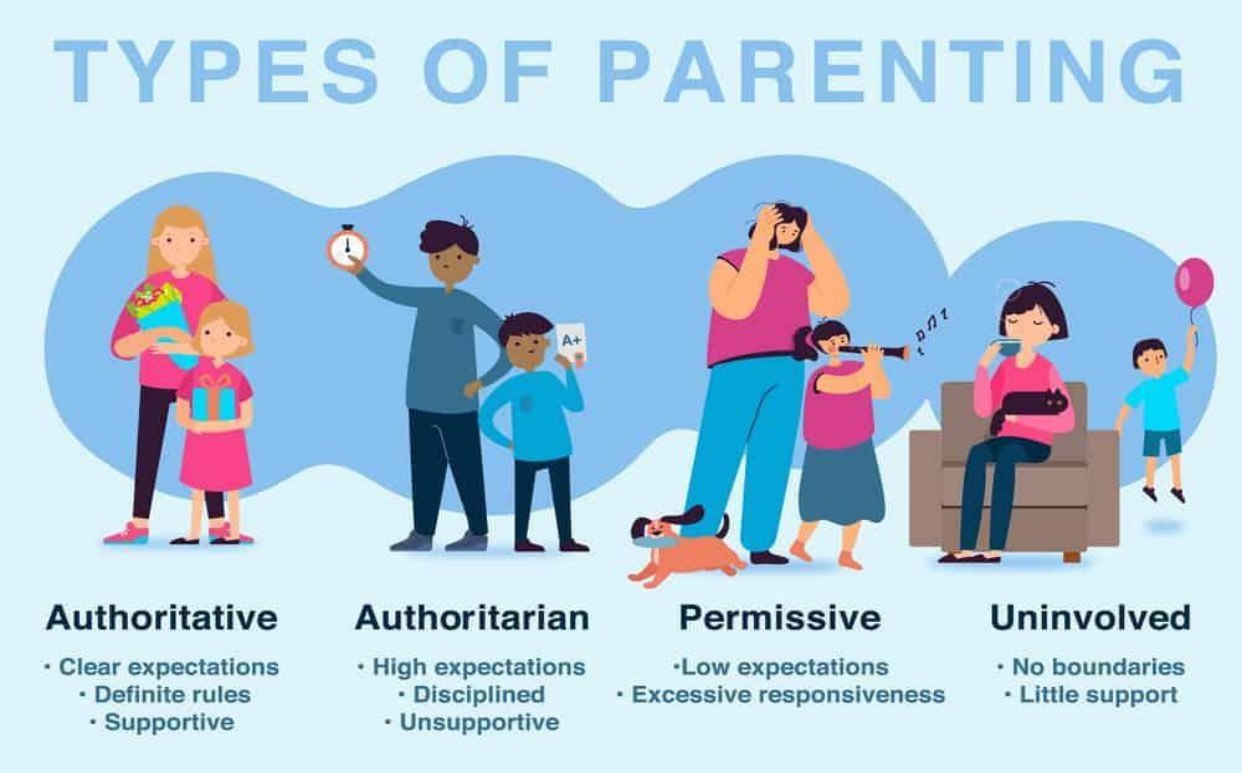
116) Executive function generally believed to be a **Dorsolateral frontal lobe** function and depends on **intact frontal-subcortical circuits.**

117) Test for abstraction... **Similarities**

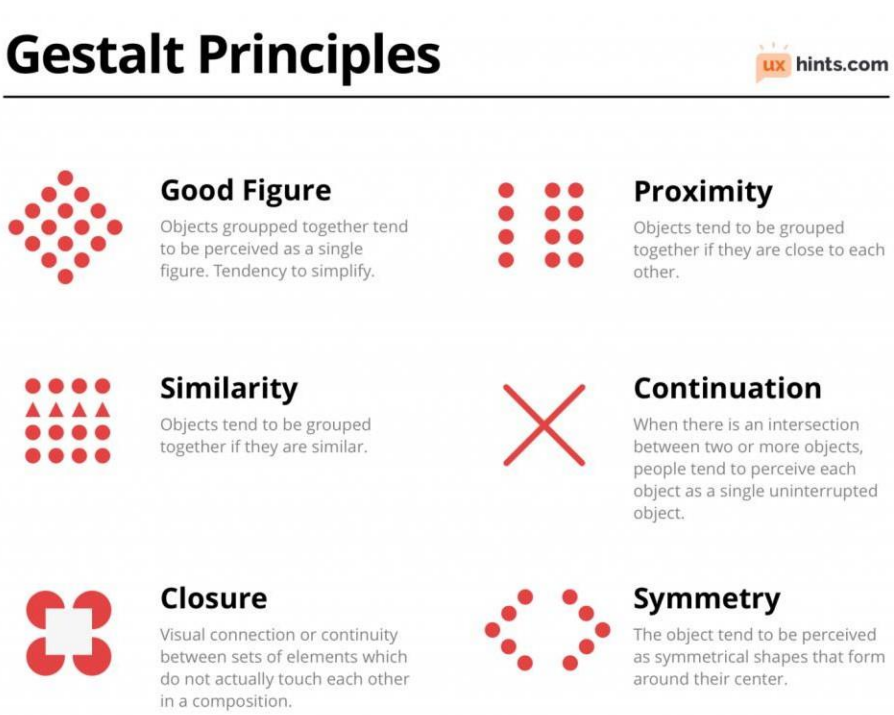
Frontal lobe function	**Test or examination**
Initiation	Verbal and categorical fluency
Abstraction	**Proverbs, similarities, cognitive estimates**
Problem solving and decision making	Tower of London, Cambridge stockings, gambling tasks
Response inhibition and set shifting	Alternating sequences, go-no-go test, Luria motor test, trail making test, Wisconsin card sorting test, Stroop test

Added during revision.
Right Answers.
Not sure about the answer.

118) Question on parenting styles... **authoritarian**, authoritative. permissive



119) Gestalt laws...**proximity**



The Gestalt principle, which refers to the way in which the human brain organizes and interprets visual information, is based on which of the following psychological laws of perception?

A. Proximity
B. Similarity
C. Continuity
D. Closure
E. Symmetry

120). Seeing a face in the sky: **pareidolia**

Some common examples are **seeing a likeness of Jesus in the clouds** or an image of a man on the surface of the moon.

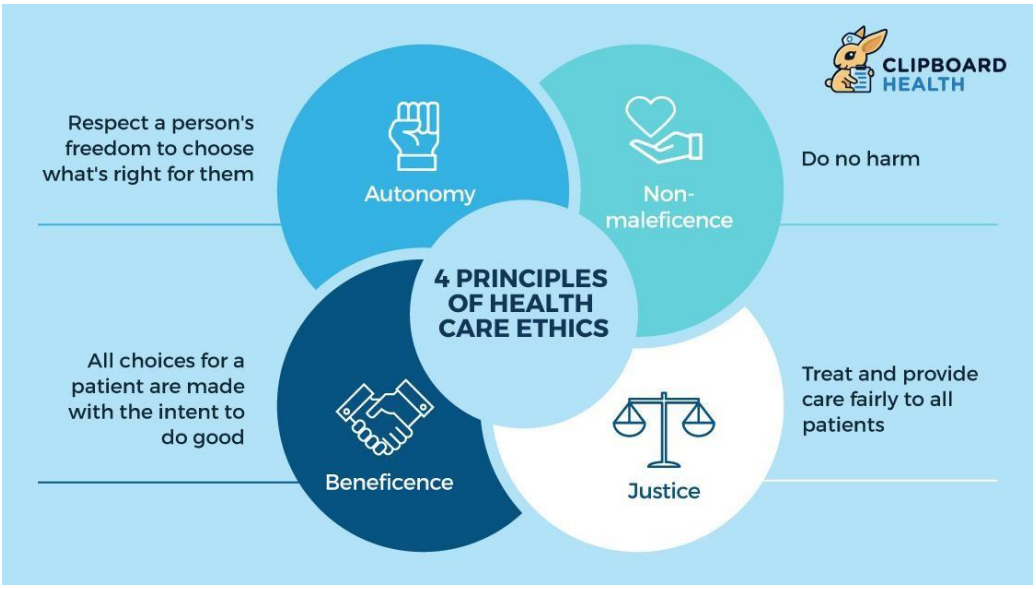
Illusion type	Description
Completion illusions	The tendency to fill in missing information in order to make sense of a stimulus
Affect illusions	Arise due to specific mood (affective) states e.g. a woman is walking home in the dark and is frightened, she mistakes a tree for a tall man in a long coat. Another example is an agitated person misinterpreting innocent gestures as threatening
Pareidolic illusions	Arise when detailed images are seen from shapes. E.g. seeing the man in the moon, or Jesus Christ on a piece of burnt toast. Pareidolic illusions tend to occur when a person is concentrating whereas affect and completion illusions occur during inattention.

Apophenia is a general term for interpreting patterns or meaning in meaningless data—this involves any kind of information, including visual, auditory, or a data set. **Pareidolia focuses on visual information**

121). Choosing group therapy over individual therapy because it will help a larger group...justice? or **utilitarianism**?

122). Withdrawal of a drug from the market because it causes cardiac event...**non maleficence**

Proposed by **Beauchamp and Childress**



113). Drug with the lowest tendency to cause hyperprolactinemia... **Aripripazole**

Effect on prolactin	**Antipsychotics**
Prolactin sparing (or very rare increase)	Clozapine Aripiprazole Asenapine Quetiapine
Low risk / minor change	Lurasidone Olanzapine Ziprasidone
High risk / major change	All the typical antipsychotics Risperidone Amisulpride Paliperidone Sulpiride

114) Basis for prions disease, **protein misfolding,**

Human Prion Diseases:

1. Creutzfeldt-Jakob Disease (CJD)
2. Variant Creutzfeldt-Jakob Disease (vCJD)
3. Gerstmann-Straussler-Scheinker Syndrome
4. Fatal Familial Insomnia
5. Kuru

In animals, examples include **bovine spongiform encephalopathy** (BSE or "mad cow disease"), **scrapie in sheep**, and **chronic wasting disease (CWD) in deer and elk.**

115) Question on reciprocal inhibition

Reciprocal inhibition (Wolpe): If stimulus with desired response and stimulus with the undesired response are presented together repeatedly, then the incompatibility leads to a reduction in frequency of the undesired response. This is evident when your dog barks at your friend; try hugging her in front of your dog every time the dog barks and slowly the dog will stop barking at your friend. This is used in relaxation therapy for anxiety and in systematic desensitisation.

116) ECT reduces ach?glutamate? Gaba?n acetyl aspartate?,

Increase Brain-derived neurotrophic factor (BDNF) and decrease 5HT2A & DECREASES BETA NORADRENERGIC AND ALPHA 2 RECEPTORS

Variables affected by ECT	Changes
Neurotrophic factors	Increase in NGF,BDNF, NF3.
Cell growth and synaptic connectivity	Increased esp. In hippocampus
Hormones	Increased cortisol, prolactin, TSH coincides with good response. TRH gene expression increased in animals. Vasopressin, ACTH, oxytocin and opioid endorphins also increase consistently.
Neurotransmitters and their receptors	5-HT-, NA-, cholinergic-, glutaminergic- and GABAergic systems, adenosine A1-receptor & 5-HT2A – all decrease in sensitivity. Activation of DA transmission and stimulation of 5-HT in hippocampus and amygdala.

117) Question on hemineglect: The definition - neglect left side of body

Visual Inattention: Hemineglect is a feature of **non-dominant parietal lesions**. Letter or star cancellation task, line bisection task, draw-a-person or draw-a-tree tasks are useful to identify hemineglect

118) Attention cause conversion from **sensory to short term**

If **attention** is paid to the sensory memories during perception, sensory memory gets consolidated or 'moves' into the short-term memory system.

Rehearsal is supposed to be the transient control process that can aid maintenance of STM and transfer to LTM.

119) Neuronal cell that aids **transfer** of glutamate: **Astrocytes**? or pyramidal cells?

Pyramidal cell uses it but Astrocyte transports or secrets it

120) Neuronal cell that aid transmission through GABAergic neurons in the caudate: **medium spiny neurons or spiny projection neurons**

121) High performing student takes alcohol be able to do a presentation in public: **Social phobia**

Social phobia is same as social anxiety(ICD11)

122) Adjustment disorder scenario

Adjustment disorder is a maladaptive reaction to an identifiable psychosocial stressor or multiple stressors (e.g. divorce, illness or disability, socio-economic problems, conflicts at home or work) that usually **emerges within a month of the stressor.** The disorder is characterized by preoccupation with the stressor or its consequences, including excessive worry, recurrent and distressing thoughts about the stressor, or constant rumination about its implications, as well as by failure to adapt to the stressor that causes significant impairment in personal, family, social, educational, occupational or other important areas of functioning. The symptoms are not better explained by another mental disorder (e.g., Mood Disorder, another Disorder Specifically Associated with Stress) and **typically resolve within 6 months, unless the stressor persists for a longer duration.** **[ICD11]**

In DSM: Adjustment disorder can starts within 3 months of the stressor

Symptoms of Adjustment Disorder may be influenced by local idioms (e.g., **Susto** or **Espanto** [fright] in Central America) that are associated with fear or subsequent preoccupation with a stressor with strong cultural connotations (e.g., experiencing intense fear when crossing an unpopulated area alone at night). **[ICD11]**

123) Question on Holmes and Rahe life event scale: **Divorce**

The first 10 of the **43** life events are listed below with their values.

	Life changing unit	Value
1	Death of spouse	100
2	Divorce	73
3	Marital separation	65
4	Jail term	63
5	Death of a close family member	63

	Life changing unit	Value
6	Personal illness	53
7	Marriage	50
8	Being fired from work	47
9	Marital reconciliation	45
10	Retirement	45

Lowest (43) in the life events is Minor law violations - value of 11

124) Man who witnessed a friend's demise 6weeks ago is unable to move limb despite absence of neurological deficit: **Conversion (Dissociative neurological symptom disorder in ICD11)**

125) Question which had this option...behaviourism, mentalization Empathy

Q. This process is in form of imaginative mental activity that lets us perceive and interpret human behaviour in terms of intentional mental states. (this question form another version of this recall)

- **Mentalization**
- Personal construct
- Mindfulness

What is an example of mentalization? For example, a father is trying to mentalize his daughter's state of mind when he is wondering about her being quiet at dinner and considering that her behaviour may be related to her not being allowed to see her friends or him not having time to watch a movie with her after dinner because of his work.

Both concepts, mentalization and the theory of mind, describe metacognitive processes.

Mentalization mainly concerns the reflection of affective mental states.

In contrast, theory of mind focuses on epistemic states such as beliefs, intentions and persuasions.

126) Valproate risk of causing congenital anomalies**: 8-12%**

Congenital anomalies (8-12%)
Developmental delays (31-40%)

127) Losses own culture at the expense of imbibing another's: **Assimilation**

127) Causes of QT prolongation. **(see Q.35)**

Added during revision.
Right Answers.
Not sure about the answer.

December 2020

Dec 2020 (1 - 15):

1) **Hypsarrhythmia**: occurs in infantile spasm (**tuberous sclerosis, west syndrome)**
   - It is diffuse giant waves (high voltage, >400 microvolts) with a chaotic background of irregular, asynchronous multifocal spikes and sharp waves.
   - Electro-decremental response: are clinical seizures associated with a marked suppression of the background.
2) **Function of MRS**:

   Also known as nuclear magnetic resonance (**NMR**) spectroscopy, is a non-invasive analytical technique (based on the same physical principles as MRI) that has been used to:
   a) Study **metabolic changes** in brain tumors, strokes, seizure disorders, Alzheimer's disease, depression and other diseases affecting the brain.
   b) Measure **the concentrations of chemical components within tissues; GABA** and **glutamate** can be detected using MRS but **Not dopamine** as it is available in a very low concentration.

MR molecule	Potential clinical uses
Hydrogen proton (1H)	**1.** Magnetic resonance imaging (MRI). **2.** Analysis of metabolism – used to quantify; N-acetyl aspartate **(NAA** to assess **neuronal integrity), creatine (**a constant element of cellular energetic metabolism**)**, and **choline** (an increase in **Cho** is considered a marker of pathological proliferation/degradation of cell membranes and demyelination) containing molecules.
Fluorine (19F)	**1.** Measurement of pO2 **2.** Analysis of glucose metabolism **3.** Measurement of PH. **4.** Pharmacokinetics: measuring concentrations of **psychotherapeutic drugs** such as lithium, fluoxetine and trifluoperazine (Stelazine).
Lithium-7 isotope (7Li)	**Pharmacokinetics**
Phosphorus-31 (31P)	**1. Analysis of bioenergetics**; measuring concentrations of phosphorus-containing compounds (e.g. ATP & GTP). **2. Measurement of pH in brain regions.**
Nitrogen (14N)	Measurement of **glutamate, urea** and **ammonia**
Carbon-13 isotope (13C), low sensitivity	1. Analysis of metabolite turnover rate. 2. Pharmacokinetics of labelled drugs
Oxygen-17 isotope (17O)	**Measurement of metabolic rate**

3) **Wilson chromosome**:

- Congenital **recessive** condition of defective copper metabolism due to a defect in **chromosome 13**.
- Characterized by: **hepatic symptoms** and later **psychiatric symptoms** and **choreoathetosis** (lenticular nuclei) as various organ systems are overwhelmed by copper.
- It can be diagnosed (once the central nervous system is affected) by finding **Kaiser-**

Fleischer rings in the cornea with a slit-lamp though these are not universally present.

- A more reliable diagnosis depends on **low ceruloplasmin** and **elevated copper** in urine and liver biopsies.

4) Alzheimer – Tau (histopathological changes)

- **NFT** (hyperphosphorelated tau protein— cytoskeletal element, basophilic).
- **Hirano bodies—** Cytoplasmic, rode-shaped and eosinophilic bodies.
- **Senile plagues—** insoluble amyloid peptide deposits, eosinophilic mass (apple green) and extracellular.
- **In addition to:** neuronal and synaptic loss, granulovascular degeneration, astrocytes gliosis and microglial activation**.**

5) FTD: Tauopathies

- In familial cases mutations in the (**MAPT**) on chromosome 17q21.
- Many cases are **tau-negative** but show ubiquitin-immunoreactive neuronal cytoplasmic inclusions (mutations in progranulin (**PGRN**) gene on chromosome 17q21)**— (MND) type.**
- **Pick cells:** abnormal swollen oval-shaped neuronal cells with loss of Nissl's substance and peripherally displaced nucleus.
- **Pick's bodies:** are argentophilic, **tau and ubiquitin** reactive filamentous inclusions.

Frontotemporal Dementia has (3) variants as per **ICD-11**:

1. **FTD behavioural variant**: is characterized by personality changes. Neurocognitive functioning may be preserved in the early stages.
2. **Primary progressive aphasia variant**: progressive impairment in language skills. **3 Subtypes**:
   - Primary deficits in word finding (logopenic subtype).
   - Word meaning (semantic subtype).
   - Word production (agrammatic subtype).
3. **FTD motoric variant:** progressive impairment in motor functioning. progressive neurocognitive impairment in; (attention, executive functioning, and visuospatial skills, with comparatively intact memory skills).
   3 Subtypes:
   - Progressive supranuclear palsy (poor balance, frequent falls, and horizontal gaze palsy),
   - Corticobasal degeneration (limb apraxia, tripping, rigidity, and dystonia).
   - Amyotrophic lateral sclerosis (e.g., muscle weakness, muscle atrophy, fasciculations, and spasticity, etc.)

6) Lewy body dementia:

- Alpha-synuclein accumulation in (temporal lobe, cingulate gyrus, frontal lobes and the dorsal motor nucleus of the vagus).
- **Lewy bodies** are weakly eosinophilic, spherical, cytoplasmic inclusions.
- Antibody to protease **ubiquitin** can be used to identify Lewy bodies.

- **Clinical features (two or more):**
  - Spontaneous motor features of Parkinsonism
  - Fluctuating cognition with notable variation in **attention + alertness**
  - Recurrent visual hallucinations (typically well-formed).
  - REM sleep behaviour disorder **(ICD-11).**
  - A progressive cognitive decline (memory loss may not be an early feature).
  - **Supportive features**: Neuroleptic sensitivity (not included in ICD-11) and history of

falls

7) **3 ques on identity theory etc..** (takes both mind and science, mental state does exist and real, it is different from personal identity).

Social identity theory (SPMM Notes):

- An individual's positive self-image depends on both **personal** and **social** identities.
- So each individual strives to improve his group's success to foster his own image. This leads to prejudice against other groups
- Has **3 Stages:** categorization, identification and comparison.

8) **Gender schema processing theory.**

- Gender identity alone provides children the motivation to assume sex-typed behaviour.
- Following this they observe and learn to be of a specific gender in the society.
- Thus, a gender schema of the particular culture gets deeply incorporated and serves as a standard for comparison

9) **Brain default network:**

Play a major role in self-processing, internally oriented thinking and processing information not relevant to a task at hand.

1. The posterior cingulate cortex (PCC),
2. Precuneus
3. Medial prefrontal cortex (mPFC)
4. Angular gyrus.

The dorsal attention network (frontoparietal or central executive network):

1. Dorsolateral prefrontal cortex.
2. Intraparietal sulcus (IPS) as well as lateral superior parietal lobule (together called posterior parietal cortex – PPC)
3. Middle frontal gyrus/premotor junction.

The salience network:

1. Anterior prefrontal cortex **(aPFC)**
2. Dorsal anterior cingulate cortex (dACC)
3. Anterior insula
4. Frontal operculum.
5. Some subcortical structures (thalamus and putamen).

10) **Gait** for; Huntington (**Lurching, chaotic gait**), upper motor neuron (**Scissoring gait**) and cerebellar disease (**Broad, unsteady gait (Drunken/sailor's gait**)

11) **Psilobin MOA**

Psilocybin (magic mushroom) and other classical hallucinogens (e.g. LSD & mescaline) exert their psychedelic actions in humans via activating 5-HT_{2A} serotonin receptors, which in turn mediates its anti-depressant and anti-anxiety effects.

12) **Contagion theory:** by scholar Gustave Le Bon (**collective behavior is irrational**)

Contagion theory assumes that people in a crowd act emotionally and irrationally because they come under the influence of the crowd's impulses.

13) **Pontine lesion:**

Locked in syndrome (ventral pontine) describes a condition of total paralysis below the level of **the third nerve nuclei**. Such patients can:

- Open their eyes voluntarily, and elevate and depress their eyes to command.
- Horizontal eye movements are usually lost.
- No other voluntary movement is possible.

14) Deindividuation.

Festinger's deindividuation theory: people in-group context act uncharacteristically more aggressive as **a sense of identity and belongingness** and **diffusion of responsibility occurs in groups.**

15) bowlby stage of grief except bargaining:

- **Bowlby and Parkes** were the first to propose a stage theory of grief for adjustment to bereavement that included **4 stages (SYDR)**: **1. S**hock-numbness (often with outbursts of anger and intense distress) **2. Y**earning-searching**, 3. D**isorganization-despair and **4. R**eorganization.
- **Erich Lindemann:** Stages of grief **(SSAR)**: **1. S**hock that lasts 10-14 days **2.** Intense **S**adness (withdraw from social contact), **3. A**nger, as the grieving person seems to '**protest**'; the unexpected death **4.** and finally, within a year or so, the grief is **R**esolved, and the person returns to normal
- **Colin Murray Parkes; 5 stages** (**A SPIDER**) of bereavement:
    1. Alarm
    2. Numbness (shock).
    3. Pining for the deceased (illusions or hallucinations of the deceased can occur)
    4. Depression
    5. Reorganisation (recovery)
- **Kübler-Ross**: adapted Bowlby and Parkes' theory to describe a **5-stage (DABDA)** response of terminally ill patients to awareness of their impending death: **1.** denial-dissociation- isolation **2.** Anger **3.** Bargaining **4**. Depression and **5.** Acceptance

Freud model for mourning: individual's life force must be fully withdrawn from its attachment to the deceased person (Decathexis).

16) computational psychiatry:

Reward prediction error: is difference between receive reward & predicted reward

17. selective d2 antagonist? **Sulpride**

18. sally anne test **theory of mind 3-4 years**

19. EEG seen as normal, what disease? **variant CJD and COMPLEX PARTIAL AND TEMPORAL NORMAL IN 50% and autosomal dominant frontal lobe epilepsy**

20. EEG centro temporal (**Rolandic**)-**benign epilepsy in children is central temporal**

Beta is found in centre position, delta is found deep sleep and any delta in awakening stage is pathology, alpha in occipitoparietal, theta is fronto temporal

21. over the counter medication cannot take with MAOI, **decongestant**

22. adhd medication cause hypotension **clonidine**

23. crisis identity emi, identity foreclosure, moraotium, diffusion

24. emi receptor : zopiclone **omega 1 alpha 1 subunit of gaba A**, corticosteroid **Glucocorticoid in the nucleus**

Added during revision.
Right Answers.
Not sure about the answer.

25. emi development how many word : 1 year old **2-3 words** 13 years old.**adult speech**

26. anorexia purging type finding : **vomiting: metabolic alkalosis/Russell sign/ hyponatremia**. laxative **metabolic acidosis, hypokalemia and hyponatremia**

27. wernicke aphasia : **fluent**? **And receptive**

28. human research : **helsinski?** or nuremberg?

29. see cloud as face : **pareidolia**

30. Fundamental attribution error- **attribution of personal factor to others behaviour**

31. Attachment- **Anxious avoidant**

32. Kohlberg- pre convetional 7-12, conventional 13-16, post- convetional 16-20

33. Reward prediction error- difference between received reward and predicted reward.

34. Neurotransmitter- fast and excitatory- **Glutamate**

35. Antipsychotic increase with absorption of food- **oral Ziprasidone** and **Lurasidone, paliperidone and quetiapine**

36. Apoe4- RISK **homozygous 9-15, heterozygous 3-4**

37. Physiology change in pregnancy-

38. At young doctors training- **metonymy**/circumstantiality

39. Landmark agnosia/ topographical agnosia- **lingual gyrus**

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Additional Notes:

Delusional disorder: is an uncommon condition in which patients present with circumscribed symptoms of non-bizarre delusions (DSM-5 allows 'with bizarre content'; ICD-11 does not specify), but with absence of prominent hallucinations and no thought disorder, mood disorder, or significant flattening of affect.

DSM-5 subtypes of delusions: include erotomanic, grandiose, jealous, persecutory, and somatic.

ICD-11 states that delusions typically persist for at least 3 months and often much longer.

Anomia due to both angular and supramarginal...if both are in the option then ans would be angular

Added during revision.
Right Answers.
Not sure about the answer.

Transcortical sensory and motor aphasia: repetition will be intact

Question number 28 can be found in fresh question A1 set 6

semantic is visual analogue scale

semantic differential scale

Osgood's semantic differential scale is used to measure verbally expressed attitudes. It allows different attitudes about a particular topic to be measured on the same scale. It includes various factors constituting an attitude; e.g. while expressing one's attitudes regarding a politician, one can rate him using an evaluative component (good ---- bad), activity component (active inactive) and potency component (powerful ---- weak) etc. With these bipolar adjectives being the two extremes. 2 polarised adjective separated by line

Non linear + zero order

frusemide is relatively safe in using with lithium..

ZOCFLAP: Zero **O**rder **C**onstant, **F**irst Order **L**inear **A**nd **P**roportional

Zero order (PES): Phenytoin, Ethanol and Salicylates

Receptors linked to G proteins, including the dopamine D2 receptor, all have the structure **of seven transmembrane regions**, meaning that they span the membrane seven times.

Adjustment disorder (onset):

According to DSM V 3 month

ICD 10/11: 1 month

PTSD within 6 months

Mentalization is the ability to understand the mental state – of oneself or others – that underlies overt behaviour. Mentalization can be seen as a form of imaginative mental activity that lets us perceive and interpret human behaviour in terms of intentional mental states.

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Alpha synucleopathies – People's Lives Matters

•Parkinson's

•Lewy Body Dementia (DLB)

Multisystem atrophy

Added during revision.
Right Answers.
Not sure about the answer.

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Added during revision.
Right Answers.
Not sure about the answer.

1) Vortioxetine action:

- **5HT1A agonist**
- **5HT1B partial agonist**
- **5HT1D, 5HT3, 5HT7 antagonist**
- **SERT inhibition**

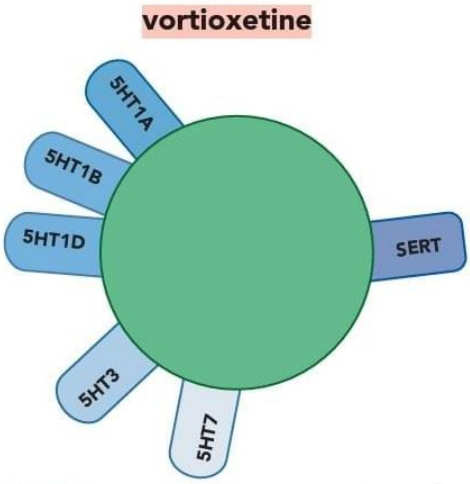


Figure 7-49 **Vortioxetine.** Vortioxetine is a serotonin reuptake inhibitor and also has actions at several serotonin receptors, including $5HT_{1A}$, $5HT_{1B}$, $5HT_{1D}$, $5HT_3$, and $5HT_7$.

Summary of Mechanisms of Action of Vortioxetine

Target	Effect	Proposed Therapeutic Activity
SERT Inhibition	• Increase in 5-HT in prefrontal cortex (PFC)	• Antidepressant Activity
5-HT1A Full agonist (Presynaptic autoreceptors)	• Acceleration of the 5-HT1A receptor desensitisation (downregulation) leading to 5-HT release	• Faster onset of action (Reduced lag in onset of action, which is attributed to 5HT1A receptor desensitisation time)
5-HT1A full agonist (Postsynaptic receptors)	• Inhibition of GABA interneurons resulting in the release of Glu, NA, DA, Ach and Hist in PFC • Enhances the activity of Ventral tegmental area (VTA) DA neurons and mesocortical DA release.	• Pro-cognitive effect • Antidepressant effect (DA and NA)
5-HT 1B Partial agonist (Presynaptic autoreceptors)	• Increase in 5-HT, Glu, ACh and Histamine. • Indirectly increases the release of DA and NA in the PFC and hippocampus.	• Antidepressant activity • Pro-cognitive effect
5-HT1D antagonism (Presynaptic autoreceptors)	5-HT increase	• Suppresses REM sleep (Sleep improvement)
• 5-HT3 antagonism Postsynaptic receptors (Heteroreceptors on GABA interneuron) • 5-HT3 Receptors on dorsal horn of spinal cord	• Increases in 5-HT, DA, NA, ACh, and Hist in PFC and hippocampus • Reduction of hyperalgesia	• Pro-cognitive effect • Antidepressant activity • Increasing NREM Sleep - Sleep improvement • Pain control
• 5-HT7 antagonism (Postsynaptic receptors) (Heteroreceptors on GABA interneuron) • Antagonist 5-HT 7 Postsynaptic receptors (Dorsal horn of the spinal cord) 20% occupancy	• Inhibition of GABAergic neurons increases 5-HT release from the raphe nucleus. • Increase in ACh and NA levels in the medial PFC • Increase in analgesia	• Pro-cognitive effect • Reduced REM Sleep - Sleep improvement • Pain control
Increase in BDNF	• Neurogenesis and neuroplasticity	• Antidepressant activity • Pro-cognitive effect
Astrocyte and microglia modulation with reduction of pro-inflammatory cytokines	• Reduction of neuroinflammation	• Antidepressant activity

Added during revision.
Right Answers.
Not sure about the answer.

2) **Diazepam**: GABA-A and GABA B / **GABA-A agonist**

- **GABA-A is an inotropic receptor**
- **GABA-B is Metabotropic receptor**

Lithium S/E: **Worsening of psoriasis**.

Topiramate S/E: **Weight loss**/ pigmentation/

- Blurred vision
- burning, prickling, or tingling sensations
- clumsiness or unsteadiness
- confusion
- continuous, uncontrolled back-and-forth or rolling eye movements
- dizziness
- double vision
- drowsiness
- eye redness or pain
- generalized slowing of mental and physical activity
- increased eye pressure
- memory problems
- menstrual changes or pain
- nervousness
- speech or language problems
- trouble in concentrating or paying attention
- unusual tiredness or weakness

Carbamazepine S/E: hair loss (rare)/**Agranulocytosis**

- dizziness
- constipation
- dry mouth

Frequency	Side-effect(s)
Very common	**leucopenia** ataxia, dizziness, somnolence vomiting, nausea urticaria fatigue
Common	**thrombocytopenia**, **eosinophilia** Oedema, fluid retention, weight increase, hyponatraemia and blood osmolarity decreased due to an antidiuretic hormone (ADH)-like effect, leading in rare cases to water intoxication accompanied by lethargy, vomiting, headache, confusional state, neurological disorders diplopia and accommodation disorders (e.g. blurred vision), headache dry mouth

Valproate: % of defect 4-7%,

Congenital anomalies (8-12%)
Developmental delays (31-40%)

3) **Genetics: EMI**
- Autism
- familial Alzheimer's
- GGSS
- Wilson
- Hunter

Multifactorial

Autism

Alzheimer's is also multifactorial

X recessive

Hunter's

Autosomal dominant conditions
- Huntington's
- Neurofibromatosis
- Marfan's
- Achondroplasia

Autosomal recessive conditions
- Homocystinuria
- Phenylketonuria
- Wilson's disease

X-linked dominant conditions
- Vitamin D resistant **R**ickets
- **R**ett syndrome

Note: Fragile X is X-linked but cannot be classified simply as dominant or recessive. Technically it is dominant because if you have a mutated FMR1 gene then you could potentially present with features of fragile X syndrome. Having said that as **it is a trinucleotide repeat disorder many people with the mutation will not have features if there are not enough repeats.**

X-linked recessive conditions
- Duchenne muscular dystrophy
- Red green colour blindness
- Hemophilia A and B
- Hunter's syndrome

Multifactorial Inheritance

Multifactorial conditions result from interaction between genes from both parents and the environment. **E.g. Schizophrenia, Autism, Alzheimer's**

nheritance pattern	Examples
Autosomal Dominant	Neurofibromatosis type 1 and 2, tuberous sclerosis, achondroplasia, Huntington disease, Noonan's syndrome
Autosomal Recessive	Phenylketonuria, homocystinuria, Hurler's syndrome, galactosaemia, Tay-Sach's disease, **Friedrich's ataxia**, Wilson's disease, cystic fibrosis
X-Linked Dominant	Vitamin D resistant rickets, Rett syndrome
X-Linked Recessive	Cerebellar ataxia, Hunter's syndrome, Lesch-Nyhan
Mitochondrial	Leber's hereditary optic neuropathy, Kearns-Sayre syndrome

4) **Disorders: EMI**

Hypochondriasis (man convinced he had cancer and refused to go to work)

- Abnormal preoccupation with one's health and a fear of having or getting a disease despite physical evidence to the contrary.
- Under **OCD and related disorder in ICD11**
- **N.B** Fear of Cancer is now under **symptoms, Signs and clinical findings not elsewhere** specified in ICD11 and not under hypochondriasis.

Somatisation (female with 5 referrals, thinks doc r missing something) - **bodily distress disorder in ICD11**

5) Stages

- Care giver - Bowlby
- Erickson
- Eisnbeg
- Kolberg

Eisenberg defined five levels in the prosocial moral reasoning that develops throughout childhood and adolescence:

1. Hedonistic reasoning,
2. Oriented to approval,
3. Oriented to the needs of others,
4. Stereotyped,
5. Internalized, which includes the reasoning based on empathy.

6) Lyonisation Is found in: Down's/ **Klinefelter**

Lyonization (also called X-inactivation) refers to **the normal phenomenon in which one of the two X chromosomes in every cell of a female individual is inactivated during embryonic development.**

7) Stimulus sensation long after removal of contact

Stimulus sensation long after removal of contact trace conditioning (anything about learning)/phantom limb (if it is about sensation) sensation after cessation of stimuli

Palinopsia: *Palin* for "again" and *opsia* for "seeing". It is a visual disturbance that causes **images to persist even after their corresponding stimulus has left.** It is seen in **LSD** use, migraine, occipital epilepsy, head trauma. It is similar to after image, but color inversion (usually shadows or distorted colors noted in afterimages) is conspicuously absent.

8) Disconnect between gesture and spoken word: **Incongruence**

Example of Paramimia: Imagine a person with paramimia who is trying to express happiness. They might be talking about something joyful but have a neutral facial expression, no smile, making it difficult for others to perceive their emotional state.

Suppose someone with **Parathymia** is discussing a tragic event like the death of a loved one. Instead of using a sad or mournful tone, they might speak with an inappropriate, cheerful tone or with a complete lack of emotional inflection.

Mimia is about gestures and expressions with face or hand, Thyme is the mood / emotion

10) Situation and psychological interpretation: **Apophony**/Anastrophe /trema/ abnormal salience in stages of delusion formation??

Conrad proposed five stages involved in the **formation of delusions:**

- **Trema**: **Delusional mood** representing a total change in the perception of the world
- **Apophony**: A search for, and the finding of **new meaning for psychological events**
- **Anastrophy**: **Heightening** of the psychosis
- **Consolidation**: Forming of a new world or psychological set based on new meaning
- **Residuum**: Eventual autistic state

Added during revision.
Right Answers.
Not sure about the answer.



10) Monitoring for methylphenidate: **LFTs/ BP**

Blood pressure should be recorded at every dose adjustment and then at least every six months; pulse should also be recorded. **Height, weight and appetite** should be recorded at least every six months.

Monitoring of Agomelatine- **LFT**

Monitor liver function at 3, 6, 12, and 24 weeks after starting treatment and regularly thereafter when clinically indicated

11) Serotonin precursor- L tryptophan

Tryptophan → 5Hydroxytryptamine → Serotonin

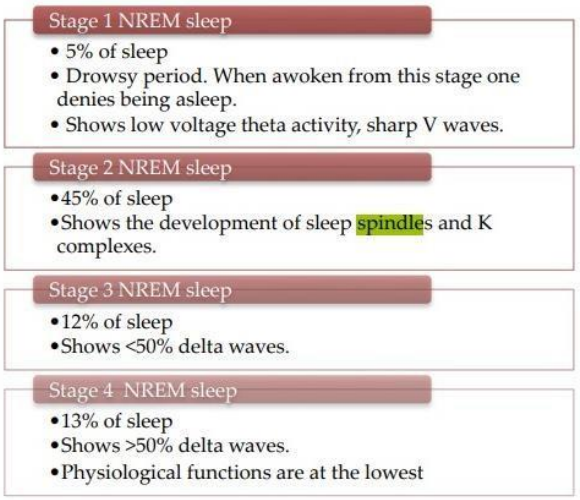
12) Group therapy instead of individual – Justice/ Utilitarianism

13) Drug taken off market- Non-maleficence

14) Which stage of sleep has more than 50% Delta – Stage 1/2/3/4/REM
Stage 4 NREM

Sleep stage	Type of sleep	Approx % of time spent in stage	EEG findings	Normal length	Comment
I	NREM	5%	Theta waves (4-7 Hz)	1-7 minutes	The dozing off stage. Characterized by hypnic jerks: spontaneous myoclonic contractions associated with a sensation of twitching or falling.
II	NREM	45%	Theta waves, K complexes and sleep spindles (short bursts of 12-14 Hz activity) **Sigma waves 12-14 hz**	10-25 minutes	Body enters a more subdued state including a drop in temperature, relaxed muscles, and slowed breathing and heart rate. At the same time, brain waves show a new pattern and eye movement stops. **Sleep spindles and K complexes are thought to play an important role in memory consolidation.**
III	NREM	15%	Delta waves (0-4 Hz) **Less than 50% Delta**	20-40 minutes	Deepest stage of sleep (high waking threshold). The length of stage 3 decreases over the course of the night. This is the stage when the body repairs and regrows tissues, builds bone and muscle and strengthens the immune system. This is also the stage when sleepwalking, night terrors, and bedwetting occurs.
IV	REM	15%	10-60 minutes	Mixed, predominately beta	High dream activity. The percentage of REM sleep decreases with age.

Characteristics of REM sleep
- Autonomic instability (variability in heart rate, respiratory rate, and BP) - Loss of muscle tone - Dreaming - Rapid eye movements - Penile erection - Deafness



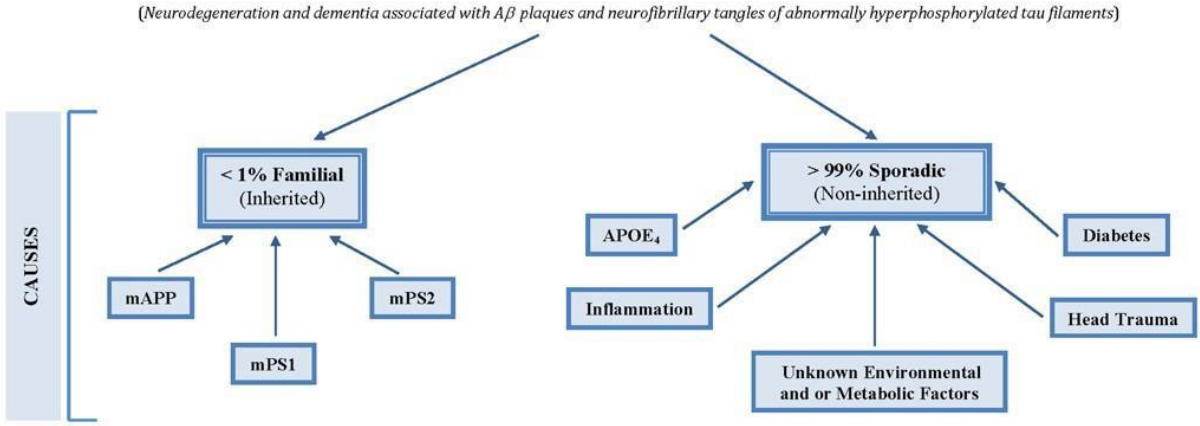
N.B : Consolidation of emotional memories occur in REM Sleep

15) GHB-B- Dissociative, **half-life 30-60min**

GABA B partial agonist
Gamma hydroxybutyrate is a designer drug. It's a Depressant, used by club goers.

Added during revision.
Right Answers.
Not sure about the answer.

16) Which one is multifactorial inheritance **Alzheimer's**/ADHD



17) Hydroxylated in process of dopamine: Tyrosine

- **Tyrosine → L-Dopa** through tyrosine hydroxylase.
- **L-Dopa → dopamine** through DOPA decarboxylase.
- **Dopamine → noradrenaline** through Dopamine Beta Hydroxylase

18) EEG normal - atypical absent seizures/Benign roland /**frontal**/

In autosomal dominant frontal lobe epilepsy, interictal EEG is often normal, and ictal EEG unhelpful or non-localising.

19) Tremors Frequency:

Type of tremor	Frequency
Intention	2-3Hz
Parkinsonian	5Hz
Essential	7Hz
Physiological	10Hz
Psychogenic	variable

Fine tremor in lithium. 10 Hz
Coarse tremors in lithium toxicity is 2-3 Hz

Tremor	Comment
Essential tremor	Otherwise known as benign essential tremor, this is the most common type of tremor. It is not associated with any underlying pathology. It usually begins in the 40's, affects mainly the hands, and is slowly progressive. It tends to worsen with heightened emotion. It usually presents with unilateral upper limb involvement then progresses to both limbs.
Parkinsonian tremor	This tremor is associated with Parkinson's disease. It is classically described as 'pill rolling' due to the characteristic appearance of the fingers.

Tremor	Comment
Cerebellar tremor	Otherwise known as **an intention tremor**. This is a slow, **coarse tremor** which gets worse with purposeful movement. This is seen in lithium toxicity (note that the tremor seen as **a side effect of long term lithium is fine and classed as physiological).**
Psychogenic tremor	Also known as a hysterical tremor. This type of tremor tends to appear and disappear suddenly and is hard to characterise due to its changeable nature. It tends to improve with distraction.
Physiologic tremor	This is a very-low-amplitude fine tremor that is barely visible to the naked eye. It is present in every normal person while maintaining a posture or movement. It becomes enhanced and visible in many conditions such as anxiety, hyperthyroidism, alcohol withdrawal, and as drug induced side effects.

The tremor in Parkinson's disease occurs at rest and is characterized by a frequency of 4 to 6 Hz and a medium amplitude. It is classically referred to as a "pill rolling" tremor of the hands but can also affect the head, trunk, jaw and lips.

20) Lithium 2nd messenger: **ITP**/Ade cycles

Inositol Triphosphate and Diacylglycerol

21) Caffeine blocked by adenosine agonist

- Caffeine is a central nervous system stimulant that belongs to a class of compounds called **xanthines**. Its primary mechanism of action is as an **antagonist of adenosine receptors**. Adenosine is a neurotransmitter that promotes sleep and relaxation by acting on its receptors, **mainly the A1 and A2A subtypes**. Caffeine works primarily by blocking these receptors, preventing adenosine from binding to them. This antagonistic action results in increased alertness and wakefulness. [Psych-Mentor]
- Toxic dose is **250 mg**, Lethal blood levels are typically above 80 to 100 mg/L, although one case occurred at 15 mg/L.
- Caffeine withdrawal is not recognized in DSM but is a likely diagnosis in a student who has just appeared for an exam (case 1). DSM research criteria lists headache, marked fatigue or drowsiness, marked anxiety or depression, nausea or vomiting as the prominent features that can be seen even at low doses (from 100mg/day), beginning within 12 hours, peaking at 24-48 hours and lasting up to 1 week.

22) Propranolol: beta 1/2/3: Beta 1 & 2 adrenergic receptor blocker Competitive

23) X lined recessive disorder? (Or autosomal) Hunter/Wilson

24) Confabulation occurs in: autobiographical memory/clouded consciousness/Korsakoff

- Confabulation is commonly seen in **Wernicke-Korsakoff syndrome, Alzheimer's disease (early stage), Traumatic brain injury, fetal alcohol spectrum disorder** (WebMD)

- Confabulation is the falsification of memory occurring in clear consciousness in association with organic pathology. It manifests itself as the filling-in of gaps in memory by imagined or untrue experiences that have no basis in fact. Some of the statements may be contradictory yet no attempt is made to correct them. The confabulation diminishes as the impairment worsens. Two broad patterns emerge (Bonhoeffer, 1901): the embarrassed type, in which the patient tries to fill in gaps in memory as a result of an awareness of a deficit, and the fantastic type, in which the lacunae are filled in by details exceeding the need of the memory impairment such as descriptions of wild adventures. Overall, the embarrassed type is much more common, and it may represent real memories displaced in time. Some schizophrenics confabulate
  and provide detailed descriptions of fantastic events that have never happened. **(fish)**

- N.B: After many discussion regarding this question we believe the most appropriate answer is Korsakoff (sometimes says occurs only), then autobiographical memory is affected should be appropriate and then there is subjective awareness

25) What about question on optogenetics.

Options
A. Single gene is introduced with single beam of light (blue light for positive and yellow for negative).
B. Single gene introduce with multiple light beams.
C. Multiple genes introduce with single light beam.
D. Multiple genes introduce with multiple light beams

[SPMM] Which of the following neuroscientific approach can be used to activate or inhibit specific neuronal projections in live animals using light exposure?
Select one:

- Photostimulation
- Flurostimulation
- **Optogenetics**
- Deep brain stimulation
- DREADD

Explanation: Genetically encoded light sensitive protein channels are used in optogenetics. Depending on the specific pigment protein that is engineered genetically to be expressed in an organism (rodents, primates), depolarization or hyperpolarization of specific neuronal populations can be achieved using specific frequency of light. Light is usually delivered very close to the target neurons due to the scattering effect of brain tissue, and this is achieved usually by implanting a lightweight fiber optic cab target neurons due to the scattering effect of brain tissue, and this is achieved usually by implanting a lightweight fiber optic cable guided by a surgical cannula.

Added during revision.
Right Answers.
Not sure about the answer.

[SPMM] When employing optogenetics to study neuronal functions, which of the following light will activate the pigment rhodopsin?

Select one:

- **Blue light**
- Red light
- Green light
- White light
- Yellow light

Channelrhodopsin-2 (ChR2) is a cation channel that opens in response to **blue** light. **Halorhodopsin**, from the halobacterium Natronomonas pharaonis, is a chloride channel that **actively moves chloride ions into the cell from extracellular zone when yellow light is presented. Archaerhodopsin,** derived from Halorubrum sodomense, **is a proton pump that actively removes hydrogen ions from the cell when a yellow light is presented.** There are two major transgenes used for optogenetic inhibition. Both halorhodopsin and archaerhodopsin serve to hyperpolarize neurons, while channel rhodopsin promotes depolarization.

26) The function of MRS – proton flow/ Metabolites. (we believe both options are correct)

- **DTI (Diffusion Tensor Imaging)** – for **white matter detection**.
- **PET** – measurement of regional brain **metabolism** by tagging organic compounds in metabolic pathway i.e. gives direct info about neuronal metabolism.
- **MRS** (Magnetic resonance spectroscopy) – dynamic measurement of resonance hydrogen and phosphate ions – odd number of protons - **Neurotransmitters**
- **SPECT** – measurement of receptor binding and distribution using radiolabeled receptor
- **fMRI** - tasks-related changes in **blood oxygenation** level - **Blood Oxygen Level Dependent (BOLD) technique**
- **Magnetoencephalography** – measurement of static magnetic activity of brain
- **DAT** (dopamine active transporter) – differentiate between LBD and Alzheimer's, **DAT is done with SPECT**

27) Functional MRI measures- structure involved in task/ ratio of deoxygenated blood to oxygenated blood.

28) Fundamental attribution error -personal /other's / common in western societies

- Overestimated dispositional factor in others behaviour and it is **also common in western societies**
- **Attributing others' mistakes to their personal dispositions**

- **Fundamental attribution error** is the tendency for people to under-emphasize situational explanations for another individual's observed behaviour while over-emphasizing dispositional and personality-based explanations for their own behaviour. It tends to be more common in the West and less so in more collectivist societies. [SPMM]

Q. Which of the following correctly describes fundamental attribution error? [SPMM]

Select one:

- Refusing to accept one's own errors
- Denying the fundamental flaws behind one's own negative behaviour
- Attributing others' mistakes to the context in which the mistakes occur
- Attributing one's own mistakes to one's character and personality
- **Attributing others' mistakes to their personal dispositions**

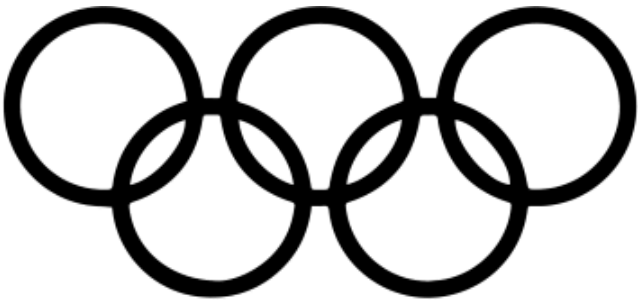
Explanation: Fundamental attribution error is a pervasive tendency to underestimate the importance of external situational pressures and to overestimate the importance of internal motives and disposition in interpreting the behaviour of others. It is more pronounced if the attributed behaviour is negative and undesirable.

29) The principle of Gestalt.

The Gestalt laws of perceptual organisation explain how we tend to organise parts into wholes. The laws are listed below

- SYMMETRY AND ORDER (also known as prägnanz) - This is the central law that states that 'every stimulus pattern is seen in such a way that the resulting structure is as simple as possible'.





- SIMILARITY - The mind groups similar elements into collective entities or totalities. These similarities may depend upon relationships of form, colour, size or brightness.



- PROXIMITY - Spatial or temporal grouping of elements may induce the mind to perceive a collective or totality.



- CONTINUITY - Points that are connected by straight or curving lines are seen in a way that follows the smoothest path.



- CLOSURE - Things are grouped together if they seem to complete a picture. When presented with a picture we tend to fill in the gaps to turn something into a meaningful image.



- COMMON FATE - Elements with the same moving direction are perceived as a collective or unit.



30) Girl asks mom if she had a baby brother -**Formal OP** (**Hypothetic deductive reasoning**)

33) Language development 13 yr and
5 yr old - 20,000 words, **understand passive speech**, understand basic grammar
5 & 13 year adult like speech

Age	Developmental abilities
0-6 months	Reacts to loud noises, vocalises pleasure and displeasure (laughs, cries)
6-12 months	Responds to own name, responds appropriately to angry and friendly tones, babbles* (says ba-ba-ba)
12-18 months	Uses single words with meaning (referred to as the 'holophrastic' stage), understands simple instructions, tries to imitate simple words
18-24 months	Has a vocabulary of up to 40-50 words (mainly nouns and pronunciation often unclear), knows simple body parts, understands simple verbs (eat, sleep), starts to combine words (e.g. More play) referred to as the telegraphic stage*
24-36 months	Able to name a number of objects, vocabulary of 150-300 words, able to combine words into short sentences, begins to use prepositions, answers simple questions
36-48 months	Has a vocabulary of 900-1000 words, begins to use plurals and past tense, handles three word sentences easily, groups objects such (e.g. Food, animals), strangers will understand most of what is said

34) Imaginative, perceiving behavior and attitude with mental conscious - **mentalization**, mindfulness

In psychology, **mentalization** is the ability to understand the mental state – of oneself or others – that underlies overt behaviour. Mentalization can be seen as a form of imaginative mental activity that lets us perceive and interpret human behaviour in terms

of intentional mental states (e.g., needs, desires, feelings, beliefs, goals, purposes, and reasons)

Mindfulness means maintaining a moment-by-moment awareness of our thoughts, feelings, bodily sensations, and surrounding environment, through a gentle, nurturing lens.

35) **EMI: Therapy**

- I accept being weak in front of drugs alcohol: **ALCOHAL ANNONYMOUS 12 steps.**
- Secure/protected environment without active use – **Cue therapy**
- Rogers's method while employing Becks cognitive behaviour model – **Motivational interview, cognitive therapy**, etc

After many discussions around that question we agreed on cognitive therapy as the answer when the question asks about Rogerian developed by INCORPORATING BECK'S.

If the question about Rogerian only then answer will be Motivational interviewing.

Cue therapy in addiction, also known as cue exposure therapy, is a type of psychological treatment used to help individuals with substance use disorders (addictions) reduce their **cravings** and **problematic behaviours** related to substance abuse. This therapy focuses on identifying and managing the specific cues or triggers that lead to substance cravings and use.

36) (Erickson??) theories were based on: Social learning/ Parents/ Education **PSYCHSOCIAL LEARNING.**

37) The big 5 factor suggest personality is **Independent of self**

It can change over time

38) Which drug has a non-linear pharmacokinetic? **Fluoxetine**

SSRI	Plasma elimination half-life		Linearity of pharmacokinetics
	Single dose	Multiple dose [active metabolite]	
Paroxetine	10h.	21h.	Nonlinear
Fluvoxamine	11h.	14h.	Nonlinear
Sertraline	26h.	26h. [36h.]	Linear
Citalopram	33h.	33h.	Linear
Fluoxetine	1.9 days	5.6 days [7-15 days]	Nonlinear

39) The half-life of Escitalopram: **30 hrs**/2 dis 0R 33

SSRI	Half life	Main side effects	Major interactions
Citalopram	33 hours	Nausea, vomiting, dyspepsia, abdo pain, diarrhoea, rash, sweating, agitation, anxiety, headache, insomnia, tremor, sexual dysfunction, hyponatremia, cutaneous bleeding disorders	Not a potent inhibitor of most cytochrome enzymes
Escitalopram	30 hours	As for citalopram	As for citalopram
Fluoxetine	4-6 days (the half-life increases with prolonged use from 1-3 days to 4-6 days due to fluoxetine and its active metabolite inhibiting its own metabolism)	As for citalopram but insomnia and agitation possibly more common Rash may occur more frequently May alter insulin requirements	Inhibits CYP2D6, CYP3A4. Increases levels of some antipsychotics, some benzodiazepines, carbamazepine, ciclosporin, phenytoin, tricyclics **Never use with MAOIs** Avoid selegiline and St John's Wort
Fluvoxamine	17- 22 hours **Shortest half-life**	As for citalopram but nausea more common	Inhibits CYP1A2/2C9/3A4. Increases levels of some benzodiazepines, carbamazepine, ciclosporin, phenytoin, some tricyclics, methadone, olanzapine, clozapine, propranolol, theophylline, warfarin
Paroxetine	22 hours **Discontinuation because of short half-life and anti-cholinergic activity**	As per citalopram but antimuscarinic effects and sedation more common	Potent inhibitor of CYP2D6/3A4. Increases levels of some antipsychotics and tricyclics Never use with MAOIs Avoid St John's Wort
Sertraline	26 hours	As for citalopram	Inhibits CYP2D6. Increases levels of some antipsychotics and tricyclics Avoid St John's Wort

40) Seizures in parietal lobe: **Somatosensory Aura**

- **Parietal lobe seizures** may begin with **a contralateral sensation, usually of the positive type (electrical sensation, tingling)**.
- **Occipital lobe seizures may** begin with **contralateral visual changes**, usually of the positive type, such as colored lines, spots, or shapes, or even a **loss of vision.**
- **Temporal aura: Autonomic** aura is the most common, causing **epigastric aura, salivation, sometimes vertigo etc**

- **Temporal-parietal-occipital** seizures may produce more formed auras.

41) Fluoxetine which should be taken with cautions: **Naproxen**

Interactions

- **NSAIDs**: NICE guidelines advise 'do not normally offer SSRIs', but if given co-prescribe a proton pump inhibitor
- **warfarin / heparin**: NICE guidelines recommend avoiding SSRIs and considering mirtazapine
- Aspirin: Increase risk of bleeding, Mirtazapine is an alternative option
- **triptans**: avoid SSRIs

42) Hypomanic patients diagnostic feature ICD or DSM:

Select one

A. functionally remains intact
B. precursor to recovery from mania ?

43) DSM structure is based on:

- **DESCRIPTIVE CATEGORICAL APPROACH, non- hierarchical**
- Operationalized approach (DSM-111)
- Standardized

ICD 10 is hierarchical and categorical, ICD 11 is life span/dimensional approach, DSM categorical non- hierarchical

[SPMM] **Operationalised approach**: In DSM-III operationalised diagnosis was first introduced. Operational criteria include the use of precise clinical description of disorders, together with predefined exclusion and inclusion criteria and details of the number and duration of symptoms required for diagnosis. It enables algorithm-based clinical diagnosis using intensity, duration of the symptoms and impairment tests. This more or less equates to using a checklist for diagnosis, but some rules are necessary while some are optional for a diagnosis.

Added during revision.
Right Answers.
Not sure about the answer.

Categorical approach	Dimensional approach
Traditionally doctors are accustomed to thinking in terms of categories – easy to understand. All existing knowledge base about the presentation, aetiology, epidemiology, course, prognosis, and treatment is based on these categories. Categories are easy to communicate with professionals Poor validity –vague categories such as 'Psychosis - not specified' are needed to include atypical cases.	More valid as most emotional and cognitive states exist as a continuum without clear cut-off point between ill and the well. Severity can be better indicated Need to entertain many comorbid diagnoses may be prevented. Research studies using dimensional scales as end points have much greater power to detect differences in groups than do studies focusing on changes in dichotomous categories Clinical utility is questionable, as dimensions cannot be directly mapped onto clinical decisions such as starting or stopping an intervention.

44) Schizophrenia in DSM 5: **All subtypes removed.**

45) classification: DISSOCIATIVE DISORDER
Ganser syndrome
Dissociative fugue?

46) Partial D2 agonist: **Aripiprazole Brexipiprazole, Cariprazine**

47) Reciprocal communication: Attitude, cooperation, and approach during the interview for the MSE assessment is called:- **Rapport**

48) Limitation to the facial and gesture expression- **blunting**/flattening

Affect [Psych-Mentor]

Affect can be described as (in order of degree):

- Within normal range
- Constricted (affect restricted in range and intensity)
- Blunted (same as constricted but a bit more so)
- Flat (virtually no signs of affective expression)

49) Wernicke - **Intact fluency**

50) The part of brain for empathy:

- DLFC
- Amygdala
- **Anterior Insula**

Added during revision.
Right Answers.
Not sure about the answer.

- Medial prefrontal cortex
- Anterior cingulate
- **Supramarginal gyrus(right)**

➢ **The consensus is to go for Insula if both are options**

51. Emotional valence for pleasure

- Entorhinal
- DLPFC
- Hippocampus

ANSWER: Hippocampus or **Sublenticular region of amygdala**, Insula, periaqueductal grey area

From another version of this recall: **Evaluation of pleasure valence occurs in which part of brain?**

A. Amygdala?
B. Cingulate cortex
C. Hippocampus
D. DLPC
E. Ventral pallidum

From Google : In the left hemisphere, the **hippocampus, amygdala, and inferior parietal lobule [Brodmann's area (BA) 40]** showed a greater response to negative arousing and negative non-arousing words than to neutral words. Thus, these regions were modulated by the presence of any emotional salience (valence or arousal).

52. Percentage of metabolic syndrome in CATIE study: **41- 51% CATIE**

The prevalence of metabolic syndrome in CATIE cohort was **40.9% and 42.7%,** respectively using the NCEP and AHA derived criteria. **In females it was 51.6% and 54.2%** using the NCEP and AHA criteria, **compared to 36.0%** (p = .0002) and 36.6% (p = .0003), respectively for males.

53. Most common epigenetic mechanism:

Select one: (only two options recalled)

- DNA Methylation
- Histone Methylation

Explanation: Epigenetics is the study of heritable phenotype changes that do not involve alterations in the DNA sequence. Epigenetics most often changes that affect gene activity and expression but the term can also used to describe any heritable phenotypic change. They include:

- DNA Acetylation
- DNA Methylation

Added during revision.
Right Answers.
Not sure about the answer.

- Histone Phosphorylation
- Histone Methylation

Non epigenetic mechanism includes:

- **Glycation**

[SPMM] Epigenetics is the study of heritable phenotype changes that do not involve alterations in the DNA sequence. Epigenetics most often involves changes that affect gene activity and expression, but the term can also be used to describe any heritable phenotypic change. Which of the following is not an epigenetic mechanism?

Select one:
- DNA acetylation
- DNA methylation
- Histone phosphorylation
- Histone methylation
- **Glycation**

[SPMM] The study of epigenetic mechanisms and their influence on the development of numerous heritable conditions is an expanding field of study, including within psychiatry. Which of the following epigenetic mechanisms is most associated with the development of schizophrenia?

Select one:

A. DNA methylation
B. Genomic imprinting
C. Micro RNA modification
D. Nucleosome positioning
E. RNA transcripts

Explanation: DNA methylation, particularly of GABA-ergic genes and brain derived neurotrophic factor, are suspected to be linked to schizophrenia.

54. The most common site of methylation:

- **Cytosine**
- Adenosine
- Uracil
- thymine etc.

Explanation: Methylation is a chemical reaction which involves a methyl group being added to molecules like DNA and proteins. The majority of DNA methylation occurs **on cytosines** that precedes a guanine nucleotide. **DNA methylation** is an epigenetic mechanism involving the transfer of a methyl group onto the C5 position of cytosine to form 5-methlcytosine.

55. Frontotemporal dementia: **Tau + Ubiquitin**

Added during revision.
Right Answers.
Not sure about the answer.

Explanation: Frontotemporal dementia is associated with intracellular accumulations of tau and ubiquitin.

- ✓ **Familial FTD: Tau -ve, Ubiquitin +ve**
- ✓ **Pick's Disease: Tau +ve, Ubiquitin +ve**
- ✓ **FTD in general: Most common is:**
  - **Ubiquitin positive , Amyloid negative** Ua tatu
  - **Tau positive , Amyloid negative THEN**
  - **Tau positive, Ubiquitin positive**

Other Tauopathies include: **APCD^2 all squared**

- Alzheimer's disease
- Argyrophilic granular dx
- Picks dx
- Progressive supranuclear palsy
- CJD
- Corticobasal degeneration
- Down syndrome
- Dementia pugilistica

Which of the following protein inclusion profiles is most common in FTD? [SPMM]

Select one:

- Tau positive, amyloid positive
- Synuclein positive, tau positive
- Ubiquitin negative, tau negative
- Ubiquitin positive, tau positive
- **Tau positive, amyloid negative**

EXPLANATION: The classification of frontotemporal dementias is based on the type of protein inclusions observed in neurons as well as the mutational status, if known. The inclusions contain predominantly or exclusively one of three markers: tau; TDP-43, a DNA/RNA-binding protein; and FUS (fused in sarcoma). Atypical frontotemporal lobar degeneration with ubiquitin-positive inclusion is a relatively rare cause of FTD characterised by FUS-positive inclusions.

56. Heritability: Phenotype in Population

Explanation : not sure what the question is asking.

Condition	Heritability Estimate
Bipolar	80-95%
Autism	90%
Schizophrenia	80-85%
ADHD	80%
OCD	25-50%
Panic disorder	45%
Anorexia	55%
Alcohol dependence	50-60%
Major depression	35-50%

[Psych-Mentor] **Heritability** is a concept that is frequently misunderstood. It is often thought of as a measure of the extent to which genes cause a condition in an individual, this is incorrect. The main point to note is that **heritability tells us nothing about individuals. Heritability is the proportion of phenotypic (phenotype is the set of observable characteristics or traits of an organism) variance attributable to genetic variance**. This means that if a condition has a heritability of 0.80 then 80% of the variance of that condition seen in a population is attributable to genetic variation. The remaining 20% is attributable to environmental factors.

Condition	Heritability estimate (approx)
ADHD	70-80% (Faraone, 2019)
Autism	**80-90% (Sandin, 2017)**
Schizophrenia	80% (Pardinas, 2018)
Bipolar	60-80% (Smoller (2003)
Anorexia	35%
Alcohol dependence	40-60% (Gupta, 2020)
Major depression	30-40% (Sullivan, 2000)
OCD	40-60% (Van Grootheest (2005)

57. Which is increased in pregnancy: **Gastric PH/GFR**

Explanation:

- Pregnancy is associated with numerous physiological changes which can impact upon the effectiveness of medications.
- Pregnancy is associated with decrease in production of gastric acid which leads to increase in overall gastric PH.

- **Glomerular Filtration Rate** is increased in pregnancy
- Gastric emptying and intestinal motility decreases
- Plasma protein binding reduces because of reduction in albumin
- Increase in cardiac output and volume of distribution

Changes in pregnancy: [SPMM]

Pregnancy is associated with several **pharmacokinetic changes**:

- ✶ Delayed gastric emptying,
- ✶ Decreased GIT motility,
- ✶ Increased volume of distribution (5%),
- ✶ Decreased drug-binding capacity,
- ✶ Decreased albumin level
- ✶ Induced liver metabolic pathway,
- ✶ Increased GFR & renal clearance.
- ✶ Increase gastric PH and decrease gastric acidity

58. Autism is associated with: **SHANK3** gene,
ADHD is associated with: **DRD4**

ANSWER: The distribution of DRD4. mRNA in the brain suggests that it plays a role in cognitive and executive functions. These functions are implicated in pathophysiology of ADHD. **Other genes associated with ADHD include**: **DAT1, DRD4, DRD5, 5HTT, HTR1B, and SNAP25**

Mutations in the **SHANK3** gene (located on chromosome **22**) are associated with autistic spectrum disorder.

59. Which of the following is not a part of grief: **Agitation**

ANS: According to Elisabeth Kubler Ross, a person facing an impending death or awful outcome undergoes the following stages of grief.

DABDA-
Denial
Anger
Bargaining
Depression
Acceptance

Erich Lindemann's Classic work on stages of grief titled Symptomatology and Management of Acute Grief described 4 stages:

1. Initial shock lasts 10-14 days.
2. intense sadness associated with withdrawal from social contact.
3. Anger; the grieving person " protests" the unexpected death.

4. The grief is resolved and the person returns to normal within a year.

60. Which of the following is a precipitating factor: **bereavement**

Explanation: The scenario of a man who lost his father 2 weeks before onset of a psychotic illness. Other options were having had a traumatic birth, and having a brother with schizophrenia.

- **Predisposing** factors: those that put a person at risk of developing a problem; may include: genetics, previous life events, or temperament
- **Precipitating** factors: a specific event or trigger to the onset of the current problem
- **Perpetuating** factors: those that maintain the problem once it has become established
- **Protective** factors: strengths that reduce the severity of problems and promote healthy and adaptive functioning

The others may be contributing factors, but most are predisposing rather than precipitating. **A precipitating factor has a strong temporal link to the onset of the episode of mental illness (e.g. it occurs just before).**

61) Patient with normal tone, left with his arms folded**:** **Waxy flexibility**

Limbs can be moved by the examiner to occupy certain postures, which are then maintained, even if these are uncomfortable and bizarre.

62) Extending and withdrawing to shake hand- **Ambitendency**/ Catalepsy

Ambitendency: It is a series of tentative, opposing alternate movements that do not reach the intended goal.

63) Which of them degrade dopamine in presynaptic – **MAO**/ Acetylcholine

MAO-B presynaptic
COMT post synaptic

64) 2nd messenger of Lithium—**ITP** / adenyl cyclase

Inositol Triphosphate and Diacylglycerol

65) What does the pro-opiomelanocortin (POMC) breakdown to in the brain?

- Enkephalin
- Substance p
- **ACTH**

Pro-opiomelanocortin (POMC) is the pituitary precursor (Pro-hormone) of melanocyte stimulating hormone (α-**MSH**), **Adrenocorticotropin** hormone (ACTH), and β-**Endorphin.**

ACTH is a 39 amino acid peptide derived from a larger precursor protein, proopiomelanocortin (POMC). This precursor molecule is synthesized principally in the corticotropic cells of the **anterior pituitary and in the arcuate nucleus of the hypothalamus**

66) Which of the following is an amino acid: Enkephalin/ substance P

Enkephalin/ Substance P are Neuropeptides
Amino acids include: Glutamic acid, Glutamine, Glycine, Alanine, Aspartic Acid.

67) What determines perception / From another version on this recall the question was: People use to organize current knowledge, images and provide a framework/match for future understanding: **Schema**

According to Piaget, Schema is a basic building block of intelligent behaviour.
It consists of organized past experience to understand future experience.
We develop perceptual schemas in order to organize impressions of people based on their appearance, social roles, interaction, or other traits; these schemas **then influence how we perceive other things in the world.**



68) The fibers that carry efferent from the cortex- **Pyramidal cells**

Which cells for myelination in the CNS- **Oligodendrocytes**

69) Adolescent brain for specialization – **synaptic pruning and myelination**

- **Synaptogenesis** occur very rapidly from the 2nd trimester through the 10 years of life. Peak- within **1st two postnatal years**. This synaptic pruning continues through the early teen years

- **Myelination** begins prenatally at around 4th gestational month. It is largely completed by 2 years. It continues till 3rd decade of life.

70) Which hormone released by post pituitary? **Oxytocin**

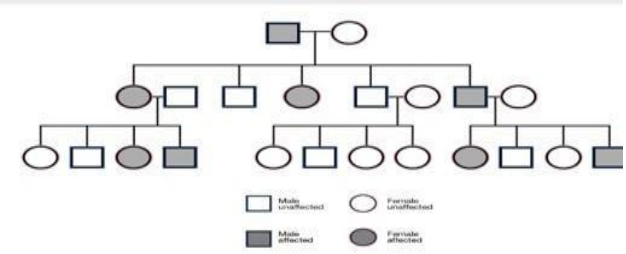
- **Posterior pituitary: Oxytocin and Vasopressin** (ADH - arginine vasopressin AVP – argipressin)
- **Anterior pituitary – GH, ACTH, TSH, Prolactin, LH, FSH**
- **Hypothalamus – Somatostatin, Prolactin Inhibiting Factor, Releasing factors.**

71) EMI: This enzyme is not present at birth.

A. Acetylcholine acetrate
B. Adenylate cyclase
C. Aldehyde dehydrogenase
D. Beta Hydroxylase
E. COMT
F. GABA
G. MOA
H. Alcohol dehydrogenase

Discussed thoroughly in the session and on the group. We believe all enzymes mentioned are present at birth however, the least enzyme present could be **Alcohol Dehydrogenase**

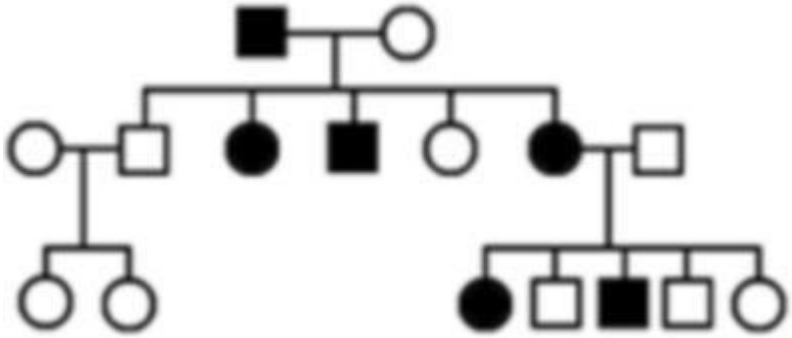
72) Genogram – X-linked recessive/ autosomal recessive



Autosomal dominant: 50% affected in each generation -

Autosomal dominant [Psych-Mentor]
Dominant conditions are expressed in individuals who have just one copy of the mutant allele. The pedigree below illustrates the transmission of an autosomal dominant trait. Affected males and females have an equal probability of passing on the trait to offspring. Affected individual's have one normal copy of the gene and one mutant copy of the gene, thus each offspring has a 50% chance on inheriting the mutant allele. As shown in this pedigree, approximately half of the children of affected parents inherit the condition and half do not.

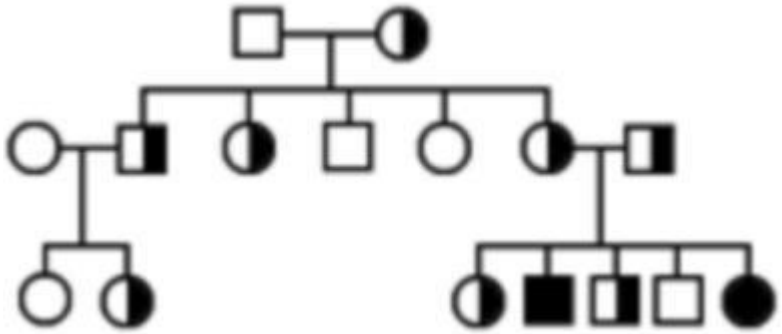
Added during revision.
Right Answers.
Not sure about the answer.



Note: Blank square = unaffected male / black square = affected male / blank circle = unaffected female / black circle = affected female / half black square or circle = heterozygous.

Autosomal Recessive

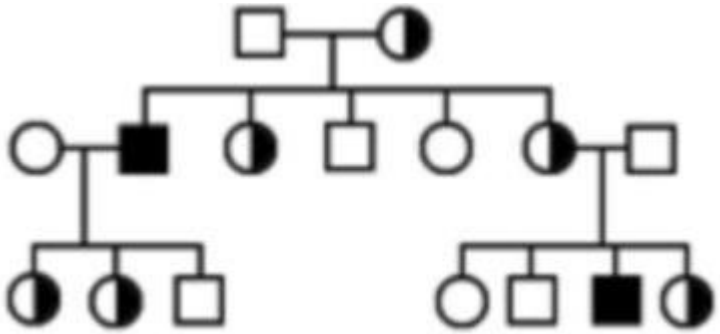
Recessive conditions are clinically manifest only when an individual has two copies of the mutant allele. When just one copy of the mutant allele is present, an individual is a carrier of the mutation, but does not develop the condition. Females and males are affected equally by traits transmitted by autosomal recessive inheritance. When two carriers mate, each child has a 25% chance of being homozygous wild-type (unaffected); a 25% chance of being homozygous mutant (affected); or a 50% chance of being heterozygous (unaffected carrier).



X-linked Recessive

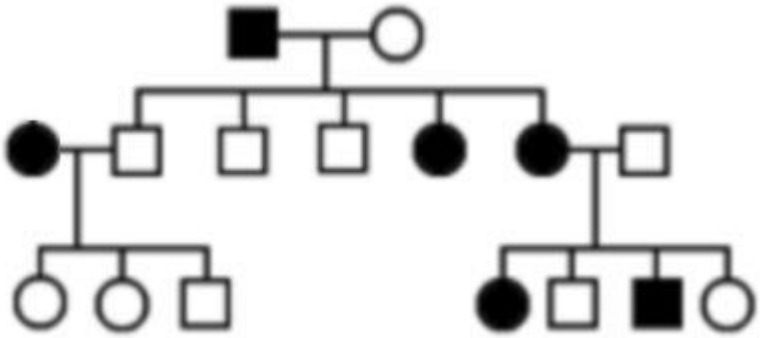
X-linked recessive traits are not clinically manifest when there is a normal copy of the gene. All X-linked recessive traits are fully evident in males because they only have one copy of the X chromosome, thus do not have a normal copy of the gene to compensate for the mutant copy. For that same reason, women are rarely affected by X-linked recessive diseases, however they are affected when they have two copies of the mutant allele. Because the gene is on the X chromosome there is no father to son transmission, but there is father to daughter and mother to daughter and son transmission. If a man is affected with an X-linked recessive condition, all his daughters will inherit one copy of the mutant allele from him.

Added during revision.
Right Answers.
Not sure about the answer.



X-linked Dominant

Because the gene is located on the X chromosome, there is no transmission from father to son, but there can be transmission from father to daughter (all daughters of an affected male will be affected since the father has only one X chromosome to transmit). Children of an affected woman have a 50% chance of inheriting the X chromosome with the mutant allele. X-linked dominant disorders are clinically manifest when only one copy of the mutant allele is present.



- ✓ **Autosomal dominant conditions = vertical transmission**
- ✓ **Autosomal Recessive = Horizontal transmission**
- ✓ **X-linked recessive conditions = knights move pattern**

PRO TIP for the Exam:
→ Look at the squares and circles:

- if you have half-filled ones in the first generation → think **Recessive** → go to the off springs to decide if its **x-linked or autosomal.**
- If you see full circles/ squares in first generation → think **dominant**. → go to off springs to decide if **autosomal or x-linked.**
- **X-linked if there is no male to male transmission**

73) Reciprocal inhibition is type of**: Systemic desensitization**/classic conditioning

Added during revision.
Right Answers.
Not sure about the answer.

74) Man who immigrated and lost his culture- **assimilation**/integration

75) Which of the rating scale used for screening? **EDRS**/ MADRAS/HAMD

76) Which is not a prodrug? Sertraline, **Desipramine**

A prodrug is a medication or compound that, after administration, is metabolised into a pharmacologically active drug. **Desipramine** is the metabolite of **imipramine**. All the others listed below are pro-drugs: [SPMM]

- Lisdexamphetamine
- Fosphenytoin
- Oxcarbazepine
- Prednisone

77) Which drug is safe to prescribe with Lithium- **Furosemide**

78) Last to mature in brain: **Anterior frontal cortex,** Mature at 25 years.

79) Southern blot for – **sq of DNA**

SNoW DRoP

Southern Blot – **DNA**

Northern Blot – **RNA**

Western Blot – **Protein / HIV**

PCR - **abnormal trinucleotide expansions**

Western blot test is **typically used to confirm a positive HIV** diagnosis. During the test, a small sample of blood is taken and it is used to detect HIV antibodies, not the HIV virus itself.

ELISA tests have greater than 98% sensitivity and specificity for HIV. Western blot analysis detects antibodies to specific HIV antigens and is best used as a confirmatory test.

80) Learning at a critical time- **imprinting**

- phase-sensitive attachment
- Lorenz
- Rapid
- Unreinforced
- occurs during a clearly defined period (known as the critical period)
- Irreversible.

81) Language processed in the brain - **inferior temporal gyrus/ inferior frontal**

- **Broca's area (Frontal lobe, Inferior Frontal Gyrus)** = language production

Broca's area, located in the left frontal lobe, which is responsible for speech production and articulation, and the Wernicke's area, in the left temporal lobe, associated with language development and comprehension.

Damage → **Expressive** aphasia

- **Wernicke's area** (junction where **temporal** & **parietal** lobes meet) = language comprehension

Someone with **Wer**nicke's can still be quite **Wer**-dy (wordy)

Damage → **Receptive** aphasia

- **Arcuate fasciculus**

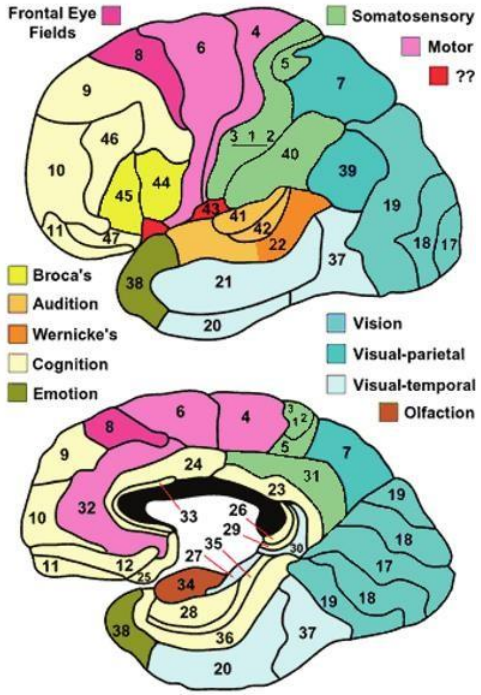
Damage → **Conduction** aphasia

Without an '**accurate**' fasciculus, you cannot **accurately** repeat what someone said

- **Anomic aphasia:**

Naming or word finding problems are the major feature of the syndrome. Grammar is unaffected and repetition is good. Auditory comprehension is relatively intact.

Anomic aphasia is localised with the least reliability of any of the aphasic syndromes. The lesion is often **temporal parietal area. The angular gyrus may also be affected.**



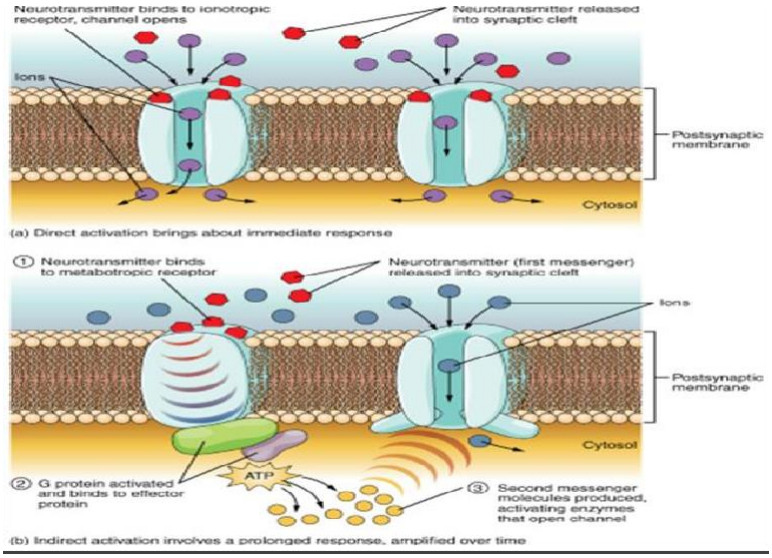
82) Alzheimer increase by APOE 4 in **heterozygote - 3 times (Risk)**
Homozygote / 10-30 times

Early onset Alzheimer's	Late onset Alzheimer's
Early onset (familial) Alzheimer's (EOFAD) is **very rare** and account for only 0.1% of all Alzheimer's cases. **(3) genes of importance:** 1. **amyloid precursor protein** (APP on Chr 21) - accounts for 10-15% of EOFAD 2. **Presenilin one** (PSEN-1 on Chr 14) - accounts for 30-70% of EOFAD	• One gene → **apolipoprotein E** (APOE) - breakdown of amyloid plaques - chromosome 19 - three alleles; APOE2, APOE3, and APOE4. - The E4 variant is the risk factor for late onset Alzheimer's. - People **homozygous for the E4 allele**

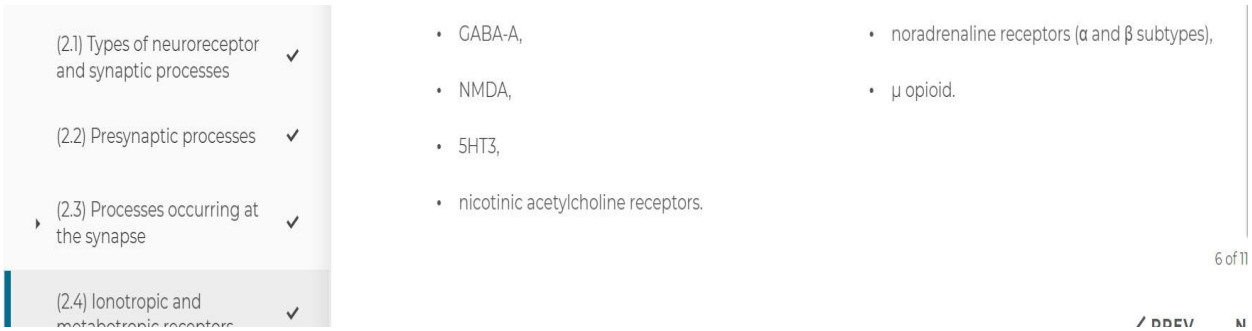
Added during revision.
Right Answers.
Not sure about the answer.

3. Presenilin two (PSEN-2 on Chr 1) - accounts for 5% of EOFAD	**have between a 10 and 30 times increased risk** for Alzheimer's compared to people not carrying any E4 alleles. **Heterozygotes for APOE4 have a 3 times increased risk**. In addition each E4 allele lowers the age of onset by 10 years. APOE2 - **protective** • APOE3 - **neutral** • APOE4 - **increases** risk

83) Which receptor does the neurons in locus coeruleus act on**:** **Noradrenergic**/ Ionotropic/ etc







84) In which is there a direct current applied using saline soaked pads? TMS / DBS/ **Transcranial direct current stimulation.(tDCS)**

85) In which of them are the electrodes implanted in the brain? **Deep B. stimulation**/ TMS

Deep brain stimulation

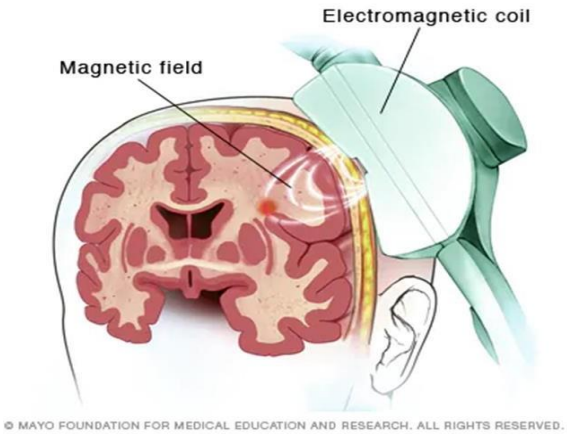
- implanting electrodes within certain areas of the brain which produce electrical impulses that regulate abnormal impulses
- amount of stimulation is controlled by a pacemaker-like device placed under the skin in the upper chest. A wire that travels under the skin connects this device to the electrodes in the brain.
- used to treat a number of conditions, such as:
  - Parkinson's disease
  - Essential tremor
  - Dystonia
  - Epilepsy
  - Obsessive-compulsive disorder

- **DBS** is also being studied as a potential treatment for:
  - Tourette syndrome
  - Huntington's disease and chorea
  - Chronic pain
  - Cluster headache



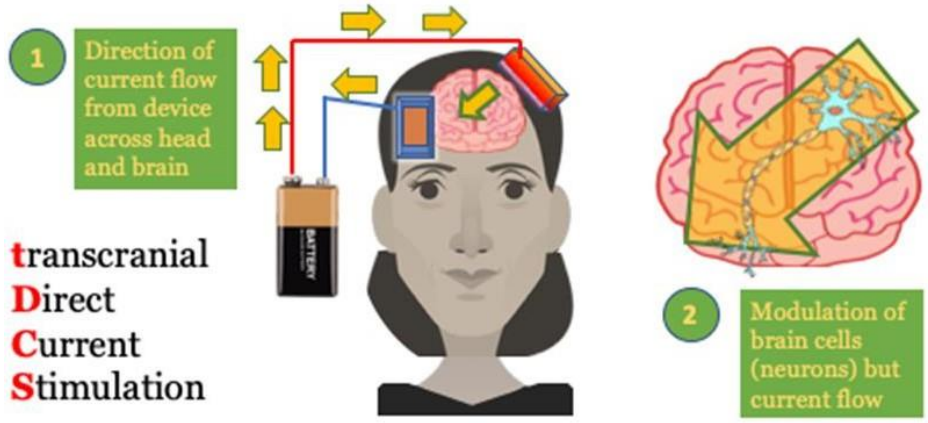
Transcranial magnetic stimulation (TMS) - non-invasive procedure that uses magnetic fields to stimulate nerve cells in the brain to improve symptoms of depression.

- used when other depression treatments haven't been effective.
- an electromagnetic coil is placed against the scalp near the forehead.
- the electromagnet painlessly delivers a magnetic pulse that stimulates nerve cells in the region of the brain involved in mood control and depression. It's thought to activate regions of the brain that have decreased activity in depression.



Transcranial direct current stimulation (tDCS)

- a portable, wearable brain stimulation technique that delivers a low electric current to the scalp
- explored in treatment of depression, schizophrenia, aphasia, addiction, epilepsy, chronic pain (migraine, fibromyalgia), attention, and motor rehabilitation.
- also used for non-medical wellness applications, for example accelerated learning, focus, relaxation, and meditation.



86) Panic state after undeveloped sense of identity in middle adulthood:

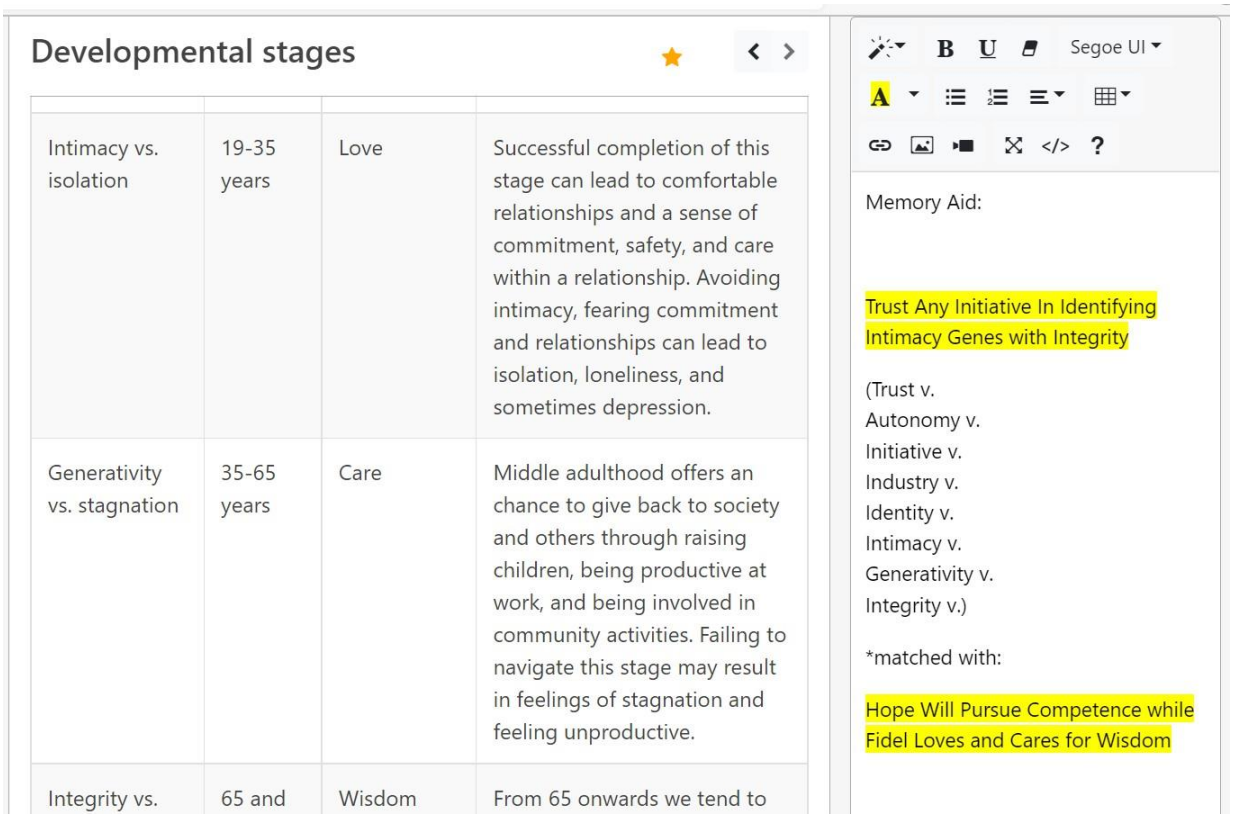
- **Generativity vs stagnation**
- identity vs role confusion

Added during revision.
Right Answers.
Not sure about the answer.

87) Which describes loss of identity formation:

- **Identity diffusion**
- foreclosure/
- Maratoriam

YouTube link: https://youtu.be/Nic8OWgCmos



Developmental stages

Intimacy vs. isolation	19-35 years	Love	Successful completion of this stage can lead to comfortable relationships and a sense of commitment, safety, and care within a relationship. Avoiding intimacy, fearing commitment and relationships can lead to isolation, loneliness, and sometimes depression.
Generativity vs. stagnation	35-65 years	Care	Middle adulthood offers an chance to give back to society and others through raising children, being productive at work, and being involved in community activities. Failing to navigate this stage may result in feelings of stagnation and feeling unproductive.
Integrity vs.	65 and	Wisdom	From 65 onwards we tend to

Segoe UI

Memory Aid:

Trust Any Initiative In Identifying Intimacy Genes with Integrity

(Trust v.
Autonomy v.
Initiative v.
Industry v.
Identity v.
Intimacy v.
Generativity v.
Integrity v.)

*matched with:

Hope Will Pursue Competence while Fidel Loves and Cares for Wisdom

Added during revision.
Right Answers.
Not sure about the answer.

Development (Erikson and Marcia)

		Individual has committed to identity: Yes	Individual has committed to identity: No
Individual has explored identity options	Yes	Identity Achievement	Moratorium
	No	Foreclosure	Identity Diffusion

Identity confusion/diffusion occurs when adolescents neither explore nor commit to any identities.

Foreclosure occurs when an individual commits to an identity without exploring options.

A moratorium is a state in which adolescents are actively exploring options but have not yet made commitments.

Individuals who have explored different options, discovered their purpose, and have made identity commitments are in a state of identity achievement.

88) Delusional disorder diagnostic criteria- 2 weeks/ **no prominent hallucination**

Home > Codes > ICD-11 viewing Tue Nov 1, 2022

Delusional disorder

International Classification of Diseases for Mortality and Morbidity Statistics, 11th Revision, v2022-02

Delusional disorder is characterised by the development of a delusion or set of related delusions, typically persisting for at least 3 months and often much longer, in the absence of a Depressive, Manic, or Mixed mood episode. The delusions are variable in content across individuals, but typically stable within individuals, although they may evolve over time. Other characteristic symptoms of Schizophrenia (i.e. clear and persistent hallucinations, negative symptoms, disorganised thinking, or experiences of influence, passivity, or control) are not present, although various forms of perceptual disturbances (e.g. hallucinations, illusions, misidentifications of persons) thematically related to the delusion are still consistent with the diagnosis. Apart from actions and attitudes directly related to the delusion or delusional system, affect, speech, and behavior are typically unaffected. The symptoms are not a manifestation of another medical condition (e.g., a brain tumour) and are not due to the effect of a substance or medication on the central nervous system (e.g. corticosteroids), including withdrawal effects (e.g. alcohol withdrawal).

sections/codes in this section (6A24-6A24)

- Delusional disorder, currently symptomatic (6A24.0)
- Delusional disorder, in partial remission (6A24.1)
- Delusional disorder, in full remission (6A24.2)

Ganser Syndrome-Other Specified Dissociative disorder in ICD 11

Alcoholic hallucinons is alcohol induced psychotic disorder with hallucinations in ICD 11

89) Reduce Group think**:** **Open debate**

Groupthink:

- coined by **Irving Janis** in 1972.
- occurs within a group of people, in which the desire for harmony or conformity in the group results in an incorrect or deviant decision-making outcome.

Symptoms of groupthink include:

- Illusion of invulnerability
- Belief in inherent morality of the group
- Collective rationalisation
- Stereotypes of 'out-groups'
- Self-censorship
- Illusion of unanimity - assuming that people who remain silent agree with the majority view
- Direct pressure on dissenters
- Self-appointed mind guards - protecting the group from outside info that challenges the groups view of itself

Several interventions can help reduce groupthink:

- Leaders should allow each member to challenge ideas and present objections
- Members should talk about and solicit ideas with people outside the group
- Outside Experts should be invited to attend meetings
- Avoid expressing opinions about the preferred outcome
- Assign 'Devil's Advocate' at all meetings to challenge any and all ideas

90) Which drug for patient with hyponatremia**?** **Mirtazapine**/venlafaxine

Mirtazapine > Bupropion > Agomelatine and MOAI

Drug / group*	Risk of hyponatremia	Level of supporting evidence
SSRIs	High	Strong
SNRIs	High	Strong
Tricyclics	Moderate	Strong
MAOIs	Low	Weak
NaSSAs	Low	Strong
Bupropion	Low	Moderate
Agomelatine	Low	Weak

91) Evaluation of giving**:** **Social exchange theory**

Old age – theories of role change: [SPMM]

1. Social disengagement theory: Mutual withdrawal of society and the individual occurs; increased individuality and shrinking life space are inevitable, moves towards death.

2. Social (non) reengagement theory (aka activity loss theory): Ageist society reduces the social interaction that older adults can have; withdrawal is not mutual but forced.

3. Social exchange theory: Age robs people of the ability to engage in reciprocal roles; retirement is a special social contract wherein productivity is exchanged for increased leisure and decreased responsibilities.

4. **Socio-emotional selectivity theory**: Wise investment of social energy in old age is to limit social interaction to those who are most familiar.

YouTube link: https://youtu.be/dmuFribrsUk)

92) Duloxetine **: used for peripheral neuropathy / half life(12 hrs)**

93. Delay gratification in a child is associated with: **Higher academic achievement Marshmallow experiment**

94. Duloxetine is used in?

- **Peripheral neuropathy**
- Enuresis

Duloxetine is used for:

- **Major depressive disorder**: 60mg once daily,
- **Diabetic neuropathy** 60mg OD can go up to 120 mg – to be discontinued after 2 months if no response,
- **Moderate to severe stress urinary incontinence** 40 mg BD

95. Which one is reward pathway?

- Meso-cortical
- **Mesolimbic**

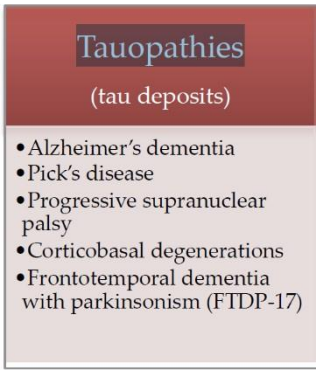
96. EMI: Familial FTD:

- Tau +ve, Ubiquitine –ve
- Tau +ve, Ubiquitine +ve
- Aplha cyneucline
- **Tau -ve, Ubiquitine +ve**

Which of the following protein inclusion profiles is most common in FTD? [SPMM]

Select one:

- Tau positive, amyloid positive
- Synuclein positive, tau positive
- Ubiquitin negative, tau negative
- Ubiquitin positive, tau positive
- **Tau positive, amyloid negative**



Added during revision.
Right Answers.
Not sure about the answer.

EXPLANATION: FTD: Tauopathies

- In familial cases mutations in the (**MAPT**) on **chromosome 17q21.**
- Many cases are **tau-negative** but **show ubiquitin-immunoreactive** neuronal cytoplasmic inclusions (mutations in progranulin (**PGRN**) gene on chromosome 17q21) **—(MND) type.**
- **Pick cells:** abnormal swollen oval-shaped neuronal cells with loss of Nissl's substance and peripherally displaced nucleus.
- **Pick's bodies:** are argentophilic, **tau and ubiquitin reactive filamentous inclusions.**

- **Conclusion**

✓ **Familial FTD: Tau -ve, Ubiquitin +ve**

✓ **Pick's Disease: Tau +ve, Ubiquitin +ve**

✓ **FTD in general: Most common is:**

- **Ubiquitin positive , Amyloid negative**
- **Tau positive , Amyloid negative THEN**
- **Tau positive, Ubiquitin positive**

97. Tyrosine hydroxylase converts

Select one:

- **L-tyrosine to L-Dopa**
- Dopamine to Adrenaline
- L-Dopa to dopamine

Explanation:

- **Tyrosine → L-Dopa** through tyrosine hydroxylase.
- **L-Dopa → dopamine** through DOPA decarboxylase.
- **Dopamine → noradrenaline** through Dopamine Beta Hydroxylase

98. Anti-social Personality is associated with?

- Increase arousal
- Callus gene
- **highest hereditability**

Added during revision.
Right Answers.
Not sure about the answer.

99. EMI: Parental figure is important in shaping the attachment behavior in children. : **Bowlby Attachment theory**

Stage	Period	Description
Preattachment	Birth to 6 weeks	Baby orientates towards mother, follows them with their eyes, and turns to mother's voice. Complete attachment not yet complete and baby is still comfortable with unfamiliar people
Attachment in the making	6 weeks to 6-8 months	Infant becomes more attached to one or more persons in the environment. Begins to show different reactions to familiar people versus strangers. Separation anxiety not yet present
Clear cut attachment	6-8 months to 18-24 months	Infant cries and shows distress when separated from the caregiver (separation anxiety). This phase can occur as early as 3 months. On being returned to the caregiver the crying stops. Child shows preference for selective caregivers and will seek to be near them
Formation of reciprocal attachment	18 months to 2 years and on	Child begins to understand that their caregiver may leave and return so separation anxiety lessens. Child may be less physically clingy but may begin to find more intelligent ways to influence their caregiver to get their needs met (bribing and sulking for instance)

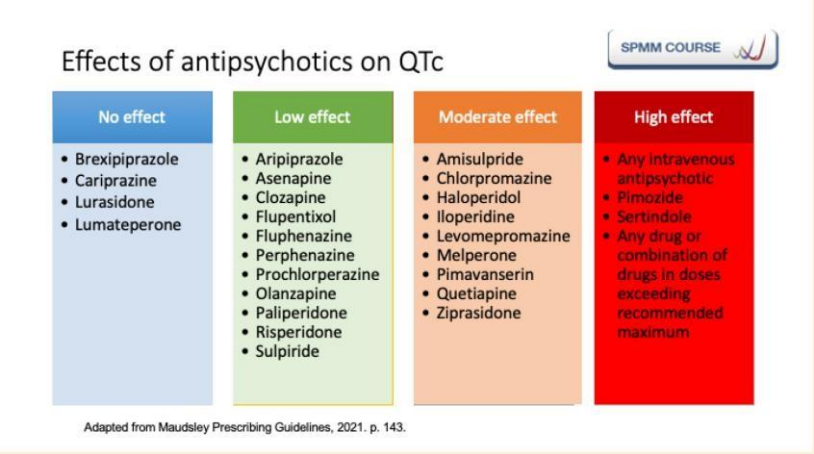
100. EMI: This theory involves the stages of development. A person moves to next stage after first is complete: **Erickson's theory**

101. EMI: main theme here is obedience, punishment, praise, and universal ethics: **Kohlberg's theory**

102. QTC prolongation occurs with?

- **Haloperidol**
- Quetiapine
- Risperidone

1st generation antipsychotics has more effect on QTc and this is why we chose Haloperidol over Quetiapine.



103. Trail making test is for?

- **Motor perceptual speed,**
- Selective Attention
- Verbal memory
- Vocabulary

Trail making test part A:

- Visual attention
- Tracking
- Processing Speed

Trail making test part B:

- Set shifting
- Attention
- Visual Sketch
- Cognitive flexibility
- Switch attention

104. EMI: Neurogline: **Schizophrenia**

Dysbnidin (chromosome 6) NRG(chromosome 8) schiz
Neur-EIGHT-gulin
SIX-bindin
Disc-ONE

105. Which one is not part of grief?

- Protest
- Depression
- Anger
- Bargain
- **Despair**

Erich Lindemann: Stages of grief **(SSAR)**: **1. S**hock that lasts 10-14 days **2.** Intense **S**adness (withdraw from social contact), **3. A**nger, as the grieving person seems to 'protest'; the unexpected death **4.** and finally, within a year or so, the grief is **R**esolved, and the person returns to normal.

Kubler-Ross in 1969. **(DABDA)**

Stage
Stage I = Denial-dissociation-isolation
Stage II = Anger
Stage III = Bargaining
Stage IV = Depression
Stage V = Acceptance

Phase	Features by **John Bowlby**	Typical time course
I	Shock-numbness (aka disbelief)	Few days
II	Yearning-searching (aka preoccupation) – referred to as **protest**	Few weeks
III	Disorganization-despair (includes acceptance of loss)	Several months
IV	Reorganization (resolution)	1-2 years

Grief and depression

A high proportion of people will meet the criteria for major depression in the first year following bereavement. These episodes usually resolve within 6 months.

Features useful for distinguishing normal grief from major depression include:-

- Generalised guilt (rather than guilt specifically related to actions taken around the time of death.
- Thoughts of death (except in relation to the deceased)
- Feeling worthless
- Psychomotor retardation
- Prolonged functional impairment
- Hallucinations (except in relation to the deceased)
- **Normal grief**

work done by John Bowlby.

Which of the following is a normal stage of grief? [SPMM]

Select one:

A. Ambivalence
B. Anger
C. Bargaining
D. Mourning
E. **Protest**

Added during revision.
Right Answers.
Not sure about the answer.

Explanation :**The classic work on stages of grief came from Erich Lindemann**, who studied 101 bereaved people and published in 1944, an article titled "Symptomology and Management of Acute Grief". In this article he described a set pattern of reaction to a loss event (grief): After an unexpected death, there is an initial shock that lasts 10-14 days. After the initial shock comes a period of intense sadness, and the grieving person may withdraw from social contact. Next comes anger, as the grieving person seems to 'protest'; the unexpected death. Finally, within a year or so, the grief is resolved, and the person returns to normal.

106. Where is Broca's area located?

- **Inferior frontal lobe**
- Superior frontal lobe
- Temporal lobe
- Parietal Lobe
- Limbic area

107. Which kind of memory loss in schizophrenia?

- **Working memory**
- Procedural memory
- Delayed memory

Episodic memory is also affected

108. EMI: What are the MRI changes in patient with **Alzheimer**: **Medial temporal lobe atrophy**

➢ **Neuroimaging findings in Alzheimer's**

- Ventricular enlargement
- Loss of **medial temporal lobe volume** – especially hippocampus
- Decreased parieto-temporal fMRI activation and SPECT blood flow

109. Aripiprazole MOA: **Partial agonist on 5HT1A and D2 and antagonist on 5- HT2A**

110. What is true regarding schizophrenia DSM 5 classification?

- Subclasses retained
- **No categories**

107. Excessive startled response occurs?

A. Hyperekplexia
B. Paramimia
C. parathymia.

Added during revision.
Right Answers.
Not sure about the answer.

E. anastrophy,
F. paramimia,
G. parathymia
H. Anastrophy is the 2nd ~~stage~~ of delusion formation (after trema) Third stage
I. apophenia,

111. EMI: **Cannabinoid receptor: Anandamide receptor? See Q. 120**

112. EMI: During old age a person give up due to Socio- economic reasons:

- **Cost benefit analysis**
- Reengagement
- Re arrangement

The midlife transition occurs around age 40 to 45. The term '**Midlife Crisis'** was coined by Elliot Jacques who distinguished it as a critical phase in development as the transition between the forties and early sixties, and is often associated with coming to realise mortality, unrealized goals, menopause or children leaving home. **Downshifting refers to voluntary opting out of a pressurized career and giving up well-paid job for more fulfilling life (anti-urbanism). Empty nest distress** refers to the feeling of loneliness when children leave home. [SPMM]

113. Which of following is not seen in DNA epigenetics?

- Phosphorylation
- DNA Acetylation
- DNA methylation
- Histone methylation
- Histone Acetylation

Epigenetics is the study of heritable phenotype changes that do not involve alterations in the DNA sequence. Epigenetics most often changes that affect gene activity and expression but the term can also used to describe any heritable phenotypic change. They include:

- DNA Acetylation
- DNA Methylation
- Histone Phosphorylation
- Histone Methylation
- Histone Acetylation can also occur

Non epigenetic mechanism includes:

- **Glycation**

114. What is the diagnostic criteria for bipolar 1 in DSM 5: **One manic episode**

In DSM-5 and ICD-11 bipolar disorders are subdivided into bipolar I and bipolar II disorder

Bipolar I requires at least **one past or present manic** or mixed episode; depressive episodes may occur but are not obligatory

Bipolar II requires at **least one hypomanic and one depressive episode** Mixed states are cases where manic and depressive symptoms occur simultaneously. Rapid

DSM5 has no mania or hypomania as separate diagnostic entity.

[ICD11] Bipolar 1: A history of **at least one Manic or Mixed Episode.** Although a single Manic or Mixed Episode is sufficient for a diagnosis of Bipolar Type I Disorder, the typical course of the disorder is characterized by recurrent Depressive and Manic or Mixed Episodes. Although some episodes may be Hypomanic, there must be a history of at least one Manic or Mixed Episode.

115. Rogerian therapy develop on the basis of Beck's cognitive distortions is?

A. Analytics
B. Behavioural
C. **Cognitive therapy**
D. DBT
E. Cue base
F. Alcohol anonymous
G. Rational emotive
H. 12 step

Motivational interview is an individualistic therapy belongs to Rogerian.

116. A chronic alcoholic is telling stories with no link. What one occurs in confabulation?

A. Clouding of consciousness
B. **Autobiographical**

117. One question was social theory on cost benefit analysis.

A. Socio-economic
B. **Social exchange theory**
C. Social deprivation theory
D. Social re-engagement

Old age – theories of role change: [SPMM]

1. Social disengagement theory: Mutual withdrawal of society and the individual occurs; increased individuality and shrinking life space are inevitable, moves towards death.

2. Social (non) reengagement theory (aka activity loss theory): Ageist society reduces the

social interaction that older adults can have; withdrawal is not mutual but forced.
3. Social exchange theory: Age robs people of the ability to engage in reciprocal roles; retirement is a special social contract wherein productivity is exchanged for increased leisure and decreased responsibilities.

4. **Socio-emotional selectivity theory**: Wise investment of social energy in old age is to limit social interaction to those who are most familiar.

118. This process is in form of imaginative mental activity that lets us perceive and interpret human behavior in terms of intentional mental states.

Select one:

A. Mentalization
B. Personal construct
C. Mindfulness

119. EMI: This help for release of serotonin neurotransmitters from vesicles.

A. 7 membrane, G protein
B. Adenylate cyclase
C. ATP base ion channel
D. Ligand ion channel ?
E. Ligand ion sodium channel
F. Ligand ion potassium channel
G. Ligand ion chloride channel
H. Ionic sodium and calcium channel ?
I.Inositol channel

[SPMM] Neurotransmitters are substances which neurons use to communicate with one another and with their target tissues. They are synthetized in and released from nerve endings into the synaptic cleft. From there, they bind to receptor proteins in the cellular membrane of the target tissue. The target tissue gets excited, inhibited, or functionally modified in some other way.

Inside the terminal bouton of the presynaptic nerve, numerous vesicles that contain neurotransmitters are produced and stored. The release of neurotransmitter is triggered by an action potential, which leads to an influx of calcium through voltage dependent calcium selective ion channels. The calcium causes vesicles packed with neurotransmitter to fuse with the synaptic membrane and release the neurotransmitter into the synaptic cleft.

After crossing the synaptic cleft, neurotransmitters bind to their receptors on the postsynaptic membrane. Once the neurotransmitter binds to its receptor, the ligand-gated channels of the postsynaptic membrane either open or close. **These ligand-gated channels are ion channels, and their opening or closing alters the permeability of the postsynaptic membrane to calcium, sodium, potassium, and chloride ions. This leads to a stimulatory or inhibitory response.**

From google: Upon electrical stimulation, transmembrane Ca2+ entry through L-type channels activated Ca2+-induced Ca2+ release. A resulting fast Ca2+ transient evoked an early exocytosis of serotonin from sparse vesicles resting close to the plasma membrane.

120. EMI: Cannabinoid receptor:

A. 7 membrane, G protein
B. Adenylate cyclase
C. ATP base ion channel
D. Ligand ion channel
E. Ligand ion sodium channel
F. Ligand ion potassium channel
G. Ligand ion chloride channel
H. Ionic sodium and calcium channel
I.Inositol channel
L. Anandamide

Anandamide is a lipid mediator that acts as an endogenous ligand of CB1 receptors. CB1 and CB2 are the best-characterized cannabinoid receptors.

121. EMI: This is for mechanism of lithium.

A. 7 membrane, G protein
B. Adenylate cyclase
C. ATP base ion channel
D. Ligand ion channel
E. Ligand ion sodium channel
F. Ligand ion potassium channel
G. Ligand ion chloride channel
H. Ionic sodium and calcium channel
I. Inositol channel

Explanation: Lithium is thought to act via the second messenger system. It putatively enhances serotonin transmission by: [SPMM]

1. Increasing tryptophan uptake into neurons
2. Enhancing serotonin release
3. Downregulation of 5HT1A, 1B and 2 receptor subtypes is also noted on chronic administration.
4. Directly inhibiting glycogen synthase kinase-3 (GSK-3) and also
5. **Competing with magnesium directly** at several important regulatory enzymes such as **inositol-monophosphatase (IMPase), which catalyzes inositol second messenger system.**

122. EMI: What receptor directly reverses effect of caffeine?

A. 7 membrane, G protein
A. Adenosine
B. Adenylate cyclase

Added during revision.
Right Answers.
Not sure about the answer.

C. ATP base ion channel
D. Ligand ion channel
E. Ligand ion sodium channel
F. Ligand ion potassium channel
G. Ligand ion chloride channel
H. Ionic sodium and calcium channel
I. Inositol channel
J. GABA A facilitator
K. GABA B facilitator
L. NAMDA
M. AMPA

Caffeine is Adenosine receptor antagonist

123. Durkheim proposed that suicide is more common in?

Select one:

- **Socially deprive people**
- unmarried lonely couple
- Living in urban area living for the long time

124. Prospective birth studies are attractive because it minimises?

Select one:

A. Minimises attrition
B. Minimises Flexibility
C. Minimises Self-reported bias
D. Minimises Retrospective bias
E. Minimises Hindsight bias

ANSWER: We are between **Retrospective** bias or Attrition Bias.

From google: The term "**hindsight bias**" refers to the tendency people have to view events as more predictable than they really are. Before an event takes place, while you might be able to offer a guess as to the outcome, there is really no way to actually know what's going to happen.

Some interesting new terms:

Parathymia an abnormal, contrary, or inappropriate mood, as seen in schizophrenia

Alloesthesia is a condition in which a sensory stimulus, given on one side of the body, is perceived to be at the corresponding area on the opposite side.

In DSM-5 and ICD-11 bipolar disorders are subdivided into bipolar I and bipolar II disorder

● Bipolar I requires at least **one past or present manic** or mixed episode; depressive episodes may occur but are not obligatory

● Bipolar II requires at **least one hypomanic and one depressive episode** Mixed states are cases where manic and depressive symptoms occur simultaneously. Rapid

DSM5 has no mania or hypomania as separate diagnostic entity.

ICD 10 needs 2 episodes (mania/depression/hypomania/mixed) for diagnosing bipolar 1

Vortioxetine causes serotonin transporter (SERT) inhibition as well as having antagonist actions at 5HT3 and 5HT7 receptors, with agonist actions at 5HT1A receptors and weak partial agonist to antagonist actions at 5HT1B/D receptors.

Topiramate S/Es: weight loss, word finding difficulties, aggression agitation, nephrolithiasis, altered acid base balance, decrease bicarb and visual disturbances also.

Eisenberg defined five levels in the **prosocial moral reasoning** that develops throughout childhood and adolescence:

1. Hedonistic reasoning,

2. Oriented to approval,

3. oriented to the needs of others,

4. stereotyped,

5. Internalized, which includes the reasoning based on empathy.

DBS at the subthalamic nucleus and the internal globus pallidus is endorsed by many guidelines for treatment of Parkinson's disease. It may be as effective as medications, but without motor side effects in many patients.

• DBS of the internal capsule has shown positive effects for OCD. STN is another target of interest for OCD.

• DBS of ventromedial thalamus can also reduce abnormal movements in patients with Tourette's disorder.

• DBS of subcallosal cingulate (Cg25 area) reduces depression.

Good luck for you all

I read about the *social identity theory.*
A person has a personal identity
And then a social identity which comes by being a part of a group or groups.

It forms in three steps
Categorize
Like suppose you group yourself into a category like I say I'm a rock music lover

Then *compare* .
Here an in group (rock music lovers) and an out group (people that love other genres) arises.
So by comparing I say that ok rock is better because this and that and maybe at some times negatively talk about the other group too.

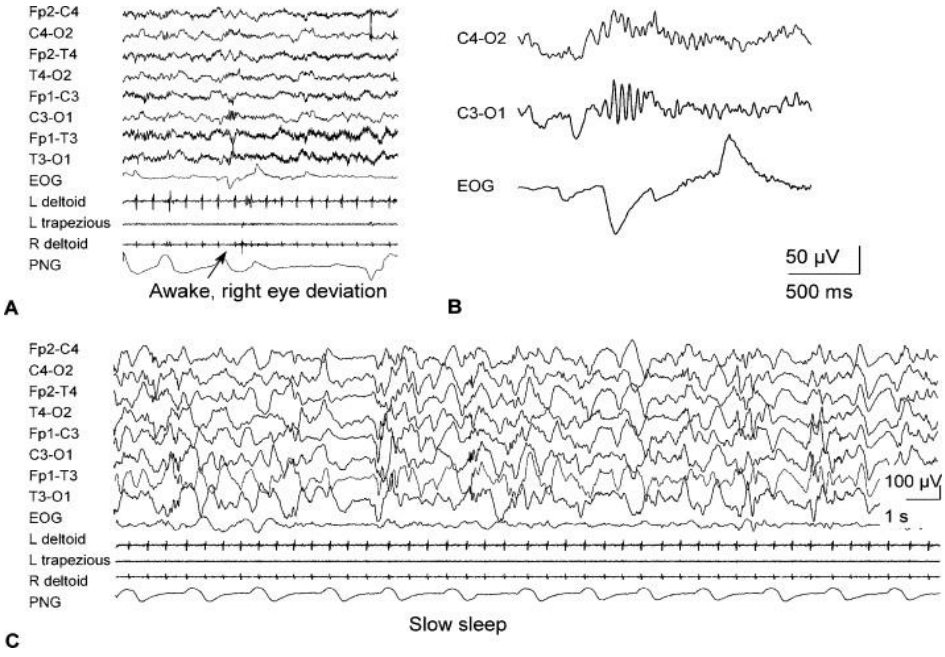
Then comes last stage *identity*
I'm finally a die hard rock fan ,I've joined social chat group and all ,and now I'm known as a rock music lover with many others as well that are not related to me in any way and just by the love for Rock music 🎶🥁

So this is how I socially identify

Added during revision.
Right Answers.
Not sure about the answer.

1. **Hypsarrhythmia**: occurs in **infantile spasm (tuberous sclerosis, west syndrome)**

- It is diffuse giant waves (high voltage, >400 microvolts) with **a chaotic background of irregular, asynchronous multifocal spikes and sharp waves.**
- Electro-decremental response: are clinical seizures associated with a marked suppression of the background.



2. **Function of MRS**:

Also known as nuclear magnetic resonance (**NMR**) spectroscopy, is a non-invasive analytical technique (based on the same physical principles as MRI) that has been used to:

a) Study **metabolic changes** in brain tumors, strokes, seizure disorders, Alzheimer's disease, depression and other diseases affecting the brain.
b) Measure **the concentrations of chemical components within tissues; GABA** and **glutamate** can be detected using MRS but **Not dopamine** as it is available in a very low concentration.

MR molecule	Potential clinical uses
Hydrogen proton (1H)	**1.** Magnetic resonance imaging (MRI). **2.** Analysis of metabolism – used to quantify; N-acetyl aspartate **(NAA** to assess **neuronal integrity), creatine (**a constant element of cellular energetic metabolism**)**, and **choline** (an increase in **Cho** is considered a marker of pathological proliferation/degradation of cell membranes and demyelination) containing molecules.
Fluorine (19F)	**1.** Measurement of pO2 **2.** Analysis of glucose metabolism **3.** Measurement of PH.

	4. Pharmacokinetics: measuring concentrations of **psychotherapeutic drugs** such as lithium, fluoxetine and trifluoperazine (Stelazine).
Lithium-7 isotope (7Li)	**Pharmacokinetics**
Phosphorus-31 (31P)	**1. Analysis of bioenergetics**; measuring concentrations of phosphorus-containing compounds (e.g. ATP & GTP). **2. Measurement of pH in brain regions.**
Nitrogen (14N)	Measurement of **glutamate, urea** and **ammonia**
Carbon-13 isotope (13C), low sensitivity	1. Analysis of metabolite turnover rate. 2. Pharmacokinetics of labelled drugs
Oxygen-17 isotope (17O)	**Measurement of metabolic rate**

3. **Wilson chromosome**:

- Congenital **recessive** condition of defective copper metabolism due to a defect in **chromosome 13.**
- Characterized by: **hepatic symptoms** and later **psychiatric symptoms** and **choreoathetosis** (lenticular nuclei) as various organ systems are overwhelmed by copper.
- It can be diagnosed (once the central nervous system is affected) by finding **Kaiser-Fleischer rings in the cornea** with a slit-lamp though these are not universally present.
- A more reliable diagnosis depends on **low ceruloplasmin** and **elevated copper** in urine and liver biopsies.

4. **Alzheimer** – Tau (histopathological changes)

- **NFT** (neuro-ribíllaíQ tangles) - hyperphosphorelated tau protein— cytoskeletal element, basophilic).(**intracellular**)
- **Hirano bodies—** Cytoplasmic, rode-shaped and eosinophilic bodies. (**intracellular)**
- **Senile plagues—** insoluble amyloid peptide deposits, eosinophilic mass (apple green) and extracellular.
- **In addition to:** neuronal and synaptic loss, granulovascular degeneration, astrocytes gliosis and microglial activation**.**

5. **FTD: Tauopathies**

- In familial cases mutations in the (**MAPT**) on **chromosome 17q21.**
- Many cases are **tau-negative** but show ubiquitin-immunoreactive neuronal cytoplasmic inclusions (mutations in progranulin (**PGRN**) gene on chromosome 17q21)**— (MND) type.**
- **Pick cells:** abnormal swollen oval-shaped neuronal cells with loss of **Nissl's** substance and peripherally displaced nucleus.
- **Pick's bodies:** are argentophilic, **tau and ubiquitin** reactive filamentous inclusions.

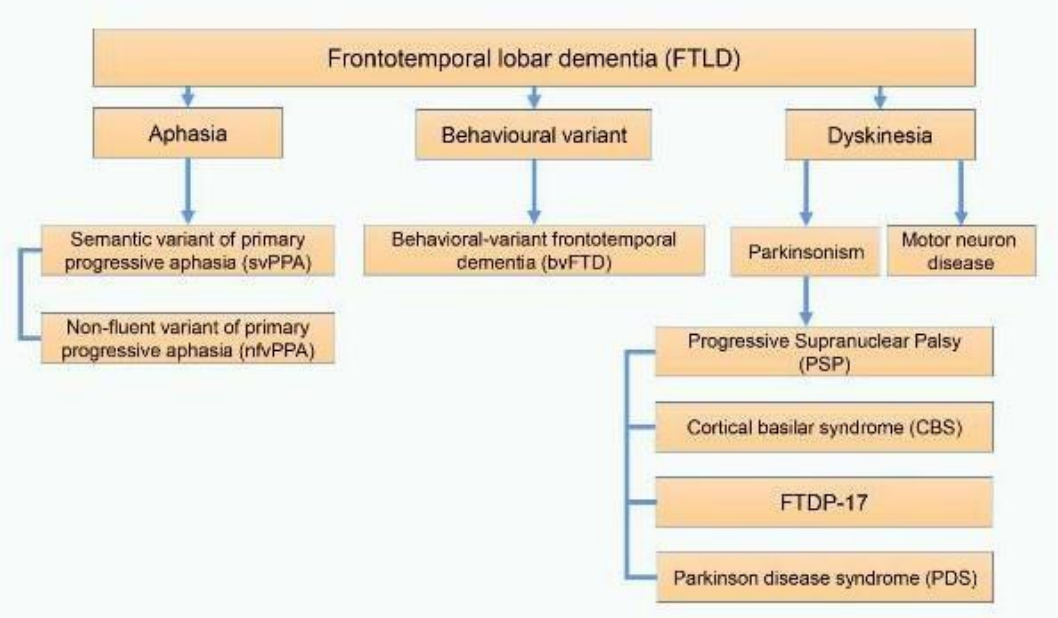
Frontotemporal Dementia has (3) variants as per **ICD-11**:

1. **FTD behavioural variant**: is characterized by personality changes. Neurocognitive functioning may be preserved in the early stages.
2. **Primary progressive aphasia variant**: progressive impairment in language skills. **3**

Added during revision.
Right Answers.
Not sure about the answer.

Subtypes:
- Primary deficits in word finding (logopenic subtype).
- Word meaning (semantic subtype).
- Word production (agrammatic subtype).

3. **FTD motoric variant:** progressive impairment in motor functioning. progressive neurocognitive impairment in; (attention, executive functioning, and visuospatial skills, with comparatively intact memory skills).
   3 Subtypes:
   - **Progressive supranuclear palsy** (poor balance, frequent falls, and horizontal gaze palsy),
   - **Corticobasal degeneration** (limb apraxia, tripping, rigidity, and dystonia).
   - **Amyotrophic lateral sclerosis** (e.g., muscle weakness, muscle atrophy, fasciculations, and spasticity, etc.)



6. **Lewy body dementia:**

- **Alpha-synuclein accumulation** in (temporal lobe, cingulate gyrus, frontal lobes and the dorsal motor nucleus of the vagus).**(Extracellular and Intracellular)**
- **Lewy bodies** are weakly eosinophilic, spherical, cytoplasmic inclusions. (**Intracellular**)
- Antibody to protease **ubiquitin** can be used to identify Lewy bodies.
- **Clinical features (two or more):**
  - Spontaneous motor features of Parkinsonism
  - Fluctuating cognition with notable variation in **attention + alertness**
  - Recurrent visual hallucinations (typically well-formed).
  - **REM sleep behaviour disorder (ICD-11).**
  - A progressive cognitive decline (memory loss may not be an early feature).
  - **Supportive features**: **Neuroleptic** sensitivity (not included in ICD-11) and history of falls

7. **Three questions on identity theory etc..** (takes both mind and science, mental state does exist and real, it is different from personal identity).

Social identity theory: [SPMM]

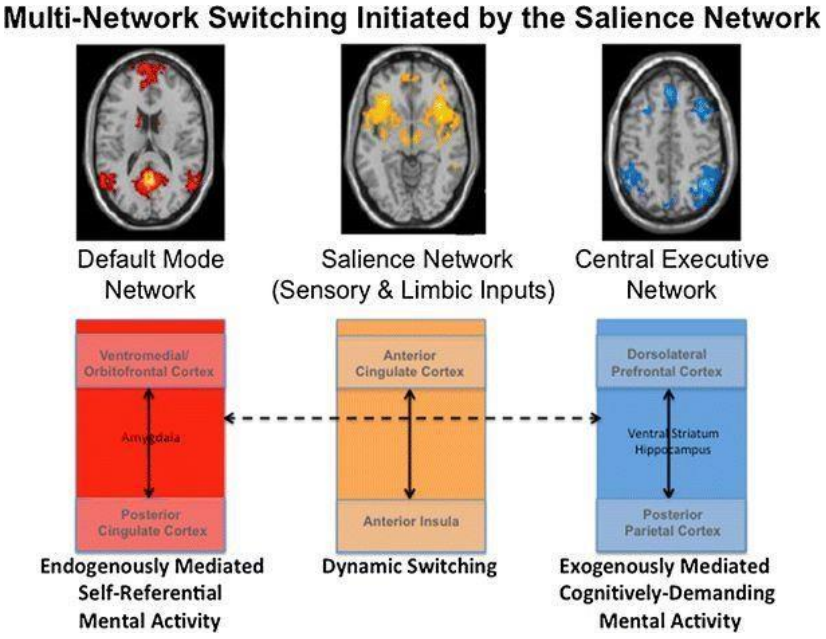
- An individual's positive self-image depends on both **personal** and **social** identities.
- So each individual strives to improve his group's success to foster his own image. This leads to prejudice against other groups
- Has **3 Stages: categorization**, **identification** and **comparison**.

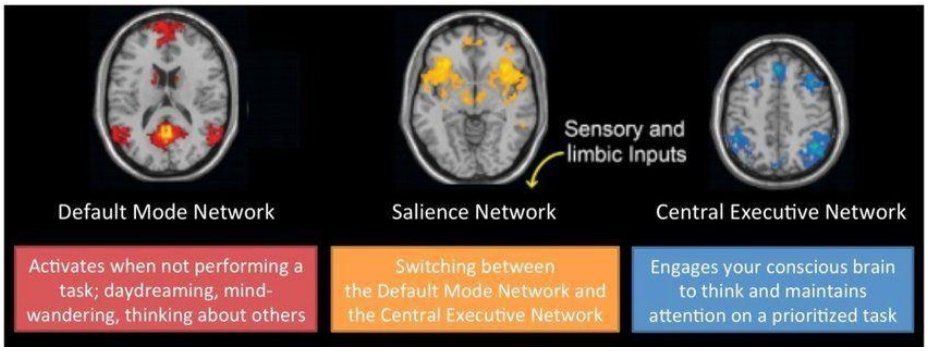
8. **Gender schema processing theory.** [SPMM]

- Gender identity alone provides children the motivation to assume sex-typed behaviour.
- Following this they observe and learn to be of a specific gender in the society.
- Thus, a gender schema of the particular culture gets deeply incorporated and serves as a standard for comparison

9. **Brain default network:**

- **Default (MAPPP)**: **M**edial prefrontal cortex, **A**ngular gyrus, **P**recuneus and **P**osterior cingulate cortex, **P**osterior parietal. **Inferior** parietal and **Lateral** parietal lobe.
- **Salience (ACTS)**: **A**nterior insular, **C**ingulate cortex (dorsal anterior), **T**halamus and **S**triatum.
- **Central executive DIP**: **D**orsolateral prefrontal cortex, **I**ntraparietal sulcus and **P**osterior parietal cortex.





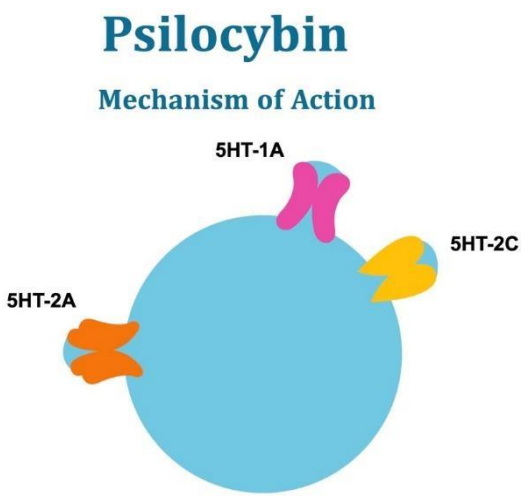
Added during revision.
Right Answers.
Not sure about the answer.

10. **Gait** for; Huntington (**Lurching, chaotic gait**), upper motor neuron (**Scissoring gait**) and cerebellar disease (**Broad, unsteady gait (Drunken/sailor's gait**)

11. **Psilocybin (Magic Mushroom) MOA**

Psilocybin (magic mushroom) and other classical hallucinogens (e.g. LSD & mescaline) exert their psychedelic actions in humans via activating **5-HT$_{2A}$** serotonin receptors, which in turn mediates its hallucinogenic effects.

Psilocybin is a strong agonist at **5-HT$_{2A}$** as well as a moderate agonist at **5-HT$_{1A}$** and **5-HT$_{2C}$**



12. **Crowed Theories:** by scholar **Gustave Le Bon** (**collective behavior is irrational**)

- **Contagion theory** assumes that people in a crowd act emotionally and irrationally because they come under the influence of the crowd's impulses. It is a theory of **collective behavior** which explains that the crowd can cause a hypnotic impact on individuals. These entire theories share a basic idea of how individual can act briefly unreasonable or inconsistent within a group and become normal when they are not in it.

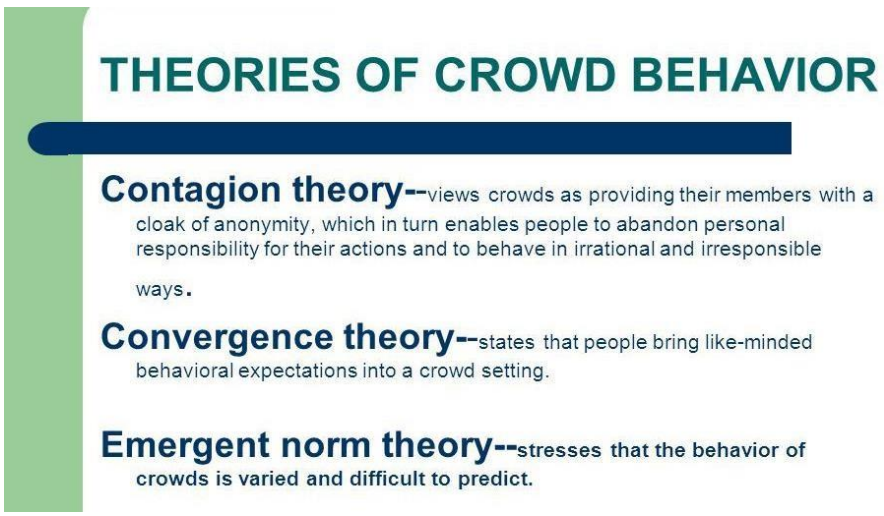
  If a people interested in country music goes to a rock concert along with friends and the friends are enjoying very much, the person will start to enjoy too due to group feeling and mass contagion. The person forgets his/her own likes and dislikes to follow the group behavior.

- **The convergence theory** is the one which postulates that all the societies as they move from the early industrial development to complete industrialization tend to move towards a condition of similarity in terms of the general societal and technological norms.

  Some examples of convergence theory include Russia and Vietnam, formerly purely communist countries that have eased away from strict communist doctrines as the economies in other countries, such as the United States, have burgeoned.

- **Emergent norm theory** hypothesizes that nontraditional behavior (such as that associated with collective action) develops in crowds as a result of the emergence of new behavioral norms in response to a precipitating crisis. The ALS ice bucket challenge is an example of collective behavior that raised money towards medical research

Contagion = contagious, convergent = industrialization, emergent = new trend



Added during revision.
Right Answers.
Not sure about the answer.

CONTAGION THEORY VERSUS CONVERGENCE THEORY

CONTAGION THEORY	CONVERGENCE THEORY
Contagion theory is a collective behaviour theory that argues that individuals act irrationally when they come under the hypnotic influence of a crowd	Convergence theory is a behaviour theory that argues the behaviour of the crowd reflects the behaviour and attitudes of individuals who join the crowd
Crowds exert an undue hypnotic influence on individuals to act in a violent way	Like-minded people come together to form a crowd, and attitudes, beliefs, and behaviour of individuals shape the crowd behaviour
Crowd affects the individuals	Individuals affect the crowd

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SOCIAL CONTAGION THEORY

Social contagion theory posits that behaviors, emotions, and ideas can spread rapidly within a social network, influencing individuals to adopt and replicate them.

OVERVIEW

There are two types of contagion:

- **Echo contagion** occurs when messages are amplified and spread rapidly through social media or other communication channels, resulting in a viral effect.
- **Hysterical contagion** occurs when people collectively experience symptoms of illness or distress, primarily due to psychological factors rather than a physical cause.

EXAMPLE

Fleeing the Scene in an Emergency

If in a large crowd where one person begins running and flailing their arms as if panicked, others in the crowd will immediately do the exact same thing. They will run in the same direction and exhibit the same sense of fright. This is a highly adaptive reflex in most mammals.

HELPFULPROFESSOR.COM

13. **Pontine lesion:**

Locked in syndrome (**ventral pontine**) describes a condition of total paralysis below the level of **the third nerve nuclei**. Such patients can:

- Open their eyes voluntarily, and elevate and depress their eyes to command.
- Horizontal eye movements are usually lost.
- No other voluntary movement is possible.

[SPMM]Locked in syndrome: According to Cartlidge (2001), the ventral pontine or locked in syndrome describes a condition of total paralysis below the level of the third nerve nuclei. Such patients can open their eyes and elevate and depress their eyes to command. Horizontal eye movements are usually lost, and no other voluntary movement is possible. The diagnosis of this state depends on the recognition that the patient can open his eyes voluntarily rather than spontaneously in the vegetative state. This generally results from infarction of the ventral pons, pontine tumours, pontine haemorrhage, central pontine myelinolysis, head injury or brain stem encephalitis.

Other pontine lesions include
Left Pontine lesion → **left** face (ipsilateral), **right** hemiparesis (contralateral),
Left Midbrain lesion → **right** face and hemiparesis, but **left** CN 3 palsy – ipsilateral)

14. **Deindividuation.**

Festinger's deindividuation theory: people in-group context act uncharacteristically more aggressive as **a sense of identity and belongingness** and **diffusion of responsibility occurs in groups.** Similarly, uniforms can reduce individuality, promoting expression of aggression (hence its use in Police and military forces). But deindividuation does not always cause aggression.

15. **Bowlby stage of grief except bargaining:**

- **Bowlby and Parkes** were the first to propose a stage theory of grief for adjustment to bereavement that included **4 stages (SYDR)**: **1. S**hock-numbness (often with outbursts of anger and intense distress) **2. Y**earning-searching**, 3. D**isorganization-despair and **4. R**eorganization.
- **Erich Lindemann:** Stages of grief **(SSAR)**: **1. S**hock that lasts 10-14 days **2.** Intense **S**adness (withdraw from social contact), **3. A**nger, as the grieving person seems to '**protest**'; the unexpected death **4.** and finally, within a year or so, the grief is **R**esolved, and the person returns to normal
- **Colin Murray Parkes; 5 stages** (**A SPIDER**) of bereavement:
  1. Alarm
  2. Numbness (shock).
  3. Pining for the deceased (illusions or hallucinations of the deceased can occur)
  4. Depression
  5. Reorganisation (recovery)
- **Kübler-Ross**: adapted Bowlby and Parkes' theory to describe a **5-stage (DABDA)** response of terminally ill patients to awareness of their impending death: **1.** denial-dissociation-isolation **2.** Anger **3.** Bargaining **4**. Depression and **5.** Acceptance

Freud model for mourning: individual's life force must be fully withdrawn from its attachment to the deceased person (Decathexis).

Bowlby (SYDR) and Lindemann = 4 stages
Ross (DABDA) and Murray Parks = 5 stages

16. Computational psychiatry: **Reward prediction error** is difference between receive reward & predicted reward

17. selective D2 antagonist: **Sulpride**

18. Sally Anne test **theory of mind 3-4 years:**

The Sally–Anne test is a psychological test, used in developmental psychology to measure a person's social cognitive ability to attribute false beliefs to others. The child passes the test if he or she knows that Sally will look for the item in the basket and not the box. Typical children struggle with this test as 3-year-olds**, but most pass it by 5 years of age.**

Reading "The mind in the eye test" – **Baron Cohen** 1997:

From google: Baron-Cohen et al (eyes test) used a natural experiment to test for **advanced theory of mind in adults with either high-functioning autism or Asperger syndrome.** This test includes 36 photographs of male and female eyes depicting emotional states. For each photograph, participants are asked to choose the emotional state that best describes the eyes, choosing between one of four possible emotions.

19. EEG seen as normal, what disease? **variant CJD and COMPLEX PARTIAL AND TEMPORAL NORMAL IN 50% and autosomal dominant frontal lobe epilepsy**

vCJD shows generalized slowing in EEG

Q. from another recall: EEG normal - atypical absent seizures/Benign roland /frontal/

In autosomal dominant frontal lobe epilepsy, interictal EEG is often normal, and ictal EEG unhelpful or non-localising.

20. EEG centro-temporal (**Rolandic**)-**benign epilepsy in children is central temporal**. Beta is found in centre position, delta is found deep sleep and any delta in awakening stage is pathology, alpha in occipitoparietal, theta is fronto-temporal

Syndrome	Notes
Infantile spasms (West's syndrome)	Brief spasms beginning in first few months of life 1. Flexion of head, trunk, limbs → extension of arms (Salaam attack); last 1-2 secs, repeat up to 50 times 2. Progressive mental handicap 3. EEG: hypsarrhythmia usually 2nd to serious neurological abnormality (e.g. TS, encephalitis, birth asphyxia) or may be cryptogenic poor prognosis
Lennox-Gastaut syndrome	May be extension of infantile spasms (50% have hx) onset 1-5 yrs atypical absences, falls, jerks 90% moderate-severe mental handicap EEG: slow spike ketogenic diet may help
Benign rolandic epilepsy	paraesthesia (e.g. unilateral face), usually on waking up
Juvenile myoclonic epilepsy (Janz syndrome)	Typical onset in the teens, more common in girls 1. Infrequent generalized seizures, often in morning 2. Daytime absences 3. Sudden, shock like myoclonic seizure usually good response to sodium valproate

Type	Frequency	Normally found	Normally seen in
Delta	**1-4Hz**	Frontally in adults and posteriorly in children	Slow wave sleep (stages III and IV) (Purves, 2001) and in babies. Should not be present when awake, when present if awake this strongly suggests pathology
Theta	4-8Hz	Generalised	Young children, drowsy and sleeping adults (stage I) (Purves, 2001), with **certain medications, meditation**. Small amount seen in awake adults, excessive amount when awake may indicate pathology
Alpha	8-12Hz	Posteriorly	When relaxed and when the eyes are closed (whilst awake). Also seen in **meditation**.
Sigma	12-14Hz	Frontal and central regions	(aka sleep spindles) Bursts of oscillatory activity that occur in stage 2 sleep. Along with k-complexes they are the defining characteristic of stage 2 sleep
Beta	12-30Hz	Frontally	When busy or concentrating
Gamma	30-100Hz	No specific areas	Seen in advanced / very experienced **meditators**

Theta, Alpha and Gamma waves appear in meditation.

21. over the counter medication cannot take with MAOI, **Decongestant**

Explanation: **Decongestant** medications taken orally have the ability to increase blood pressure by themselves, especially in patients with pre-existing hypertension or those whose hypertension is poorly controlled. Therefore, when such medications are combined with MAOIs the likelihood of elevated blood pressure significantly increases as well as the potential for a hypertensive crisis. **Decongestants** to be avoided mainly include the over-the-counter **α1-adrenoceptor agonists, phenylephrine and oxymetazoline. Ephedrine, phenylpropanolamine, and pseudoephedrine are also contraindicated with MAOI use.** [ncbi]

22. ADHD medication cause hypotension **clonidine(alpha 2 agonist)**

22. crisis identity emi, identity foreclosure, moratorium, diffusion

		Individual has committed to identity: Yes	Individual has committed to identity: No
Individual has explored identity options	Yes	Identity Achievement	Moratorium
	No	Foreclosure	Identity Diffusion

24. EMI receptor :

Zopiclone: **Omega 1 alpha 1 subunit of gaba A**,
Corticosteroid: **Glucocorticoid in the nucleus**

25. EMI development how many word : 1 year old **2-3 words** 13 years old.**adult speech**

26. anorexia purging type finding:

Hypochloremia

- **Vomiting: hypokalemic and ~~hyperchloremic~~ metabolic alkalosis/Russell sign/ hyponatremia**.
- **Laxative: metabolic acidosis, hypokalemia and hyponatremia**

27. wernicke aphasia : **fluent**? **And receptive / Broca's broken and expressive non-fluent.**

28. human research : **Helsinki - identifiable human data**

- **Nuremberg → Code of Ethics**
- **Declaration of Malta → hunger strike,**
- **Declaration of Geneva → revision of the Hippocratic oath,**
- **Tokyo → not to participate in torture of prisoners or detainees,**
- **Lisbon → international statement of rights of pts,**
- **Ottawa → principles for optimal child health**

29. See cloud as face : **Pareidolia:** Arise when detailed images are seen from shapes. E.g. seeing the man in the moon, or Jesus Christ on a piece of burnt toast.

Pareidolic illusions tend to occur when a person is **concentrating** whereas **affect and completion illusions occur during inattention.**

30. Fundamental attribution error- **attribution of personal factor to others behaviour**

31. Attachment- **Anxious avoidant**

32. Kohlberg- pre convetional 7-12, conventional 13-16, post- convetional 16-20

Regarding Kohlberg stages Age:

- SPMM says **7-12, 13-16yr and 16-20**
- Psych-Mentor it says 0-9, 9-20 and >20,

It was reported that **TRONs** uses Psych-Mentor values

33. Reward prediction error- **difference between received reward and predicted reward.**
34. Neurotransmitter- fast and excitatory- **Glutamate**

Group	Neurotransmitter	Function
Acetylcholine	Acetylcholine	Excitatory
Amines	Epinephrine	Excitatory
Amines	Norephinephrine	Excitatory
Amines	Dopamine	Excitatory and Inhibitory
Amines	Serotonin	Excitatory
Amino Acids	Glutamate	Excitatory
Amino Acids	Glycine	Mainly inhibitory
Amino Acids	g-Aminobutiric acid (GABA)	Inhibitory

35. Antipsychotic increase with absorption of food- **oral Ziprasidone** and **Lurasidone,**

- **Ziprasidone**: These results confirm that **ziprasidone** should be taken with food and that a meal equal to or greater than **500 kcal**, irrespective of fat content, is required for optimal and reproducible bioavailability of the administered dose
- **Paliperidone(recommended to be taken with food as it improves absorption)**
- **Quetiapine(food initially delays absorption but over all increases the amount absorbed)**
- **Food decrease Diazpam absorption**

36. Apoe4- RISK **homozygous 10-30, heterozygous 3-4**

Added during revision.
Right Answers.
Not sure about the answer.

37. Physiology change in pregnancy- **increase GFR, increase renal elimination** (Only **APE decrease** = absorption, protein binding, emptying)
Old age - **FTFT increases** = fat, toxicity, free drugs, T-half life (others reduce, liver metabolism unaffected)

38. At young doctors training- **metonymy**/circumstantiality

> **Metonymy** is a speech disturbance common in schizophrenia in which the affected person uses a word or phrase that is related to the proper one but is not the one ordinarily used (for example, the patient refers to losing the 'piece of string' of the conversation rather than the 'thread' of the conversation)

39. Landmark agnosia/ topographical agnosia/ Topographanosia - **lingual gyrus on occipital lobe.**

> **Visuospatial agnosia** is a neurological condition characterized by a person's inability to recognize or understand the spatial relationships between objects and the physical world. It typically affects an individual's perception and understanding of the spatial layout of their environment, objects, and their own body. There are different types of visuospatial agnosia, including:
>
> 1. **Simultanagnosia**: In this type, individuals have difficulty perceiving and recognizing more than one object at a time. They might only be able to focus on one object or aspect of a scene, making it challenging to process complex visual information.
>
> 2. **Topographical Disorientation**: People with this form of visuospatial agnosia have difficulty with spatial orientation and may struggle to navigate familiar or unfamiliar environments.
>
> 3. **Constructional Apraxia:** This type involves difficulties in building or copying complex geometric shapes or structures, even though the person may have intact perception of the individual components.

Added during revision.
Right Answers.
Not sure about the answer.

Additional Notes:

Delusional disorder: is an uncommon condition in which patients present with circumscribed symptoms of non-bizarre delusions (DSM-5 allows 'with bizarre content'; ICD-11 does not specify), but with absence of prominent hallucinations and no thought disorder, mood disorder, or significant flattening of affect.

DSM-5 subtypes of delusions: include erotomanic, grandiose, jealous, persecutory, and somatic.

ICD-11 states that delusions typically persist for at least 3 months and often much longer.

Anomia due to both **angular and supramarginal**...if both are in the option then ans would be angular

Transcortical sensory and motor aphasia: **repetition will be intact**

Question number 28 can be found in fresh question A1 set 6

Semantic is visual analogue scale

Semantic differential scale

> Osgood's semantic differential scale is used to measure verbally expressed attitudes. It allows different attitudes about a particular topic to be measured on the same scale. It includes various factors constituting an attitude; e.g. while expressing one's attitudes regarding a politician, one can rate him using an evaluative component (good ---- bad), activity component (active ----- inactive) and potency component (powerful weak) etc. With these bipolar adjectives being the two extremes.
>
> **2 polarised adjective separated by line**

- Non-linear → zero order

Frusemide is relatively safe in using with lithium..

- **ZOCFLAP: Z**ero **O**rder **C**onstant, **F**irst Order **L**inear **A**nd **P**roportional
- **Zero order (PES)**: Phenytoin, Ethanol and Salicylates
- **Receptors linked to G proteins**, including the dopamine D2 receptor, all have the structure **of seven transmembrane regions**, meaning that they span the membrane seven times.

Added during revision.
Right Answers.
Not sure about the answer.

Adjustment disorder (onset):
According to DSM V 3 month
ICD 10/11: 1 month

Adjustment disorder is a maladaptive reaction to an identifiable psychosocial stressor or multiple stressors (e.g. divorce, illness or disability, socio-economic problems, conflicts at home or work) that usually **emerges within a month of the stressor**. The disorder is characterized by preoccupation with the stressor or its consequences, including excessive worry, recurrent and distressing thoughts about the stressor, or constant rumination about its implications, as well as by failure to adapt to the stressor that causes significant impairment in personal, family, social, educational, occupational or other important areas of functioning. The symptoms are not better explained by another mental disorder (e.g., Mood Disorder, another Disorder Specifically Associated with Stress) and **typically resolve within 6 months, unless the stressor persists for a longer duration.** **[ICD11]**

- **PTSD within 6 months**

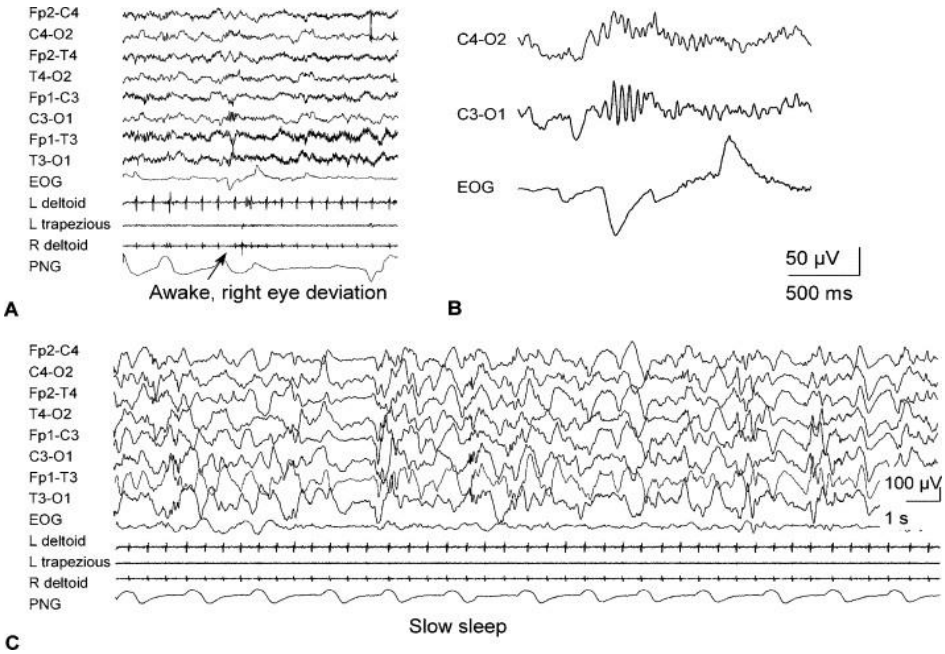
Mentalization is the ability to understand the mental state – of oneself or others – that underlies overt behaviour. Mentalization can be seen as a form of imaginative mental activity that lets us perceive and interpret human behaviour in terms of intentional mental states.

- **Alpha synucleopathies – People's Lives Matters**
  - **Parkinson's**
  - **Lewy Body Dementia (DLB)**
  - **Multisystem atrophy**

1. 3Hz – **Typical absence seizure**

Seizure type	Pattern during the seizure (ictal EEG)
Typical absence (aka petit mal)	Generalised 3 Hz spike-wave
Atypical absence	Slow (<2.5 Hz) generalized spike-and-wave
Focal (previously known as partial)	Focal spikes
Myoclonic	Generalized 3-6 Hz polyspike and wave discharge
Generalised tonic-clonic	EEG often obscured by artifact (movement). Generalized fast rhythmic spikes are seen in the tonic stage. Bursts of spikes and after-coming slow waves are synchronous with clonic jerks. A postictal period of irregular slow activity follows
Atonic (drop attack)	Generalized spike-and-wave is typical, with atonia at the time of the slow wave

2. Hypsarrthmia occurs in infantile spasm (**tuberous sclerosis, west syndrome)**
   - It is diffuse giant waves (**high voltage, >400 microvolts**) with **a chaotic background of irregular, asynchronous multifocal spikes and sharp waves.**
   - Electro-decremental response: are clinical seizures associated with a marked suppression of the background.



3. EMI : chromosome number specifically involve in;
   - Chr 4q
   - Chr22
   - Chr17
   - Chr5

- **Huntington**: **Chromosome 4, 4p16.3**

Huntington's disease is an **autosomal dominant** disorder with complete penetrance. The Huntington's disease gene, IT15, is located on chromosome 4p, contains CAG-trinucleotide repeats, and codes for a protein called huntingtin. The protein is found in neurons throughout the brain

- **Lewy body**: **SNCA on Chromosome 4**

Lewy bodies is caused by SNCA or SNCB gene mutations, it is inherited in an **autosomal dominant** pattern.

- **The SNCA gene is located on the long arm of chromosome 4 (4q21)**
- The SNCB gene is located on the long arm of chromosome 5 (5q35)

- **FTD** is genetically complex and can present as familial or sporadic disorder. Most familial FTD cases are linked to **chromosome 17q21**. In 1994, linkage of the autosomal dominant inherited form of FTD with parkinsonism on chromosome 17q21.

4. EEG is use because high?

A. sensitivity
B. reliability
C. validity
D. positive predictive
E. specificity Increased

5. True of Hypnotic suggestibility = associated with reduced metacognitive agency (i.e. less robust awareness of how effective an individual affects the world).

6. Group **polarization**: refers to the tendency for a group to make decisions that are more **extreme** than the initial inclination of its members.

7. Trail making test test on? = **processing speed and executive functions**

Trail making test part A:

- Visual attention
- Tracking
- Processing Speed

Trail making test part B:

- Set shifting
- Attention
- Visual Sketch
- Cognitive flexibility
- Switch attention

Response inhibition and **set shifting**	Alternating sequences, go-no-go test, Luria motor test, trail making test, **Wisconsin card sorting test**, Stroop test

If more than one test in the options go for **WCST**

8. Theory of mind test?

   A. Reading mind in the eyes test
   B. penn
   C. Eysenck

   Children with autism often have difficulty understanding the beliefs of others. Traditional tests of this ability, called theory of mind, require children to have good enough language skills to understand what they're being asked to do. / **Mind in the eyes test is for ADULTS with high functioning Autism and Asperger.**

9. Caffeine reversal: **Adenosine Agonist**

   - Caffeine is a central nervous system stimulant that belongs to a class of compounds called **xanthines**. Its primary mechanism of action is as an **antagonist of adenosine receptors**. Adenosine is a neurotransmitter that promotes sleep and relaxation by acting on its receptors, **mainly the A1 and A2A subtypes**. Caffeine works primarily by blocking these receptors, preventing adenosine from binding to them. This antagonistic action results in increased alertness and wakefulness. [Psych-Mentor]

   Caffeine is an ANTAGONIST / Revered by Adenosine AGONIST .

10. Higher protein bound SSRI is **Sertraline.**

    Sertraline is highly protein-bound in the blood (98%)

11. Antipsychotic associated with epileptic seizure: Clozapine

    Explanation: Clozapine is the highest risk antipsychotic for seizures, with a cumulative risk **10%** for seizure activity. **Chlorpromazine is the highest risk first generation antipsychotic for seizures.**

12. EMI. Restlessness, felt like swallow his tongue, couldn't breathe: Answer – **Akathisia**

13. Bowlby stage of grief except (**SYDR)**

    - despair
    - yearning
    - numbing
    - **Bargaining**
    - Reorganization

    Bowlby (SYDR) and Lindemann = 4 stages
    Ross (DABDA) and Murray Parks = 5 stages

14. Attachment theory: **Oxytocin release**

15. Attachment theory - Due to mom failure, child develop his own strength and weakness

16. Utilitarian – **best outcome**

17. Durkheim suicide attribution?

A. **More social force than individual** Fatalistic suicide
B. Social capital

Explanation: Durkheim's theory posits two core principles: **(1)** that the structure of suicide rates is a positive function of the structure of a group or class of people's social relationships and those **(2)** that social relationships vary according to their level of integration and (moral) regulation.

18. Alzheimer neuroimaging: **Medial temporal lobe atrophy**

Neuroimaging findings in Alzheimer's disease include: **ventricular enlargement, reduced medial temporal lobe volume** - especially hippocampus and a reduction in parieto-temporal fMRI activation and SPECT blood flow

19. Delay gratification in child is the ability to postpone an immediate gain in favor of greater and later reward. It leads to **higher academic performance.** (marshmallow experiment)

20. EMI: Neuroimaging in:

Schizophrenia:

- Ventricular enlargement
- Loss of grey matter – especially insular cortex, anterior cingulate (medial prefrontal cortex) and medial temporal lobe
- Progressive loss of brain volume in first few years of diagnosis
- fMRI reveals poor DLPFC activation in executive tasks
- Decreased NAA (N-Acetyl aspartate) in PFC (neuronal loss) in MRS
- Widespread reduction in DTI (diffusion tensor) – fractional anisotropy: frontal and corpus callosum – more in chronic treated patients

Auditory hallucination : activation primary auditory cortex

The most consistent finding of structural neuroimaging studies of patients with auditory hallucinations is **reduced grey matter volume in the left superior temporal gyrus, including the primary auditory cortex**

Working memory: DLPFC

Dementia in elderly : white matter hyperintensities ?

21. Lewy body imaging = **DAT Scan**

DAT scan shows Hypopigmentation in substantial nigra

Q. Mark Coombes is a 78-year-old man who has been diagnosed with Dementia with Lewy Bodies following a period of variable cognitive impairment, falls and intermittent visual hallucinations. He has recently received a scan and his children want to know about the result. Which of the following are true regarding imaging findings in Dementia with Lewy Bodies? [SPMM]

Select one:

- **Pathological nigrostriatal uptake on DAT scan**
- DAT scans can distinguish it from Parkinson's disease

Added during revision.
Right Answers.
Not sure about the answer.

- More pronounced posterior parietal grey matter loss than in Alzheimer's disease
- It preferentially affects frontotemporal white matter tracts
- There is more diffuse cerebral atrophy than in Alzheimer's disease

[SPMM] **Explanation** : Dementia with Lewy Bodies is associated with profoundly reduced radiotracer uptake in the substantia nigra and striatal region on DAT scans. This is also found in Parkinson's disease, although there is less generalised grey matter loss in the latter condition. Dementia with Lewy Bodies tends to show less diffuse cerebral atrophy and grey matter loss than Alzheimer's disease. The medial temporal lobe and entorhinal cortex tends to be preserved. Neuromelanin MRI shows decreased hyperintensity in the lateral part of the substantia nigra and locus coeruleus indicating dopamine neuronal loss. In perfusion SPECT, a bilateral occipital hypoperfusion is frequently observed in DLB. Outside the brian, markedly decreased uptake of MIBG- 123I-metaiodobenzylguanidine on myocardial scintigraphy is also observed in DLB..

Q. Recent studies on Lewy body dementia suggest hypoperfusion and hypometabolism in which part of the brain areas?

Select one:

- Prefrontal cortex
- **Occipito-parietal**
- Tempero-parietal
- Cerebellum

Explanation: Single photon emission computerised tomography (SPECT) and fluorodeoxyglucose positron emission tomography (PET) studies in dementia with Lewy bodies have demonstrated occipital and occipitoparietal visual association area hypoperfusion and hypometabolism respectively. However, occipital deficits are not consistently reported in dementia with Lewy bodies. Higher visual areas, particularly occipito-parietal, appear abnormal in dementia with Lewy bodies, while there is a preservation of function in lower visual areas (V1 and V2/3).

Lewy body - since there is memory impairment and visual hallucination - changes in imaging- Occipito(visual hallucination), parietal(memory)

22. Which antiepileptic causes aggressive behavior?

LVT = Levetiracetam > Vigabatrin > Topiramate (according to SPMM)
Vigabatrin was used in a child in the SPMM question (26% risk)

Table 2. Psychosis With AEDs

Drug	Incidence (%)
Zonisamide	2.2-18.9
Topiramate	1.3-11.7
Ethotoin	8.7
Vigabatrin	0.7-6.7
Rufinamide	0.6-4.0
Levetiracetam	0.3-1.6
Primidone	1.4
Gabapentin, lacosamide, lamotrigine, perampanel	<1.0
Phenytoin	Toxic effect
Carbamazepine, clobazam, clonazepam, ethosuximide, ezogabine, oxcarbazepine, phenobarbital, pregabalin, tiagabine, valproate	NR

AED: antiepileptic drug; NR: none reported.
Source: References 2, 21, 29, 35, 50-62.

Table 1. Aggressive-Type Behavior With AEDs

Drug	Incidence (%)
Topiramate	1.9-81.8
Tiagabine	1.0-36.4
Clobazam	1.3-31.9
Levetiracetam	2.3-27.0
Vigabatrin	3.4-23.0
Perampanel	<1.0-20.0
Primidone	3.0-18.2
Gabapentin	9.4-13.0
Valproate	0.0-12.0
Rufinamide	3.0-11.6
Zonisamide	2.6-9.0
Ethotoin	8.7
Lacosamide	8.0
Lamotrigine	0.0-7.0
Carbamazepine	2.0-7.0
Oxcarbazepine	3.3
Ethosuximide	3.0
Ezogabine	1.0-2.0
Clonazepam, phenobarbital	CR
Phenytoin, pregabalin	NR

AED: antiepileptic drug; CR: case reports; NR: none reported.
Source: References 2, 21-50.

Table 3. Hyperactivity/Restlessness With AEDs

Drug	Incidence (%)
Phenobarbital	10.0-42.0
Clobazam	4.0-27.0
Vigabatrin	6.1-25.7
Clonazepam	0.0-12.0
Carbamazepine	7.0-12.0
Lacosamide	11.8
Gabapentin	0.0-11.0
Ethosuximide	0.0-11.0
Valproate	0.0-10.0
Lamotrigine	0.0-10.0
Rufinamide	3.0-4.7
Primidone	4.5
Topiramate	2.7-4.4
Levetiracetam	2.5-3.8
Ethotoin, ezogabine, oxcarbazepine, perampanel, phenytoin, pregabalin, tiagabine, zonisamide	NR

AED: antiepileptic drug; NR: none reported.
Source: References 21, 22, 25, 29, 30, 32, 33, 35, 38, 40, 43, 53, 58, 61, 63-69, 70-72.

23. **Q** Sort test or methodology = **subjective opinion/viewpoint (in research)**

Added during revision.
Right Answers.
Not sure about the answer.

Q Methodology is a research method used in psychology and other social sciences to study people's "**subjectivity**" - that is, their viewpoint. Q was developed by psychologist William Stephenson. It has been used both in clinical settings for assessing patients, as well as in research settings to examine how people think about a topic.

The Q sort test is used in psychology and social sciences. Which of the following does it measure? [SPMM]

Select one
- Executive functioning
- **Subjective opinion**
- Attention
- Distractibility
- Bias

Explanation: Q methodology uses a form of factor analysis to find correlations between variables across a number of subjects. It is used in research to understand how people think about an issue (**subjective viewpoints**), and in clinical practice to assess patients. It often uses the method of individuals, either patients or clinicians, ordering statements about a condition or patient in order of relevance.

24. Frontal lobe consist of how many percent of our total brain volume?

Answer: **Percentage of total cerebral cortex volume (human):**

- **Frontal lobe = 41%;**
- Temporal lobe = 22%;
- Parietal lobe = 19%;
- Occipital lobe = 18%.

What proportion (in percentage) of the brain is occupied by prefrontal cortex? [SPMM]

Select one:
- 20
- 70
- 50
- **30**
- 10

Total brain Volume of prefrontal cortex : 15 %.
Percentage/proportion of prefrontal cortex :30%

The prefrontal cortex (PFC) is the anterior part of the frontal lobes of the brain, lying in front of the motor and premotor areas, which occupies **nearly 30% of the brain**.

Q. The prefrontal cortex is associated with executive functioning, personality, language and planning. Which of the following best represents the percentage of the total brain volume made up by the prefrontal cortex? [SPMM]

Select one:
- 35
- 30
- **15**
- 25
- 20

Explanation: The percentage of prefrontal cortex relative to total brain volume was 12.51 for humans and 10.68 for baboons. Similarly, the proportion of both dorsal and orbital prefrontal cortex volumes is larger inhuman brains. Relative to total brain volume, the

percentages of dorsal and orbital prefrontal cortex was 8.22% and 4.29% respectively in humans and 7.21% and 3.47% in baboons

N.B. Verbal working memory tasks mainly activated the left DLPF and visual working memory tasks mainly activate DLPF

25. **EMI: apraxia/agnosia:**
Landmark agnosia called?

Answer: Landmark agnosia, also known as **topographical agnosia** or **topographagnosia**, is marked by the inability to recognize salient environmental stimuli such as landmarks. This is usually due to lesions in the **lingual gyrus**. Patients are able to draw detailed maps and visualize places familiar to them **before** the illness.

[SPMM] **Q.** Mr Liam Feeley is a 58-year-old man who has suffered a significant head injury in a road traffic collision. He now finds that he cannot identify landmarks near his home, despite having been an avid local historian.

Explanation: Visuospatial agnosia is a loss of the sense of "whereness" in the relation of oneself to one's environment and in the relation of objects to each other.

Unable to use hammer upon command but able to use it at home

Answer: the areas of the brain most often seen to be damaged when **ideomotor apraxia** is present: the **left parietal lobe and the premotor cortex.**

Unable to do successive task – ideational apaxia (Multi-step tasks)

26. Callous unemotional trait?

Callous-unemotional traits are a specific set of personality characteristics that include a lack of empathy, lack of guilt or remorse, shallow or deficient affect, and a disregard for the feelings, rights, and welfare of others. These traits are often associated with a distinct subtype of antisocial behavior and are considered a risk factor for the development of more severe and persistent conduct problems.

The term is most often used in relation to children and adolescents, where it has been incorporated into the diagnosis of conduct disorder (CD) in the Diagnostic and Statistical Manual of Mental Disorders (DSM-5). The DSM-5 includes a "with limited prosocial emotions" specifier for the diagnosis of conduct disorder, which is meant to identify individuals who display these callous-unemotional traits.

Research has shown that children and adolescents who display these traits often have more severe conduct problems, are more likely to engage in aggressive and antisocial behavior, and are less responsive to typical interventions. They may also have different neurobiological characteristics and respond differently to treatment compared to those with conduct disorder who do not display these traits.

However, it's important to note that not every child or adolescent who displays callous-unemotional traits will develop conduct disorder or engage in antisocial behavior. These traits can be influenced by a variety of factors, including environmental factors such as parenting

practices and exposure to violence, as well as genetic factors. Early intervention can also help to mitigate the impact of these traits and reduce the risk of future conduct problems.

[ICD11] The core feature of the Dissociality trait domain is disregard for the rights and feelings of others, encompassing both self-centeredness and lack of empathy. Common manifestations of Dissociality, not all of which may be present in a given individual at a given time, include: self-centeredness (e.g., sense of entitlement, expectation of others' admiration, positive or negative attention-seeking behaviours, concern with one's own needs, desires and comfort and not those of others); and lack of empathy (i.e., indifference to whether one's actions inconvenience hurt others, which may include being deceptive, manipulative, and exploitative of others, being mean and physically aggressive, callousness in response to others' suffering, and ruthlessness in obtaining one's goals).

27. **Not** FTD variant?

A. psp
B. ftd
C. spp

Answer: Clinically, FTD can be categorized into 3 broad subtypes—the behavioral variant of FTD (bvFTD), semantic dementia (SD), and progressive non-fluent aphasia (PNFA).

[SPMM] There are various types of fronto-temporal dementia characterised by different pathological features. Which of the following is not a form of frontotemporal dementia?

Select one:
A. Pick's disease
B. Progressive non-fluent aphasia
C. Progressive supranuclear palsy
D. Huntington's disease
E. Semantic dementia

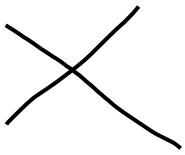
Huntington's disease typically affects the basal ganglia in the early phases of the disorder but then generally spreads to affect all cortical areas. Progressive supranuclear palsy was thought to be related to Parkinson's disease, but is now thought to be more related to frontotemporal dementia.

Types of Frontotemporal Disorders	
Diagnostic Terms	Main Early Symptoms
Progressive Behavior/Personality Decline	
• Behavioral variant frontotemporal dementia (bvFTD) • Temporal/frontal variant FTD (tvFTD, fvFTD) • Pick's disease	• Apathy, reduced initiative • Inappropriate and impulsive behaviors • Emotional flatness or excessive emotions • Memory generally intact
Progressive Language Decline	
• Primary progressive aphasia (PPA) • Progressive nonfluent aphasia • Semantic dementia	• Semantic PPA (also called semantic dementia): can't understand words or recognize familiar people and objects • Agrammatic PPA (also called progressive nonfluent aphasia): omits words that link nouns and verbs (such as to, from, the) • Logopenic PPA: trouble finding the right words while speaking, hesitation, and/or pauses in speech
Progressive Motor Decline	
Corticobasal syndrome (CBS)	• Muscle rigidity • Difficulty closing buttons, operating simple appliances; difficulty swallowing • Language or spatial orientation problems
Progressive supranuclear palsy (PSP)	• Progressive problems with balance and walking • Slow movement, falling, body stiffness • Restricted eye movements
FTD with parkinsonism	• Movement problems similar to Parkinson's disease, such as slowed movement and stiffness • Changes in behavior or language
FTD with amyotrophic lateral sclerosis (FTD-ALS)	• Combination of FTD and ALS (Lou Gehrig's disease) • Changes in behavior and/or language • Muscle weakness and loss, fine jerks, wiggling in muscles

Source: National Institute on Aging (NIA)
https://www.nia.nih.gov/alzheimers/publication/frontotemporal-disorders/types-frontotemporal-disorders

28. Schizophrenia type not either in ICD10 or DS]

A. Simple
B. disorganized
C. residual
D. paranoid



All the options were subtypes in ICD10

No subtypes of Schizophrenia in ICD11

No subtypes in DSM 5 or ICD 11

29. Over the counter medicine cannot be taken wit

A. NSAD
B. Decongestant

C. **Antihistamine**

Explanation: **Decongestant** medications taken orally have the ability to increase blood pressure by themselves, especially in patients **with pre-existing** hypertension or those whose hypertension is poorly controlled.

Drugs to avoid with MAOI
SSRIs SNRIs TCAs (clomipramine & imipramine) Opioids Clomipramine St John's Wort Cold remedies (dextromethorphan and chlorpheniramine) Triptan migraine medications (e.g. sumatriptan)

30. Which cannot be paired with ACE inhibitor? **Lithium**

31. Try to shake hand, reach out, pulled back till 24 times still x shake hand.
    A. ambivalent
    B. **Ambitendency**

32. antipsychotic Drug least likely cause orthostatic hypotension:

    Aripiprazole – **A**misulpride – **A**senapine
    Anti-depressnats : Venlafaxine
    Alpha one adrenergic receptors antagonist and Alpha 2 (central) agonist cause the hypotension in antipsychotics, for example: Clonidine and Lefoxidine

Drug	Sedation	Weight gain	Diabetes	EPSE	Anti-cholinergic	Postural Hypotension	Prolactin elevation
Amisulpride	-	+	+	+	-	-	+++
Aripiprazole	-	+/-	-	+/-	-	-	-
Asenapine	+	+	+/-	+/-	-	-	+/-
Clozapine	+++	+++	+++	-	+++	+++	-
Flupentixol	+	++	+	++	++	+	+++
Fluphenazine	+	+	+	+++	++	+	+++
Haloperidol	+	+	+/-	+++	+	+	+++
Olanzapine	++	+++	+++	+/-	+	+	+
Paliperidone	+	++	+	+	+	++	+++
Pimozide	+	+	-	+	+	+	+++
Quetiapine	++	++	++	-	+	++	-
Risperidone	+	++	+	+	+	++	+++
Zuclopenthixol	++	++	+	++	++	+	+++

33. Drug causing - interstitial nephritis: **Lithium**

34. Pregnancy physiological changes

A. increase PH
B. increase gastric emptying
C. increase GFR

If both option (PH and GFR) there choose GFR

35. EMI: What do we monitor if giving methylphenidate Or atomoxetine

A. BP
B. LFT
C. Rp

36. EMI: What do we monitor if giving ~~Atomoxetine~~ agomeLatine

A. BP
B. LFT
C. Rp

37. which is a Screening tools?

A. Hamilton Depression Scale
B. positive negative Syndrome Scale
C. yale brown OCD
D. MADRS
E. EPDS

Screening tools include:

- The Edinburgh Postnatal Depression Scale,
- PHQ-9 and
- AUDIT

ondition / area	Self rated	Clinician rated
Depression / anxiety	BDI (Beck depression inventory) GDS (geriatric depression scale) ZSRDS (Zung self rated depression scale) HADS (Hospital Anxiety depression scale) EPDS (Edinburgh postnatal major depression scale) **PHQ-9 (9-question Patient Health Questionnaire)**	MADRS (Montgomery-Asberg depression rating scale) HAMD (Hamilton depression rating scale) BASDEC (Brief Assessment Schedule Depression Cards) HAMA (Hamilton anxiety rating scale) CSDD (Cornell Scale for Depression in Dementia)
Bipolar		YMRS (Young mania rating scale)
Schizophrenia		BPRS (Brief psychiatric rating scale) PANSS (Positive and negative syndrome scale)
OCD		Y-BOCS (Yale-Brown Obsessive compulsive scale)
Medication	LUNSERS (The Liverpool University Neuroleptic Side Effect Rating Scale) DESS (Discontinuation-Emergent Signs and Symptoms scale)	AIMS (Abnormal involuntary movement scale) SAS (Simpson-Angus scale)
Global function	GHQ (General health questionnaire)	CGI (Clinical global impression) GAF (Global assessment of functioning)
Dementia		CAMDEX (Cambridge Mental Disorders of the Elderly Examination)

38. which one scale use for quality of life

A. **SF36 (short form 36)**

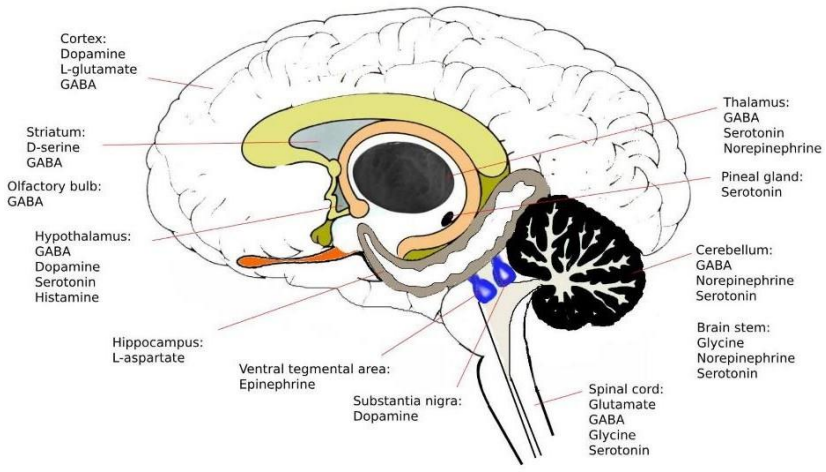
39. Which one full clinical assessment scale: no options

40. Reward pathways.

A. Meso-cortical
B. Mesolimbic
C. Nigrostriatal

41. Locus Cerulous secretes: **Noradrenergic**

Neurotransmitter	Site of synthesis	Increased in	Decreased in	Comments
Acetylcholine	Basal nucleus of Meynert	Parkinson's disease	Alzheimer's Huntington's	Involved in learning and memory
Dopamine	Ventral tegmentum Substantia Nigra pars compacta Arcuate nucleus (tubuloinfundibular pathway)	Huntington's disease Schizophrenia	Parkinson's Depression	Also known as prolactin-inhibiting factor (hence antipsychotics can increase prolactin secretion
GABA	Nucleus accumbens	-	Huntington's disease Anxiety	Major inhibitory neurotransmitter of the brain
Norepinephrine	Locus ceruleus	Anxiety	Depression	Involved in mood control and sleep-wake cycle
Serotonin	Raphe nucleus	-	Depression Anxiety	Serotonin re-uptake inhibitors (SSRIs) are effective in treating depression, anxiety disorders, obsessive compulsive disorder



42. APOE4 **hete**rozygous how many fold increase risk Alzheimer

A. One
B. **Three**
C. Six

D. Five
E. Nine

Homozygous = 10-30 times.

43. Constant, dream-like perceptions:

A. Oneiroid state,
B. catatonic schizophrenia
C. twilight

Oneiroid (dream-like) state can occur, and patients may experience visual hallucination. Transient catatonic features can be seen in all schizophrenia types

44. Recall false memory?

A. Retrospective falsification

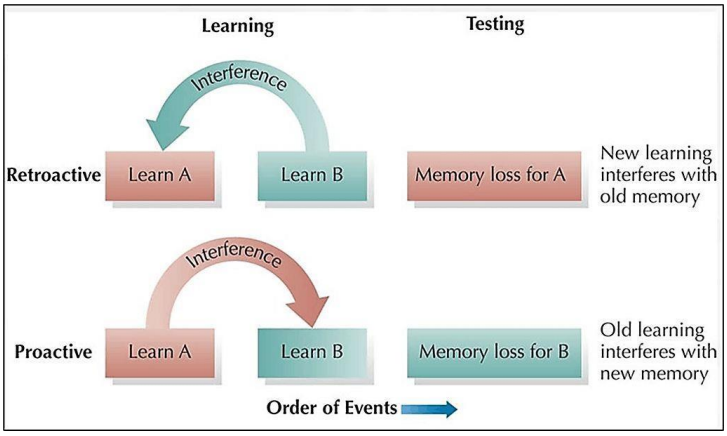
Retrospective Falsification: An unconscious distortion of past experiences to align to a person's need in the present. Retrospective falsification of memory, also known as memory distortion or memory malleability, refers to the phenomenon where our memories of past events can be altered or influenced by various factors, leading to inaccuracies or false recollections. Memory is not a perfect record of our experiences, but rather a reconstruction that can be influenced by a range of factors, including external suggestions, personal biases, and the passage of time.

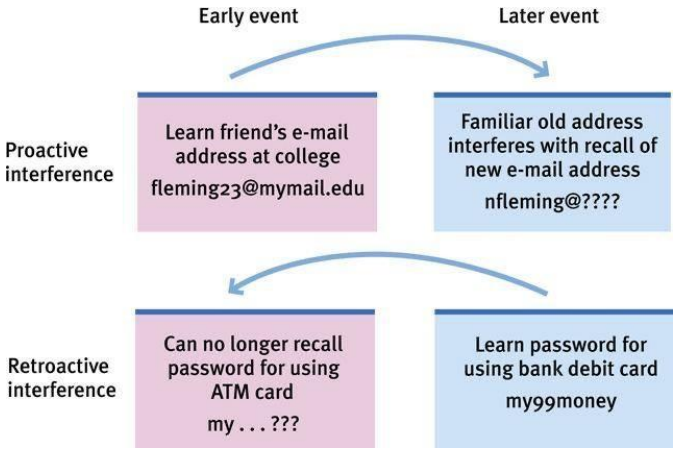
Proactive interference (pro=forward) occurs when you **cannot learn a new task** because of an old task that had been learnt. When what we already know interferes with what we are currently learning - **where old memories disrupt new memories.**

Retroactive interference (retro=backward) occurs when **you forget a previously learnt task** due to the learning of a new task. In other words, later learning interferes with earlier learning – **where new memories disrupt old memories.**

Retroactive is not being able to remember your old phone number because of your new UK number

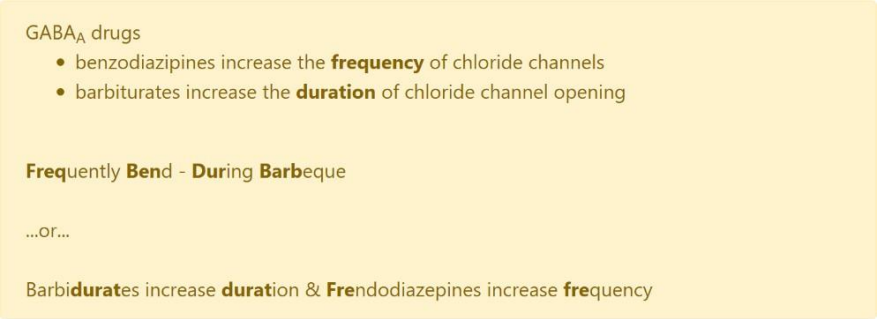
Proactive is not being able to memorise your new UK phone number because you keep mixing it with your old number.





45. Classes mechanism of action of propranolol, diazepam.

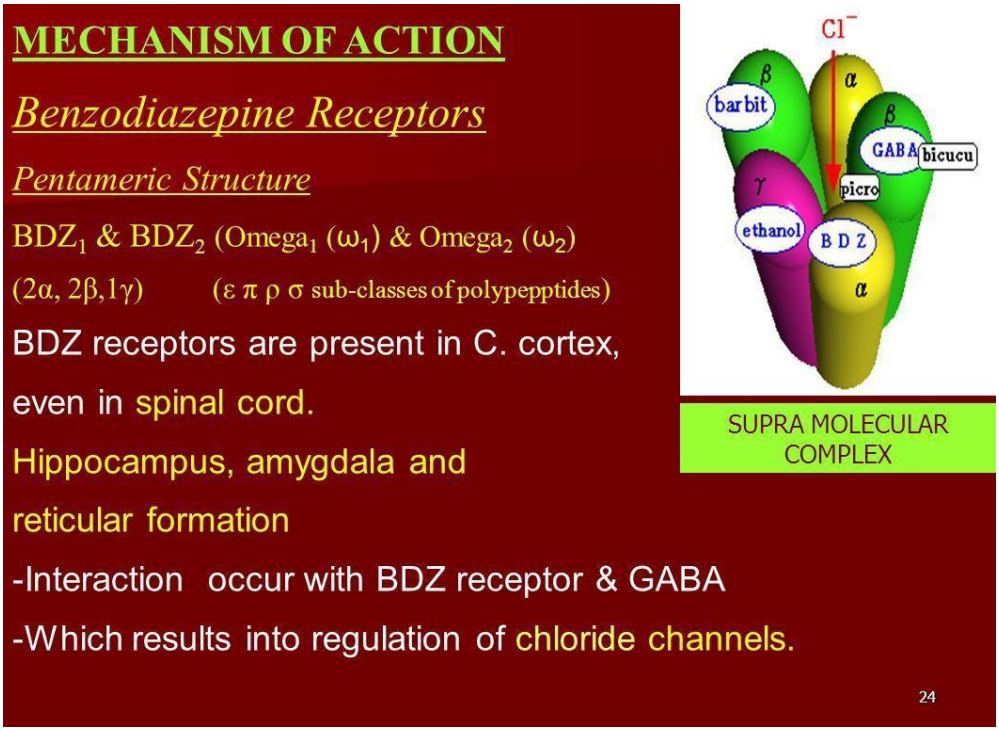
Propranolol: non-selective Beta1 and Beta2 Blocker
Diazepam: GABA-A – Omega 1,2,3



[SPMM] Benzodiazepines are important class of medications that have application as sedatives and hypnotics. Which of the following best describes their mechanism of action?

Select one:

A. GABA-B agonist
B. GABA-A positive allosteric modulator
C. GABA-A antagonist
D. GABA-B positive allosteric modulator
E. GABA-A agonist



46. EMI: Equivalent dose diazepam 10mg with :

- Chlordiazepoxide **25mg**
- Lorazepam **1m**

- **Diazepam 5 mg is appíoximately equivalent to** [Ashton, 2002c; BNF, 2023]**:**
  - Zolpidem 10 mg.
  - Zopiclone 7.5 mg.
  - l'emazepam 10 mg.
  - **Oxazepam 10 mg.**
  - **Nitíazepam 5 mg.**
  - Loímetazepam 0.5 mg to 1.0 mg.
  - **Loíazepam 0.5 mg.**
  - Lopíazolam 0.5 mg to 1.0 mg.
  - **Clonazepam 0.25 mg.**
  - Clobazam 10 mg.
  - **Chloídiazepoxide 12.5 mg.**
  - Alpíazolam 0.25 mg.

47. Drive theory of motivation based on:
A. Homeostasis

48. Eysenck's theory **[Extrovert decreased RAS]**

Eysenck's approach: Eysenck used second order analysis (orthogonal factor analysis) that identified small number of powerful independent factors. This method yielded 3 dimensional traits. These are neuroticism (vs. stability), psychoticism and extraversion (vs. introversion). **Eysenck's personality questionnaire** contains a lie scale. Biologically, extraversion is related to arousal and ascending reticular activating system; neuroticism may be related to sympathetic system reactivity. Introverts are said to be easily aroused. Hence they are also more easily conditionable than extraverts; this may explain why introverts stay indoors more often. Extraverts have low arousal state; hence they are not easily conditionable. Eysenck maintained that no fundamental discontinuity exists between normal and abnormal personalities (dimensional view)
PEN = **P**sychoticism, **E**xtraversion and **N**euroticism

Which of the following statements about Eysenck's three-factor theory is true?
Select one:
A. Extraversion is related to autonomic lability
B. Introversion is associated with autonomic lability
C. **Introverts have a more reactive reticular activating system**
D. Extraverts have a more reactive reticular activating system
E. Extraversion is related to cortisol levels

In introverts, the ascending reticular activating system (ARAS) is over-aroused and excited. In extraverts, RAS is under-aroused and inhibited.

49. AESOP study black has high fold of _____compared to white

Aetiology & Ethnicity in Schizophrenia and other Psychoses (AESOP) study was conducted in London, Bristol and Nottingham. It reported 2-fold higher rate of incidence of psychosis in London compared to the other 2 centers. Afro-Caribbeans had a **9-fold** increase in rates of

psychosis. In addition, minority ethnic groups had a far higher likelihood to be detained on first presentation, accessing health often via police than GPs.

[SPMM] The ÆSOP study was a multicentre case control study intended to investigate psychosis in the UK. Which of the following factors was observed by the ÆSOP study to be associated with a 9-fold increase in the risk of schizophrenia amongst immigrants?

A. ack of employment
B. Emigrating from the Caribbean
C. Emigrating from India
D. Separation from parent during childhood
E. Social isolation

The ÆSOP study was meant to investigate the increased rate of psychosis amongst AfricanCaribbean populations in the UK. It identified that immigrants of Afro-Caribbean origin were 9 times more likely to develop schizophrenia than the general population

Much research has focused upon risk factors for schizophrenia, including those related to an individual's surroundings or circumstances. Which of the following has been best established as increasing the risk of developing schizophrenia?
Select one:
A. Socioeconomic deprivation
B. Summer birth
C. Maternal alcohol use in pregnancy
D. **Migration**`
E. Rural living

50. EMI: Personality disorder

A. Avoids responsibility / **Consistent irresponsibility**, as indicated by repeated failure to sustain consistent work behavior or honor financial obligations → avoidant or **anti-social.**
B. Self-admiration → **Narcissists**
C. Transient → **psychosis**
D. Appears indifferent to the praise or criticism of others → **Schizoid**
E. Transient, stress-related paranoid ideation or severe dissociative symptoms." → **Borderline personality disorder**)

[SPMM] **EMI: What is the most likely diagnosis for each of the following patients?**

1. Tod is a 25-year-old male who works in a family-run cornershop. He forgot to lock-up after work, which resulted in the store being burgled and a financial loss for the business. Tod appeared to demonstrate no concern for his actions. This is not the first

2. Sabine is a 19-year-old fashion student. She holds the self-perception of being unique and superior to her peers. She told her supervisor that she felt she was exceptional and that was going to be the 'next Chanel', despite scoring a satisfactory grading

3. Seema is a 33-year-old doctor who irritates her friends by speaking over them at dinner parties. Her friends complain that she 'always has to be the centre of attention.'

4. Jason is a 45-year-old courier. He has a poor rapport with his work colleagues and tends to withdraw socially. Despite working for his company for fifteen years, he has never attended an out-of-work social gathering. Jason's **colleagues suspect that he is into some cult worship and carries unusual ideas.**

5. A 21-year-old history student, Raphael, met up with his dissertation supervisor, who was baffled to learn that Raphael did not appear to be remotely bothered with the feedback he received

Explanation:

1. In Antisocial personality disorder, individuals show a lack of concern for others, irresponsibility, irritability, aggression, inability to maintain enduring relationships and disregard and violation of others' rights.

2. In narcissistic personality disorder, grandiosity, lack of empathy and the need for admiration are common traits

3. In histrionic personality disorder, individuals show a high-degree of self-dramatization, shallow affect, egocentricity, craving attention, excitement and manipulative behaviour

4. Interpersonal discomfort with **peculiar ideas**, perceptions, **appearance and behaviour are common traits** in **schizotypal** personality disorder

5. Emotionally cold attitude, detachment, lack of interest in others, indifference to praise or criticism, excessive introspection and fantasy are common traits of schizoid personality disorder.

EMI: Disorder of thought:

A. fusion
B. omission
C. substitution
D. condensation
E. asyndesis
F. driveling

51. Replace major thought with minor: **substitution**

Explanation: Carl (not Kurt) Schneider proposed a different set of 5 elements of Formal Thought Disorder:

Substitution: one thought – often inappropriate, fills the gap between other appropriate, more consistent thoughts.
Omission: A chunk of thought goes missing from stream of conversation, patient being unaware – best analysed when written,

Fusion: various thoughts fuse together, leading to loss of goal direction.

Driveling: disordered intermixture of constituent parts of one complex thought

Derailment (aka entgleisen); In derailment normally flowing track of thoughts suddenly change. The determining tendency is preserved but is misdirected. Schneider also described desultory thinking, sometimes considered along with driveling. In desultory thinking, speech is grammatically correct but sudden ideas force their way in from time to time. Each one of these ideas is a simple thought that, if used at the right time would be quite appropriate.

Cameron proposed 4 characteristic formal thought disorders and one of them is Asyndesis:

Asyndesis: This refers to the lack of genuine causal links in speech. For example, 'I got up at eight this morning as well as few birds of different colours on the painting shrinking all the time to drop few coins. On the floor. All the time.'

[SPMM] Various forms of formal thought disorder have been described which each having distinct characteristics. Which of the following are characterised by a lack of causal links in the content of speech?

Select one:
A. Metonymy
B. Derailment
C. Drivelling
D. Asyndesis
E. Fusion

52. heterogenous thought interwoven each other to 1: **Fusion**

EMI on Social Group:

53. Individuals work better when they are being observed in a group than when they are alone - **Hawthorn**
54. The chances of being in a group increases when the members discuss and reach an understanding: **Group cohesion**

EMI on Groups

55. One person join group throw object to police: **deindividuation**

56. A group who have same crises show same behaviour
Emergent norm theory is the answer
57. Zopiclone acts on which receptor: **Alpha 1 subunit of omega 1 receptor**

58. Pharmacokinetic of elderly **hydrophilic** vs hydrophobic drugs? **Increased volume of distribution of lipophilic drugs**

59. Which is most consistently resilient in normal ageing:

A. Working memory
B. Selective attention
C. Executive function
D. Spatial orientation
E. information processing speed

Procedural or non-declarative memory functions appear to be well preserved over the lifespan. Semantic memory or knowledge is also relatively preserved compared to episodic memory, although semantic knowledge often has a late life decline. Processing speed tends to decline in later life. [SPMM]

In normal aging, working memory is often considered the most consistently resilient among the options you mentioned. While there can be some decline in working memory abilities with age, it tends to be less affected compared to selective attention and spatial orientation. Working memory is the system that allows us to temporarily hold and manipulate information in our minds, and it tends to maintain its functionality relatively well in many individuals as they age. However, it's essential to remember that individual differences exist, and some people may experience different patterns of cognitive changes in aging. [ChatGPT]

60. Neurotransmitter which has fast excitatory effect? **Glutamate (most abundant fast excitatory)**

We believe **Aspartate and nitric oxide are also fast excitatory**

61. Lurasidone:

A. Food calorie increases absorption
B. Food calorie decreases absorption
C. No effect of food
D. Used sublingually

N.B. Asenapine can be gived sublingual

62. Antipsychotic with duration >50H: **Aripiprazole**

EMI theme was **Side effects:**

63. change in antipsychotic following which wife kicks husband before going to sleep, **Akathisia**

[SPMM] Movement disorders
Choose the movement disorder that matches the description.
1. A man whno kicks his wife once as he falls asleep.
Explanation: : Sleep myoclonus/hypnic jerk is an involuntary twitch that occurs just as a person is beginning to fall asleep

[SPMM] Mrs Paula Taylor is a 42-year-old woman with a diagnosis of paranoid schizophrenia. She has recently changed her medication and her husband now reports that she often complains of an unquiet feeling in her legs and continuous movement of her legs preventing her from falling asleep

1) Among antipsychotics, olanzapine and quetiapine have been the most frequently reported agents associated with restless leg syndrome (RLS) symptoms. The mechanism of antipsychotic-induced RLS is suspected to be dopaminergic dysfunction. This may be attributed to the medications' variable affinity for the D2 receptor as well as serotonin receptor activity.

64. young female come to AE with restlessness, ankle clonus, **Serotonin syndrome**

65. Old man comes with poor sleep, daytime drowsiness takes more than 20 mins to get up from sleep: **Diazepam**

[SPMM] Mr Simon Hughes is a 70 old man who has attended his GP after taking medication for poor sleep but now reports daytime drowsiness and taking more time than usual to get up in the morning due to sedation

[SPMM] Diazepam generally has an elimination half-life of 20-50 hours. Older adults have more body fat, less body water, and less albumin than younger adults. This leads to an increased volume of distribution and a longer duration of action for some fat-soluble drugs (e.g. diazepam), higher concentrations of drugs at the site of action (e.g. digoxin), and a reduction in the amount of drug bound to albumin (increased amounts of active 'free drug'; e.g. warfarin, phenytoin) in the elderly.

66. SSRI questions: post MI, switch to mania:

A. Bupropion,
B. Fluvoxamine,
C. **Sertraline**,

67. Side effect of Venlafaxine**: Hypertension**

68. Duloxetine MOA: **Serotonin and noradrenaline reuptake inhibitor (SNRI)(Dual anti-depressant)**

69. Dorso lateral Pre-frontal cortex (DLPFC) used in:

A. Mentalization,
B. Execution,
C. Working memory

Important functions of the DLPFC are executive functions such as working memory, cognitive flexibility, planning, inhibition, and abstract reasoning. However, DLPFC is not solely responsible for executive functions. All complex mental activities require additional cortical and subcortical circuits to which the DLPFC is connected. The DLPFC is also the highest cortical region involved in planning, organizing and controlling movement.

The **dorsomedial prefrontal cortex** (dmPFC) is part of the mentalizing network, a set of brain regions consistently engaged in inferring mental states

70. Which of these is (? underactive) in autism spectrum disorder:

A. Amygdala
B. Insula
C. Orbito-parietal
D. Parieto-temporal

Autism shows **atrophy of cerebellar vermis** on scan
SPMM: Also **Decrease in volume of corpus callosum**, Increase caudate/amygdala

71. Antipsychotic with the lowest risk of hypotension - **Aripiprazole**

72. Cause of weight gain with antipsychotics - **5HT2C antagonism**

Explanation: Suggested mechanisms for antipsychotics weight gain include:

- 5HT2a and 5-HT2C antagonism
- D2 and D3 antagonism
- **H1** and M3 antagonism
- Hyperprolactinemnia
- Increased serum leptin (leading to leptin desensitisation)
- Ghrelin

5HT1A antagonism weight loss
Olanzapine causes weight gain via M3 antagonism

H1 is the most important for Antipsychotics

- **High risk of weight gain**: Clozapine and Olanzapine
- **Moderate risk of weight gain**: Quetiapine, Risperidone and Paliperidone

Antipsychotic	Risk of weight gain
Clozapine	High
Olanzapine	High
Chlorpromazine	Moderate
Quetiapine	Moderate
Risperidone	Moderate
Paliperidone	Moderate
Amisulpride	Low
Asenapine	Low
Aripiprazole	Low
Haloperidol	Low
Sulpiride	Low
Trifluoperazine	Low
Ziprasidone	Low
Lurasidone	Low

Miss Jane Sessions is a 28-year-old woman with a diagnosis of first episode psychosis. She has agreed to start an antipsychotic but is worried as she has heard that they are associated with weight gain. By which of the following mechanisms can antipsychotics lead to an increase in weight?

Select one:

A. 5HT2C agonism
B. H1 antagonism
C. 5HT1A antagonism
D. M1 antagonists
E. M3 agonists

Added during revision.
Right Answers.
Not sure about the answer.

Explanation: Weight gain associated with antipsychotics is thought to be linked to multiple mechanisms, including antagonism at H1 receptors, which increases appetite by impairing hypothalamic functioning and falsely indicating negative energy balance. It is also associated with 5HT2C antagonism, which delays the onset of satiety, and 5HT1A agonism, increasing food intake. Interesting, dopamine blockade may also increase weight by impairing limbic systems and leading to more reward-seeking behaviour, such as also increase weight by impairing limbic systems and leading to more reward-seeking behaviour, such as eating.

N.B. 5HT2C is the most important receptor causing weight gain in **anti-depressants**

73. Glutamate decarboxylase is involved in synthesis of? **GABA**

74. Which of the following is most common in human brain

A. D1
B. D2
C. D3
D. D4
E. D5

D4 is the lowest in the brain D4 is second most abundant in brain.

75. which of the following is a metabotropic receptor: **CB1**

Metabotropic receptors include **muscarinic acetylcholine, Metabotropic glutamate, GABA-B receptors as well as most serotonin receptors (except 5HT-3)**. The **endocannabinoid CB(1) or CB(2) receptors are also metabotropic,** though the endocannabinoids anandamide has been observed to have activating properties at certain voltage gated calcium ion channels.

76. Differences in mentalisation between adolescents and adults?

- **Area of Mentalisation in Adolescent is Dorsomedial prefrontal cortex**
- **Area more active in adolescent : DLPFC and then DMPFC**

77. What type of receptor do corticosteroids act on in the brain: **Nuclear receptors**

78. Disorders of consciousness:

A. Global impairment of consciousness,
B. disturbed sleep-wake cycle
C. Delirium

79. Questions on symptoms such as feeling of limbs twisting?
A. **paraschemazia** (body image distortion) This is kinaesthetic hallucinations

Paraschemazia is described as a feeling that parts of the body are distorted or twisted or that they are separated from the rest of the body. It can occur in association with hallucinogenic use, with an epileptic aura, and with migraine. [SPMM]

A young man attends A&E complaining that his leg feels like it's being twisted. It appears normal on examination. He is noted to have blood shot eyes and tar stained fingers.Options Akathisia,Reduplicative paramnesia, PARASCHEMEZIA,Anosognosia,Dysmorphic delusions-Mentor

80. Patient on depot antipsychotic (as well as an antidepressant) complains of back stiffness, takes 10 minutes to get out of bed in the morning:

A. **parkinsonism,**
B. Axial rigidity

What is axial rigidity? Rigidity is the term used to define the stiffness of an object. And Axial Rigidity is used to define the stiffness of the object

81. person throws stones – **Deindividualiziation**

The term "deindividualization" can be associated with situations where individuals in a group may engage in behaviors they might not normally do as individuals. For example, in a large and anonymous crowd, some people might feel less personally accountable for their actions. This can lead to behaviors like throwing stones or engaging in other forms of aggression, which they might not do if they were alone or in a smaller, more identifiable group.

It's worth noting that deindividualization doesn't justify or excuse harmful actions. Instead, it's a concept used to understand how group dynamics can influence individual behavior.

82. Daily activities: **MOHOST**, ?Vineland social maturity scale,

For assessment of activities of daily living, the OT department have many tools. The association don't endorse one specific tool above all in Alzheimer's but note that some aspects of each tool may need to be modified. From this list, the MOHOST is the only assessment of ADLs included

83. Lewy Body Dementia EEG findings ?

EEGs in LBD often reveal a **slowing of brain wave activity**, particularly in the alpha and theta frequency ranges. This slowing is consistent with the cognitive impairments seen in LBD.

FIRDA: Frontal intermittent rhythmic delta activity

84. **Blood supply of the brain:**
Inferior temporal region PCA
Medial temporal region ACA
Anterior temporal region

Anterior temporal region supplied by anterior temporal artery branch of middle cerebral artery

The anterior cerebral artery supplies the medial and superior strip of the lateral aspect of the cerebral cortex up to the parietal/occipital border.

The middle cerebral artery supplies most of the lateral aspect of the cerebral cortex. This includes the Broca's and Wernicke's areas in the dominant hemispheres.

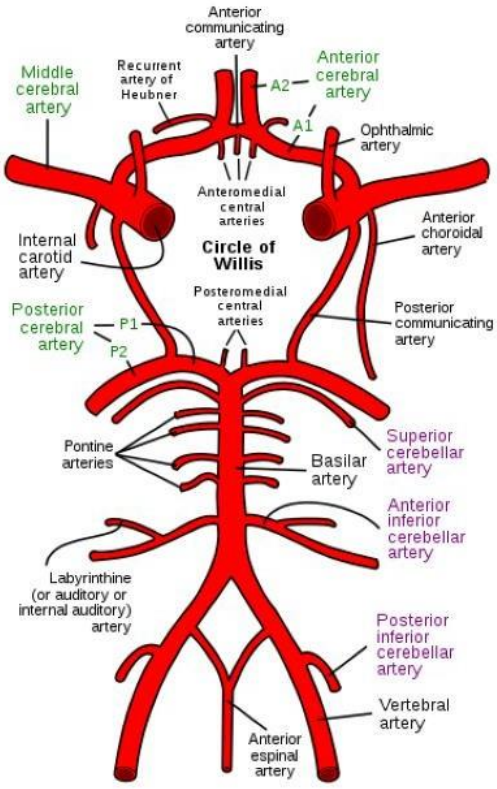
The posterior cerebral artery arises from basilar artery and supplies the inferomedial temporal lobe and the occipital lobe.

The medulla is supplied by posterior inferior cerebellar arteries and anterior spinal branches of vertebral arteries. Pons is supplied by the basilar artery that runs along the midline of the pons.

EEG-studies, DLB patients showed decreased reactivity of the background activity and pronounced slow-wave and paroxysmal activity, such as frontal intermittent rhythmic delta activity (FIRDA).

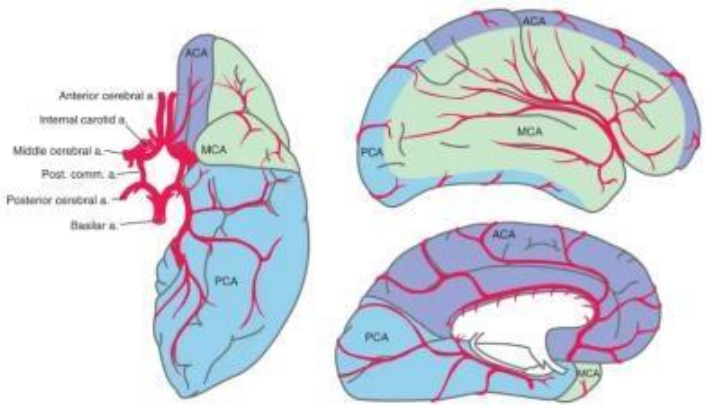
Added during revision.
Right Answers.
Not sure about the answer.

[Psych-Mentor] The body supplies blood to the brain via the internal carotid and vertebral arteries. These vessels come together to form a ring called the circle of Willis. The function of the circle of Willis is to provide a shunt system should any of the vessels become damaged.



Arising from the circle are the three main vessels that supply the brain with blood; the anterior cerebral artery (ACA), middle cerebral artery (MCA), and the posterior cerebral artery (PCA).

It is important to be able to recognise the consequences of the occlusion of the three main vessels supplying the brain. The image below provides a topographical indication of the areas supplied.



Anterior cerebral artery (ACA) occlusion
Occlusion of the anterior cerebral artery may result in the following defects (suspect if frontal signs)

- Hemiparesis of the contralateral foot and leg (more severely than the arm)
- Sensory loss of the contralateral foot and leg
- Transcortical motor aphasia (lack of fluency with intact comprehension and repetition)
- Abulia, disinhibition, executive dysfunction, anosognosia (lack of insight), emotional lability, frontal release signs (re-emergence of primitive reflexes)

If stroke occurs prior to the anterior communicating artery it is usually well tolerated secondary to collateral circulation

Middle cerebral artery (MCA) occlusion

This is the most common artery affected.

Occlusion of the middle cerebral artery may result in the following defects (suspect if motor and speech issues)

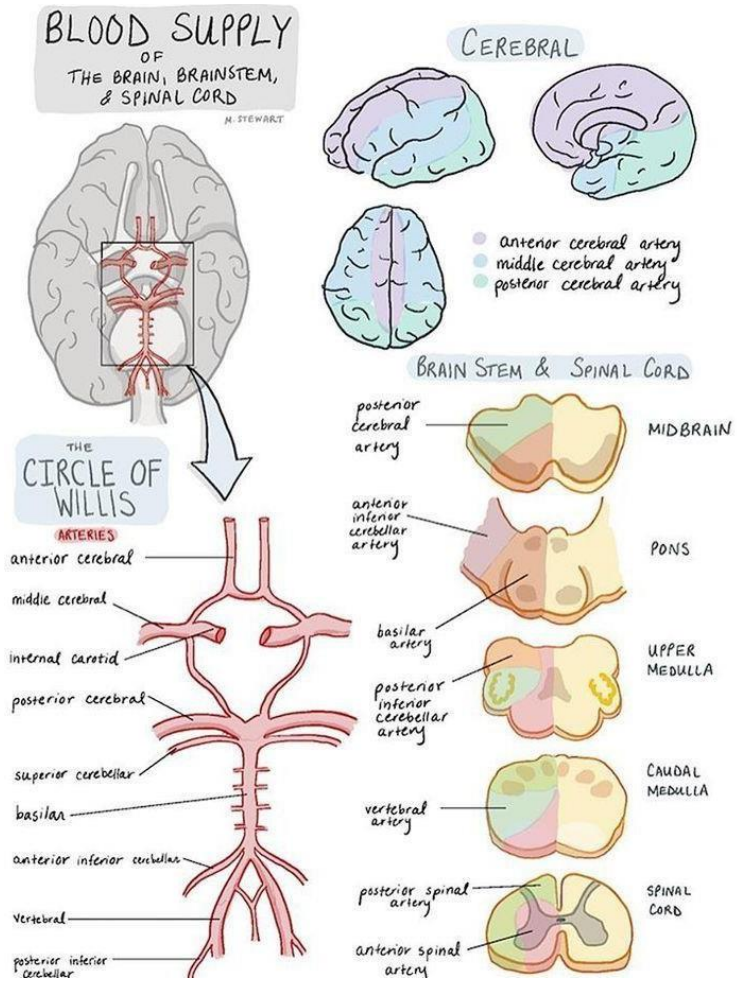
- Hemiparesis of the contralateral face and limbs
- Sensory loss of contralateral face and limbs
- Dysphasia / aphasia (when dominant hemisphere affected) (Broca and Wernicke)
- Contralateral neglect
- Homonymous hemianopia or quadrantanopia without macular sparing (macular vision not preserved)
- Dorsolateral prefrontal dysfunction

Posterior cerebral artery (PCA) occlusion

Occlusion of the posterior cerebral artery may result in the following defects (suspect if visual field issues)

- Alexia without agraphia (left PCA)

- Contralateral loss of pain and temperature sensation
- Contralateral hemianopia (with macular sparing)
- Prosopagnosia
- Ipsilateral cranial nerve defects (V, VIII, IX, X, & XI)
- Horner's syndrome



85. success of person internal attribute, Self-serving bias

Self-serving bias is an attributional bias that leads people to ascribe their own successes to internal personal factors, such as ability, and their failures to external situational causes, such as bad luck. This bias is a mechanism for individuals to protect or enhance their own self-esteem.

86. temperament poorly stable over developmental years,

In this schema, the choleric and melancholic are characterized by strong emotion (**unstable**) in contrast to the sanguine and phlegmatic that are characterized by weak emotions (**stable**).

Temperamental Differences

Infants vary in their autonomic reactivity and temperament. Chess and Thomas identified nine behavioral dimensions in which reliable differences among infants can be observed (Table 32-7).

In studies of temperament, most temperamental dimensions of individual children remain stable through adulthood, but some do not persist. This is likely due to genetic and environmental effects on personality. A complex interplay exists among the initial characteristics of infants, the mode of parental interactions, and children's subsequent behavior. Observations of the stability and plasticity of certain temperamental traits support the importance of interactions between genetic endowment (nature) and environmental experience (nurture) in behavior.

Table 32-7
Temperament—Newborn to 6 Years

Dimension	Description
Activity level	Percent of time spent in activities
Distractibility	Degree to which stimuli are allowed to alter behavior
Adaptability	Ease moving into change
Attention span	Amount of time spent on attending
Intensity	Energy level
Threshold of responsiveness	Intensity required for response
Quality of mood	Amount positive compared to amount negative behavior
Rhythmicity	Regulation of functions
Approach/withdrawal	Response to new situations

87. panic disorder = **pregabalin**
social phobia = **phenelzine**

Generalized anxiety disorder
Social anxiety and social phobia are same

[SPMM] Social anxiety is characterised by avoidance of and anxiety over social situations where an individual will be observed by others. Which of the following drugs has proven effective for social anxiety?

Select one:

A. Pregabalin
B. Quetiapine
C. Carbamazepine
D. Lamotrigine
E. Risperidone

Explanation: Some research has found Pregabalin to be effective in the treatment of social anxiety.

[SPMM] **Q.** Phenelzine has been shown to be superior to placebo in the treatment of

Select one:

A. Arachnophobia
B. Zoophobia
C. Agoraphobia
D. Social phobia
E. Obsessional symptoms

Explanation: The monoamine oxidase inhibitors (MAOIs) have shown superior efficacy to other antidepressants in the treatment of generalized social phobia. Phenelzine in particular, has extensive placebo controlled trial evidence in support of its use in social anxiety with nearly 67% patients showing significant improvement during acute treatment.

Box 3.8 Social phobia (social anxiety disorder)

First-line drug treatment[128] (In order of preference)	
SSRIs (Up to maximum licensed dose)	If no response to the first SSRI, try an alternative SSRI Supporting meta-analyses for fluvoxamine[129] and citalopram[130] Emerging data for vilazodone[131]
Venlafaxine modified release 75–225mg/day	
Second-line drug treatment (Less well tolerated or weaker evidence base, no order of preference)	
Olanzapine 5–20mg[132]	Few studies with antipsychotics. Most evidence with olanzapine
Atenolol 25–100mg/day	Reduces autonomic symptoms in performance situations[132]
Benzodiazepines Clonazepam 0.3–6mg/day[132] Sertraline plus clonazepam up to 3mg/day[133]	Benzodiazepines are helpful on PRN basis. Most evidence for treatment with clonazepam and bromazepam Switching an SSRI to venlafaxine no more effective than adding clonazepam to SSRI[133]
Gabapentin 900–3600mg/day[132]	
Levetiracetam 300–3000mg/day in divided doses[134]	

(Continued)

Box 3.8 (*Continued*)

Moclobemide 600mg/day in divided doses	Initiate at 300mg/day in divided doses. Moclobemide has a UK licence for Social Phobia. Recommended by NICE[128]
Phenelzine 15–90mg/day[135]	Avoidance of tyramine-rich food important Recommended by NICE[128]
Pregabalin 150–600mg/day[132]	600mg/day superior to placebo[132]
Experimental	
Ketamine 0.5mg.kg IV	One good RCT[136]
Topiramate 25–400mg/day[137]	Small open label study of 23 patients suggests efficacy but poorly tolerated
Valproate 500–2500mg/day[138]	Small open label study of 17 patients suggests efficacy

Box 3.5 Panic disorder

Crisis management

Drug	Comment
Benzodiazepines	Rapid effect although panic symptoms return quickly if the drug is withdrawn.[61] NICE do not recommend.[6] Cochrane lukewarm[62]
First-line drug treatment (In order of preference)[6,63]	
SSRIs (Up to maximum licensed dose)	Therapeutic effect can be delayed (this applies to all antidepressants[64]) and patients can experience an initial exacerbation of panic symptoms.[6] Use supported by Cochrane[65]
Venlafaxine MR 75–225mg[63]	Initiate at 37.5mg for 7 days
Second-line treatment (Less well tolerated or weak evidence base, no order of preference)	
Mirtazapine 15–60mg/day[66]	A meta-analysis suggests that mirtazapine does not help with panic symptoms but with the anxiety associated with this disorder.[63] Rather limited data overall[67]
Moclobemide 300–600mg/day[68]	One fixed-dose study of 450mg/day and one flexible-dose study suggest efficacy[68,69]
MAOIs Phenelzine 10–60mg/day[64]	No long-term studies, reserve for treatment-resistant cases due to poor tolerability[64]
Tricyclic antidepressants Clomipramine 25–250mg/day[64] Imipramine 25–300mg/day[64] Lofepramine 70–140mg/day in divided doses[70]	Start with a low dose and increase dose according to response and tolerability

88. HPA dysregulation relapse = ? Increased cortisol -major depression, PTSD, anxiety disorder, anorexia, schizophrenia, bipolar

89. minimum district trial before launching drug = preclinical, phase 0, 1-4 phases during med use

[SPMM] New medications must undergo testing prior to being approved for general use to ensure their efficacy and safety. How many stages of approval must a medication pass through prior to general use?

Select one:
A. 2
B. 5
C. 3
D. 4
E. 6

Explanation: There are **5** stages of clinical trials related to a new medication, with the last stage (phase 4) comes after general use begins (post-marketing). The stages are: Preclinical - Animal or in vitro studies to determine toxicity, pharmacokinetics and initial indications of efficacy.

- **Phase 0** - Small sample of human subjects exposed to low dose to further investigate pharmacokinetics .
- **Phase 1** - Larger sample of healthy volunteers with increasing doses used to determine appropriate ranges.
- **Phase 2** - Testing on group of patients to determine efficacy in clinical practice
- **Phase 3** - Testing on larger group of patients to establish efficacy, effectiveness and safety
- **Phase 4** - Post-marketing monitoring after the medication is released for general use.

Added during revision.
Right Answers.
Not sure about the answer.

Any drug must undergo the following steps before approval is granted by regulatory agencies such as FDA in the US and MHRA in the UK.

1. **Preclinical Animal Studies:** The pathway a drug must undergo before approval and marketing start with animal studies where the molecule is demonstrated to have specific actions. These extensive preclinical animal studies must be carried out at least on two different animal species. Mutagenicity, carcinogenicity and organ system toxicity are studies at this phase.

2. **Human trials – volunteers *phase 1*:** (safety) An investigational new drug then enters human trials. The first phase consists of determining if the drug is safe for human subjects. It is administered to a small group of volunteers and safety; tolerability and pharmacokinetics of the drug are ascertained. They are usually open or uncontrolled studies.

3. **Human trials – patients *phase 2*:** (effectiveness) In phase 2 effectiveness is studied in hundreds of patients with target disease in comparison to placebo to see if it works at all against the disease. The main methods are controlled trails or small randomized controlled trials.

4. **Human trials – patients *phase 3*:** (superiority or equivalence to standard looking for comparative efficacy) and tolerance profile) In phase 3 the drug undergoes extensive double-blind RCT to determine how well does it work and what are the common side effects.

5. **Human trials – post-marketing surveillance *phase 4***: Phase 4 takes place if all the previous phases are successfully crossed – the drug undergoes an approval process by regulatory bodies and post-marketing surveillance ensues. Less common side effects, which sometimes could lead to drug withdrawal, can be picked up when large scale prescribing takes place during post-marketing surveillance observations.

This is the end of this recall
I wish you all the best
Good luck

Not recall:

William = FISH

Cell types using glutamate as a NT = Granular and Pyramidal cells

Soon after start of antipsychotic, arm and neck muscles rigid and twisted in a painful way = Dystonia
Long hx of schizo, strange involuntary movts in face, particularly puckering and smacking of lips = TD
25yr old, IM haloperidol, unwell since, confused, sweating and rigid muscles = NMS

Perturb and measure techniques
Extracellular recording = Neuronal spiking, Local field potentials
Intracellular recording = Inhibitory Postsynaptic potential, Excitatory postsynaptic potential
Patch-clamp technique = specific ion channel function

How does an ion channel react to pharmacological manipulation of the neuron? = Patch clamp technique
How does a neurotransmitter affect the action potentials of a single neuron? = Intracellular recording
How is neuronal spiking activity coordinated = Extracellular recording

Animals Models
Schizophrenia = Maternal Immune activation, Subacute ketamine administration, Dysbindin knockout, Neonatal ventral hippocampal lesion
OCD = Aromatase knockout, Neonatal clomipramine
Depression = Chronic restraint, maternal deprivation, social defeat, olfactory bulbectomy
Bipolar = BDNF haploinsufficient mice, Circadian Rhythm Disruption, ERK1 knockout

Advanced Neurosciences – 2018
hiPSC – human induced pluripotent stem cells = have properties similar to embryonic stem cells ("brain grown in a dish")
Which mutations in Retts studied through hiPSC = **MECP2 Mutation**
Advantage of hiPSC over embryonic stem cells = **Ethically acceptable**
Homology directed repair = mechanism highly conserved across species and serves to repair damage to DNA.
gRNA refers to = guide RNAs
Endonucleases used for gene editing result in = double-strand break (DSB)
In scalp electromagnetic fields, what's most prominent source of recorded activity = post-synaptic electrical flow along the pyramidal dendrites
Technique requires a helium filled tank to enhance superconductivity = MEG (Magnetoencephalography)
True about MEG = signal is dominated by currents flowing tangential to the skull
Neuroscientific approach used to activate or inhibit specific neuronal projections in live animals using light exposure = Optogenetics
In Optogenetics, ChannelRhodopsin (pigment rhodopsin) = Blue light, Halorhodopsin and Archaerhodopsin = Yellow light.
Most common adverse effects of rTMS is = headaches (Scalp pain is commonest)
CLARITY is term used to describe = Hydrogel tissue clarification
DBS in resistant OCD targets = **anterior limb of the internal capsule**

DBS in Parkison = **Globus pallidus interna, subthalamic nucleus**
DBS in OCD = **ventral internal capsule and subthalamic nucleus**
DBS in Resistant depression = **Subcallosal cingulate cortex**
DBS = pacemaker implanted under collarbone and clicked during visits.
In resting state fMRI, the frequency of BOLD signal captured = <0.1Hz
tDCS refers to = Transcranial direct current stimulation (modulate neural activity in brain in a non-invasive manner)
when applied to DLPFC, can reduce GABA to glutamate conc ratio = intermitten theta-burst stimulation.
Gyrification refers to development process of = **Cortical folding**
Completed by age 2 in normally developing children = **Cortical gyrification**
Functional connectivity = **refers to correlation in the time course of activation between 2 brain regions**
Frequency of transcranial magnetic stimulation effective and used in depression = **10 Hz (targeted at DLPFC as against Anterior cingulate cortex in DBS)**
Frequency of stimulation considered to have excitatory effect on neuronal cells = **5-20Hz**
Estimating subject specific motor threshold before prescribing tx dose required in **= TMS**
Clozapine-N-oxide bind to **= M3 Muscarinic**
Endogenous process used to regulate gene expression and to degrade foreign mRNA from phages **= RNA interference**

Suicide types

Ego - I'm alone and the only one

Fatalistic the rules are killing me -fatal

Altruistic-im dying for the people -remember from altruisim you do it for others.

anomaly(anomic) - something is abnormal-lack of normal rules

Hope this will help with making some concepts.
🥳

Added during discussion-Green

Right answers – Yellow

Not sure about the answer- Red

Nov 2023 recall

1. EMI.

a. Test before starting Atomoxetine: Blood pressure done prior to initiation as it can increase the BP and heart rate.

b. Test before starting Agomelatine: Liver function test is done prior to starting Agomelatine as it can cause liver function abnormalities.

2. EMI- 7-8 rat model tests options were mentioned. Options were elevated maze, morris water maze, forced swim test….

a. Rat model for schizophrenia- Prepulse inhibition abnormalities . this measures sensorimotor gating

b. Rat model for Alzheimer's- Morris water maze. This measures learning and memory

c. Rat model for anxiety : Elevated maze test(Open field,light dark box)

d. Rat model for depression:_ Forced swim test (Tail suspension, Chronic mild stress model)

Others

a. Novel object recognition test: - Evaluates recognition memory and cognition	Rats exposed to two identical objects and then tested with one familiar and one novel object. Preference for the novel object indicates intact memory
b. Morris Water Maze: - Assesses spatial learning and memory	Rats placed in a water pool with a hidden platform, learning to navigate using spatial cues. Impaired performance suggests deficits in spatial memory
c. Pre-pulse inhibition test; -measures sensorimotor gating	Rats exposed to weak pre-pulse followed by a startle stimulus. Reduction in startle response due to pre-pulse reflects normal sensorimotor gating. Deficient in schizophrenia
d. Y- Maze test: - Evaluates spatial working memory	Rats freely explore a Y-shaped maze with three arms. Alternation between arms indicate intact spatial working memory
e. Social interaction test: - Assesses social behavior and cognition	Rats placed in a controlled environment with a conspecific. Duration and frequency of social interactions, such as sniffing and grooming are measures.
f. Passive avoidance test: - measures associative learning and memory	Rats learn to avoid a chamber where they previously received an aversive stimulus. Impaired retention suggests memory deficits.
g. Elevated Plus Maze: _ Evaluates anxiety like behavior	Rats placed on an elevated plus-shaped maze with open and enclosed arms. Time spent in open arms reflect anxiety levels
h. Radial Arm Maze:- Test spatial learning and memory	Rats navigate a maze with multiple arms, some containing rewards. Ability to remember visited arms indicates intact spatial memory.

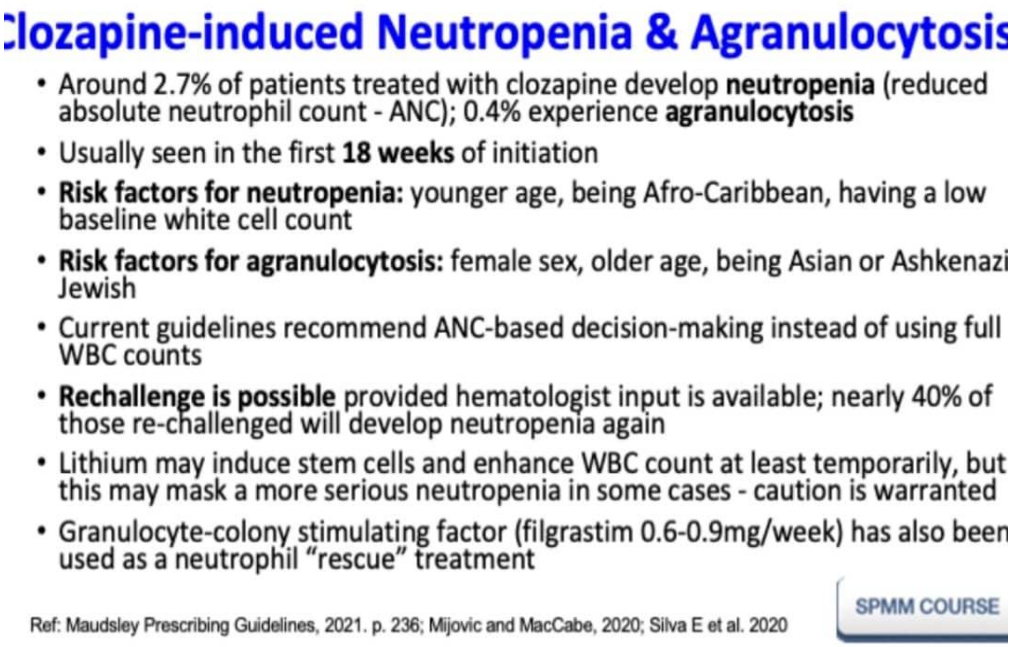
6. Clozapine and antibiotics. Antibiotic that interacts with clozapine? A. minocycline, b. metronidazole, c. penicillin, d. amoxicillin e. Erythromycin.

Erythromycin (macrolide) inhibits clozapine metabolism causing increased clozapine levels and potentially lead to toxicity/severe neutropenia

Other antibiotics:- Ciprofloxacin (a fluoroquinolone) Inhibits CYP1A2 enzyme which ↑clozapine levels, increasing the risk of toxicity.

Metronidazole-Inhibits CYP2C19 and CYP3A4 enzymes. May affect clozapine metabolism.

Rifampicin- Induces liver enzymes:- Decreases clozapine levels, reducing its efficacy.



7. Chess and Thomas childhood temperament:-Difficult child

Chess and Thomas identified 3 childhood temperament categories:_

They found that 65% of children fall into one of the following three groups:-

- Easy child (40%). These obtained high scores on regularity (degree to which they fall into a schedule), approachability, adaptability, mild intensity of reactivity, and predominance of positive mood.
- Difficult child (10%). These children were the polar opposite of easy children.
- Slow to warm up child (15%). These children are somewhere in the middle. They often have moderately negative reactions to new things but warm to the idea eventually.

8. Is Khat classified as an amphetamine?YES

Proposed question:-The drug 'Khat' (Catha edulis)has effects comparable to?

a. Cannabis
b. Benzodiazepines
c. Alcohol
d. Amphetamine
e. LSD

JOFAA
JEW, OLD, FEMALE, ASIAN
-AGRANULOCYTOSIS

YAWN
Young, afro, WBC count low-NUETROPENIA

Khat contains cathinone, a natural amphetamine like stimulant which produces similar effects to synthetic amphetamine.

9. Linguistic milestone stones which hold true for for 1,2 & 5 years:- Speech milestones.EMI a. 1 yr, b. 2yrs, c. 5yrs. Options were: babbles. 20,000 words, saying first words, Sentences made of at least 2-3words. saying 1-2 words. Passive speech. Uses basic grammar. Understands grammar, phoneme expansion, phoneme reduction, understanding basic grammar,

   0-6 months: Reacts to loud noises, vocalizes pleasure and displeasure (laughs, cries)
   6-12 months: Responds to own name, responds appropriately to angry and friendly tones, babbles* (says ba-ba-ba)
   12-18 months: Uses single words with meaning (referred to as the 'holophrastic' stage), understands simple instructions, tries to imitate simple words
   18-24 months: Has a vocabulary of up to 40-50 words (mainly nouns and pronunciation often unclear), knows simple body parts, understands simple verbs (eat, sleep), starts to combine words (e.g. More play) referred to as the telegraphic stage
   *24-36 months: Able to name a number of objects, vocabulary of 150-300 words, able to combine words into short sentences, begins to use prepositions, answers simple questions
   36-48 months: Has a vocabulary of 900-1000 words, begins to use plurals and past tense, handles three word sentences easily, groups objects such (e.g. Food, animals), strangers will understand most of what is said

10. question about the ideomotor and ideational apraxia. A person uses hammer at home, But fails to imitate the movement?. What type of agnosia is characterized by the inability to imitate movements despite functional use of object. Ideomotor Apraxia/,

    It occurs in extensive left hemisphere damage, Dementia, Delirium

    Ideational Apraxia is the inability to carryout set of instructions in the correct order.

11. Which SSRI has the most 5HT2C receptor antagonism. Fluoxetine.

    Fluoxetine exhibits 5HT2C antagonism in addition to its primary action as a serotonin reuptake inhibitor. Sertraline also has some degree of binding to 5-HT2C receptor.

    5HT2C antagonism:- weight gain

12. Early onset Alzheimer, At what age is early-onset Alzheimer's typically diagnosed:-

    a. before45
    b. before 50
    c. before 55
    d. before 60
    e. before65

13. Which gene is most particular for schizophrenia; a.Neuregulin 1 -NRG1. .b. DISC 1

Other Schizophrenia genes -Disc1 , DRD2 (dopamine D2 receptor), COMT (Catechol-O-Methyltransferase), CNTNAP2, MHC, ZNF804A, BDNF (Brain derived neurotrophic factor), G72/G30 gene complex. ON TRON examples given for schizophrenia susceptibility genes include- COMT, Neuregulin 1 and DTNBP1 (dysbindin)

14. Bias .Advantage of Birth prospective cohort study?.a.Hindsight bias b. minimizes Attrition/attribition bias c. Flexible study population d. minimizes recall bias e. Minimizes selection bia

    Birth prospective cohort studies collect data from the beginning of the child's life, minimizing recall compared to retrospective study.

15. Hindsight bias ?a. Arrogance. Hindsight bias is the tendency to perceive events as more predictable after they have occurred. The tendency to assume knowing the outcome of an event after the outcome has already been determined.

    b.Lack of insight. C.Absence of selection bias?

16. Child suffers from bullying at school. And not wanting to go to school anymore. Rephrased- a child's refusal to attend school due to bullying is an example of

a. Positive reinforcement
b. Negative reinforcement.
c. Positivepunishment
d. Negative punishment

Attending school was the behavior and he stops doing that because bullying was added

Punishment reduces behavior, Reinforcement increases behavior. Question may be different/incomplete but positive punishment fits best according to the information available.

17. Released from adrenal gland a. Cortisol b. ACTH

18. POMC : ACTH. What hormone is derived from proopiomelanocortin? ACTH

    Proopiomelanocortin is cleaved to produce various pepetide, including ACTH, which stimulates cortisol production in the adrenal glands.

19. Vortioxetine : 5HT1 Agonist. What type of receptor does vortioxetine primarily act on ? 5HT1 agonist.

    MOA of Vortioxetine:- It is an agonist of 5HT1A and partial agonist of 5H1B partial agonist. Antagonist at 5HT3 AND 5HT7.

20. Esketamine : NMDA. What receptor does esketamine primarily act on? NMDA receptor

    Esketamine is an NMDA receptor antagonist, which modulates glutamate. It is used in the treatment of depression.

21. Wilson's disease : chromosome 13. What chromosome is associated with Wilson's disease?

    Wilson's disease is caused by mutations in the ATP7B gene located on chromosome 13 leading to impaired copper metabolism and toxicity.

    Wilson's disease (otherwise known as hepatolenticular degeneration), is a disease of copper storage. Failure to excrete copper results in very high levels in the liver and brain. In the brain there is degeneration of the lenticular nucleus (putamen and globus pallidus).

1	Presenilin 2 (Alz), DISC-1 (Scz& BAD), DISC-2 (Scz), RGS-4 (Scz- interacts with neuregulin)
2	2q- assoc with autistic spectrum disorder
3	Von Hippel-Lindau disease
4	Huntington's Disease (CAG trinucleotide repeat-4p), Wolf-Hirschhorn 4p
5	Cri-du Chat (terminal deletion 5p)
6	Dysbindin(Scz-6p), PREP (mood disorder 6q)
7	7q assoc with ASD (autistic spectrum disorder), Williams syndrome (microdeletion)
8	Neuregulin-1 (Scz& BAD- 8p)
9	Tuberose Sclerosis I
10	Nothing
11	Brain-Derived Neurotrophic Factor (BAD-rapid cycling 11p)
12	Assoc with trait neuroticism
13	Patau's syndrome (trisomy), Wilson's disease (ATP7B), DAO-A (Sch& BAD- 13q)
14	Presenilin 1 (Alz D)
15	Prader-Willi (paternal microdeletion), Angelman's (maternal microdeletion)
16	Assoc with ASD (autistic spectrum disorder)-16p, Tuberose Sclerosis II
17	Familial frontotemporal dementia, Smith-Magenis-17p, Neurofibromatosis 1 -17q
18	Edward's syndrome (trisomy)
19	ApoE (Alz&Vasc), Notch 3 (CADASIL), assoc with ASD (autistic spectrum disorder)
20	PrP (inherited CJD)
21	Amyloid Precursor Protein (Alz), Down's Syndrome (trisomy)
22	Neurofibromatosis 2, DiGeorge Syndrome (velocardiofacial), COMT (Scz& BAD)
X/Y	Fragile X, Lesch-Nyan (purines), Turners XO, Klinefelter's XXY

22. DNA to RNA,thymine : In RNA, what base is thymine replaced by ? Uracil

    DNA strands, composed of a sugar phosphate backbone and attached bases, is connected to a complementary strand by hydrogen bonding (non- covalent) between paired bases, adenine (A) with thymine (T) and guanine (G) with cytosine (C). In RNA, thymine (T) is replaced by uracil (U). This can be remembered as GCAT.

23. EMI neuroanatomical :- What brain structure connecting mammillary body to hippocampus in the papez circuit (medial limbic circuit)Fornix: brain structure connected to septum by stria terminalis.?

septal nuclei

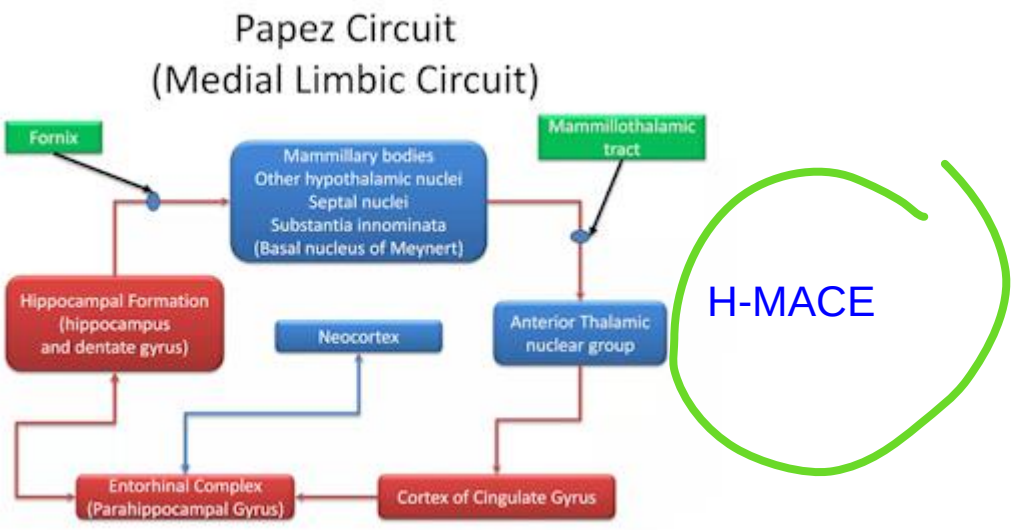
According to Papez, emotional information begins in the hippocampus and follows a specifictrajectory:Hippocampus â†’ Mamillary Bodies: Information passes from the hippocampus to the mamillary bodies via the fornix.

Mamillary Bodies â†’ Anterior Nucleus of the Thalamus: From the mamillary bodies, the circuit then progresses to the anterior nucleus of the thalamus via the mammillothalamictract.

Anterior Nucleus of the Thalamus â†’ Cingulate Cortex: Information is relayed to the cingulatecortex.

Cingulate Cortex â†’ Entorhinal Cortex: The cingulate cortex forwards the information to the entorhinal cortex, which acts as a hub for inputs and outputs to thehippocampus.

Entorhinal Cortex (contained within the parahippocampal gyrus) â†’ Hippocampus: Finally, the circuit closes by sending the information back to the hippocampus, thus completing the loop.



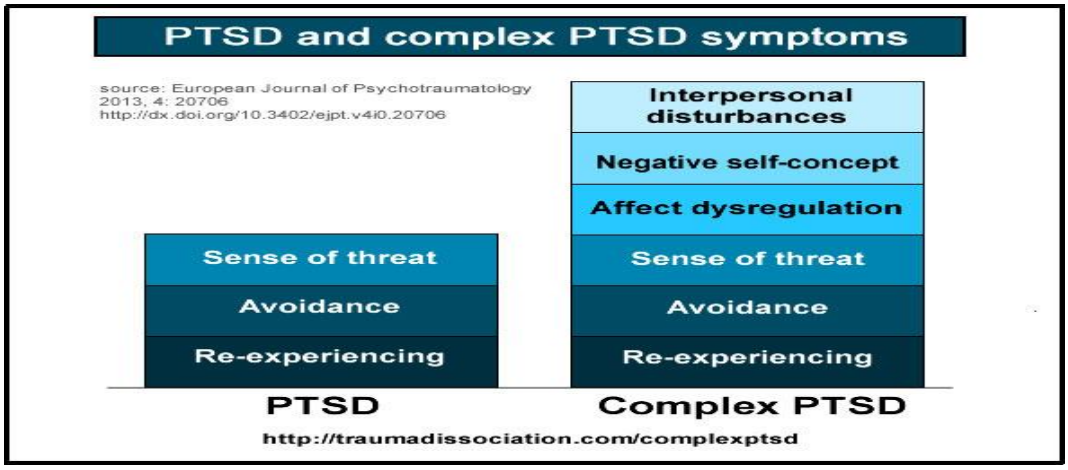
24. Q about biomarkers test in Alzhiemer:-What test to detect it? a. FISH; b. ELISA; c.Western blot ; d. Immunochemical stain ; Forgotten the last option (Options different from given in SPMM).

    Immunochemical stain detects Amyloid P, heparin sulphate

25. Difference between PTSD &complex PTSD . ; A.hopelessness b. Affect dyregulation c. prolonged repeated stressors

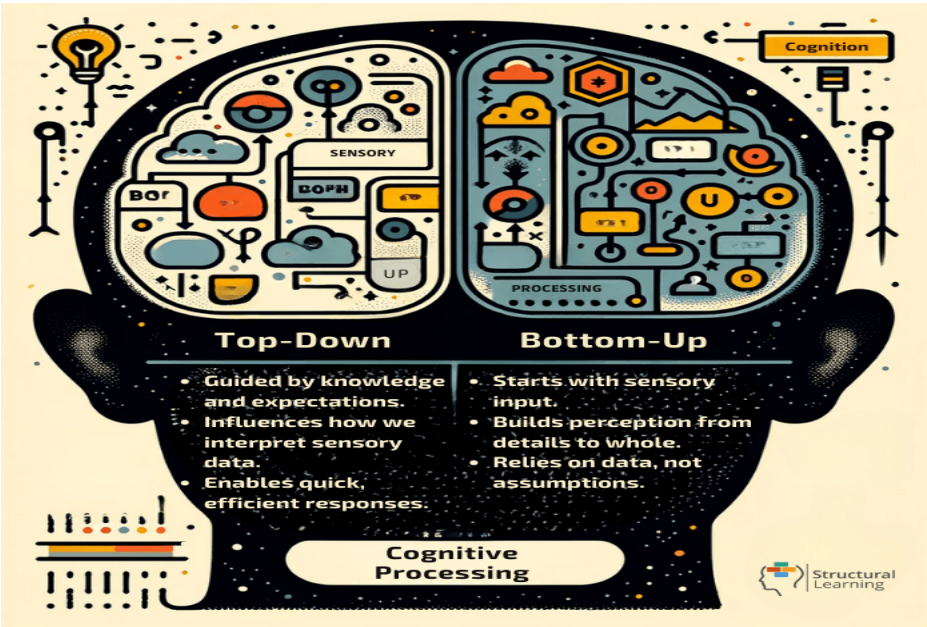
26.

Typical PTSD can arise after a traumatic episode, such as a car collision, an earthquake, or a sexual assault. It is generally related to a single traumatic event. Complex PTSD, on the other hand, is related to a series of traumatic events over time or one prolonged event.



27. Question about top-down theory? Top down . A. Perceptional and conceptual b. Cognitive and conceptual c. Perceptional and sensory processing

Top-down processing is the idea that our brains do not simply take in information from the environment and respond to it directly. Instead, our brains actively interpret and make sense of the information we receive based on what we already know.



28. Q about intellectual disabilities according to WHO? Which of these is true about Intellectual

disability according to WHO:

a. More in Asians,

b. More in women,

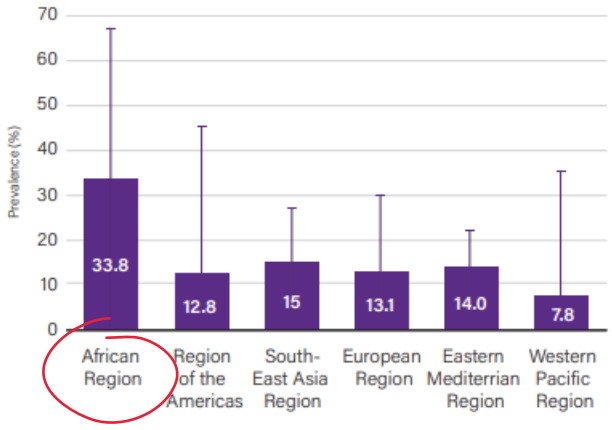
c. Most in ages 0 to 10yrs

D. More in older population .

E. More in females than males

The prevalence is higher among children (< 10 years), at 18.3% (ranging from 0.9% to 67.3%). Only two studies provide prevalence estimates specifically for adolescents (age range: 14–18 years), with prevalence estimates of 3.2% and 12.8% respectively. Developmental disabilities are more prevalent among boys than girls, with a median male-to-female prevalence ratio of 1.5 (range 0.9–2.0). The median of prevalence estimates of developmental disabilities in children and adolescents identified from review of peer-reviewed literature across WHO regions varies from 7.8% (ranging from 1.6% to 35.2%) in the Western Pacific Region to 12.8% (ranging from 1.5% to 44.0%) in the Region of the Americas, 13.1% (ranging from 3.0% to 30.0%) in the European Region, 14.0% (ranging from 8.4% to 21.0%) in the Eastern Mediterranean Region, 15.0% (ranging from 0.8% to 27.3%) in the South-East Asia region and 33.8% (ranging from 0.9% to 67.3%) in the African Region.

Fig. 2.1. Estimated median prevalence (and range) of developmental disabilities in children and adolescents in the six WHO regions, based on the peer-reviewed literature



29. Q on Lecanemab?It Is only given for patients with evidence of amyloid beta deposition.

A. Given intranasal

B. Prevent formation of tau & amyloid protein

C. Plaque formation is independent of dose

D. Given only in severe AD .

E. Given only if there is evidence of amyloid beta deposition

Lecanemab is a monoclonal antibody medication used for the treatment of Alzheimer's disease. Mechanism of action a humanized IgG1 monoclonal antibody that targets soluble aggregated Aβ proteins

Lecanemab is an amyloid beta-directed antibody.[2] It is given via intravenous infusion. Lecanemab may cause amyloid-related imaging abnormalities (ARIA).Treatment was initiated in participants whose disease was in the stage of mild cognitive impairment or mild dementia and who had confirmed presence of amyloid beta pathology.[3] Participants receiving the treatment showed significant dose- and time-dependent reduction of amyloid beta plaque: ARIA is often asymptomatic, but serious and life-threatening events rarely may occur. ARIA most commonly presents as temporary swelling of the brain that usually resolves over time and may be accompanied by small spots of bleeding in or on the surface of the brain, though some people may have symptoms such as headache, confusion, dizziness, vision changes, nausea and seizure.[3][4] Compared to placebo, all doses of the drug caused accelerated brain shrinkage. According to a phase III clinical trial (n = 1795), lecanemab has been associated with both ARIA-E (cerebral edema) and ARIA-H (microhaemorrhages, or small haemorrhages, and hemosiderosis) sub-types.[11] Mild to moderate infusion-related reactions may also occur

30. Criteria of depression found in ICD 11 and not in DSM 5? Hopelessness.
    - Anomic question about suicide- Emile Durkheim Anomie suicide cause : a. Lack of social norms b. Lack of support c. Intense dysfunction
31. Question about some results of CATIE study (had different options from previous recalls). CATIE STUDY :- The time to treatment discontinuation for all causes was longest with olanzapine, compared with quetiapine (p<. 001) and risperidone. Olanzapine was the answer

The NIMH-funded Clinical Antipsychotic Trials of Intervention Effectiveness (CATIE) Study was a nationwide clinical trial that compared the effectiveness of older (first available in the 1950s) and newer (available since the 1990s) antipsychotic medications used to treat schizophrenia.

CATIE is the largest, longest, and most comprehensive independent trial ever done to examine existing therapies for schizophrenia.

It consisted of two phases.

Phase I

Phase I compared old and new antipsychotics.

CATIE compared four of the newer medications to one another, and to an older medication. Participants in CATIE were followed for 18 months so that investigators could evaluate longer-term

patient outcomes. There were more than 1400 participants in the study. CATIE was conducted at many different treatment sites, broadly representative of the real life settings where patients receive their care (pragmatic study). The results from CATIE will be applicable to the wide range of people with schizophrenia in the United States.

In the first phase of CATIE, patients were randomly assigned to one of four newer, 'atypical' antipsychotics:

- olanzapine
- quetiapine
- risperidone
- ziprasidone

or to the older, 'typical' medication:

- perphenazine

Overall, the medications were comparably effective but were associated with high rates of discontinuation due to intolerable side-effects or failure to adequately control symptoms.

One new medication, olanzapine, was slightly better than the other drugs but also was associated with significant weight-gain as a side-effect. Patients assigned to olanzapine had the longest successful treatment time, and the fewest hospitalizations as a result of exacerbation of schizophrenia.

Surprisingly, the older, less expensive medication (perphenazine) used in the study generally performed as well as the four newer medications.

Contrary to expectations, movement side effects (rigidity, stiff movements, tremor, and muscle restlessness) primarily associated with the older medications were not seen more frequently with perphenazine than with the newer drugs.

Phase II

Phase II sought to provide guidance on which antipsychotic to try next if the first failed (either due to ineffectiveness or intolerability).

Participants who discontinued their first antipsychotic medication because of inadequate management of symptoms were encouraged to enter the efficacy (clozapine) pathway, while participants who discontinued their first treatment because of intolerable side effects were encouraged to enter the tolerability (ziprasidone) pathway.

Ziprasidone was added later in the trial.
Perphenezine had highest EPSE related discontinuation

Clozapine was remarkably effective and was substantially better than all the other atypical medications.

Metabolic syndrome

The CATIE study also looked at the risk of metabolic syndrome (MS) using the NCEP (US National Cholesterol Education Program Adult Treatment Panel) criteria which are as follows:

- Central obesity: waist circumference 102 cm or 40 inches (male), 88 cm or 36 inches(female)
- Dyslipidaemia: TG 1.7 mmol/L (150 mg/dl)
- Dyslipidaemia: HDL-C < 40 mg/dL (male), < 50 mg/dL (female)
- Blood pressure 130/85 mmHg
- Fasting plasma glucose 6.1 mmol/L (110 mg/dl)

The prevalence of MS at baseline in the CATIE group was 40.9%. By gender this was 51.6% in females and 36% in males. Male patients were twice as likely to have MS than matched controls, and female patients were three times as likely compared to matched controls.

32. Question about when a person own beliefs/views become the same as the group' *internalizing* group ideas. question regarding conformity??. A. Compliance b. Obedience. C. Identification d. internalization e. individuation

The term groupthink was coined by Irving Janis in 1972.
Groupthink is a phenomenon that occurs within a group of people, in which the desire for harmony or conformity in the group results in an incorrect or deviant decision-making outcome.
Group members try to minimise conflict and reach a consensus decision without critical evaluation of alternative ideas or viewpoints.
Several conditions must take place for groupthink to occur . The group must be isolated from outside influences. Group loyalty prevents individuals from raising controversial issues of alternative solutions. There is a loss of individual creativity and independent thinking. The group experiences the 'illusion of invulnerability,' an inflated certainty that the right decision has been made. Typically the group is under a high level of pressure to make a decision, and it lacks an impartial leader. These factors can lead a group to make a catastrophically bad decision.
Symptoms of groupthink include:



- Illusion of invulnerability
- Belief in inherent morality of the group
- Collective rationalisation
- Stereotypes of 'out-groups'
- Self-censorship
- Illusion of unanimity - assuming that people who remain silent agree with the majority view
- Direct pressure on dissenters
- Self-appointed mind guards - protecting the group from outside info that challenges the groups view of itself

nonconscious process characteristics, beliefs, feelings, or attitudes of other individuals or groups are assimilated into the self and adopted as one's own(Internalization)

Several interventions can help reduce groupthink:

- Leaders should allow each member to challenge ideas and present objections
- Members should talk about and solicit ideas with people outside the group
- Outside Experts should be invite to attend meetings
- Avoid expressing opinions about the preferred outcome
- Assign 'Devil's Advocate' at all meetings to challenge any and all ideas

Groupshift

Group shift is the phenomenon in which the initial positions of individual members of a group are exaggerated toward a more extreme position. For example, when people are in groups, they assess risk differently from when they are alone. In the group, they are likely to make riskier decisions as the shared risk makes the individual risk seem to be less.

Deindividuation

Deindividuation is a concept in social psychology that is generally thought of as the loss of self-awareness in groups. Theories of deindividuation propose that it is a psychological state of decreased self-evaluation and a decreased evaluation of apprehension causing abnormal collective behaviour, such as violent crowds and lynch mobs. Conversely it also explains people's tendency to donate more readily to charity when in groups.

33. Q about prion disease has pulvinar sign- . In which condition is bilateral FLAIR hyperintensities of the posterior thalamic nucleus seen? Variant CJD

    Pulvinar sign can also be seen in status epilepticus, acute disseminated encephalomyelitis (ADEM), Bilateral thalamic infarcts

34. Psilocybin;-5HT2A (Hallucinogen). What is the mechanism of action of psilocybin?

    Psilocybin has agonist action at 5HT2A, 5HT2C and low affinity at 5HT1.

    Other 5HT drugs – LSD, DMT, Mescalline

35. Something about development. chimera(?) Which of the following is the term given to an organism that has more than one genetically-distinct population of cells that arise from a single zygote?

    a. Plasmid
    b. Chimera
    c. Morula
    d. Mosaic
    e. Clone

    Mosaics and chimeras are animals that have more than one genetically-distinct population of cells

In mosaics, the genetically different cell types all arise from a single zygote, whereas chimeras genetically different cell types originate from more than one zygote.

A plasmid is an autonomously replicating, extrachromosomal circular DNA molecule, distinct from the normal bacterial genome and nonessential for cell survival under non selective conditions. Some plasmids are capable of integrating into the host genome. A number of artificially constructed plasmids are used as cloning vectors

A clone is an organism that is generally identical to the unit or individual from which it was derived.

36. Which of the following is a long-acting antagonist targeting opioid receptors.

    a. Methadone (full agonist on mu receptor)
    b. Naloxone (short acting)
    c. Naltrexone
    d. Buprenorphine (partial agonist on mu receptor)

37. Partial opioid agonist: Buprenorphine. A drug user is trying to stop heroin and to be prescribed medication to help reduce cravings and it is decided a partial agonist is preferred. Of the following medications. Which will be prescribed ?

    a. Methadone
    b. Buprenorphine
    c. Naloxone
    d. Naltrexone

38. Seeing everything brighter than normal a. Visual Hyperesthesia. brighter color usual than normal

    Hyperesthesia refers to increased sensitivity of any of your senses, such as sight, sound, touch, and smell. It can affect just one or all of the senses. Often, the heightening of an individual sense is referred to by a separate name

39. GABA mechanism ; what Gaba receptor subunit causes sedation effect - Gaba A ionotropic and Gaba B metabotropic

    a. Alpha 1
    b. alpha 2
    c. alpha 3
    d. alpha 4
    e. alpha 5

Alpha 1 subunit receptor cause amnesia, sedation and anticonvulsant effects of BZDS

More than 90% of the GABAA receptors in the brain contain α1-, α2-, and α3-subunits (6), and GABAA receptors containing the α1-subunit (α1GABAA receptors) recently have been implicated in the sedative effects of benzodiazepines, whereas GABAA receptors containing α2- and α3-subunits (α2GABAA and α3GABAA receptors)

40. EMI:- separation anxiety, selective mutism. What makes you distinguish it from other anxieties. Some answers fear of Social meetings, fear of dying, onset over 5yo, lasting at least 2years, different fears in other options

Childhood fear (normal development)

Fear and anxiety are a normal part of child development. Normal fear and anxiety in youths follows a clear developmental pattern described by Marks in 1987 as the â€˜ontogenetic paradeâ€™, which refers to the rise and disappearance of certain fears in a predictable sequence during children,s development.

In preschool years, children's fears are focused on imaginary creatures (e.g. ghosts, witches), animals, strangers, and their natural environment (e.g, the dark, thunderstorms). In middle childhood, fears of physical danger, bodily injury, and school performance become more prominent, whereas during adolescence, youths more often report fears about social evaluations and interactions.

Fear(s)	Age of prominence (Gullone, 1999)
Strangers, heights, and emergence of separation anxiety	Towards the end of the first year
Being alone, the dark, animals	Preschool (3-4)
Bodily injury, illness, social situations, supernatural phenomenon, failure and criticism	School years (4-12)
Death, economic and political concerns	Begins early and persists through to late adolescence

41. Which of the following has an Autosomal Dominant inheritance pattern:-

a. Prader Willi syndrome (PWS) . life expectancy is 30years

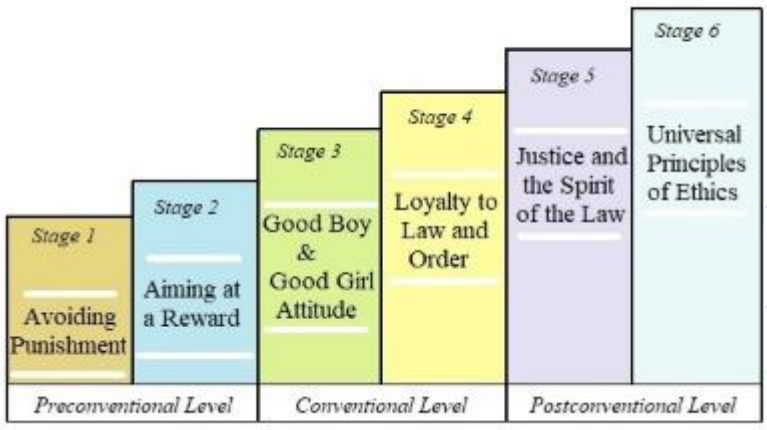
b. DiGeorge's syndrome

42. EMI on ethics - options- utilitarianism, paternalism, justice, Non maleficence, deontology, beneficence
    a. Dr decided to not prescribe clozapine due to agranulocytosis - non maleficence.
    b. Dr found out pt has a lung cancer, will not tell him, will monitor with chest xray - paternalism
    c. Group therapy is chosen over individual therapy to see overall benefit for everyone:- In between Justice or utilitarianism.
43. EMI of Freud, Kohlberg.

Free association allowing the patient to discuss thoughts, dreams, memoriesregardless of coherency with no pause in a stream of words. Freud believed this non-linear way of thinking would lead to the unconscious mind.

Sigmund Freud - Free association, transference, ego, super-ego, id, eros, thanatos, defense mechanisms, oedipus Complex, the unconscious.

KOHLBERG



Transference of feelings or desires for another person to an entirely different person. One example of transference is when you observe characteristics of your father in a new boss. You attribute fatherly feelings to this new boss. They can be good or bad

Instrumental relativist orientation/self interest orientation/ aiming at reward all mean the same stage

Good boy and good girl attitude/ interpersonal accord and conformity.

44. Personality disorder EMI:- Antisocial and Schizotypal , and schizoid. Someone Callous & reckless in his job, not responsible and does not listen to authority : Antisocial PD; Individual who is Indifferent to criticism : schizoid PD

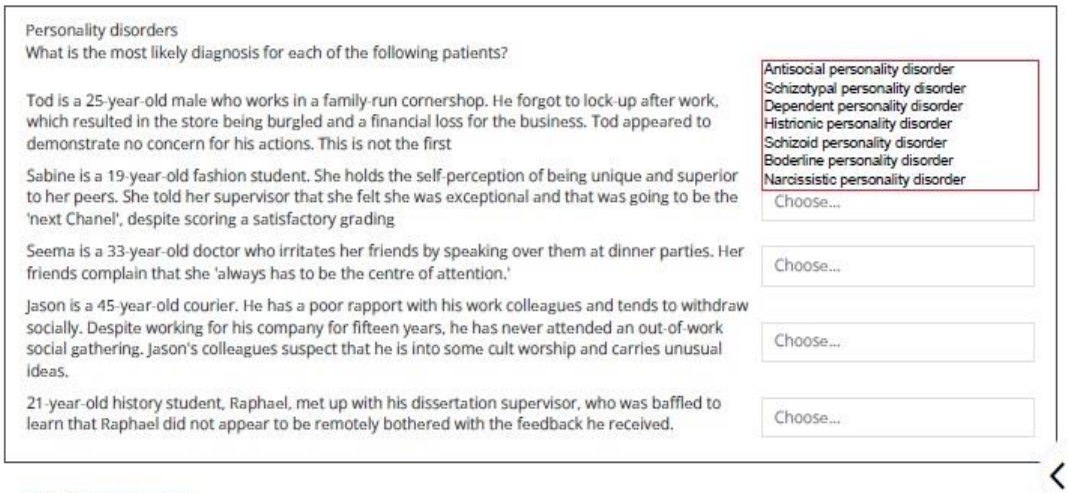
Obsessive-compulsive (anankastic):-
Avoidant:-
Schizoid:- Reserved and indifferent to criticism
Paranoid:-
Borderline:- EUPD, difficulty maintaining r/s
Antisocial:- No regard for law and order
Dependent:-
Schizotypal:- Magical thinking, day dreaming

3) In histrionic personality disorder, individuals show a high-degree of self-dramatization, shallow affect, egocentricity, craving attention, excitement and manipulative behaviour

4) Interpersonal discomfort with peculiar ideas, perceptions, appearance and behaviour are common traits in schizotypal personality disorder

5) Emotionally cold attitude, detachment, lack of interest in others, indifference to praise or criticism, excessive introspection and fantasy are common traits of schizoid personality disorder.

The correct answer is: Tod is a 25-year-old male who works in a family-run cornershop. He forgot to lock-up after work, which resulted in the store being burgled and a financial loss for the business. Tod appeared to demonstrate no concern for his actions. This is not the first → Antisocial personality disorder, Sabine is a 19-year-old fashion student. She holds the self-perception of being unique and superior to her peers. She told her supervisor that she felt she was exceptional and that was going to be the 'next Chanel', despite scoring a satisfactory grading → Narcissistic personality disorder, Seema is a 33-year-old doctor who irritates her friends by speaking over them at dinner parties. Her friends complain that she 'always has to be the centre of attention.' → Histrionic personality disorder, Jason is a 45-year-old courier. He has a poor rapport with his work colleagues and tends to withdraw socially. Despite working for his company for fifteen years, he has never attended an out-of-work social gathering. Jason's colleagues suspect that he is into some cult worship and carries unusual ideas. → Schizotypal personality disorder, 21-year-old history student, Raphael, met up with his dissertation supervisor, who was baffled to learn that Raphael did not appear to be remotely bothered with the feedback he received. → Schizoid personality disorder



Your answer is incorrect.

1) In Antisocial personality disorder, individuals show a lack of concern for others, irresponsibility, irritability, aggression, inability to maintain enduring relationships and disregard and violation of others' rights

2) In narcissistic personality disorder, grandiosity, lack of empathy and the need for admiration are common traits

45. EMI options were- Onerioid, delirium, stupor, coma, agitation,
    a. Attention disturbances with sleep wake cycle - delirium
    b. twilight state - stupor

Dissociative states can lead to twilight state hence dissociative stupor might be a possible answer.

c. Dreamlike state- onerioid

Oneiroid syndrome (OS), or dream-like fantastic delusional derangement of consciousness, is characterized by a kaleidoscopic quality of psychopathological experiences, wherein reality, illusions and hallucinations are merged into one. It is typically accompanied by motor and, in particular, catatonic disturbance

46. Stria terminalis something. Connects Amygdala to hypothalamus controlling autonomic, neuroendocrine.behavioral responses.

The **stria terminaliages** (or **terminal stria**) is a structure in the brain consisting of a band of fibers running along the lateral margin of the ventricular surface of the thalamus. Serving as a major output pathway of the amygdala, the stria terminalis runs from its centromedial division to the ventromedial nucleus of the hypothalamus.

47. Holmes and Rahe:-Friendship breakdown,

Which was not included in holmes rahe scales. A. Divorce b. Friendship Breakdown c. Death of a spouse d. Child leaving home e. pregnancy

In 1967, Holmes and Rahe studied the effect of stress on illness. They surveyed more than 5,000 medical patients and asked them to say whether they had experience any of a series of 43 life events in the previous two years.

Each event, called a Life Change Unit (LCU), had a different 'weight' for stress. The more events the patient added up, the higher the score. The higher the score, and the larger the weight of each event, the more likely the patient was to become ill.

The first 10 of the 43 life events are listed below with their values.

1. Death of spouse	100	
2	Divorce	73
3	Marital separation	65
4	Jail term	63
5	Death of a close family member	63
6	Personal illness	53
7	Marriage	50
8	Being fired from work	47
9	Marital reconciliation	45
10	Retirement	45

48. Pregnancy Physiological changes:- lithium plasma level decreases in pregnancy, why? Increased gfr

Maternal physiological changes begin early in gestation and are most pronounced in the third trimester. Plasma volume ↑↑and Renal blood flow and the glomerular filtration rate increase,

Ph increased

leading to enhanced elimination of some drugs (GFR increases 50%).↑↑

The volume of distribution increases for lipophilic drugs↑↑

The concentration of plasma albumin ↓↓ decreases.. The total plasma concentrations of albumin-bound drugs decrease ↓↓ during pregnancy due to hemodilution. Elevation of progesterone levels leads to a delayed gastric emptying and reduced small intestine motility. This may lead to altered absorption of drugs, but the effects on total bioavailability are probably relatively small.

The activity of hepatic drug-metabolizing enzymes changes during pregnancy because estrogens and progesterone induce some CYP enzymes and inhibit others.

49. Autoimmune encephalitis. 25year male with mild cognitive deficits and some movement abnormality?LgI1.

Antibodies against neuronal cell surface or synaptic receptors: These are usually treatable and include the most common subtype, anti-N-methyl-D-aspartate receptor (NMDAR) encephalitis. Others include antibodies against the Î±-amino-3-hydroxy-5-methyl-4-isoxazolepropionic acid receptor (AMPAR), gamma-aminobutyric acid receptors (GABAR), and Leucine-rich glioma-inactivated protein 1 (LGI1), amongst others.
Antibodies against intracellular antigens: These tend to be less responsive to treatment and are often associated with tumours. Examples include anti-Hu, anti-Ma, and anti-GAD antibodies.

50. Catatonia- a. Passivity, b. Negativitism

Catatonia is the psychiatric syndrome of disturbed motor functions amid disturbances in mood and thought.

Behaviour	Description
Stupor	combination of immobility and mutism
Posturing	maintaining the same posture for long periods. A classic example is the 'crucifix'. An extreme version of posturing is catalepsy
Waxy flexibility (cerea flexibilitas)	patient can be positioned in uncomfortable postures, which are maintained for a considerable period of time
Negativism (Gegenhalten)	patient resists the attempts of the examiner to move parts of their body and, according to the original definition, the resistance offered is exactly equal to the strength applied
Automatic obedience	exaggerated cooperation, automatically obeying every instruction of the examiner
Mitmachen	a form of automatic obedience whereby the body of the patient can be put into any posture, even if the patient is given instructions to resist. The body part immediately returns to the original position once the force is removed (unlike in waxy flexibility)
Mitgehen	an extreme form of mitmachen in which the examiner is able to move the patient's body with the slightest touch (anglepoise lamp sign), like in mitmachen, the body part returns to the original position once the force is removed

*Perseveration can be motor or verbal, you keep on doing the act or talking about something even when the purpose is served.

Behaviour	Description
Ambitendency	the patient alternates between resistance to and cooperation with the examiner's instructions; for example, when asked to shake hands, the patient repeatedly extends and withdraws the hand
Psychological pillow	the patient assumes a reclining posture, with their head a few inches above the bed surface, and is able to maintain this position for prolonged periods
Forced grasping	the patient forcibly and repeatedly grasps the examiner's hand when offered
Obstruction	the patient stops suddenly in the course of a movement and is generally unable to give a reason. This appears to be the motor counterpart of thought block
Echopraxia	the patient imitates the actions of the interviewer
Aversion	the patient turns away from the examiner when addressed
Mannerisms	these are repetitive, goal-directed movements (e.g. Saluting a door or combing hair without a brush in the hand) General Manager
Stereotypies	these are repetitive, regular movements that are not goal-directed (e.g. rocking)
Motor perseveration	the patient persists with a particular movement that has lost its initial relevance
Echolalia	involuntary repetition of language and sounds
Echopraxia	involuntary imitation or repetition of someone else's actions
Logorrhoea	incoherent talkativeness
Logoclonia	meaningless repetition of a syllable, particularly an end syllable of a word
Verbigeration	meaningless, repetition of words or phrases

Palilalia is repeating last uttered word.

Look

51. Stereotyped non goal directed movement after rx?? The one with some bizarre options . A. Perseveration b. Punding
52. Mdma act on which receptors - serotonin vesicles. It blocks SERT
53. Nmda act on what receptors - glutamate receptor
54. first order kinetics ans ? A. Same fraction of drug is elimated Constant fraction b. Constant amount of drug eliminated per unit time.
55. Question about Sigmund Freud stages:- Freud. Which of the following is correctly matched A. 3 -5 -phallic b. 11 above -genital c.5-7 -latency

Old age parrots love grapes- Psychosexual development

Stage	Range	Description
Oral	0-18 months	Pleasure and needs explored though sucking, swallowing and biting
Anal	18-36 months	Pleasure and need explore through bowel and bladder, through both elimination and retention
Phallic	3-5 years	During this phase, boys pass through the Oedipal complex and girls the Electra complex. Girls are also said to develop penis envy in this stage

Punding:stereotypical motor behavior, intense fascination with repetitive, excessive and non-goal oriented handling, and examining of objects.It is generally precipitated by drug use and alleviated by a reduction or change

Stage	Range	Description
Latency	5-puberty	The sexual drive remains latent during this stage
Genital	Puberty-adulthood	During this stage a person achieves independence from their parents and forms intimate relationships with others

56. Modafinil and student usage.

Modafinil is a psychostimulant that inhibits dopamine reuptake by binding to dopamine reuptake pump, which enhances wakefulness, attention, and vigilance. It is similar to amphetamines apart from that it tends to lack the euphoric effects, does not seem to be associated with dependence or tolerance, and does not tend to precipitate psychosis .It is licensed for narcolepsy, obstructive sleep apnea, and chronic shift work, and is also suggested as an adjunctive treatment for depression (in the Maudsley). More recently it has also been used as a 'smart drug' in the hope that it will boost normal cognitive functioning.

57. Piaget friend got hurt question?

Understands pain of freind -concrete operational

58. Which of the following is a central H1 antagonist drug. Central histamine acting activity. Mirtzapine

59. Antidepressant	Mechanism
Mirtazapine	Noradrenaline and serotonin specific antidepressant (NaSSa) 5HT2 antagonist, 5HT3 antagonist, H1 antagonist, alpha 1 and alpha 2 antagonist, moderate muscarinic antagonist
enlafaxine	Serotonin and noradrenaline reuptake inhibitor (SNRI)
Duloxetine	Serotonin and noradrenaline reuptake inhibitor (SNRI)
Reboxetine	Noradrenaline reuptake inhibitor (NaRI)
St John's Wort	Weak MAOI and weak SNRI (also considered by some to be a weak SSRI)
Trazodone	SARI (Serotonin antagonist and reuptake inhibitor). Blocks SERT (the serotonin reuptake pump) and also antagonist at 5HT2A and 5HT2C. SERT blockage at 5HT1A is thought to be responsible for the antidepressant effect of SSRI but they also block SERT at 5HT2A and 5HT2C (thought to cause the side effects). So trazadone has simultaneous effect of blocking SERT and also antagonising those receptors responsible for the SSRI side-effects such as insomnia, sexual dysfunction, and anxiety
Moclobemide	Reversible inhibitor of monoamine oxidase type A
Agomelatine	Melatonergic agonist (MT1 and MT2 receptors) and 5-HT2C antagonist
Bupropion (Zyban)	Norepinephrine-dopamine reuptake inhibitor (NDRI), and nicotinic acetylcholine receptor antagonist

Antidementia drug	Mechanism
Donepezil	Reversible acetylcholinesterase inhibitor
Tacrine	Reversible acetylcholinesterase inhibitor
Rivastigmine	Reversible acetylcholinesterase inhibitor and butyrylcholinesterase inhibitor
Galantamine	Selective, competitive and reversible inhibitor of acetylcholinesterase. In addition, galantamine

Antidementia drug	Mechanism
	enhances the intrinsic action of acetylcholine on nicotinic receptors, probably through binding to an allosteric site of the receptor.
Memantine	Non-competitive NMDA antagonist

Note: Acetylcholine is at significantly lower levels in Alzheimer's. Acetylcholine is metabolised by cholinesterase of which there are two forms, acetylcholinesterase and butyrylcholinesterase. Rivastigmine, but not its competitors, inhibits both acetylcholinesterase and butyrylcholinesterase.

Mood stabiliser / anti epileptic	Mechanism
Valproate	GABA modulation (increases GABA in the brain), sodium channel inhibition and NMDA antagonist
Gabapentin	Binds to the alpha-2-delta subunit of voltage-gated calcium channels, which reduces the release of certain neurotransmitters, including glutamate and substance P, and can help reduce neuronal excitability.
Topiramate	GABA modulator, NMDA antagonist, and Na channel stabiliser
Carbamazepine	Stabilises Na channels
Phenytoin	Stabilises Na channels
Lamotrigine	NMDA receptor modulation, stabilises Na channels and blocks calcium channels, some GABA modulation
Pregabalin	Binds to the alpha-2-delta subunit of voltage-gated calcium channels, which reduces the release of certain neurotransmitters, including glutamate and substance P, and can help reduce neuronal excitability.

Anxiolytic/hypnotic drug	Mechanism
Benzodiazepines	Enhance action of GABA by action of non-specific positive allosteric modulation of GABA-A, in the absence of GABA they have no effect on GABA functioning
Z-drugs	Enhance action of GABA by action of specific positive allosteric modulation of GABA-A (high affinity for Î±1-containing GABA-A receptors)
Buspirone	5HT1A partial agonist
Flumazenil	Benzodiazepine antagonist

Antipsychotic	Mechanism
Amisulpride	D2/D3 selective antagonist (low affinity selective antagonist of 'D2 like' receptors (D2=D3>D4) it has little affinity for D1 like' receptors (D1 and D5) or non dopaminergic receptors (serotonin, histamine, adrenergic, and cholinergic)
Olanzapine	Dopamine and 5HT2 antagonism
Aripiprazole	Partial agonist at 5HT1A and D2, and 5HT2A antagonist

Antipsychotic	Mechanism
Clozapine	Dopamine and serotonin antagonist (5HT2A) High affinity for D4, (to a lesser extent D1, D2, D3, D5) which may explain less EPSEs 5-HT 1A partial agonist which may explain reduction of negative symptoms Muscarinic M1, M2, M3, M5, histamine, and alpha-1 adrenergic-receptor antagonist Norclozapine, the metabolite of clozapine, actively works on the M1 and M4 receptors.
Lurasidone	Dopamine D2, 5-HT2A, 5-HT7, alpha2A- and alpha2C- adrenoceptor antagonist, and is a partial agonist at 5-HT1a receptors. Lurasidone does not bind to histaminergic or muscarinic receptors.

ADHD drug	Mechanism
Dexamfetamine	Inhibiting DA (dopamine) and NA (noradrenaline) reuptake
Methylphenidate	Inhibiting DA and NA reuptake
Lisdexamfetamine	Prodrug, absorbed by GI tract, converted to dexamfetamine which inhibits the reuptake of NA and DA. Converted by enzymes in red blood cells into dexamfetamine and L-lysine (amino acid). The activation of the prodrug results in longer action and a reduced abuse potential
Atomoxetine	Targets the NET (noradrenaline transporter), inhibiting the reuptake of NA, therefore increasing NA levels in the synaptic cleft. Selective NA reuptake inhibitor
Guanfacine	Selective agonist of Î±2A-adrenergic receptors. Binds to postsynaptic Î±2A-adrenergic receptors, mimicking NA
Clonidine	Agonist of Î±2-adrenergic receptors, mimic NA

Other drug	Mechanism
Lofexidine	Alpha 2 agonist
Buprenorphine	Partial agonist at the mu-opioid receptor
Naloxone	Pure opioid antagonist (will reverse mu, delta, and kappa)
Naltrexone	Long acting (compared to naloxone) opioid antagonist, targeting endogenous opioid receptors (mu, delta, and kappa receptors), but particularly the mu-opioid receptor
Atomoxetine	Noradrenaline reuptake inhibitor
Varenicline (Champix)	Nicotinic receptor partial agonist
Disulfiram	Binds irreversibly to enzyme aldehyde dehydrogenase (ALDH) and causes acetaldehyde to accumulate following ingestion of alcohol. Elevated blood acetaldehyde causes facial flushing, severe headache, palpitations, tachycardia, hypertension, respiratory distress, nausea and vomiting. These symptoms commence within 15â€“30 min of ingesting alcohol and persist for several hours
Acamprosate	NMDA glutamate receptor antagonist and positive allosteric modulator of GABAA receptors (effect on both receptors is indirect)
Selegiline	selective, irreversible inhibition of monoamine oxidase type B (also inhibits MAO-A at higher doses
Sildenafil	Inhibits cGMP-specific phosphodiesterase type 5 (PDE5)

60. Treatment of narcolepsy which is an H3 blocker? a. Pitolisant, b. Clomipramine c. sodium oxybate d. modafinil

H3 antagonist used in Narcolepsy. Ans. Pitolisant.-pitolisant:(not modafinil) , Votioxetine 5HT1A agonist, Doluxetine- peripheral neuropathy

61. Heinz and stealing a. Not stealing due to punishment, b. Stealing to save a life - post convention; Not stealing because it's wrong and will be in trouble – Not stealing due to punishment. preconventional- Obedience and punishment.

    Preconventional. To avoid punishment is pre-conventional. Not stealing because its wrong is conventional

62. EMI. Wilson chromosome 13- Huntington chromosome 4 - frontotemporal chromosome 17- dementia chromosomes

NA, DA reuptake inhibitor(amphetamine, cocaine , ecstasy, modafinil)

63. Modafinil isn't antihistamine. Refer to q58

64. Ubiquitin 11. Genes for Ubiquitin are located on chromosome 1,2, 13 and 17 Chromosome 17

65. Question on interleukins in a. OCD- IL 6 IL-1ß, IL-6, TNF-α and IL17 high in OCD

    b. Generalized anxiety disorder

    Options are IL 8,10, 6

66. 5HT2 antagonist? . mostly 5HT1A therapeutic effect of antidepressant. 5HT2A stimulation causes Anorgasmia (5HT2A AND 5HT2C stimulation due to their role in inhibiting dopamine release), hallucinogenic effect, serotonin syndrome and therapeutic effect of antipsychotics. 5HT2C antagonism cause weight gain. 5HT3 stimulation cause nausea and antagonism has antiemetic effect. Anti psychotics are 5Ht2a antagonist

    5-HT1A receptors when stimulated can potentially enhance sexual behaviours as they play a role in inhibiting serotonin release, thus allowing dopamine to act freely. However, in the context of elevated serotonin levels due to SSRIs, the role of 5-HT1A receptors in mitigating anorgasmia might be subdued.

67. The defense mechanism to remove threatening thoughts? Superstition? Repression, Denial. Not sure Anxiety provoking or unpleasant thoughts

68. Relationship between Depression and Infectious disease. Which condition when treated can be complicated with depression. A. Hiv b. neurosyphilis c. leptospira d. Hep C e. Tuberculosis

Hep C as they asked which treatment cause depression commonly in Hep C we give interferons which cause depression

69. Long-term Anti-retroviral therapy (ART)changes ? HIV patient with *cognitive disorder* on long

Post hep C treatment causes depression.

time ART; what *neuropathological* change you expect? HIV BRAIN SCAN a. Increased cd4 b. Decreased cd8 c. Microglial activation d. Neuronal death. Ans. Evidence of Neuronal loss.

So the question had the specifier for the option to choose and that was neuropathology so the option was most likely the neuronal loss from the long term ART

70. Which of the following medication has the most affinity to SERT? A . Tramadol, b. Mirtazapine c. lithium d. Trazodone

71. Reciprocal reinforcement. ??

Reciprocal inhibition is a technique that aims to replace an undesired response (e.g., anxiety) with a desired one by counterconditioning. It relies on the gradual substitution of a response that is incompatible with the original one and is potent enough to neutralize the anxiety-evoking power of the stimulus

72. The receptor for memory loss in benzos:-Alpha 1,2,3...? .

73. Clinical scenario of patient presenting with Auditory hallucination. What part of the brain is responsible for presentation-: Superior temporal

74. Default mode network - parietal

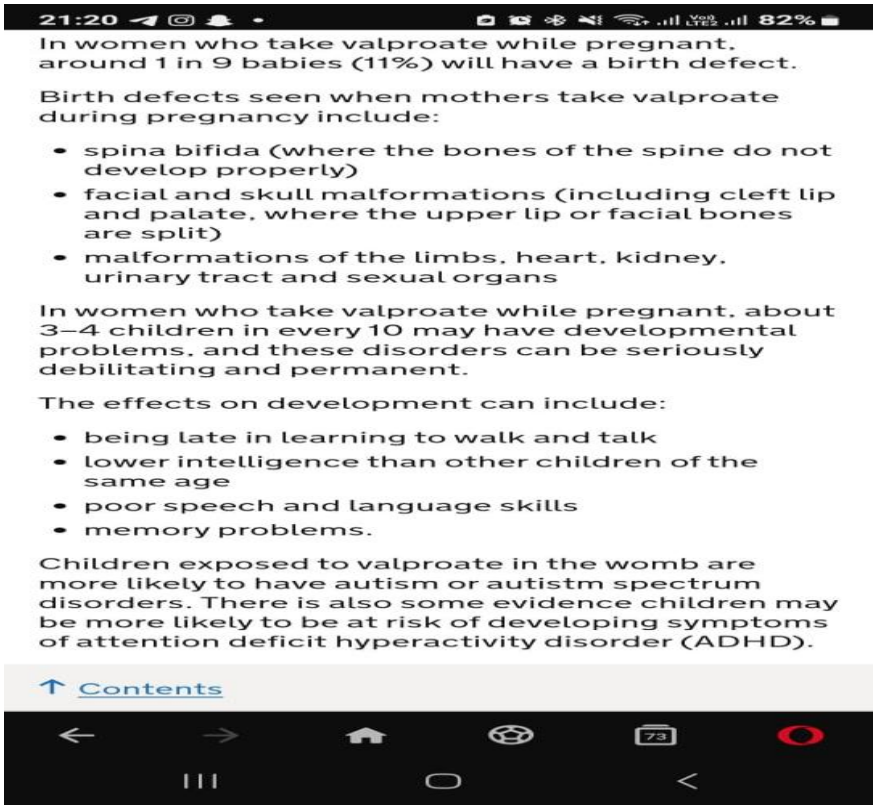
75. The percentage of sodium valproate with congenital anomaly. A. 8-12, b. 4-7 c. about 10%

Approximately 10% of children exposed to valproate monotherapy during pregnancy suffer from congenital malformations (the general population risk is 2-3%). The risk is dose dependent.

Autism due to

- Valproate
- Carbamazepine

Congenital Anomaly 1 to 10%
Developmental Delay 31 to 40 %



76. Visuospatial construction? Ray-Osterrieth Complex Figure

77. Set shifting- Wisconsin Card Sort Test. Test for abstract and cognition- WCST

Wisconsin is a test for executive function

78. Benzo occupies all w subunits of GABA A, Z drugs occupies one w subunit. Benzo increases duration and frequency of Cl channel so. Barbiturates increase duration only.; Antidote for Gaba A receptor drugs a. Flumazenil

79. Screening tool for post-natal depression- Edinburg post natal depression scale.

80. Scale for Activites of daily living??- Rivermead scale

81. EMI- Cck- released by intestine, increase meal frequency, and reduce portions?? b. Neuropeptide y - hypothalamus. (this increases appetite) C.Leptin- adipose tissues. d. ACTH - Pro opiomelanocortin-

82. Alzheimer's with better response to rivastigmine? The options were wild butyryl, k butyryl, ps1 , homozygous apoe, Heterozygous apoe, Wild buteryl, K buteryl

Wild type butyryl Cholinesterase

83. Williams genetics. =- AD autosomal dominant

84. Which anti dementia drug should be used with caution in stage 4 CKD ? a. Memantine Refer to Q61 for other antidementia medications. B. Rivastigmine

85. 7 year old Erikson psychosocial development? Industry vs inferiority

The Rivermead Mobility Index (RMI) is a hierarchical mobility scale used in neurological rehabilitation.

86. Regarding Tetrabenazine, Which of the following is incorrect-a. interacts with CYP 2D6 b. half life 10-12 hours, c. VMAT 1 inhibitor, d. Parkinsonism is a side effect e. orthostatic hypotension

    Tetrabenazine acts primarily as a reversible high-affinity inhibitor of mono-amine uptake into granular vesicles of presynaptic neurons by binding selectively to VMAT-2.

87. Q was about studies on adverse childhood events. Brain changes in childhood abuse include the following except ? a. Reduced size of amygdala b. reduced hippocampus c. reduced anterior cingulate cortex (ACC) d. Adult with smaller prefrontal cortex

88. Question on pathological gambling brain changes;:Increased activity of nucleus accumbens

    - a question about MRI findings in pathological gambling . A. Increased metabolism in DLPFC , b. Decreased Metabolism in DLPFC, c. Increased Metabolism in Orbitofrontal cortex , d. Decreased metabolism in orbitofrontal Corterx, e. increased nucleus accumbens .Phasic dopaminergic something cycle with gambling? Increased Orbitofrontal

89. Sequential order of planets. Which memory is used. SEMANTIC MEMORY, Explicit Memory-declarative

90. Resting tremor: Parkinsonism 4 to 6 hz , Essential is 2 to 3hz

    Resting tremor in Parkinson's disease is often 4-7hz and often asymmetrical and stops with volitional movement.

    Essential tremor is 6-12hz, only present when exerting effort.

91. Dopamine agonist in parkinsonism, with non goalrelated

92. Prenatal stress associated mechanism ? a. DNA methylation

93. Which of the following is a risk factor for QTc prolongation. ? A. Females b. Tachy, c. hypercalcemia

94. Question on pro-opiomelanocortin ACTH

95. what Is true about placebo- Naloxone can block placebo effect. Placebo - naloxone .Makes sense. Placebo analgesia works via opioids receptor. So Naloxone should decrease this

96. functional somatic type question – a. nociceptor b. meta c. interoception.

97. illusion top down

98. perception and sensory signaling

99. door picture – shape. my argument was they did not colored the door ! better not color it then That door with opened, and close.....shape, size, location, depth. 3 diagrams of a door progressively shutting was shown, and what form of constancy is shown was the question- a. Size b. Shape c. Colour d. Location e. Depth

Door When it is closed, we see it as rectangular, but when it is open, we see only its edge and it appears as a line. But we never perceive the door as changing shape as it swings — perceptual mechanisms take care of the problem for us by allowing us to see a constant shape.

Interoception means sensing internal signals from your body, like when you are hungry, when your heart is beating fast, or when you need the toilet.

Shape constancy means that we perceive objects as staying the same shape, even when they're rotated or viewed from different angles. Colour constancy means that we perceive objects as staying the same colour, even when the lighting changes.

100. EMI : they found Healthcare professionals complied with handwashing more than 50#% when someone is watching. Hawthorne effect. B. Police officer sees well dressed man doesn't think he is criminal halo effect

101. A All are good prognostic factors for cognitive remediation therapy in schizophrenia, except? Cognitive remediation therapy in schizophrenia not affected in

A. Younger age, b. Low Premorbid Low IQ , c. low school education, d.. Psychotic symptoms less severe e. low global functioning
OR

100b. Not a predictor of low response in cognitive remediation therapy for schizophrenia- (framing of Q was confusing) a.low premorbid IQ b. low school education c. younger age d. low global functioning d. symptoms severity

102. EMI questions: NMDA which is for g protein?- NMDA, MDMA, seperation anxiety- 10months , selective mutism.

EMI There was questions on family dynamics and family trying to reach equilibrium (like family trying to navigate situations when one family is going through some problems) there was, triangulation , detouring, circular causality, homeostasis, triangulation, enmeshment a. making one family member a scapegoat :- Detouring b. if something happens to few family members it effects others – Enmeshment c. one more family Q cant remember

Circular causality:- A concept that people's interaction are influenced by others which inturn are affected by others. A sequence of causes that leads back to the original cause and either alters

Triangulation : Is when a toxic person brings a third party in order to maintain control

Detouring:- Parents form alliance against the child so the parents are closer and the child is distant from them because they focus the negativity on the child

Enmeshment: Two people feel each others emotion. Teenage daughter gets anxious and depressed, her mum gets anxious and depressed

103. Q: migrant family successfully adapt to new culture at work while maintaining traditional norms at home , Options : a. Accommodation; b. Assimilation ; c. Acculturation; d. Enculturation; e. Melting pot

Enculturation is learning the appropriate behavior of one's own culture
Acculturation is learning the appropriate behavior of one's host culture while maintaining one's own culture.

Cultural accommodation happens when a person accommodates to a new culture but stays true to their home culture, give impression that they fit in.

acculturation are almost same

A Canadian psychologist, John Berry, identified four paths to acculturation:

Assimilation: This refers to the partial adaptation of a new culture (seen in migrants or refugees) without retaining or giving up all of one's culture of origin completely. Historically, assimilation was seen as individuals giving up their home culture and embracing the dominant culture. From Seminars in Gen Adult Psych 1e. Pg. 783. Integration: Maintaining one's home culture but also embracing the dominant culture. Separation: Maintaining one's home culture and being isolated from the dominant culture. Marginalization: Giving up the home culture and failing to relate properly to the dominant culture.

104. Q: What is the term used to describe when one forgets specific painful events:- a. retrospective falsification; b.Anterograde amnesia; c.retrograde amnesia; d. Catathymic amnesia

Catathymic amnesia is the inability to recall specific painful memories, and is believed to occur due to the defense mechanism of repression

105. Stereotypic non goal directed movements seen in Parkinson's disease. A. Perseveration b.Punding— for the non goal directed seen in Parkinson.

Punding has been described as "a stereotypical motor behavior denoted by intense fascination with repetitive, excessive and non-goal oriented handling, and examining of objects."1 It is generally precipitated by drug use and alleviated by a reduction or change in medications. Punding is a compulsive need to carry out a repetitive motor behavior such as sorting materials that you are no longer using, counting small objects, lining them up and then counting them again, or taking everything out of a drawer, examining it, and then doing it all over again. There is no apparent or gainful purpose associated with punding activities.

The development of such impulsive and compulsive behaviors is believed to be complications of dopamine replacement therapy. Like so many Parkinson's traits, punding will not be experienced by everyone, but it can be disruptive and thus worth knowing about.

106. Question of self-serving bias: attributes success to internal attributes

107. Question about Bereavement in children:More in boys than girls

108. Circadian rhythm - Bright light therapy-suprachiasmatic nucleus

109. Bright/Blue light therapy

Question **20**

Not answered

Marked out of 1.00

Flag question

When employing optogenetics to study neuronal functions, which of the following light will activate the pigment rhodopsin?

Select one:

- Blue light
- Red light
- Green light
- White light
- Yellow light

Channelrhodopsin-2 (ChR2) is a cation channel that opens in response to blue light. Halorhodopsin, from the halobacterium Natronomonas pharaonis, is a chloride channel that actively moves chloride ions into the cell from extracellular zone when yellow light is presented. Archaerhodopsin, derived from Halorubrum sodomense, is a proton pump that actively removes hydrogen ions from the cell when a yellow light is presented. There are two major transgenes used for optogenetic inhibition. Both halorhodopsin and archaerhodopsin serve to hyperpolarize neurons, while channelrhodopsin promotes depolarization.

The correct answer is: Blue light

Bright light therapy involves exposure to artificial light sources that mimic natural sunlight.

71. Bright light therapy is used in – **circadian rhythm**, diurnal rhythm,

Bright Light Therapy

Bright light therapy is used to manage circadian rhythm disorders such as delayed sleep phase syndrome (DSPS), a condition which shifts the normal sleeping pattern outside what is considered the social norm. People who have DSPS generally don't fall asleep until several hours after midnight and have trouble waking up in the morning, making adhering to a normal work or school schedule difficult. However, if allowed to follow their own schedules, their quality of sleep is otherwise normal.

circadian rhythm controlled by **5HT7 and Suprachiasmatic Nucleus**

110. Delirium-fluctuating consciousness. Kindly refer to mock 7.

	Delirium	Alzheimer's Disease (AD)	Lewy Body Dementia (LBD)	Depression
Onset	Sudden	Insidious	Insidious	Recent
Alertness	Fluctuating	Normal except in late phases	Fluctuating	Preserved
Duration	Hours to weeks	Months to years	Months to years	Variable
Orientation	Disorientation with onset	Disorientation occurs late in course	Fluctuating	Intact
Hallucinations	From onset	May occur late in course	From onset; well- formed visual hallucinations	Can occur in depression with psychotic features
Cognitive functioning	Fluctuating with awareness	Progressive deterioration	Progressive deterioration	Initially intact; may deteriorate
Mood	Fluctuating	Labile	Labile	Usually sad
Sundowning	Present	Present	Present	Absent; mood improves as day progresses
Course	Usually reversible with treatment	Irreversible with progressive deterioration	Irreversible with progressive deterioration	Reversible
Adapted from Tampi et al 2019				

SPMM COURSE

Ref: Tampi RR et al. Psychotic disorders in late life: a narrative review. Therapeutic advances in psychopharmacology. 2019 Oct;9:1-13.

The correct answer is: Lewy body dementia

111. The duloxetine question did not have snri as an option, what it had was selective noradrenaline not snri .Which was very tricky😅.

Task that require concentration need low arousal n task that need stamina need high arousal. Generally optimal arousal needed for high performance

Duloxetine a. half life 24 hours b.Sari c.Ssri d. used in Peripheral neuropathy

Half life of Duloxetine is 12hours

112. Robust evidence of increased Interleukin in Generalized anxiety disorder :Ans. IL-6

113. Flumazenil acts to manage toxicity caused by which of the following –a. alcohol, b. benzodiazepine and zopiclone (Does not act on barbiturates) Non- benzodiazepine agonist like zopiclone is also blocked by flumazenil.

114. Camberwell scale? The Camberwell Family Interview (CFI; Vaughn and Leff 1976) is a semi-structured interview that is considered the gold standard measure of expressed emotion (EE) in families of individuals with mental health problem

115. long-term stress vs performance? Yerkes dodson A. high demanding task performance decrease with high arousal, b. Highly emotionally regulated kids-less reward sensitization. C. Performance and arousal have inverse relationship d. Constant relationship e

An inverted U-shaped curve relates to the level of arousal with the performance of an act. Optimum arousal which is moderate is required for the best performance. A simple task is performed best when arousal levels are relatively high and complex tasks are best performed when arousal levels are lower.

116. Genotype causing Phenotypic variance in population. :- Heritability

The same gene mutation (i.e. genotype) exhibit the same degree of observable characteristics (i.e. phenotype). This phenomenon is described using two terms - penetrance and expressivity. Penetrance describes the proportion of a population of individuals who carry a disease-causing allele who express the related disease phenotype. Expressivity describes the extent to which a genotype shows its phenotypic expression in an individual. These phenomena are thought to be due to a number of factors, including:

- Modifier genes
- Environmental factors
- Allelic variation

heritability what it means. The main point to note is that heritability tells us nothing about individuals. Heritability is the proportion of phenotypic (phenotype is the set of observable characteristics or traits of an organism) variance attributable to genetic variance. This means that if a condition has a heritability of 0.80 then 80% of the variance of that condition seen in a population is attributable to genetic variation. The remaining 20% is attributable to environmental factors.

117. Goffman's work about institutions – a. mortification of the self, b. double bind, c. voluntary adoption of sick role

Moral career of a mentally ill patient:betrayal funnel(freinds and family report you), role stripping(institutionalization),mortification(civil death,self image lost)and privilege system(reduction in institution's control over patient)

118. EMI – options- Metamorphosia, dysmegalopsia, macropsia, micropsia, palindopsia, visual agnosia, hemisomatoagnosia, simultognosia, prosopagnosia

a. seeing things/people upside down- Metamorphopsia.

Visual distortion where objects, (including people) appear distorted or inverted.

b. Half body cant be seen - hemisomatoagnosia

c. not being able to see landmarks- visual agnosia. (refers to the inability to recognize landmarks or objects.)

d. Inability to recognize faces. - Prosopagnosia

Hemisomatognosia is a neurological condition characterized by a lack of awareness or recognition of one side of the body. It typically occurs as a result of damage to the brain, particularly the parietal lobe, which is involved in processing sensory information and spatial awareness.

119. There's also a question of Preoptic and thermoregulation- answer is Hypothalamus

Anterior preoptic hypothalamus is hypothermic center, Posterior hypothalamus is hyperthermic center.

120. Question to choose which test is for visuospatial -a. clock drawing, b. Tower of london,

The test examines a range of different cognitive functions simultaneously, including:-

- Visuospatial ability
- Motor function
- Attention
- Comprehension

Weak social bonds not only makes communities less safe, but also more open to crime.development of criminal subcultures and groups as a means to deal with harsh living conditions and fulfill unmet needs

121. Question about Hayling scale and what it assesses.: b. executive function, b. language, ...

The Hayling and Brixton tests are designed to assess executive function. The Hayling Sentence Completion Test consists of two sets of 15 sentences each having the last word missing. In the first section the examiner reads each sentence aloud and the participant has to simply complete the sentences, yielding a simple measure of response initiation speed.

122. Question about increase chances of criminality. a. decreased parental care b. Marked Social disorganization, c. being put in social care

123. Question about mechanism of action of ketamine;- NMDA antagonist Noncompetitive

124. Which of the following is a endocannabinoid a. Anandamide b. arachidonic acid c. enkephalin

Example of exogenous endocannabinoid is THC (active substance in cannabis)

Endorphins- works at Mu receptors ; Enkephalins- Works at delta receptors ; Dynorphins- Works at

Kappa receptors

125. Maslow original need- Esteem

126. Which of the following is responsible for Mother attachment epigenetics? A. Oxytocin, b. prolactin

127. Limbic encephalitis question ..focal limb and face seizures..

128. Alcohol metabolism rate limiting step. a. Alcohol dehydrogenase b. aldehyde dehydrogenase

129. Apophany Q

130. Lyonization occurs in which of the following conditions.- a. Klinefelter

131. Sudden realization- first rank symptom

Sudden delusional idea primary delusion

132. Muscarinic receptor action are found in a. peripheral parasympathetic b. Synapse c. parasympathetic post synaptic d. parasympathetic ganglia e. Neuromuscular junction

133. EMI drug that act on serotonin?vesicle?

134. Side effect is stress incontinence

Antagonism at muscarinic receptors

There are several antipsychotics that have been associated with urinary incontinence

Chlorpromazine, Thioridazine, Fluphenazine, Haloperidol and Pimozide

135. Question on Old age pharmacokinetics? Absorption less than 30year old

136. 72year old male, Dystonia movement. a. CASPR b. LG1

137. Constant stress something...a. Homelessness b. Job loss

138. Locus coeruleus – Norepinephrine

139. which of the following is a first rank symptom of schizophrenia-

- man sees clock strike 12 and believes he is Messiah
- voices commanding him
- believing his neighbors are poisoning him

140. which of the following is a daily hassle? options-

- A child getting kidnapped
- Serious illness
- Loosing car keys

141. Dopamine precusor- L tyrosine

142. Q on Flynn effect- Intelligence increases with successive generations

143. At what stage in Conrad's model does the individual start to find new meaning for psychological processes: a. Trema b. Apophany c. anastrophe d. residuum

144. Innate tendency to attach to particular individual- Monotropy

145. 2 cases are compared- a. LOD b. Linkage c . Association d. Recombination

It's association if the statement says cases and controls are compared

146. 112. MAOi interaction with a. nasal decongestant b. antihistamines

147. AUC denotes- drug Exposure over time, bioavailability.

148. Protective factor in childhood- sensitive mother

149. Bereavement in children, true is . a. young children more affected than older b. boys

more affected than girls, c. reduced cortisol response in later life d. school going children revert to

150. Pharmacokinetics of old age – a. Above age 80, 15% kidney stops working b. Free drug more in plasma c. Gastric motility increased d. Albumin increases.
151. Parent enters and leaves, the child has no response- anxious avoidant
152. True about building superstition- a. learnt from family b. builds by chance association
153. Most common epigenetic mechanism? a. DNA acetylation b. DNA methylation c. Histone phosphorylation d. Histone methylation e. Glycation
154. some Q on cerebral development
155. Absorption constant is independent of a. elimination constant b. form of drug c. composition of drug d. extravascular route of administration e. food taken with drug

Presence of psychotic symptoms during manic episode is a poor prognosis factor for bipolar disorder

The zone of proximal development in Vygotsky's theory is the range of tasks a child can do with guidance but not independently.